

GE Healthcare

Signa EXCITE to HDx 1.5T System Upgrade

Technical Publication



5166005
Rev 10

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

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GE Medical Systems, a General Electric Company, going to market as GE Healthcare
3200 N. Grandview Boulevard
Waukesha, Wisconsin 53188
USA

www.gehealthcare.com



WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・ このサービスマニュアルには英語版しかありません。
- ・ GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・ このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・ この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV DATE PRIMARY REASON FOR CHANGE

1	...	Mar 2, 2006	...	Initial Release.
2	...	Jul 11, 2006	...	Section 2–3: Added CAUTION note to upgrade flowchart that refers to the IIS Key and SaveINFO operation. Revised s/w load instructions and added LCD monitor adjustments (MRIhc15805, MRIhc16251). Added blocks for restoreINFO, ICW, and update of eLicense (MRIhc15873, MRIhc15892, MRIhc16209). Section 6–5–10–1: Updated the type of fastener used to attach the 16 Channel Switch mounting plate for a welded or bolted rear pedestal. Section 7: Added note informing the installer not to connect additional monitors to host computer prior to operating system software load (MRIhc15870). Clarified the installation of jumpers when setting up the DVD module is the SCSI tower.
3	..	Oct 22, 2006	..	Section 6: updated PAC replacement instructions with information on removing the PAC Control Housing (MRIhc17607).
4	..	Feb 22, 2007	..	Section 2: Reversed flowchart procedures <i>Software Option Install w/eLicense</i> and <i>Install and Calibration Wizard</i> . Section 4: Added note for instructions on storing Service Cable Kit (CQA 13094026). Section 6: Removed PAC2B part number. Section 7: Added information on cable disconnection from existing GOC. For installing Run 1323, added notes for routing with Run 1322 and that CAT6 cable must replace existing CAT5 cable. Added note that Humidity iButton can be installed in either direction. Section 8: Updated MGD chassis install information. Revised installation location of ICN brackets (CQA 13078452). Updated board installation instructions for upgrading to 16 channels. Added instructions for storing Service Cable Kit.
5	..	May 24, 2007	..	Section 7: Added iButton installation and Environmental Sensors Block Diagram to Section 7–6, Temperature and Humidity Sensor Kit Installation. Section 8: Updated APS Board part numbers from 2294300–14 to 2294300–16 (ECO 2044030). Adding existing cables connected to Term Server (CQA13112624).
6	..	Aug 10, 2007	..	Section 6: Updated the Hybrid Splitter upgrade, Section 6–4–7, by adding the Netcom Hybrid Splitter which has different Isolation Filter mounting brackets than the existing Teledyne Hybrid Splitter.
7	..	Jun 17, 2008	..	Section 2: Added True Mobile flowchart (MRIhc37270). Section 4: Updated cable map to show OW component and cable upgrades. Updated cable maps to show new Temperature and humidity Sensor installation and cables. Corrected the J3 and J4 connection designations at the hybrid splitter (PQR 13172263). Section 6: Added instructions for installing gradient lead board (PQR13162916) and rear pedestal stabilization. Section 7: Revised entire section for replacing the PC computer in the GOC, and other components, if required. Also revised temperature/humidity sensor installation to reflect the new design.
8	..	Sep 18, 2008	..	Section 1: Added HDxt (Release 15.x) information to Introduction. Section 8: Revised section to incorporate the new SUN ICN modules for upgrades.
9	..	Jun 17, 2009	..	Section 4: Added Subsection 4–4 which provides information about discarding Runs 1189, 1190 and 1082 for Mobile sites. Added note on cable maps and in Subsection 4–3 that indicates a new Run 1329 is provided for Mobile sites (PQR 13255521). Section 6: Removed instructions for installation of rating plates (PQR 13246367). Added instructions on applying laser labels on WideOpen and Horizon style enclosures (PQR 13246299).
10	..	Jul 27, 2010	..	Section 2: Updated eLicense model type from Signa HDx to MR per Direction 2397845). Section 4: Updated cable maps for removal of OW components and installation of Z400 GOC (ECO 2096124). Section 7: Updated instructions for removal of obsolete components and GOC. Added instructions for installing Z400 GOC (ECO 2096124). Section 8: Removed second set of ICN brackets in cabinet because new Sun 4170 ICN replaces two ICN's installed in previous upgrades (ECO 2101939).

LIST OF EFFECTIVE PAGES

PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV
Title Page	10	TAB 2		TAB 4		TAB 7		TAB 9	
Damage in Trans.	–	UPGRADE		SYSTEM UPGRADE		OPERATOR		SRF OR SRFD2	
Direction 2128126	1*	INSTALLATION		CABLES		WORKSPACE		CABINET	
A	10	PROCESS		2164904–4	–	2164904–7	–	2164904–9	–
i	1	2164904–2	–	4–1 to 4–2	9	7–1 to 7–26	10	9–1 to 9–2	1
		2–1 to 2–5	7	4–3 to 4–18	10				
TAB 1		2–6	10	4–19 to 4–20	1	TAB 8		TAB 10	
GETTING STARTED		2–7	9			SYSTEM CABINET		ACGD/HFD CABINET	
2164904–1	–	2–8 to 2–10	7	TAB 5		2164904–8	–	2164904–10	–
1–1 to 1–6	8	TAB 3		PENETRATION PANEL		8–1 to 8–7	8	10–1 to 10–9	1
		POWER OFF		2164904–5	–	8–8 to 8–10	10		
		2164904–3	–	5–1 to 5–2	1	8–11 to 8–13	8	TAB 11	
		3–1	1	TAB 6				COMPLETION	
				MAGNET ROOM				2164904–11	–
				2164904–6	–			11–1 to 11–2	1
				6–1 to 6–64	9				

* This revision number/letter corresponds to the indicated document's revision control system.

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
SECTION 1 – GETTING STARTED	1-1
SECTION 2 – UPGRADE INSTALLATION PROCESS	2-1
SECTION 3 – POWER OFF	3-1
SECTION 4 – SYSTEM UPGRADE CABLES	4-1
SECTION 5 – PENETRATION PANEL	5-1
SECTION 6 – MAGNET ROOM	6-1
SECTION 7 – OPERATOR WORKSPACE	7-1
SECTION 8 – SYSTEM CABINET	8-1
SECTION 9 – SRF OR SRFD2 CABINET	9-1
SECTION 10 – ACGD/HFD CABINET	10-1
SECTION 11 – COMPLETION	11-1

SECTION 1 – GETTING STARTED

1-1 UPGRADE INTRODUCTION

This publication documents the mechanical installation information for upgrading a Signa® 1.5T EXCITE (Release 11.x, EXCITE II) to Signa HDx 1.5T (Release 14.x) or HDxt (Release 15.x). Refer to illustration 1-1 for pictorial overview of upgraded system.



FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

1-2 SITE READY CHECK

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion.

Tools and test equipment to install and calibrate the Signa HDx1.5T system are listed in the *Signa HDx 1.5T Upgrade Pre-Installation* manual, Section 10, TOOLS AND TEST EQUIPMENT.

Service installation tools used in this manual that may not be included in above references lists include:

- 46-301450G1 Fiber Optic Cable Connector Repair kit

1-3 PRE-DELIVERY FDO CHECKS FOR SIGNA EXCITE

Before the shipment is made, Be sure that FDO review has been completed by consultation with the Signa 1.5T Upgrade Specialist. Also verify the Surface coil compatibility issues have also been resolved.

1-4 UPGRADE SCHEDULE CONSIDERATIONS

FMI's, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man-hour and system down-time total. The IIS key will need to be re-generated. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1-5 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The "Shipping Document" lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short. Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with "Technical Publication" boxes for a list of delivered documents.

1-6 DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1-800-548-3366 and use option 8.
- Contact your local service coordinator for more information on this process.

1-7 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1-7-1 MATERIAL POLICY OVERVIEW

1-7-1-1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly either to GoldSeal or Renewable Resources. Do not remove or swap any parts from the material to be returned.

1-7-1-2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1-7-2 MATERIAL TO BE RETURNED

Return all parts, cables, modules, and equipment deinstalled for the HDx Release 14.x upgrade to GoldSeal which includes:

- GOC and Operator Workspace components
- Driver Module
- SRI 2
- 8 Channel Switch
- LPCA
- Front and Rear Split Bridge
- All replaced Circuit Boards
- RF Body Coil (TRM only)

1-7-3 RETURN KIT

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal. The following material may have been provided to facilitate return of material.

- Shipping labels
- Waybill
- Address labels
- Reuse appropriate containers that new hardware and modules were shipped to site for return of above material.
- Packing material

1-7-4 DEINSTALLATION PROCEDURES

Refer to instructions in this manual for deinstallation procedures of site.

1-7-5 REGION SPECIFIC RETURN INFORMATION**1-7-5-1 Sites in Americas**

A GoldSeal return pre-printed BAX waybill and mailing label(s) may be included with the upgrade kit. The upgrade returns kit (p/n 2352804) consists of:

- 2358604 – Waybill
- 2357240 – GoldSeal Return Labels
- 2357241– Upgrade Return Envelope (waybill and return labels)

Gold Seal Address and contact for coordination of transportation arrangements and packaging:

GE Medical Systems – GoldSeal
3114 North Grandview Blvd
Waukesha, WI 53188
Phone: 1-877-251-4468 or 262-513-4147

1-7-5-1 Sites in Americas (Continued)

For non-GoldSeal Returns, utilize GEMS Renewable Resources. For coordination of transportation arrangements and packaging:

Phone: (414)–747–6997 or Dial Comm 8*579–6997

Fax: (414)–747–6855

Field ASPEN: 1–800–525–1516, Box #26041

1-7-5-2 Sites in Asia

At this time, system components are not returned for systems in Asia, but whole systems must be returned from complete system (forklift) upgrades.

Return procedure for GEMS–Asia countries:

Please contact your Regional Service Headquarters to get instructions.

1-7-5-3 Sites in Europe

The GoldSeal Returns Coordinator contacts each site individually and reviews the material to be returned and the logistics for the return. A list of material and a printable shipping label is sent to the site's primary FE, service providers, and forwarders.

GoldSeal Europe return information:

GEMS–E – GOLD SEAL OPERATION & RECYCLING CENTER

GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL

Quai 45 à 49 – 91090 LISSES – FRANC

Tel: (+33) 1–60–91–57–35 or (+33) 1 64 97 55 41

Fax: (+33) 1–60–86–12–31

For non-Gold Seal returns in Europe:

European Renewable Resources Center

GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL2

Quai 45 à 49 – 91090 LISSES – FRANCE

TEL = (+33)–1–64–97–55–41 FAX = (+33)–1–60–86–12–31

1-7-6 PRODUCT LOCATOR INFORMATION

Process product locator information for all serialized components removed from the system according to policy. Refer to Section 1–8.

1-8 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at:
<http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following:
[Product Locator Support Central Page](http://supportcentral.ge.com/15563) – <http://supportcentral.ge.com/15563>

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-9 UPGRADE SEQUENCE

Follow the system Upgrade Flow chart in SECTION 2 for best installation efficiency.

Note

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

The general sequence of events is:

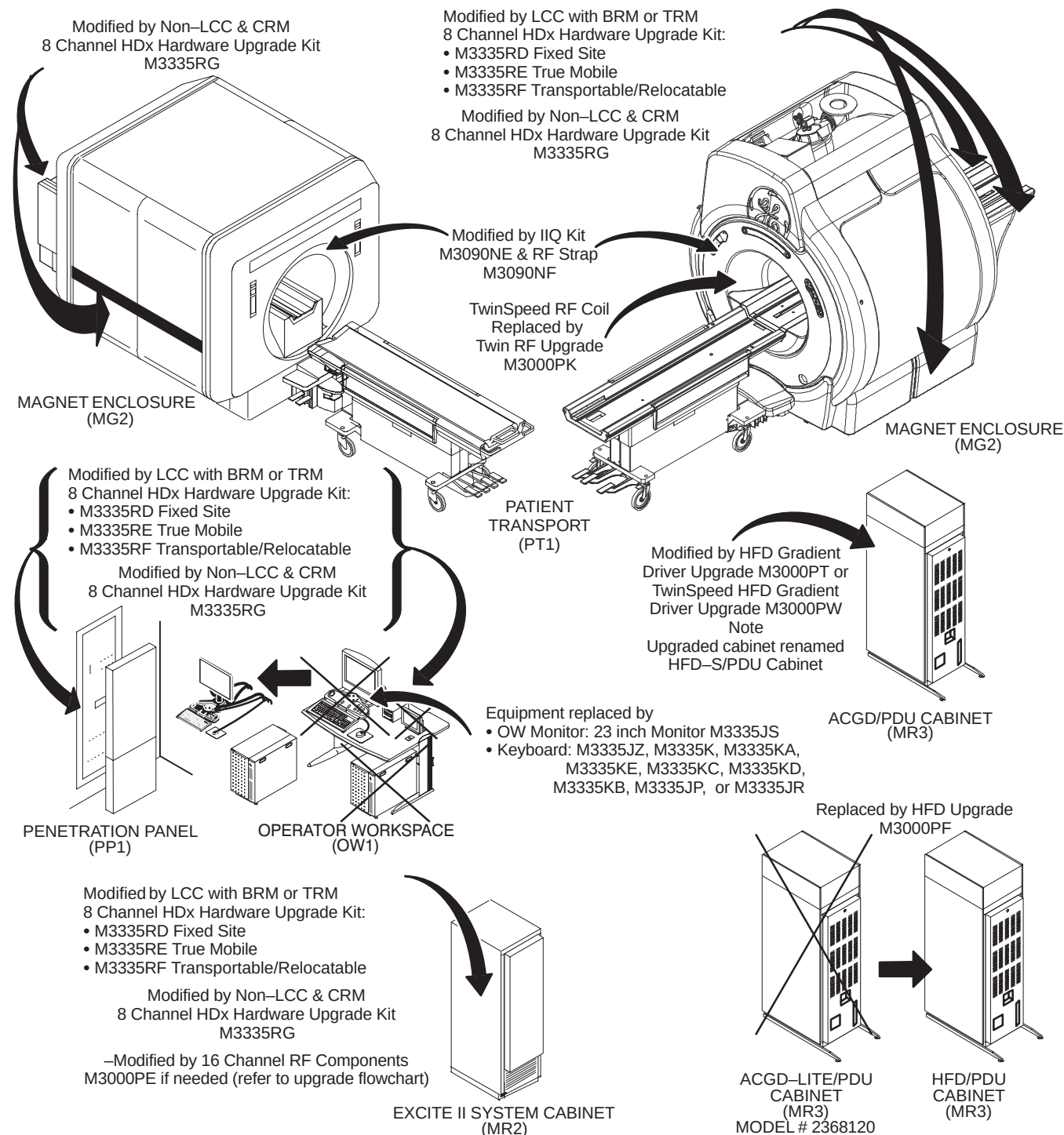
- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Shut site down.
- Remove obsolete equipment and cables.

Aside: Removing and installing cables is more efficient and accurate when the equipment room is un-cluttered. It is strongly recommended that this be done after old equipment is removed and before new equipment is positioned. This house cleaning will make the installation tasks much easier.

- Route new cables
- Install new equipment and modify existing equipment.
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Load software.
- Perform Functional Checks, System calibrations, and System performance checks
- Replace covers, complete final tasks, and return site to customer.

1-10 UPGRADE SYSTEM CONFIGURATION

Illustration 1-1 shows the major equipment replaced and modified by Signa 1.5T EXCITE 11.x to HDx 14.x or HDx 15.x Upgrade.



1.5T Release 11.x to Release 14.x Upgrade Catalogs Without Magnet Upgrade Modified Equipment

ILLUSTRATION 1-1

SECTION 2 – UPGRADE INSTALLATION PROCESS

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
2-1 SITE READINESS	2-2
2-2 UPGRADE PROCEDURE	2-2
2-3 FIXED SITE SYSTEM MECHANICAL UPGRADE FLOWCHART	2-3
2-4 TRUE MOBILE SYSTEM MECHANICAL UPGRADE FLOWCHART	2-7

2-1 SITE READINESS

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to upgrade the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

**WARNING!**

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS UPGRADE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

2-2 UPGRADE PROCEDURE

The following upgrade flow charts contain information for upgrading to Signa HDx 1.5T, Release 14.x. Select the appropriate flowchart for the site being upgraded:

- Fixed Site, Section 2-3
- True Mobile, Section 2-4

The upgrade flow chart should be followed for an orderly and efficient installation. Note that procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

This upgrade flowchart has been developed assuming that all Signa EXCITE HD equipment has been delivered together. **Make sure that every part required is available before starting.**

2-3 FIXED SITE SYSTEM MECHANICAL UPGRADE FLOWCHART

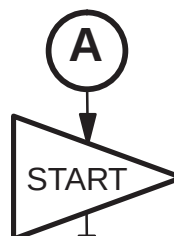
 Procedures to be Completed by Field Engineer

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS

CAUTION

Do not proceed without an IIS Key on the
SaveINFO media. Installation calibration
wizard will not permit the exit from
Installation mode in 14.x without IIS Key.



Signa EXCITE II to EXCITE HDx
1.5T UPGRADE CHECKLIST

VERIFY THAT THE FOLLOWING HAVE BEEN INSTALLED
(If applicable mobile van)
IIQ KIT (Radial & Bridge Support and RF Ground Strap)
BRM ISOLATION KIT
REAR PEDESTAL STABILIZATION

☐ RE-GENERATE IIS KEY.
This must be done before upgrade starts to
make sure that sytem meets specifications.

☐ PERFORM "saveINFO" PROCEDURE
using DVD if possible.

 Procedures to be Completed by FE and/or Mechanical Installers

Section 1
☐ Section 1-5
PRODUCT DELIVERY INSTRUCTIONS

Section 1
☐ Section 1-6
DAMAGE IN TRANSPORTATION

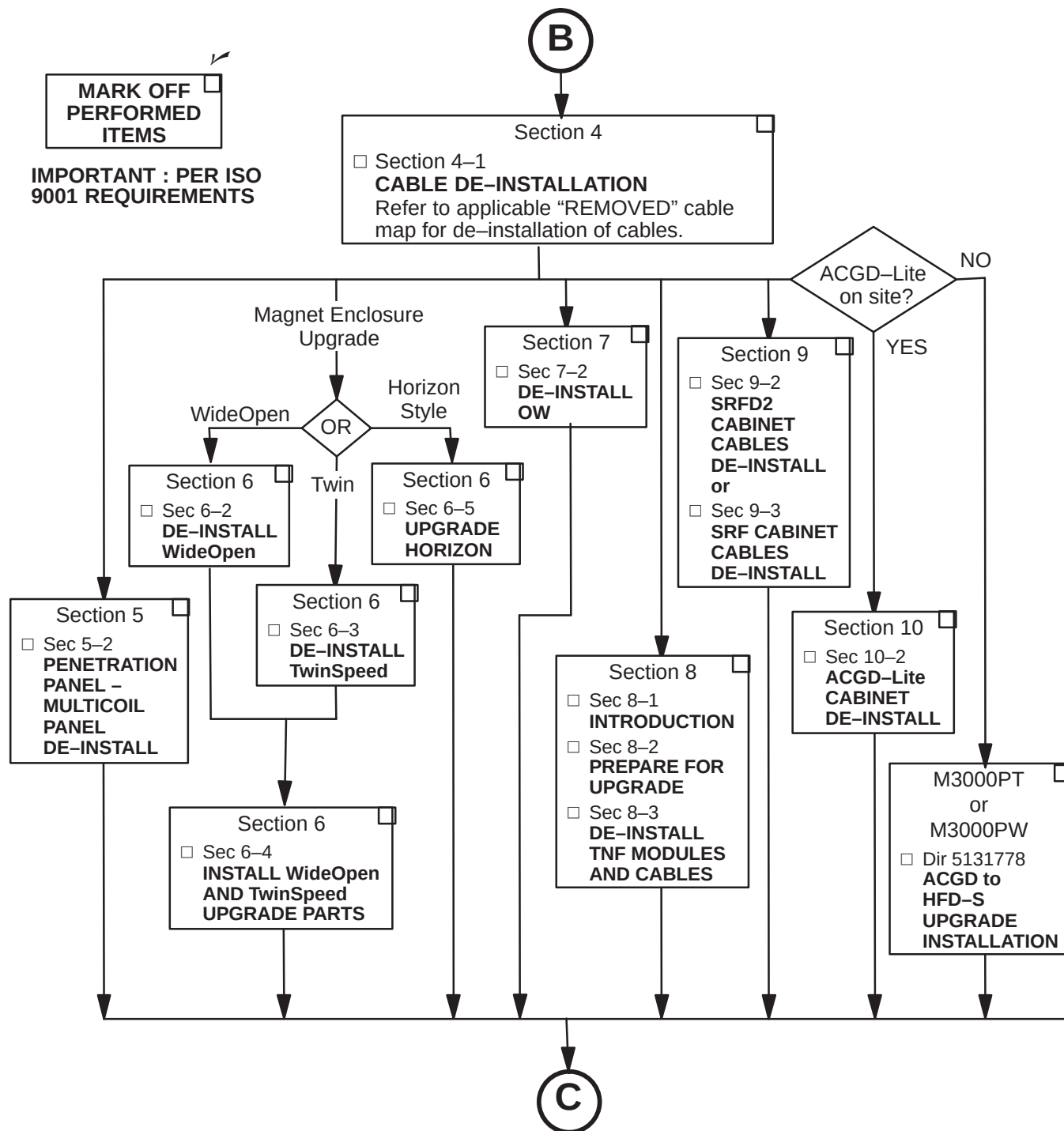
Section 1
☐ Section 1-8
PRODUCT LOCATOR

Section 3
☐ Section 3-1
SYSTEM POWER SHUTDOWN
☐ LOCKOUT & TAGOUT MAIN DISCONNECT

(B)

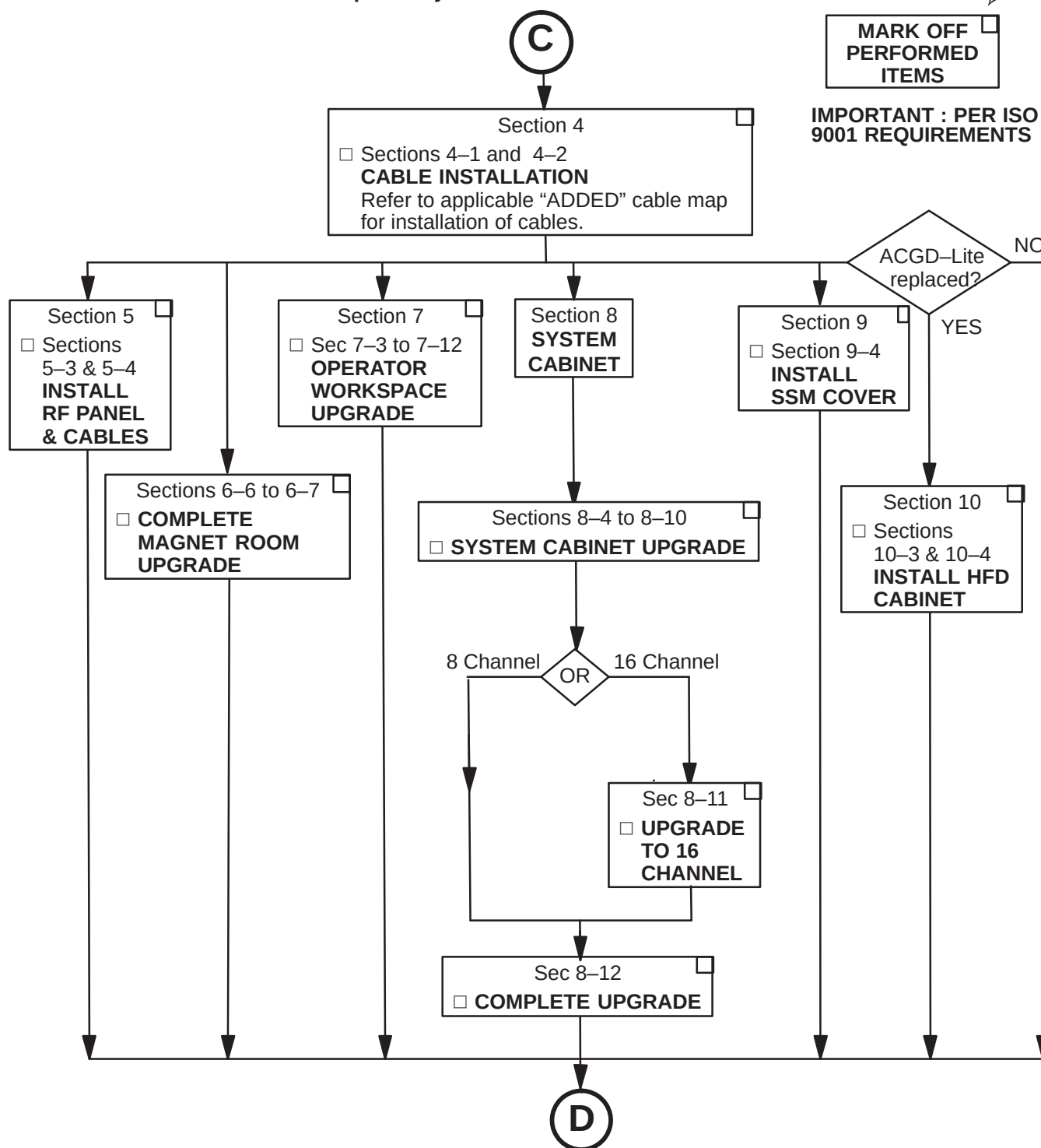
2-3 FIXED SITE SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

De-Installation Procedures to be Completed by FE and/or Mechanical Installers



2-3 FIXED SITE SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Installation Procedures to be Completed by FE and/or Mechanical Installers



2-3 FIXED SITE SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and/or Mechanical Installers

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS

D

Section 11

- ☐ Section 10-1
**MECHANICAL INSTALLATION
COMPLETION CHECKLIST**
- ☐ Section 11-2
RETURNING SHIPPING EQUIPMENT
- ☐ Section 11-3
MATERIAL DISPOSITION GUIDANCE
- ☐ Section 11-4
FINAL COORDINATION

MECHANICAL INSTALLATION
COMPLETE

Procedures to be Completed by FE and Applications

Section 10

- ☐ Section 10-5
POWER ON

MR Service Methods CD-ROM
Direction 5155500

- ☐ Install/Cals>System>Global Operator Console>
LCD Monitor Screen Adjustments

If Host Software is NOT loaded?

If Host Software is loaded?

OR

MR Service Methods CD-ROM
Direction 5155500

- ☐ Software>System Host Software>
Host Operating System Software Load

MR Service Methods CD-ROM
Direction 5155500

- ☐ PERFORM "restoreINFO" PROCEDURE
Software>System Host Software>
MR Apps Software Load
STEPS 16 TO 20

MR Service Methods CD-ROM
Direction 5155500

- ☐ Installation>Options>
Software Option Install with eLicense
When performing this procedure the
following must change:
HOST ID (See eLicense manual)
MODEL TYPE (change to MR)
- ☐ Installation>System>
Install and Calibration Wizard

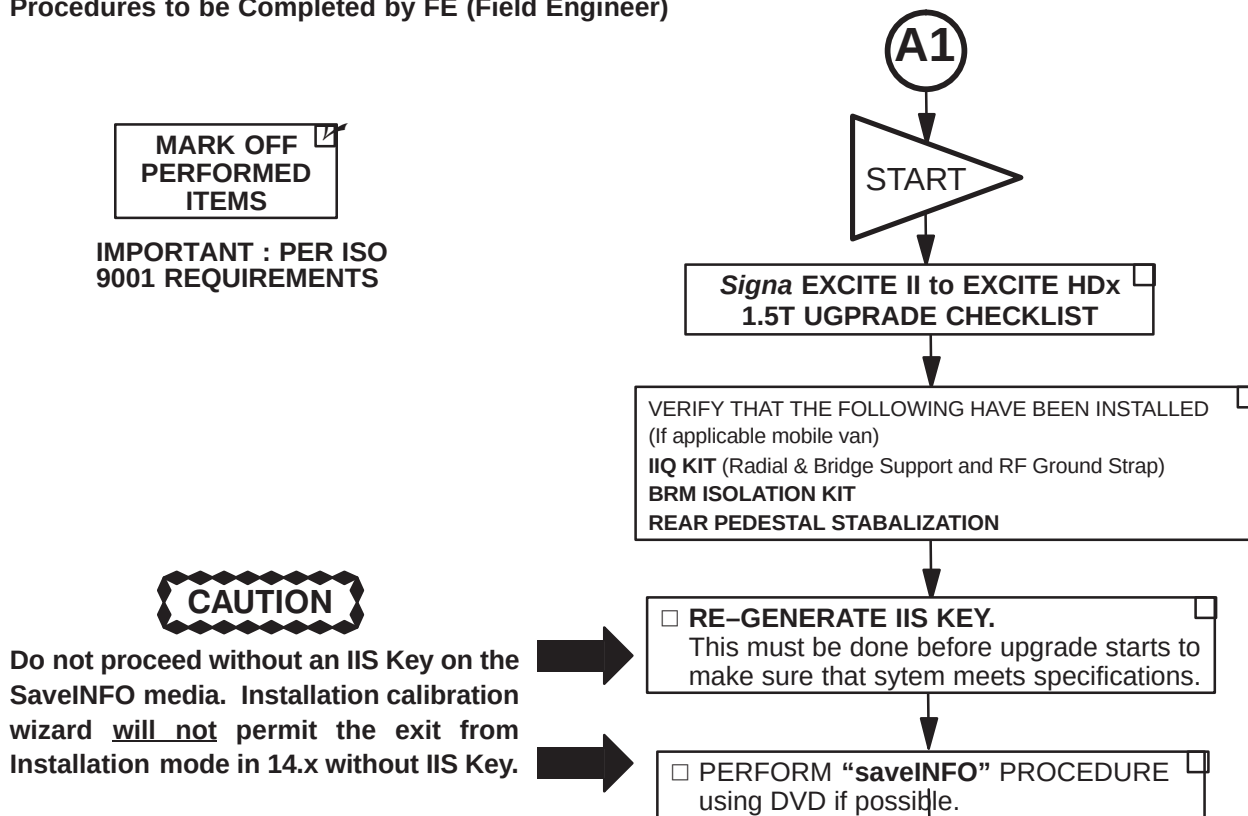
Applications/Training

- ☐ **Turn site over to Applications
who will instruct users.**

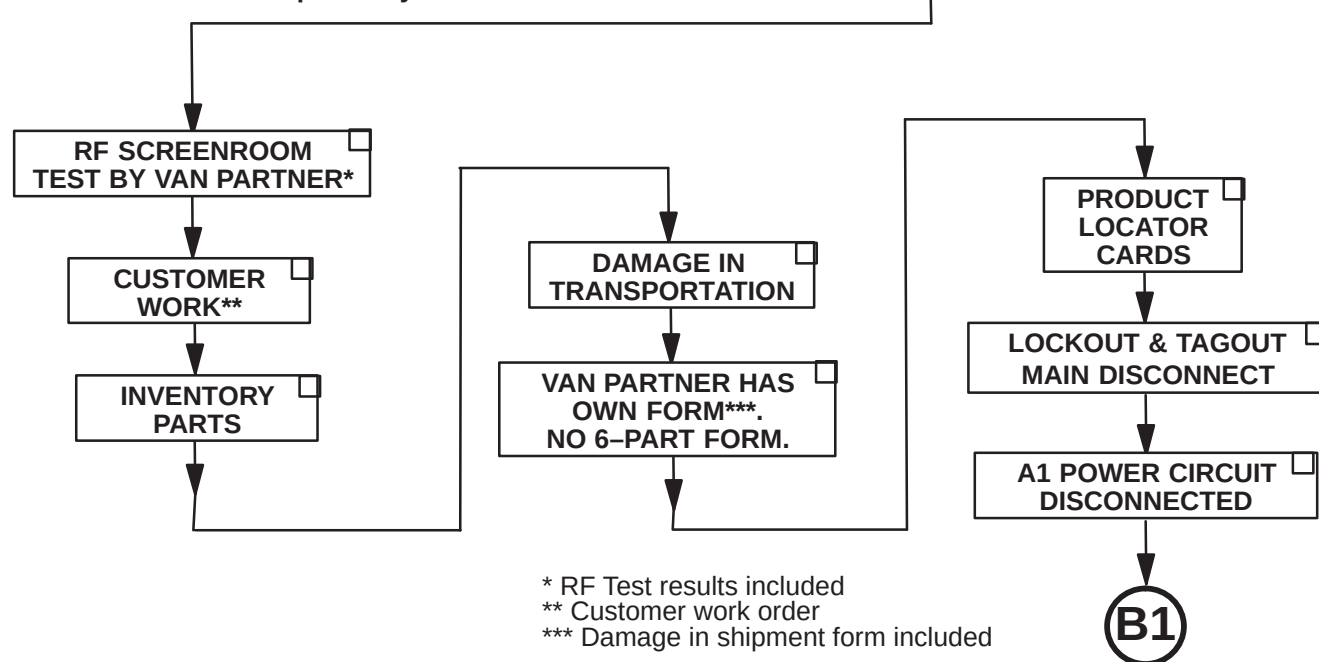
TURN OVER TO CUSTOMER

2-4 TRUE MOBILE SYSTEM MECHANICAL UPGRADE FLOWCHART

Procedures to be Completed by FE (Field Engineer)

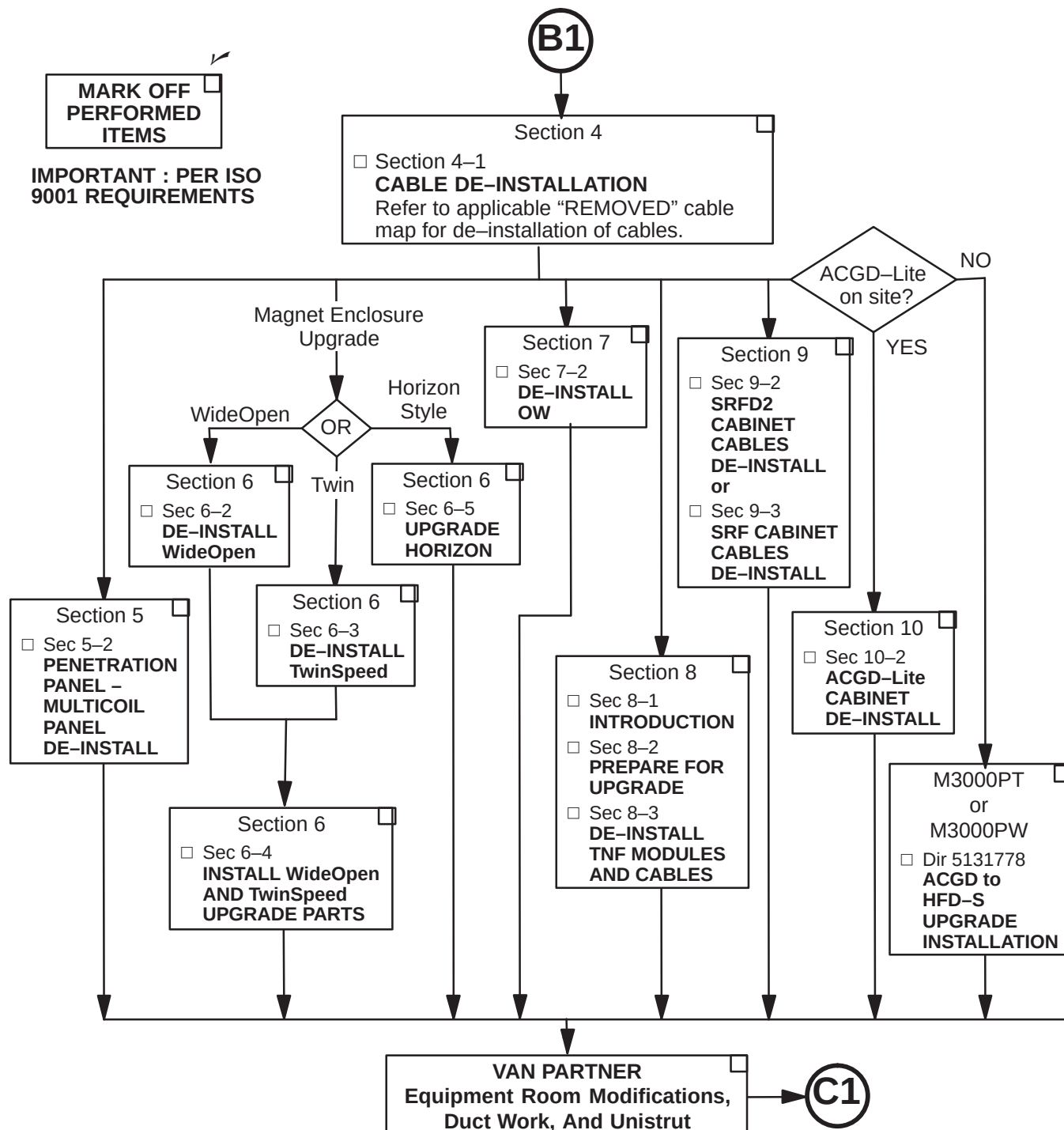


Procedures to be Completed by FE and/or Van Partner



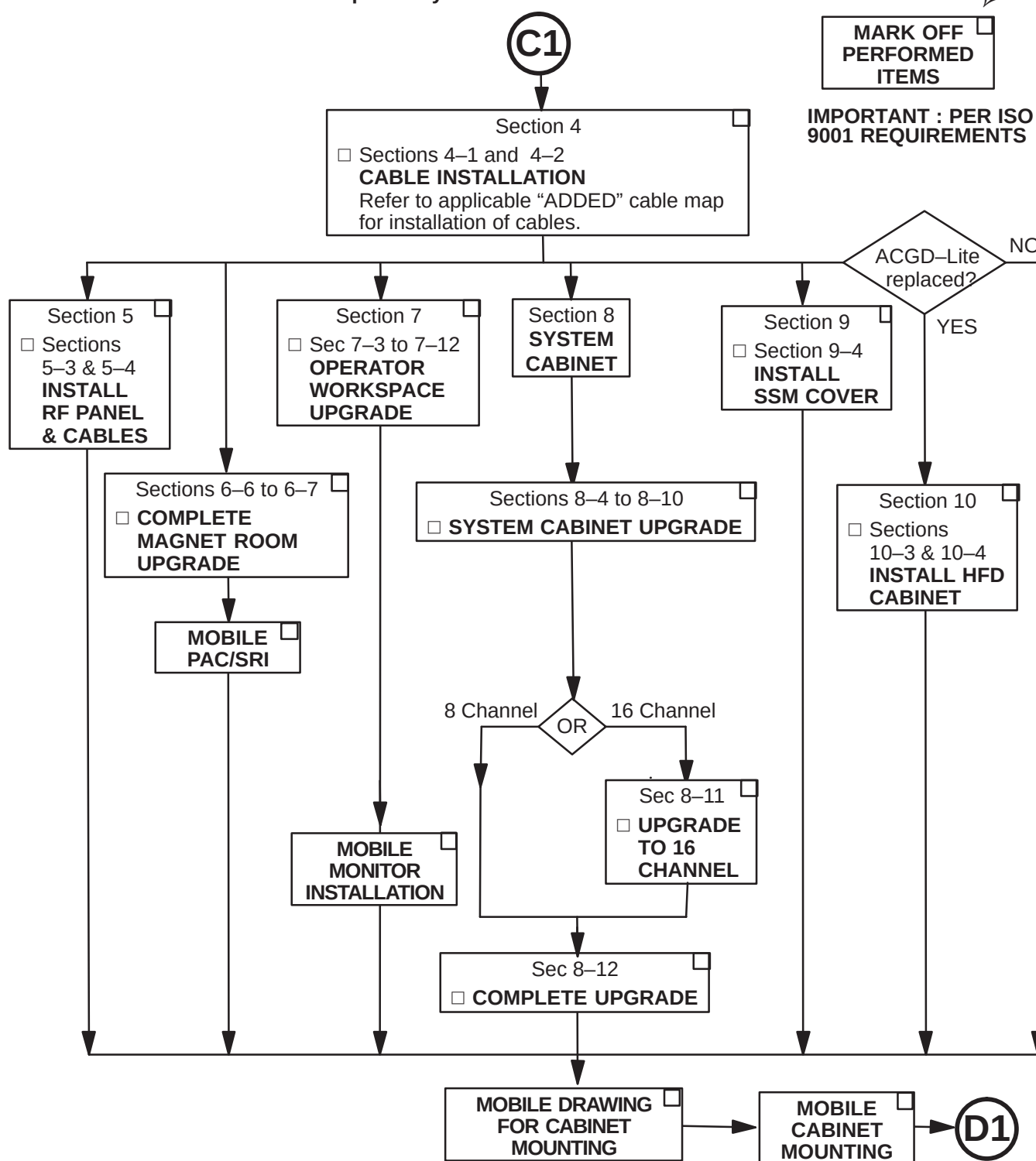
2-4 TRUE MOBILE SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

De-Installation Procedures to be Completed by FE and/or Van Partner



2-4 TRUE MOBILE SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Installation Procedures to be Completed by FE and/or Van Partner

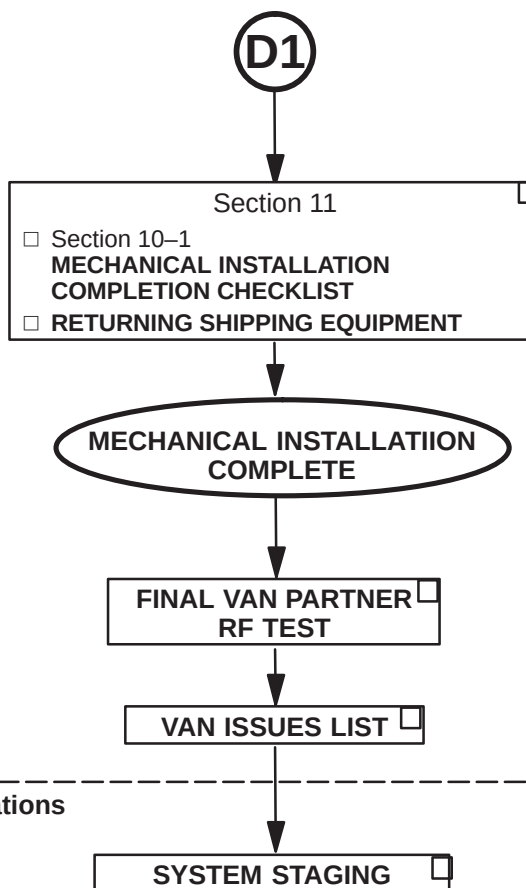


2-4 TRUE MOBILE SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and/or Van Partner

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS



SECTION 3 – POWER OFF

3-1 SYSTEM POWER SHUTDOWN



FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Make sure that site customer has archived all patient information. Execute "Save Info".
2. Verify that System performance baseline measurements have been obtained, especially:
 - SPT Full Test Mode
 - White Pixel Test
 - LV Shim
 - IIS
3. Record center frequency.
4. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
5. Press the FULL OFF button on the PDU front control panel.
6. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
7. Locate and shut off main disconnect supplying power to the PDU.
8. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
9. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring

SECTION 4 – SYSTEM UPGRADE CABLES

4-1 INTRODUCTION

This section provides information on removing, relabeling and installing system cables for upgrading to Signa HDxt 1.5T Release 15.x. Refer to the applicable “Removed” and “Added” cable map for the system that is being upgraded.

The “Affected” (removed) Cable maps are located on the following pages.

- Page 4-3 1.5T EXCITE with ACGD Cabinet
- Page 4-5 1.5T EXCITE SmartSpeed with ACGD-Lite Cabinet
- Page 4-7 1.5T EXCITE TwinSpeed with ACGD Cabinet
- Page 4-9 1.5T EXCITE Non-LCC Magnet with ACGD Cabinet

The “Affected” (added) Cable maps are located on the following pages.

- Page 4-11 1.5T EXCITE with ACGD Cabinet
- Page 4-13 1.5T EXCITE SmartSpeed with ACGD-Lite Cabinet
- Page 4-15 1.5T EXCITE TwinSpeed with ACGD Cabinet
- Page 4-17 1.5T EXCITE Non-LCC Magnet with ACGD Cabinet

Tape map in a convenient location to check off routed cable. Additional information on each cable is located in the Appendix. Upon completion of connecting cables, return fold-out map to the binder for future reference.

4-2 SORTING AND ROUTING CABLES



Ty-wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

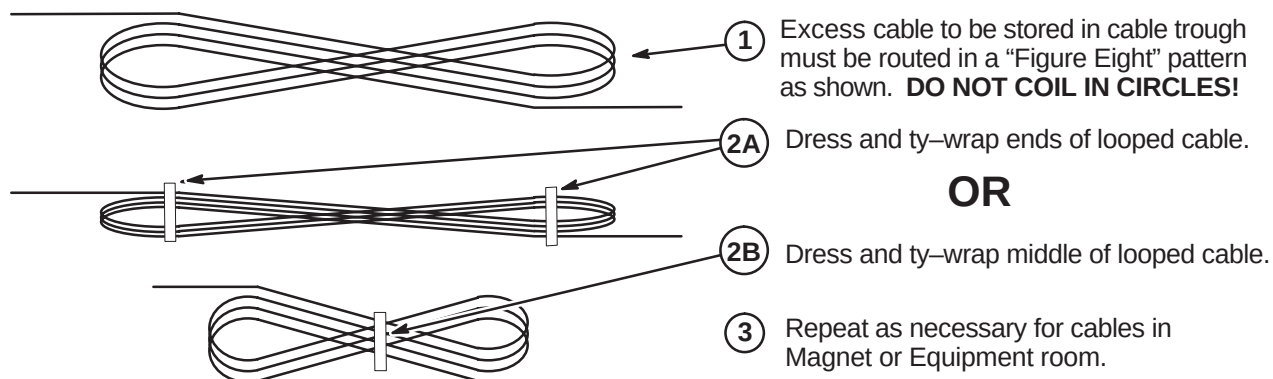
Note: The Service Cable Kit (5160683) is a collection of service cables delivered to site in a plastic bag. Refer to Section 8-12 for instructions on storage of this kit within System cabinet.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

Note: Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.
3. Plan for storage location of coiled cable excess length if cables are to remain at delivered length. See Illustration 4-1. Some cables are usually cut to length such as gradient cables, power cables, Head and Body RF, and Fiber-optic cables.

4-2 SORTING AND ROUTING CABLES (Continued)



PROPER STORAGE OF EXCESS CABLES
ILLUSTRATION 4-1

4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.
5. Rear Pedestal Cables – It is desired to keep the gradient cables separated from the signal and bias cables.

4-3 CABLE RE-LABELING

Relabel existing cables as described below:

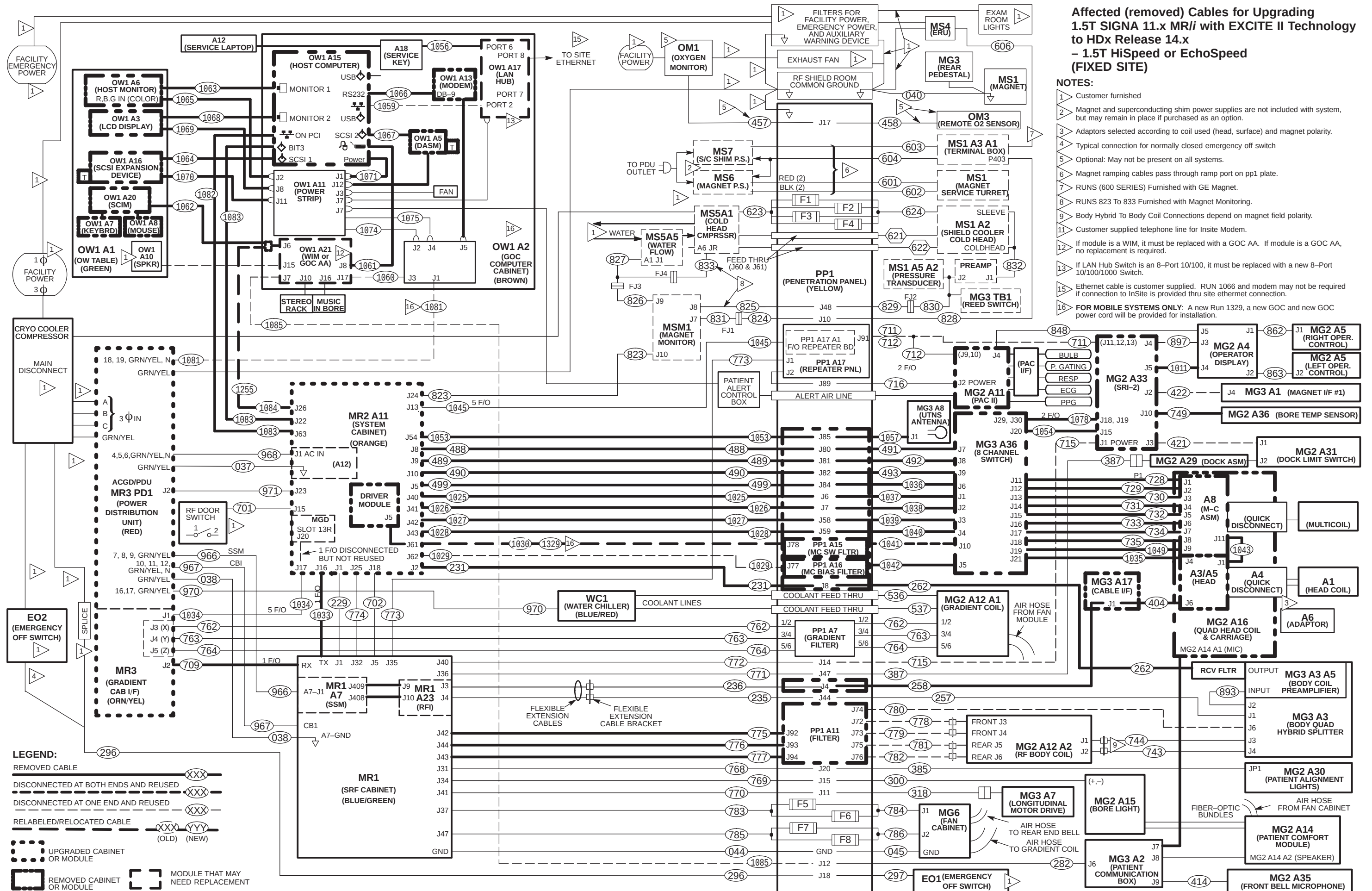
- The new GOC upgrades the WIM to a GOCAA. This requires that Run 1084 be relabeled as Run 1255. Attach blank label to each end of cable and mark with information below.
- Swap and relabel existing equipment room cable Run 1030 as instructed below.

ONE END		OTHER END	
OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
RUN #1084 MR1-A11-J26 2354601	RUN #1255 MR2-A11-J26 5120763-3	RUN #1084 OW1-J21-J6 VIA RUN #1062 2354601	RUN #1255 OW1-A21-J18 5120763-3
RUN #1030 (See Note) PP1-J78 2329624	RUN #1329 (See Note) PP1-J78 5160765	RUN #1030 (See Note) MR2-A11-J61 2329624	RUN #1329 (See Note) MR2-DVR-J10 5160765

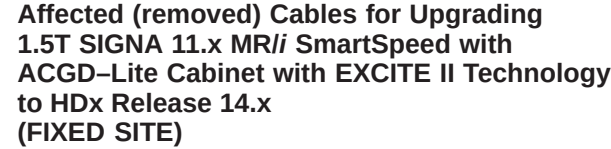
Note: Relabeling of Run 1030 to 1329 is for Fixed sites only. Mobile system upgrades will be provided a new Run 1329.

4-4 EXTRA MOBILE UPGRADE CABLES

For Mobile upgrades, Runs 1189, 1190, and 1082 are shipped to site, but not installed. Discard Runs 1189, 1190, and 1082 for Mobile upgrades.



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- 1 Customer furnished
- 2 Magnet and superconducting shim power supplies are not included with system but may remain in place if purchased as an option.
- 3 Adaptors selected according to coil used (head, surface) and magnet polarity
- 4 Typical connection for normally closed emergency off switch
- 5 Optional: May not be present on all systems.
- 6 Magnet ramping cables pass through ramp port on pp1 plate.
- 7 RUNS (600 SERIES) Furnished with GE Magnet.
- 8 RUNS 823 To 833 Furnished with Magnet Monitoring.
- 9 Body Hybrid To Body Coil Connections depend on magnet field polarity.
- 10 Customer supplied telephone line for Insite Modem.
- 11 If module is a WIM, it must be replaced with a GOC AA. If module is a GOC AA no replacement is required.
- 12 If LAN Hub Switch is an 8-Port 10/100, it must be replaced with a new 8-Port 10/100/1000 Switch.
- 13 Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.
- 14 **FOR MOBILE SYSTEMS ONLY:** A new RUN 1329, a new GOC and new GOC power cord will be provided for installation.

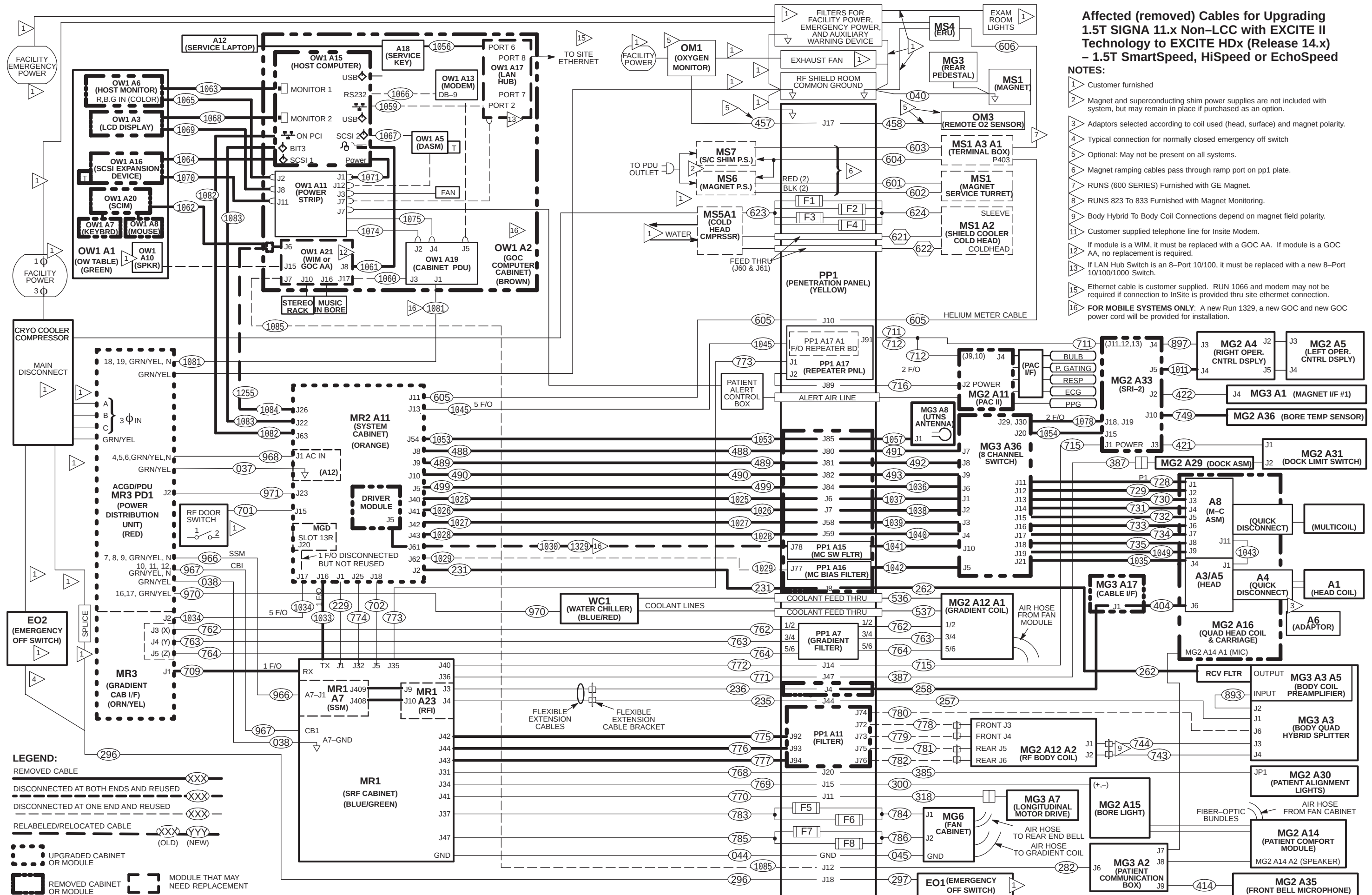
REV 10

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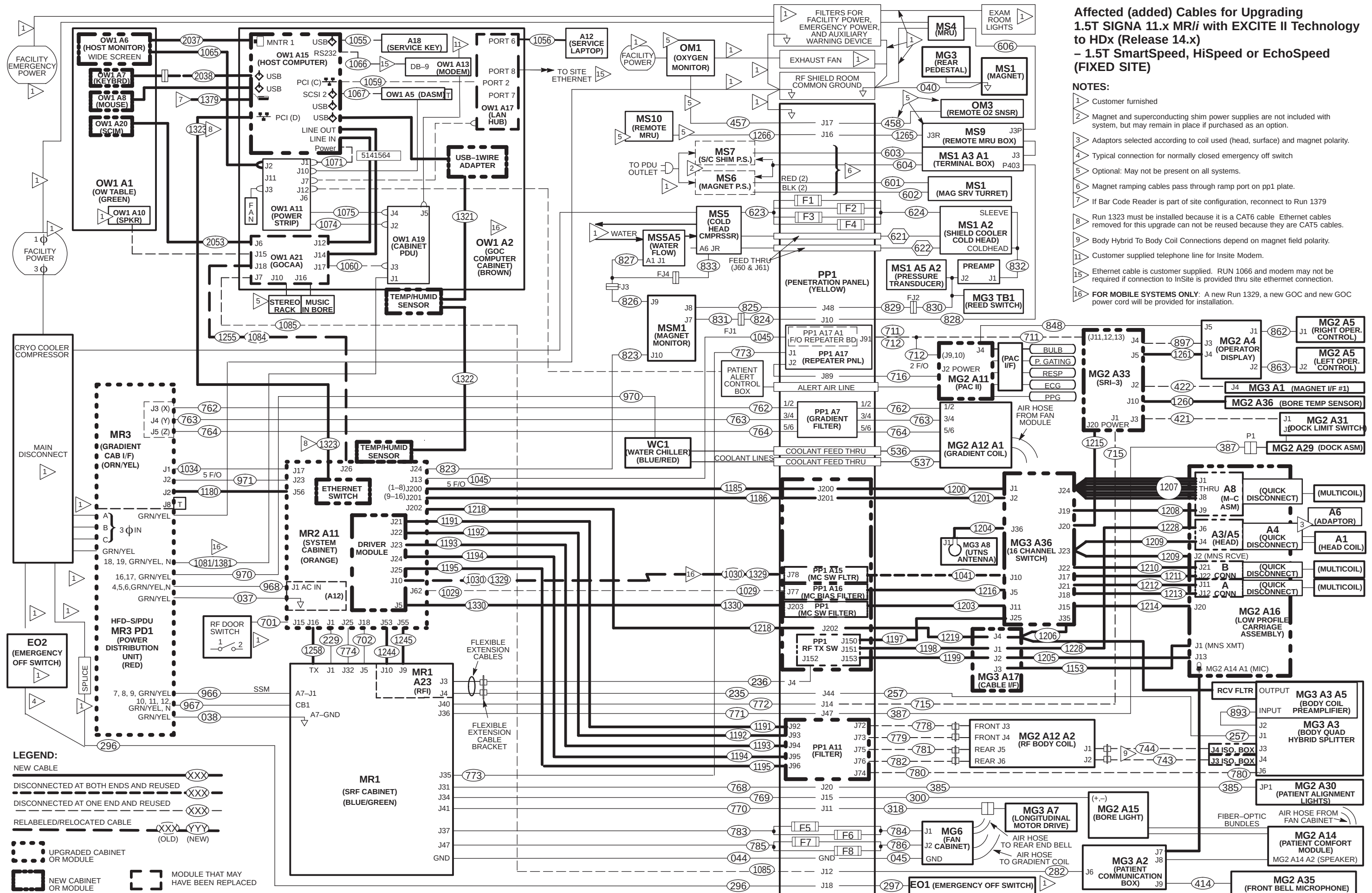


-

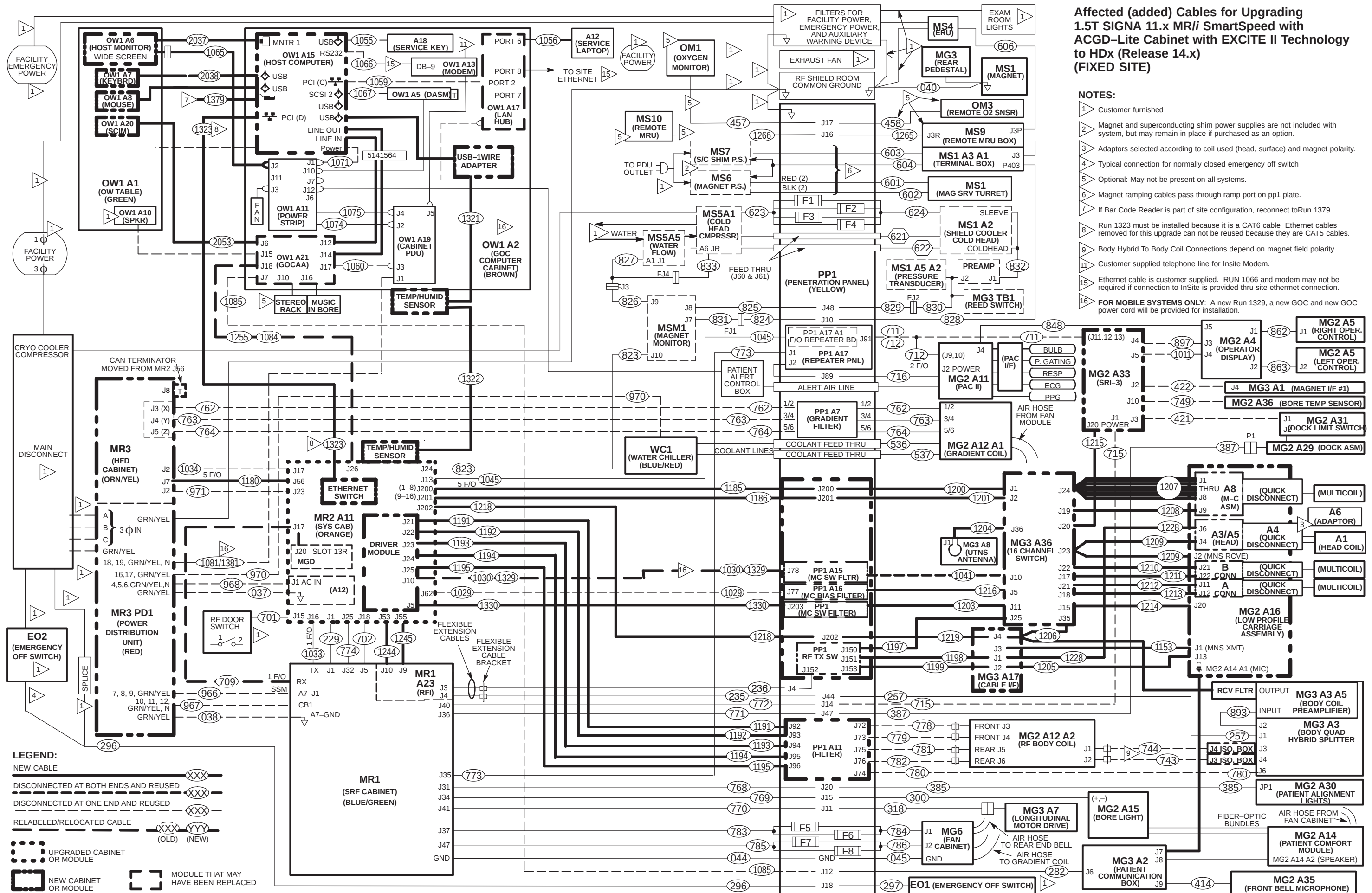
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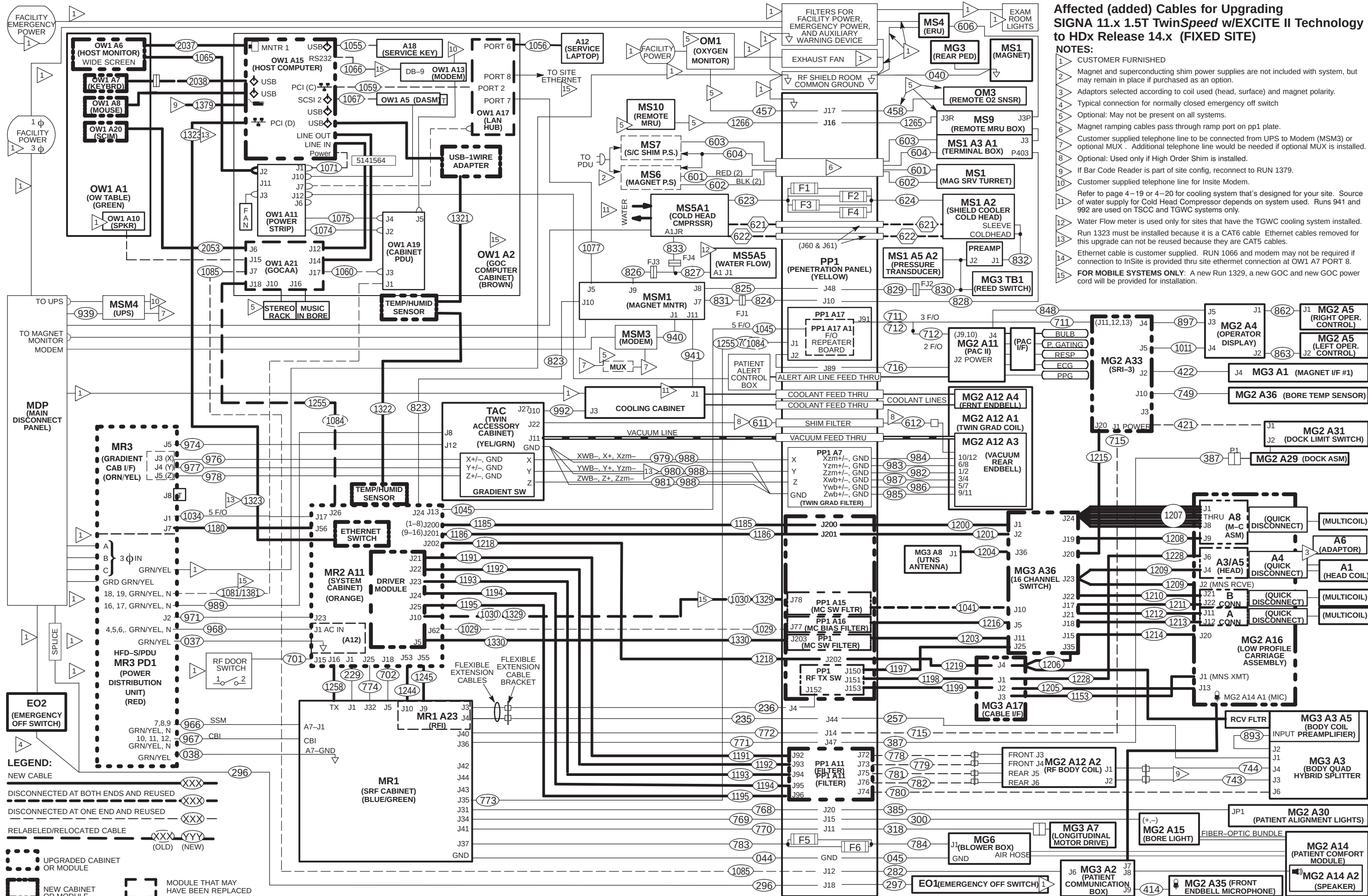
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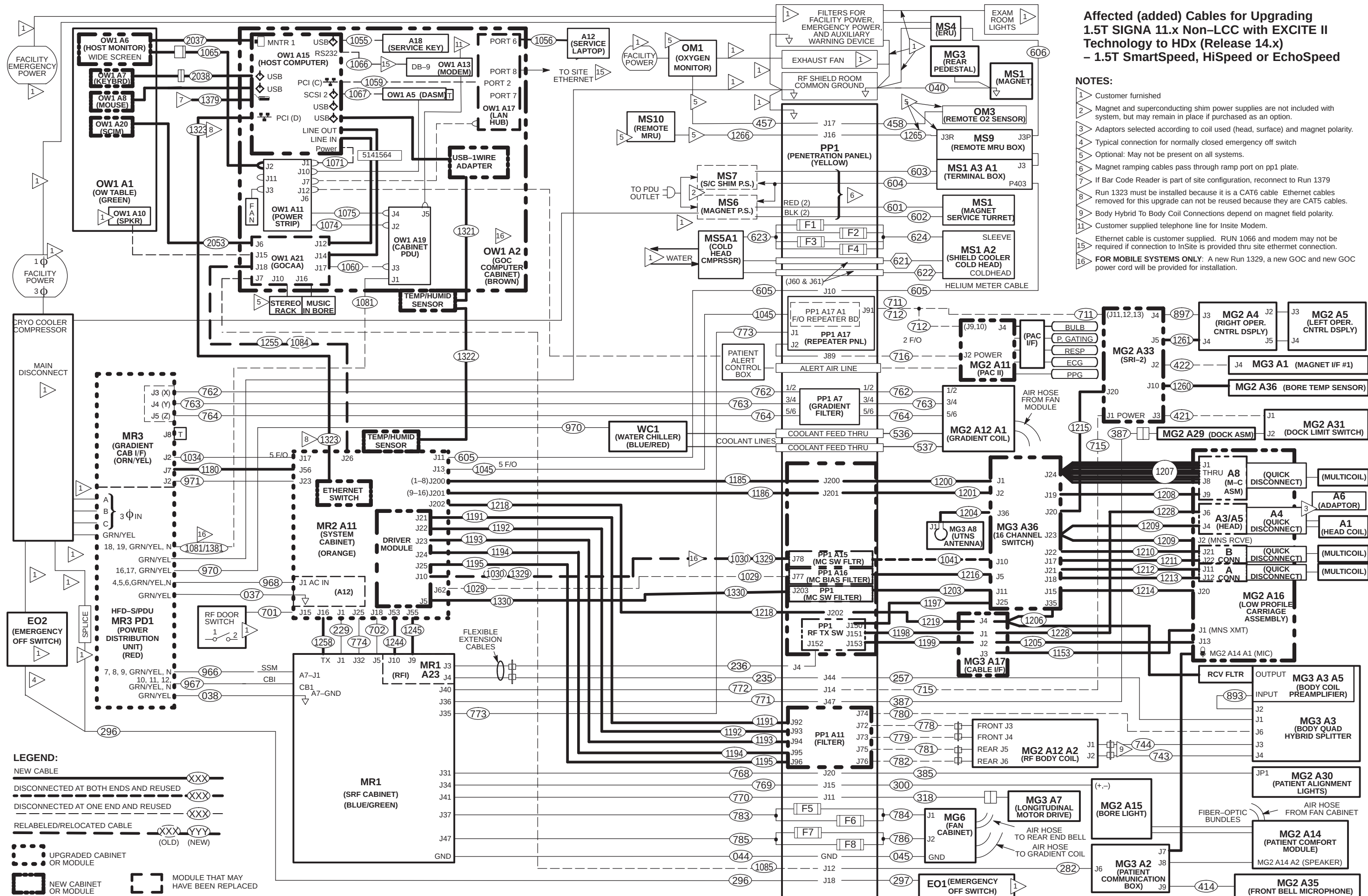
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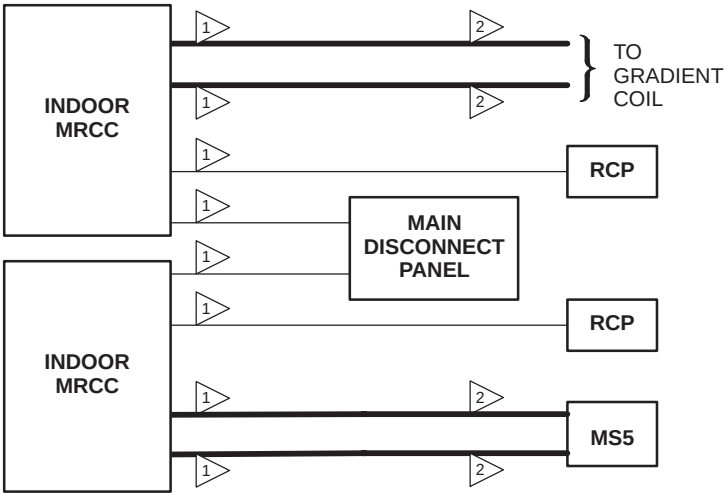
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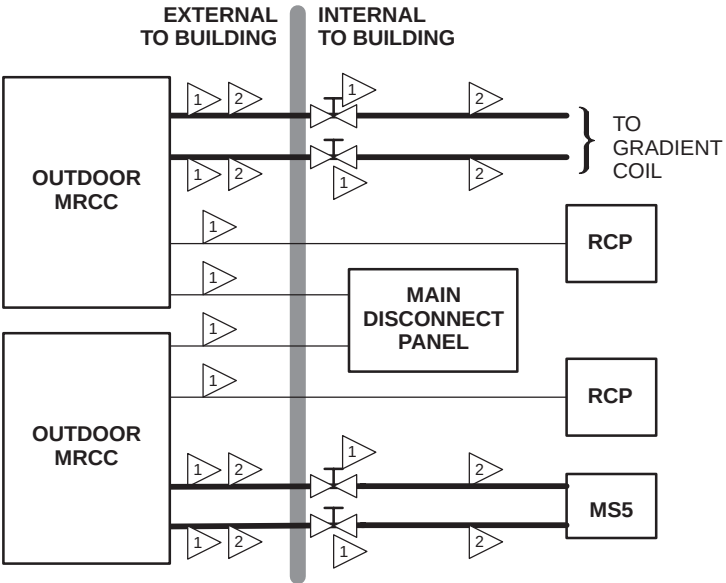


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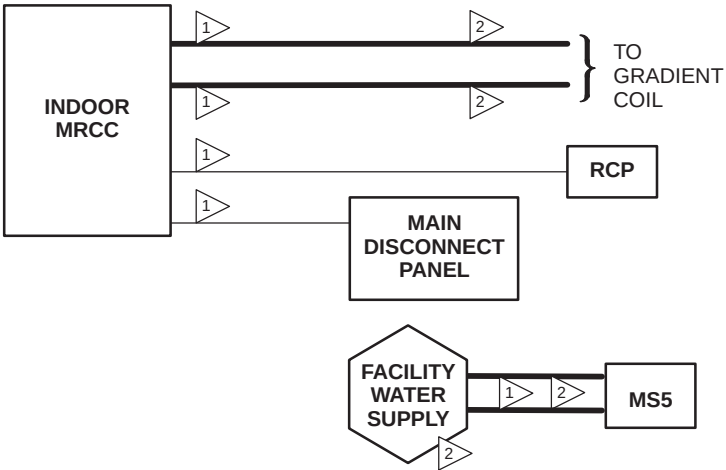
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

DUAL INDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



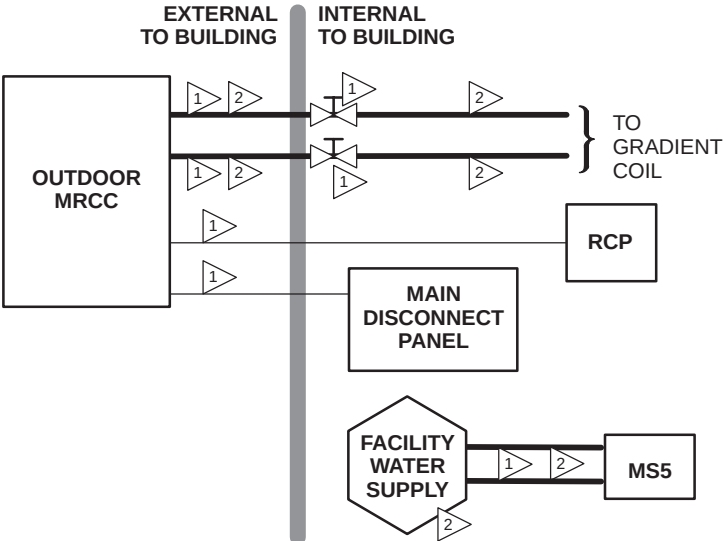
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

DUAL OUTDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



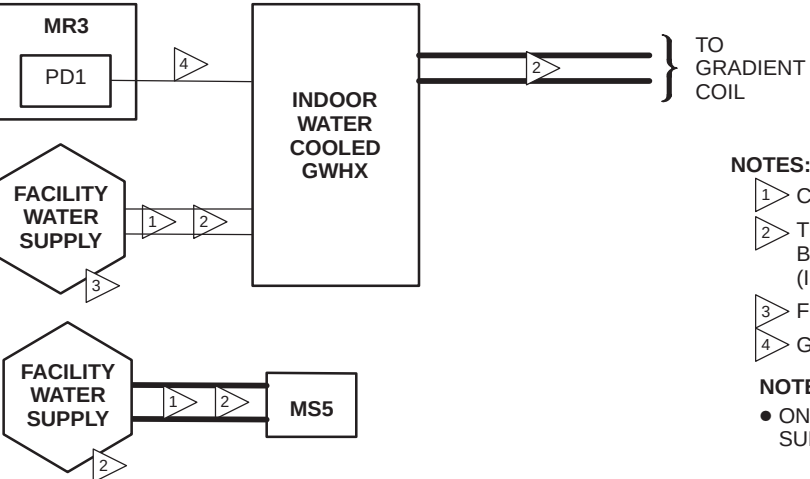
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

INDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



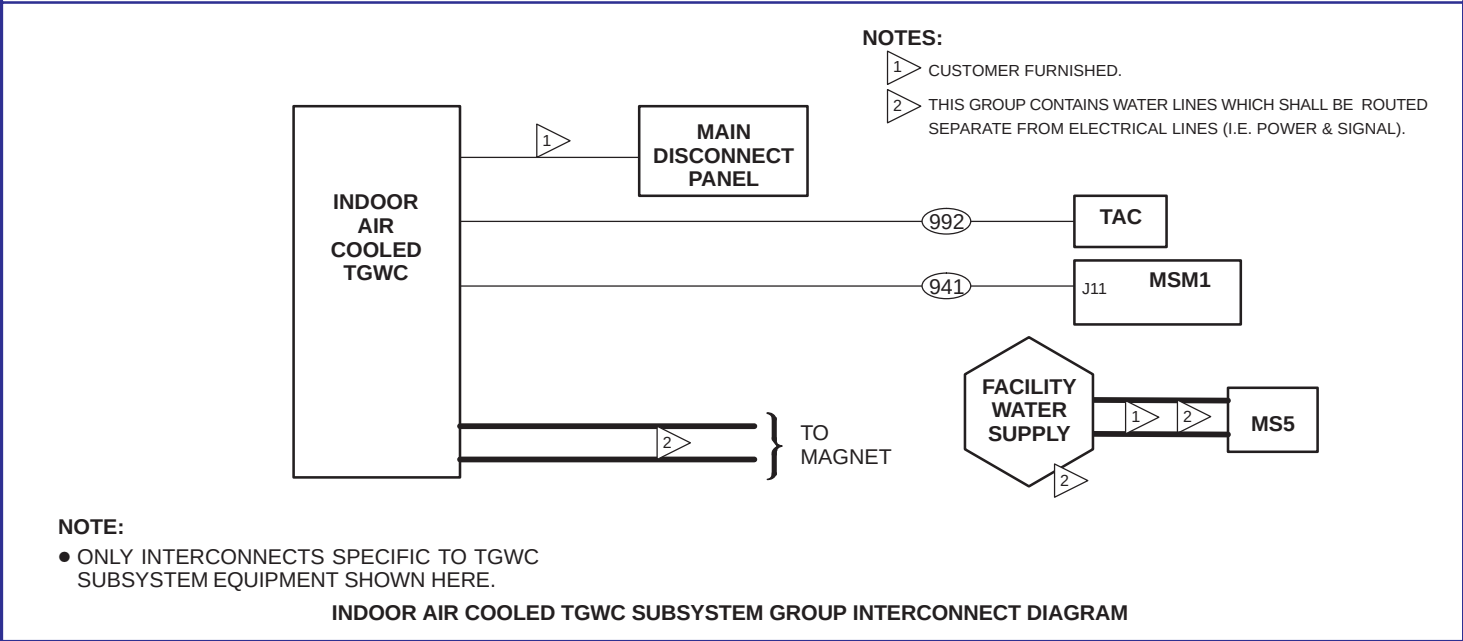
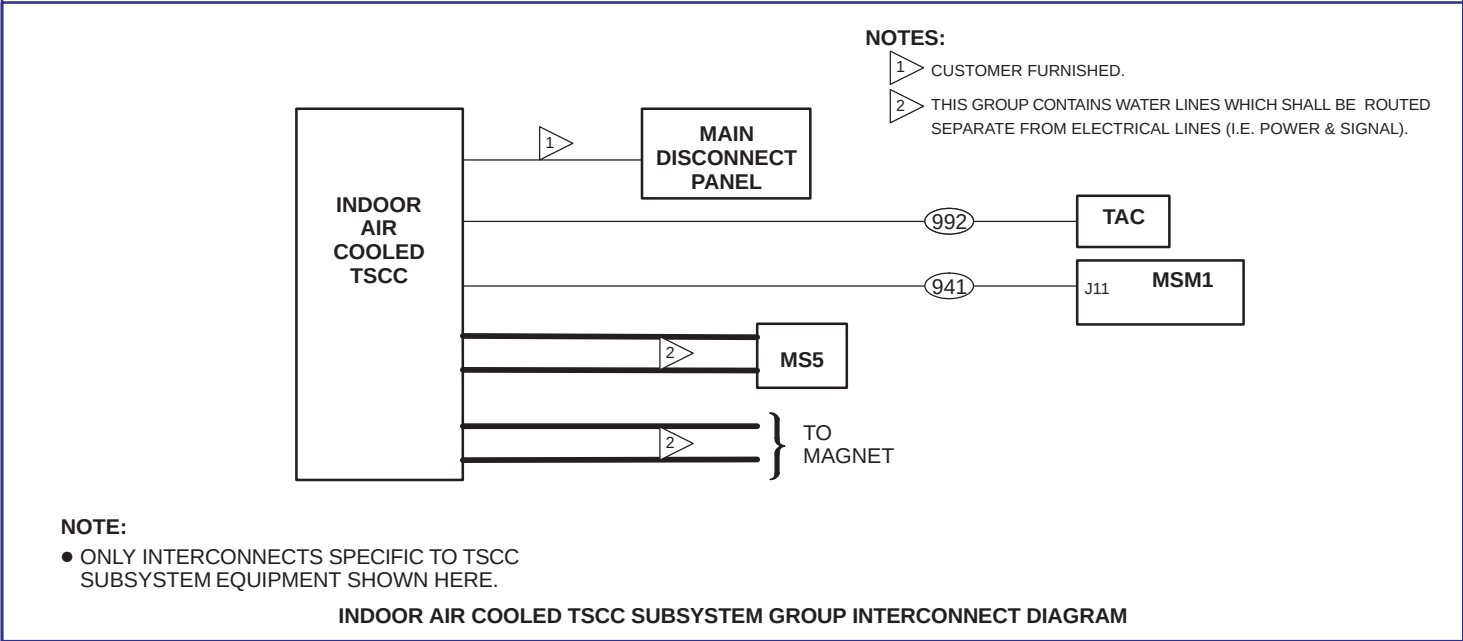
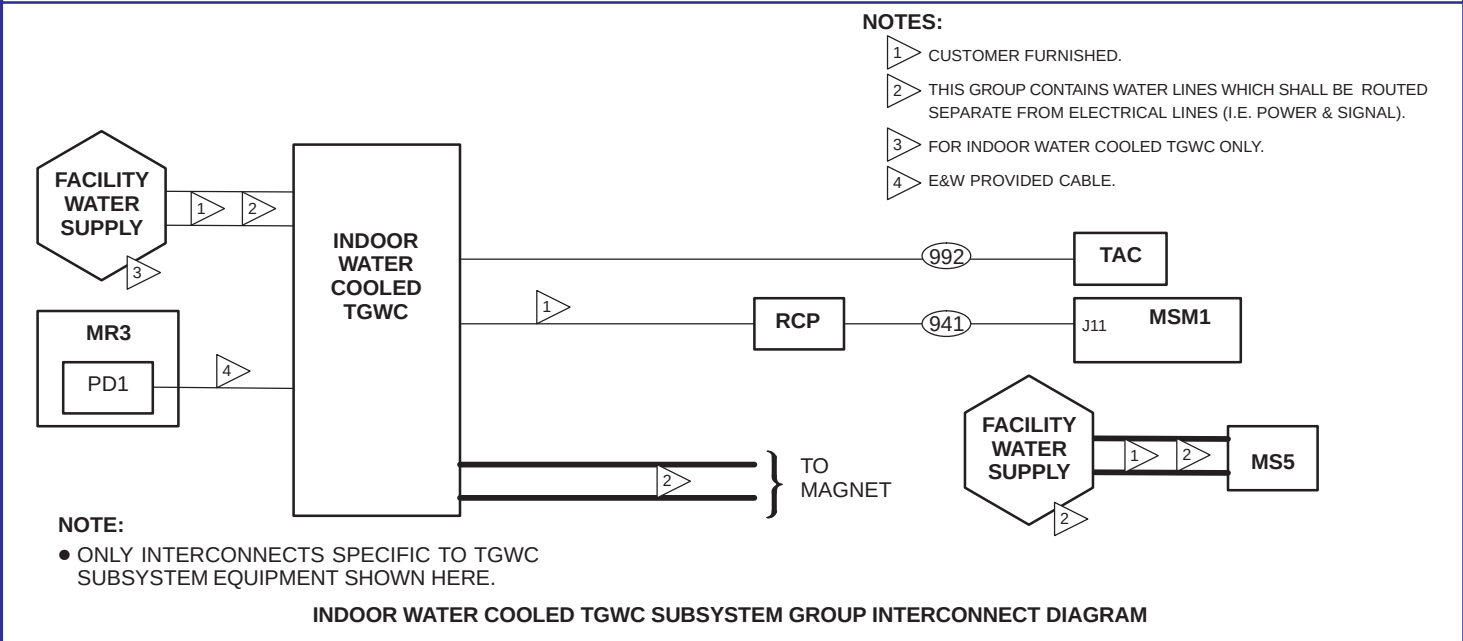
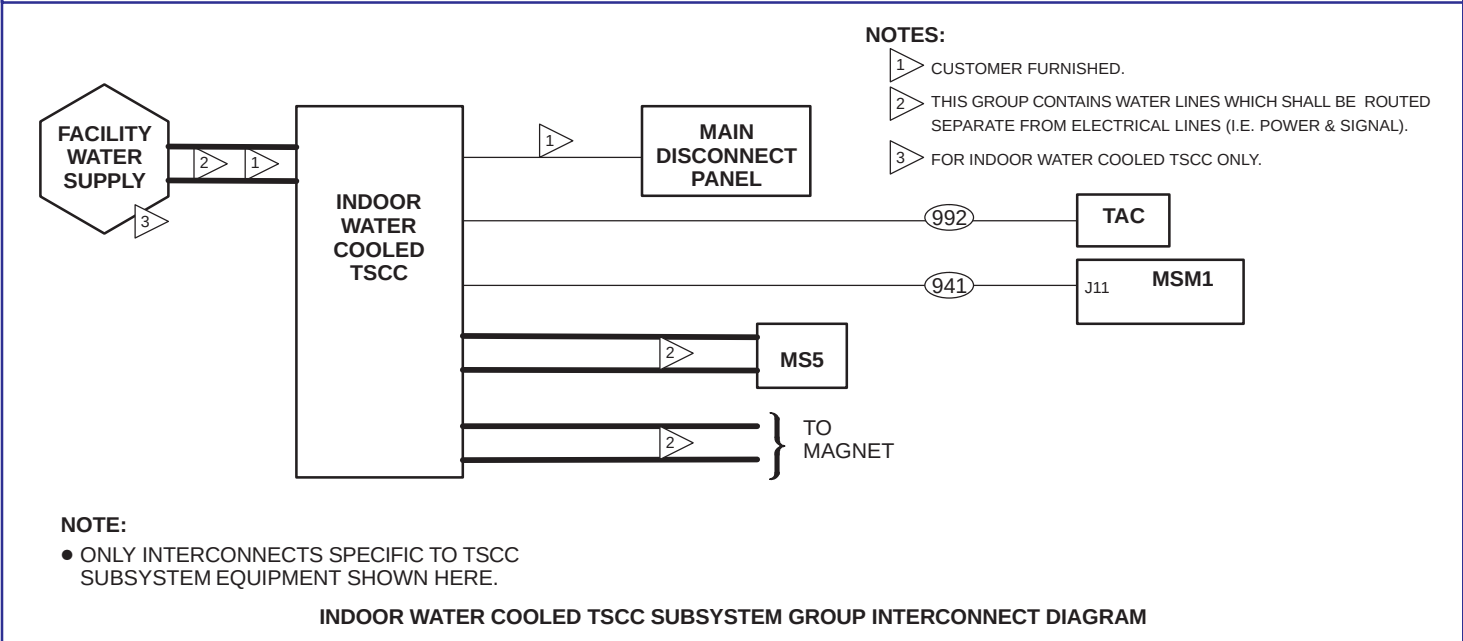
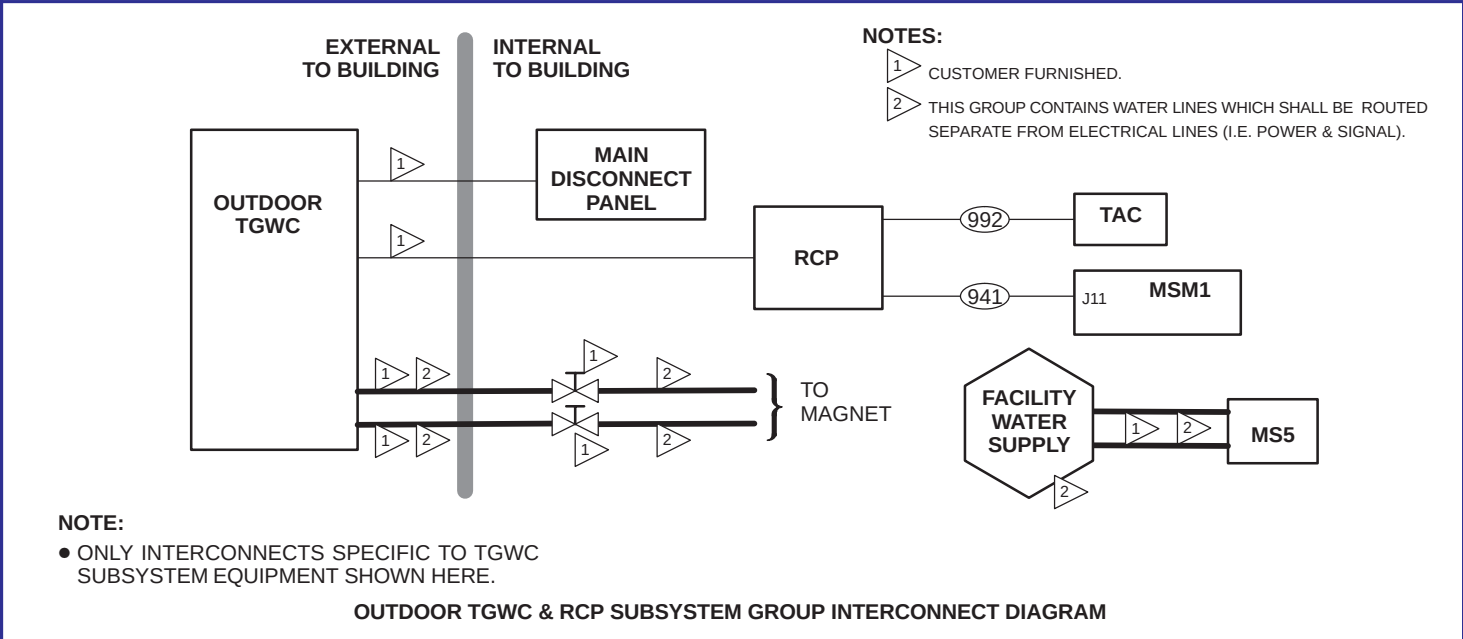
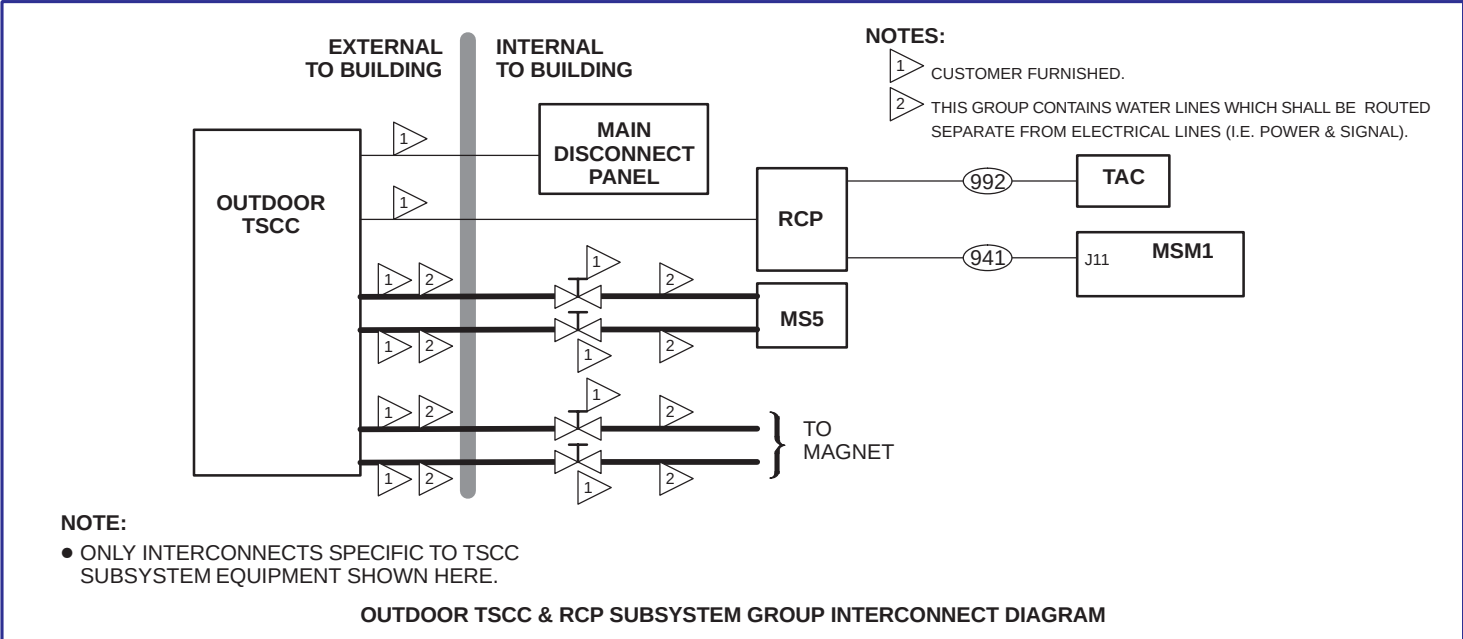
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

OUTDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
 - 3 FOR INDOOR WATER COOLED GWHX ONLY.
 - 4 GWHX PROVIDED CABLE.
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO GWHX SUBSYSTEM EQUIPMENT SHOWN HERE.

INDOOR GWHX SUBSYSTEM GROUP INTERCONNECT DIAGRAM



SECTION 5 – PENETRATION PANEL

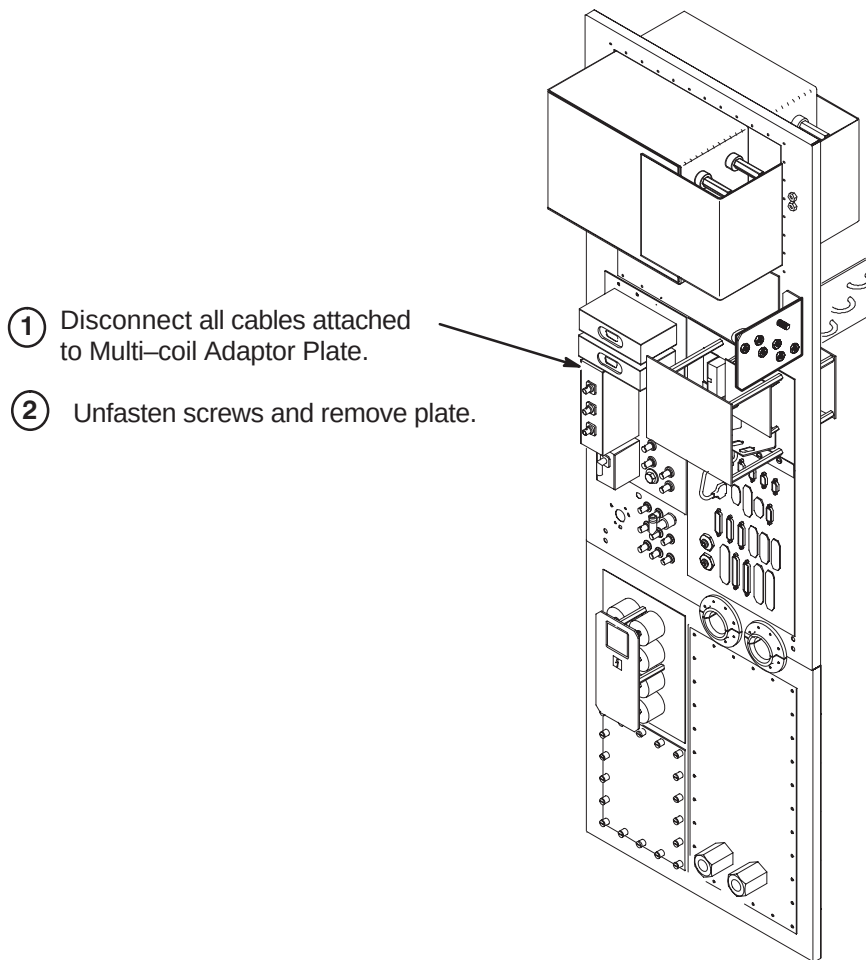
CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

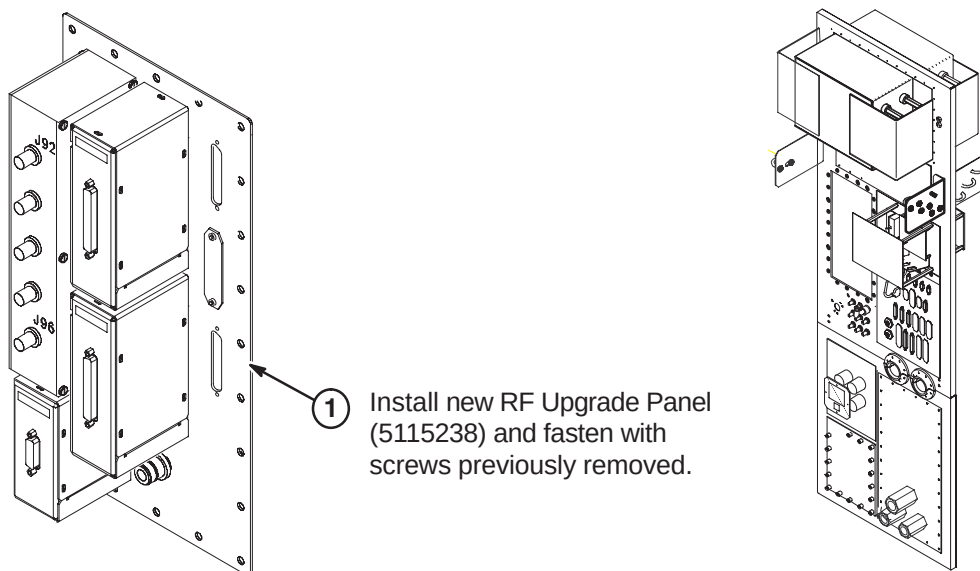
5-1 INTRODUCTION

The Multi-coil Adaptor Plate will be replaced by the RF Upgrade Panel (5115238).

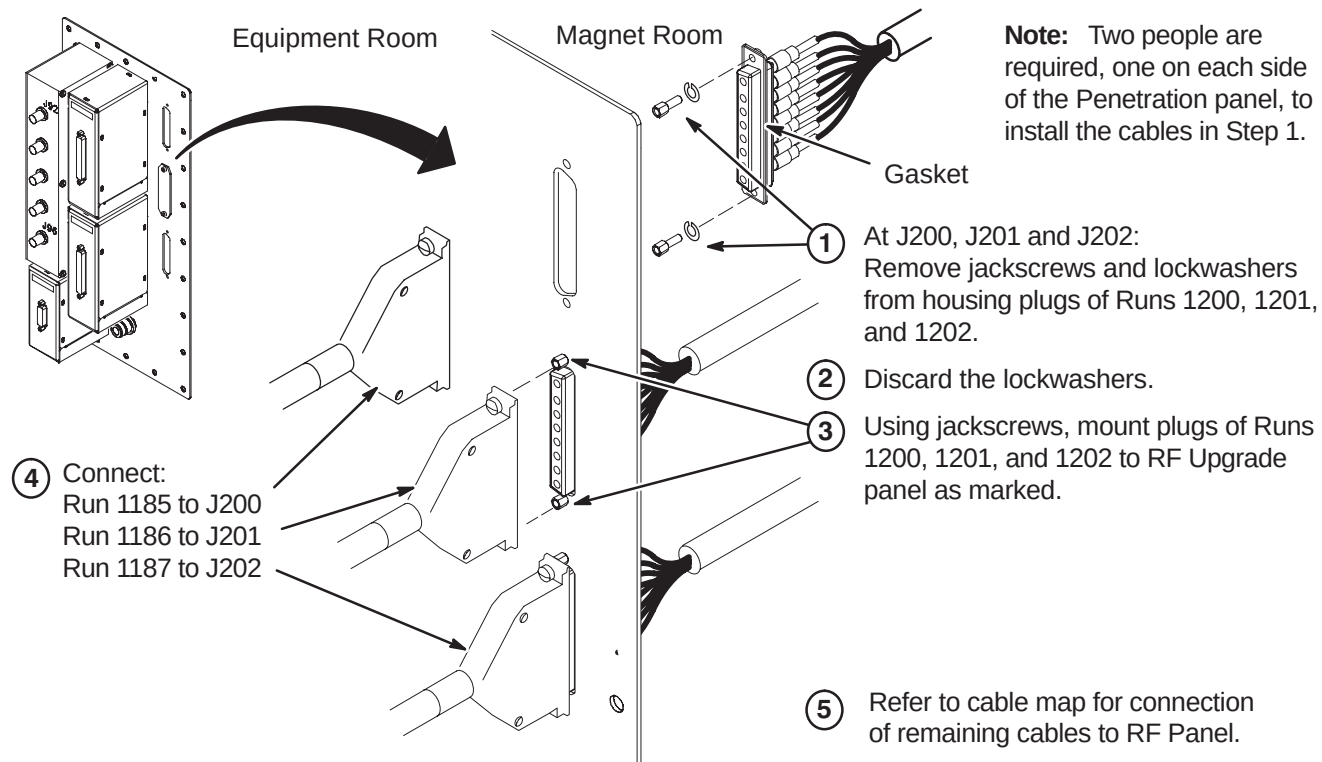
5-2 REMOVE MULTI-COIL ADAPTOR PLATE FROM PENETRATION PANEL



5-3 INSTALL NEW EXCITE RF PANEL



5-4 INSTALLATION OF CABLES AT J200, J201, AND J202



SECTION 6 – MAGNET ROOM

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
6-1 INTRODUCTION	6-3
6-2 PREREQUISITE UPGRADE PROCEDURES FOR WideOpen ENCLOSURE	6-3
6-2-1 DE-INSTALL EXISTING LPCA & CABLE TRACK	6-3
6-2-2 DE-INSTALL SPLIT BRIDGE FOR WideOpen	6-5
6-2-3 DE-INSTALL PAC MODULE FOR WideOpen	6-7
6-2-4 DE-INSTALL SRI MODULE FOR WideOpen	6-7
6-2-5 DE-INSTALL FRONT END BELL	6-8
6-2-6 INSTALL IIQ KIT AND RF GROUND STRAP (If Required)	6-9
6-2-7 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE, RUN 1260	6-10
6-2-8 INSTALL BRM GRADIENT CABLE LEAD BOARD (MR/i or CV/i Systems Only) ..	6-12
6-2-9 REPLACE GRADIENT CABLE FOAM TUBES WITH CLAMP BLOCKS	6-13
6-3 PREREQUISITE UPGRADE PROCEDURES FOR TwinSpeed ENCLOSURE	6-14
6-3-1 DE-INSTALL EXISTING LPCA & CABLE TRACK	6-14
6-3-2 DE-INSTALL BRIDGE FOR TwinSpeed	6-16
6-3-3 DE-INSTALL PAC MODULE FOR TwinSpeed	6-19
6-3-4 DE-INSTALL SRI MODULE FOR TwinSpeed	6-19
6-3-5 1.5T TwinSpeed RF BODY COIL REPLACEMENT	6-20
6-3-6 INSTALL MANIFOLD PROTECTOR PLATE	6-20
6-3-7 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE, RUN 1260	6-20
6-3-8 REPLACE GRADIENT CABLE FOAM TUBES WITH CLAMP BLOCKS	6-21
6-4 INSTALLATION OF NEW WideOpen AND TwinSpeed PARTS	6-22
6-4-1 ATTACH EDGE FOAM TO BRIDGE	6-22
6-4-2 INSTALLATION OF FRONT SPLIT BRIDGE For WideOpen	6-22
6-4-3 INSTALLATION OF FRONT SPLIT BRIDGE For TwinSpeed	6-23
6-4-4 INSTALL REAR SPLIT BRIDGE	6-24
6-4-5 INSTALL BULKHEAD INTERFACE	6-26
6-4-6 INSTALL 16 CHANNEL SWITCH	6-26
6-4-7 HYBRID SPLITTER UPGRADE	6-27
6-4-8 INSTALL LPCA AND CABLE TRACK(S)	6-29
6-4-9 CONNECT REMAINING CABLES TO REAR PEDESTAL	6-34
6-4-10 INSTALL SRI-3 MODULE	6-35
6-4-11 INSTALL PAC MODULE AND ECG CABLE	6-36
6-4-12 INSTALL COVERS	6-36
6-4-13 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS	6-37

(Continued on next page)

TABLE OF CONTENTS (Continued)

<u>SECTION</u>	<u>PAGE</u>
6-5 UPGRADE PROCEDURES FOR HORIZON STYLE MAGNET ENCLOSURES	6-38
6-5-1 DE-INSTALL EXISTING LPCA & CABLE TRACK	6-38
6-5-2 DE-INSTALL PAC MODULE	6-39
6-5-3 DE-INSTALL SRI MODULE	6-40
6-5-4 PREPARE END BELL TO INSTALL NEW DUAL THERMAL SENSOR CABLE	6-41
6-5-5 INSTALL IIQ KIT AND RF GROUND STRAP (If Required)	6-47
6-5-6 INSTALL GRADIENT LEAD BOARD & RESTRAINT BLOCKS (If Required)	6-47
6-5-7 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE, RUN 1260	6-48
6-5-8 RE-INSTALL FRONT END BELL, COMFORT MODULE, AND BRIDGE	6-48
6-5-9 INSTALL SRI-3 MODULE	6-49
6-5-10 INSTALL NEW PAC MODULE AND ECG CABLE	6-51
6-5-11 INSTALL NEW COMPONENTS ON REAR PEDESTAL	6-51
6-5-12 INSTALL LPCA AND CABLE TRACK	6-55
6-5-13 CONNECT REMAINING CABLES TO REAR PEDESTAL	6-60
6-5-14 INSTALL COVERS	6-60
6-5-15 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS	6-61
6-6 TIMES MICROWAVE RF CABLES INSTALLATION	6-62
6-6-1 CABLE INFORMATION	6-62
6-6-2 ROUTE, TERMINATE, AND CONNECT RF CABLES	6-62
6-7 PATIENT TABLE CRADLE RELEASE UPGRADE	6-63
6-7-1 PREPARE CRADLE FOR UPGRADE	6-63
6-7-2 REMOVE EXISTING RELEASE MECHANISM	6-63
6-7-3 INSTALL NEW RELEASE MECHANISM	6-64
6-7-4 ATTACH CORD	6-64

CAUTION

MAKE SURE POWER TO SYSTEM IS OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

6-1 INTRODUCTION

Modules on the Rear Pedestal, the front and rear Split Bridge on WideOpen style enclosures, LPCA and Cable Track, SRI, and PAC will be replaced.

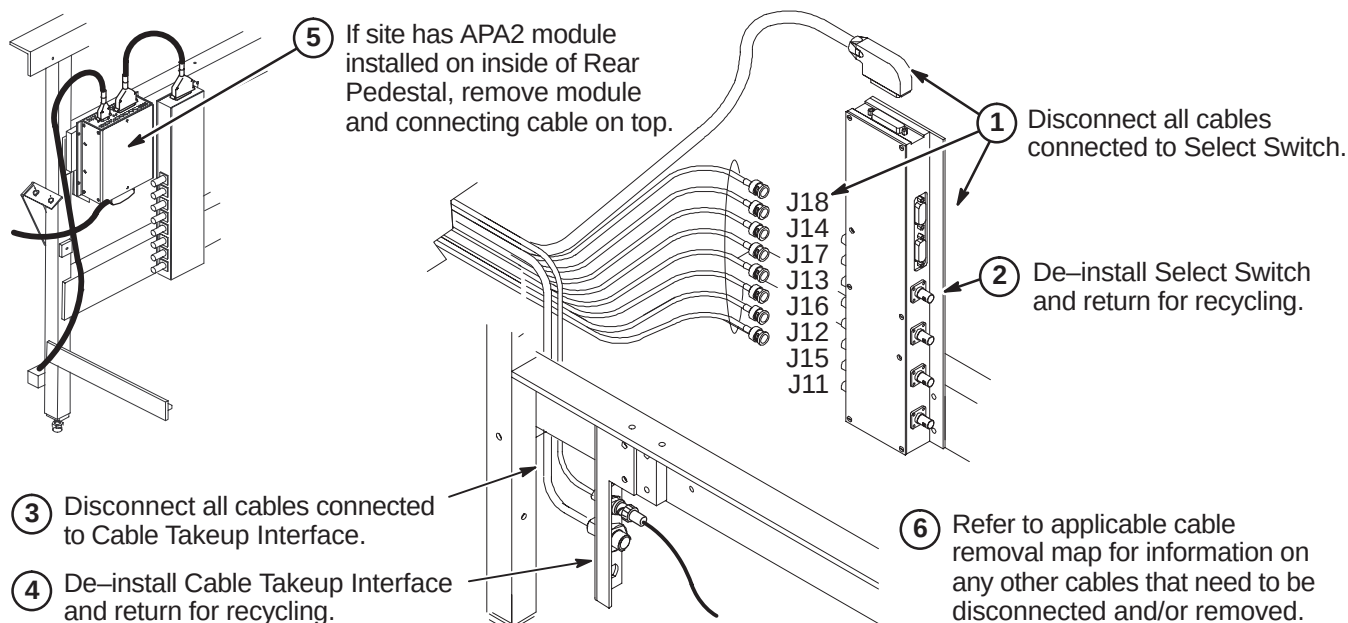
A new Dual Thermal Sensor cable, Run 1260, will be installed along with other new cables that will be routed and connected. Several other cables will also be removed and replaced with new cables. Refer to Section 4 cable maps for information on removing and installing cables

- Refer to Sections 6-2 and 6-4 for upgrading a system with a LCC magnet and a WideOpen Enclosure
- Refer to Sections 6-3 and 6-4 for upgrading a system with a LCC magnet and a TwinSpeed Enclosure
- Refer to Section 6-5 for upgrading a system with a Non-LCC magnet and a Horizon style Enclosure
- Complete Section 6-6 for all systems
- Also included are the instructions for upgrading the Cradle Release on the Patient Table. Refer to section 6-7 if this upgrade has not already been performed at this site.

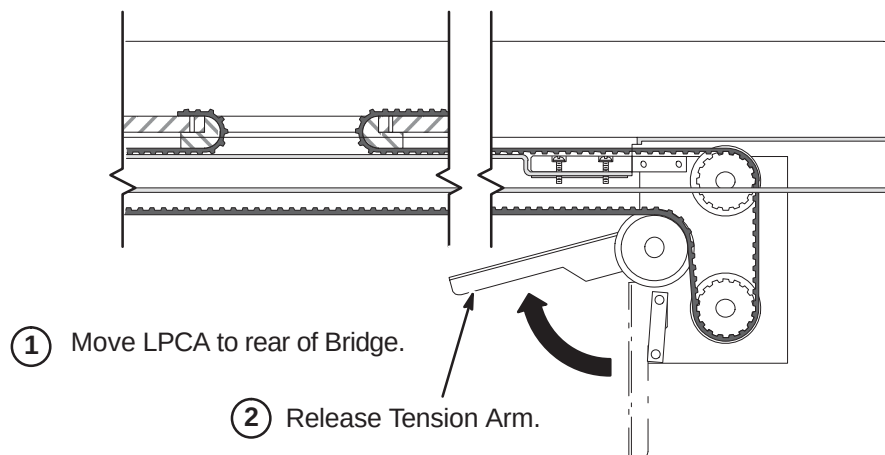
6-2 PREREQUISITE UPGRADE PROCEDURES FOR WideOpen ENCLOSURE

6-2-1 DE-INSTALL EXISTING LPCA & CABLE TRACK

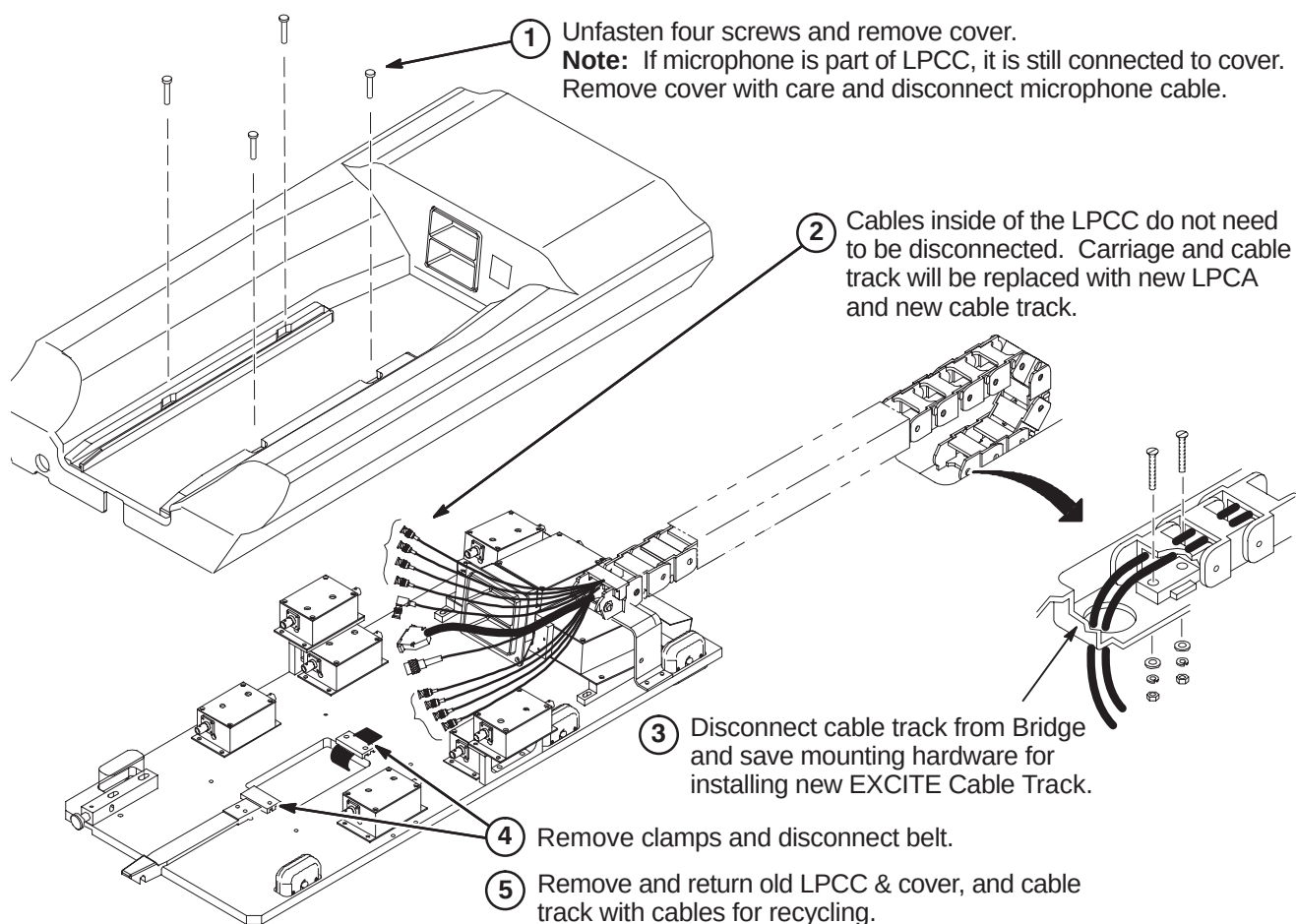
6-2-1-1 Disconnect Cables In Rear Pedestal



6-2-1-2 Release Tension Arm Under Rear Pedestal



6-2-1-3 Disconnect Low Profile Carriage Cover & Cable Track



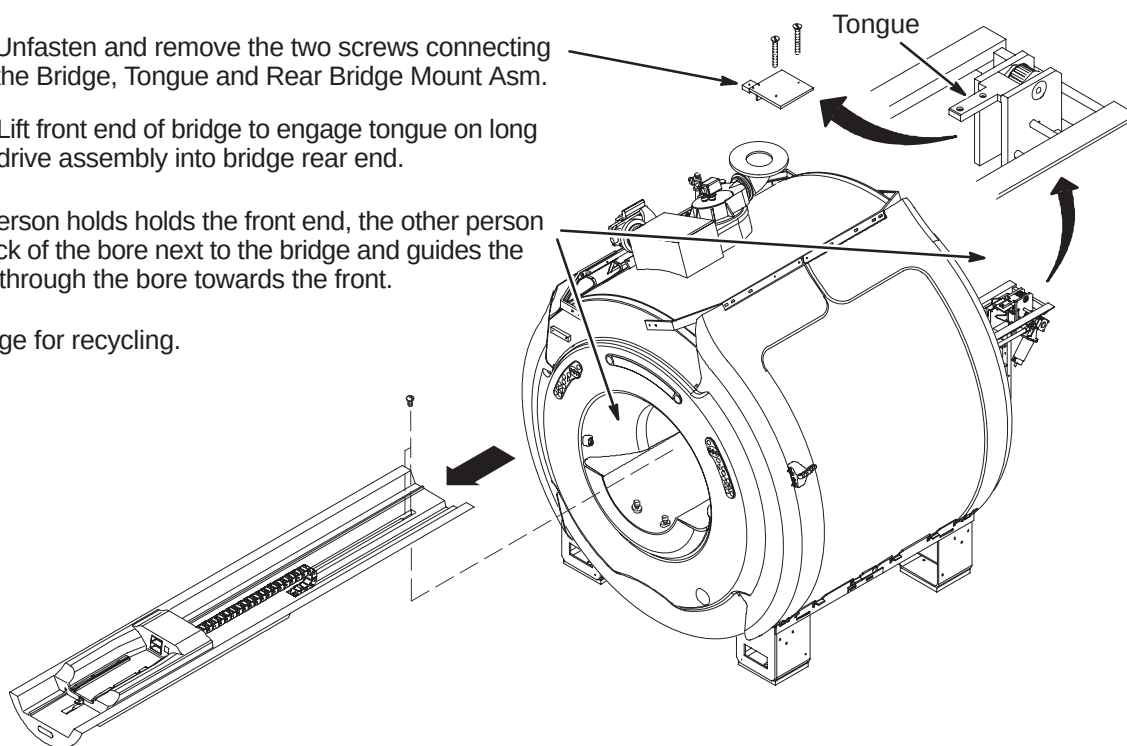
6-2-2 DE-INSTALL BRIDGE FOR WideOpen

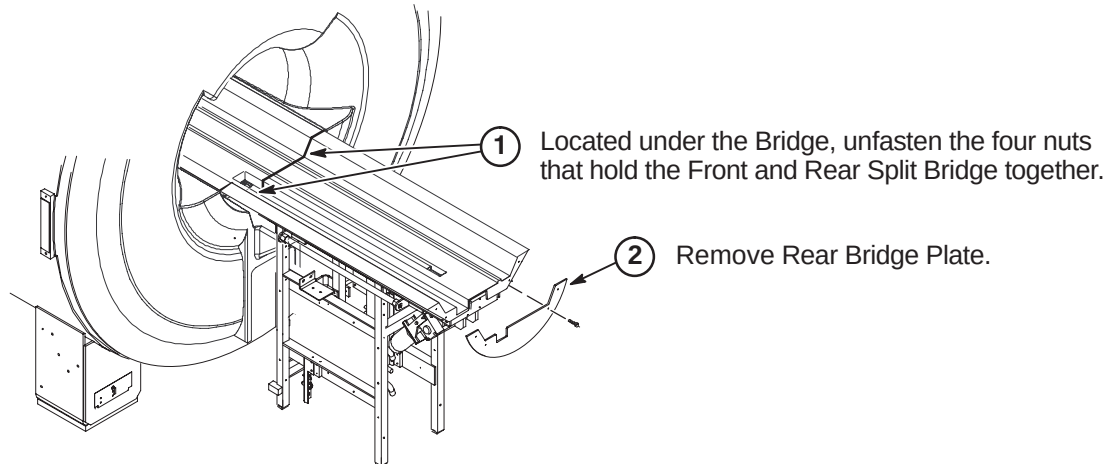
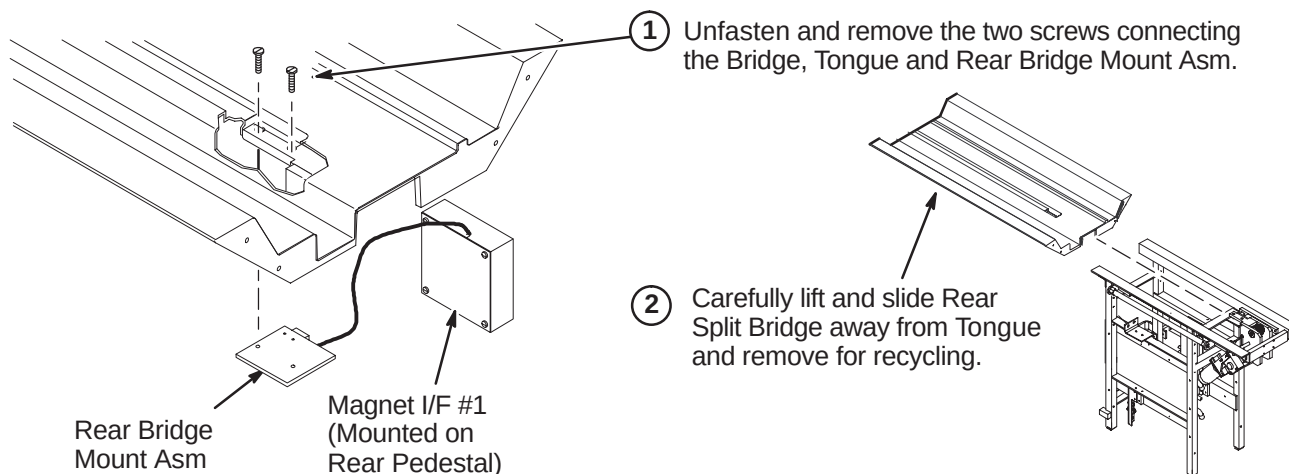
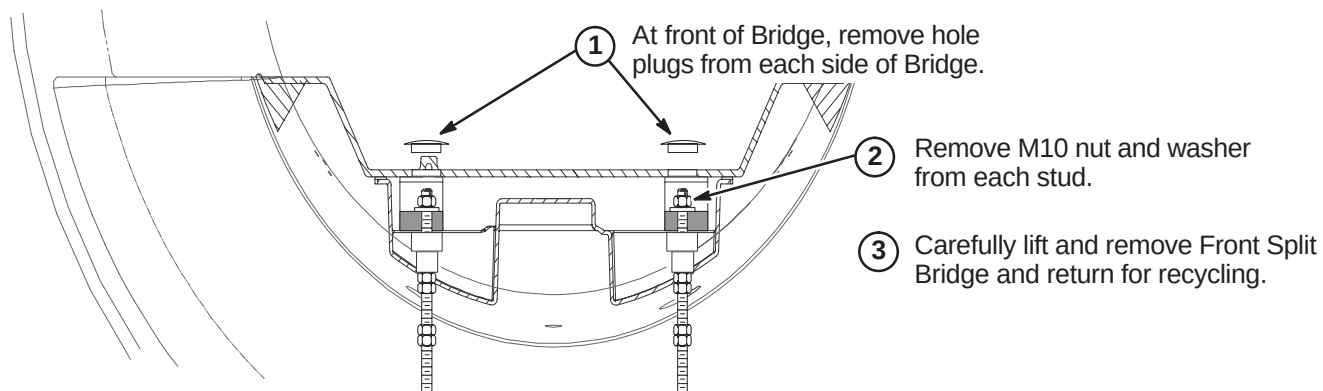
Upgrading to HDx requires a new Front and Rear Split Bridge. Early production versions of the WideOpen enclosure were installed with the long single-piece Bridge. Later versions used the Split Bridge. Either existing bridge type will need to be removed for installation of the new dual-track Split Bridge. Refer to one of the following for instructions of removing the bridge at site being upgraded.

- Procedure **6-2-2-1** for the de-installation of the long single-piece bridge.
- Procedures **6-2-2-2** thru **6-2-2-4** for the de-installation of the single-track split bridge.

6-2-2-1 Remove Long Bridge

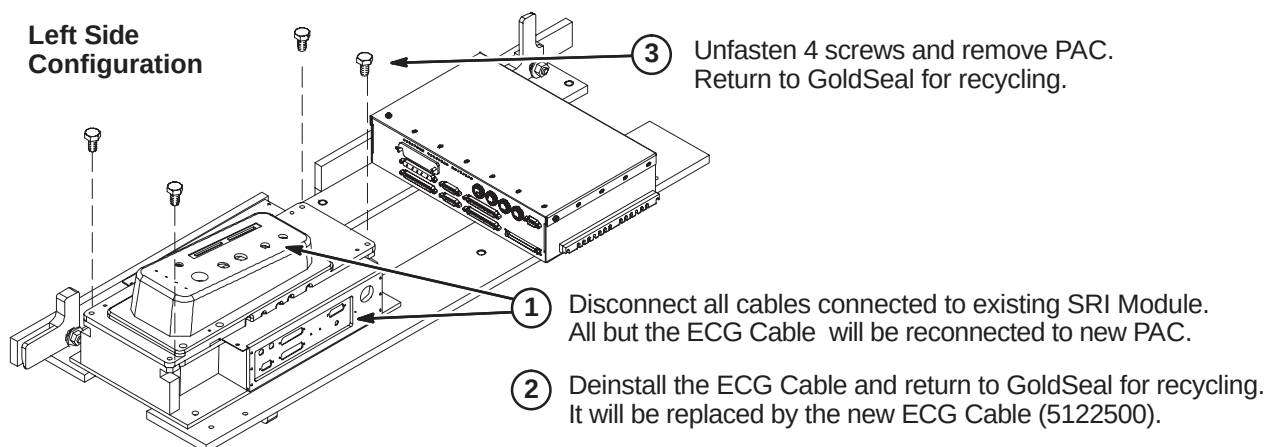
- ① Unfasten and remove the two screws connecting the Bridge, Tongue and Rear Bridge Mount Asm.
- ② Lift front end of bridge to engage tongue on long drive assembly into bridge rear end.
- ③ While one person holds the front end, the other person stands in back of the bore next to the bridge and guides the Bridge Asm through the bore towards the front.
- ④ Return Bridge for recycling.



6-2-2-2 Unfasten Rear Split Bridge**6-2-2-3 Remove Rear Split Bridge****6-2-2-4 Remove Front Split Bridge**

6-2-3 DE-INSTALL PAC MODULE FOR WideOpen

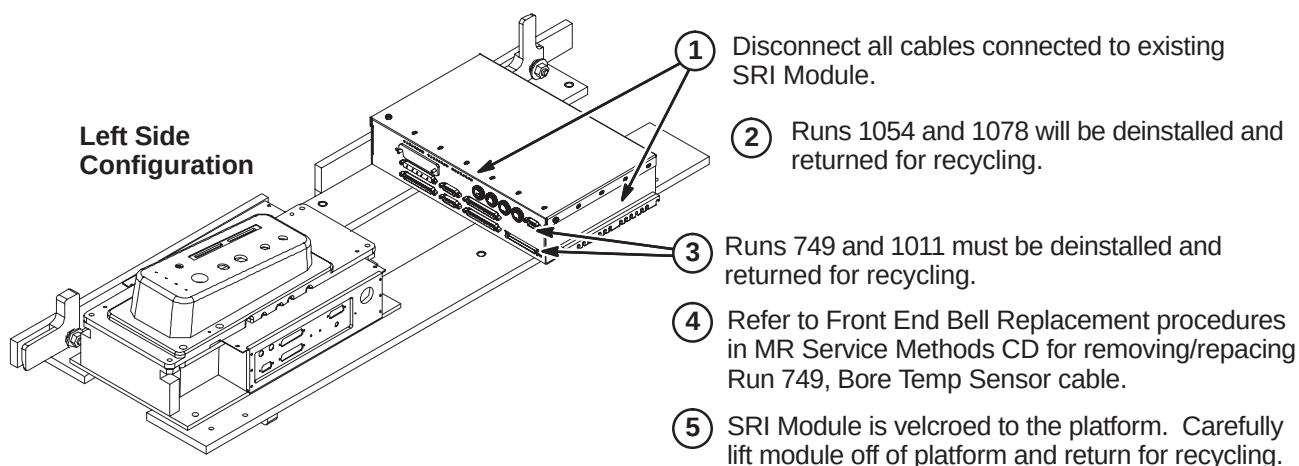
The existing PAC Module will be replaced by a new PAC Module with independent vector gating.



6-2-4 DE-INSTALL SRI MODULE FOR WideOpen

The existing SRI Module will be replaced by the SRI-3 Module.

NOTE: Existing Runs 749 and 1011 must be removed and replaced with new Runs 1260 and 1261. There is a functionality difference.



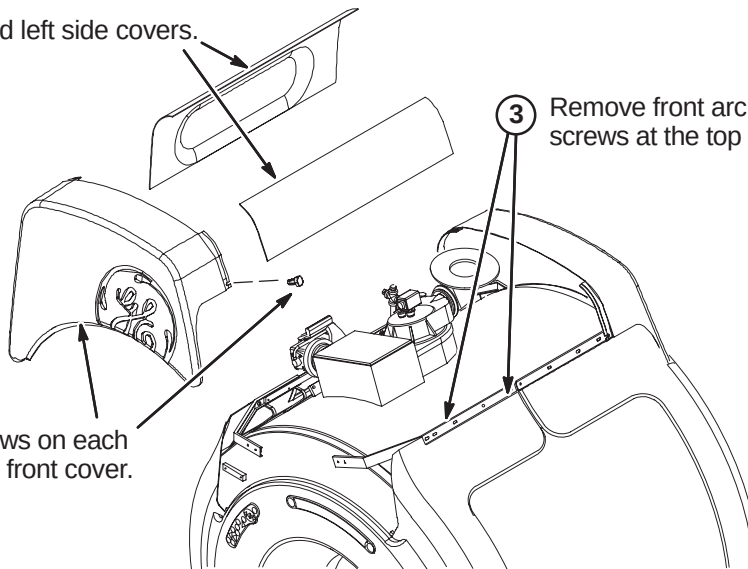
6-2-5 DE-INSTALL FRONT END BELL

6-2-5-1 Remove WideOpen Enclosure Covers

- ① Remove tight and left side covers.

- ② Remove screws on each side, then top front cover.

- ③ Remove front arc covers by removing two screws at the top and sliding cover to the side.



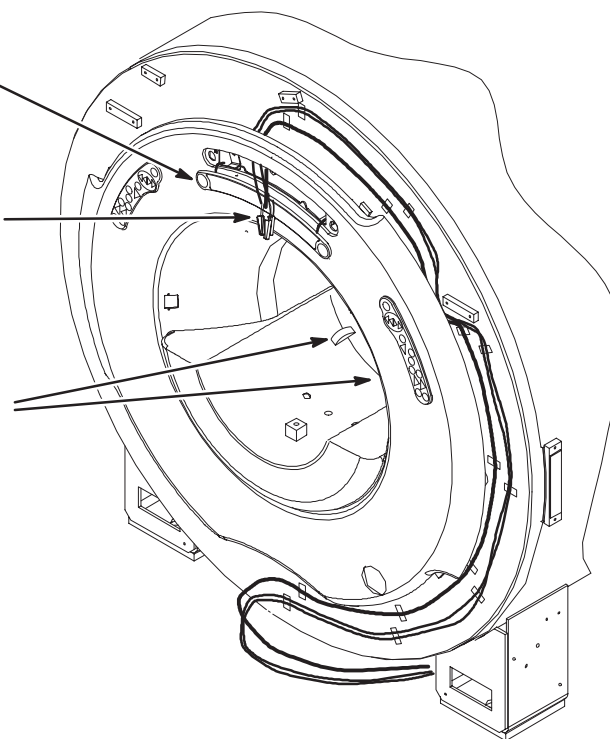
6-2-5-2 Disconnect Operator Control Panel

- ① Insert 3mm allen wrench into hole in Control Panel and turn counter clockwise 1/2 turn.

- ② Carefully, using a small screwdriver, pry the Operator Display away from the front cover.

- ③ Disconnect the three cables attached to the Operator Display.

- ④ Lift open the two latches connecting the front end bell to the RF Coil.

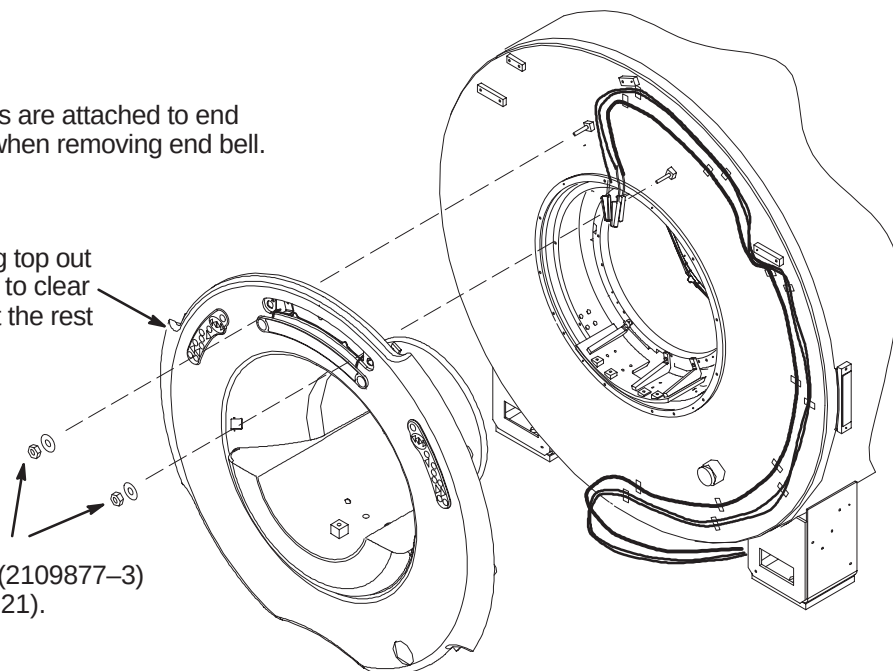


6-2-5-3 Remove Front End Bell**CAUTION:**

Bore Light inserts are attached to end bell. Take care when removing end bell.

- ② Remove front end bell by pulling top out past studs, then lifting vertically to clear end bell inserts, then pulling out the rest of the way.

- ① Unfasten the M10 Lock Nuts (2109877-3) and Rubber Washers (2167321).

**6-2-6 INSTALL IIQ KIT (IF REQUIRED)**

Prior to this upgrade, the site should have been audited to determine if the IIQ Kit (M3090NE) and the RF Ground Strap (M3090NF) have been installed.

The IIQ Kit includes parts and instructions for the installation of the Gradient Radial Support and the Bridge Support. It would be required for earlier versions of the WideOpen enclosure. If this upgrade has not been performed on the existing enclosure, then M3090NE should have been ordered and must be installed at this point of the EXCITE HD upgrade.

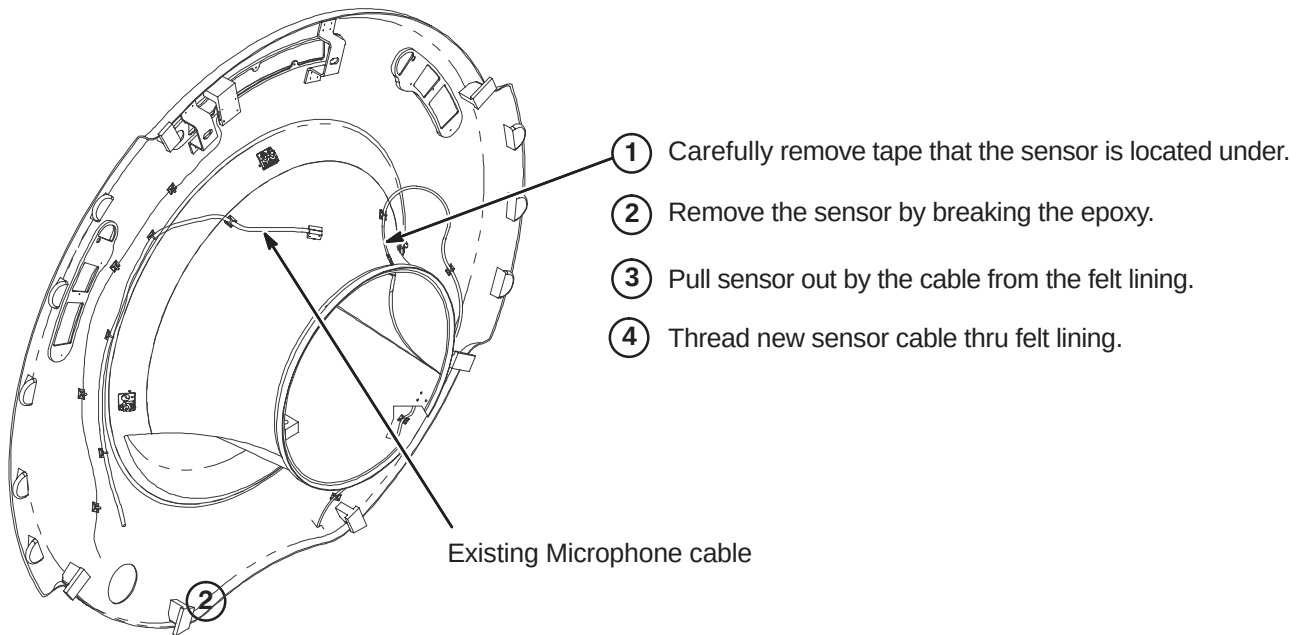
Refer to steering Direction 2401983, included in M3090NE, for guidance. Directions 2318581, *Signa Cx/K4 Magnet WideOpen Enclosure Bridge Support* and Direction 2319693, *Signa Gradient Radial Support Upgrade*, along with associated parts, are also included in M3090NE.

Also, the RF Ground Strap (M3090NF) must be installed if this upgrade hasn't been performed previously. This kit includes parts and instructions for installation of the RF Ground Strap. Refer to Direction 2319685, *BRM of CRM RF Shield Grounding Upgrade*, for instructions on installing this kit.

Note: If this procedure is not completed by Mechanical Installer or Van Partner, make sure that the GE Field Engineer has been notified that this upgrade procedure needs to be completed. For Van Partner, refer to the Design Review Checklist to report status of completion.

6-2-7 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE, RUN 1260**6-2-7-1 Temperature Sensor Cable Removal**

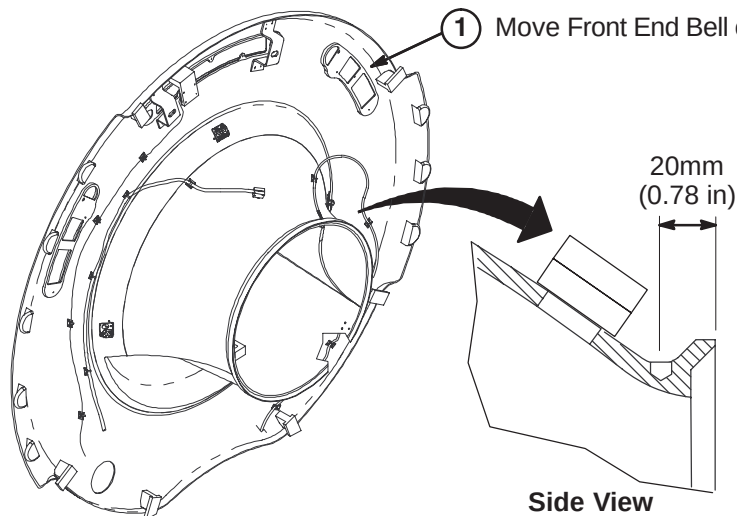
The site may have upgraded to a Dual Thermal Sensor Run 749 recently, but this cable has to be removed and replaced with new Dual Thermal Sensor cable, Run 1260 (5123812). EXCITE HD requires Run 1260 to be installed.



6-2-7-2 Prepare End Bell for Installation of Dual Thermal Sensor Cable

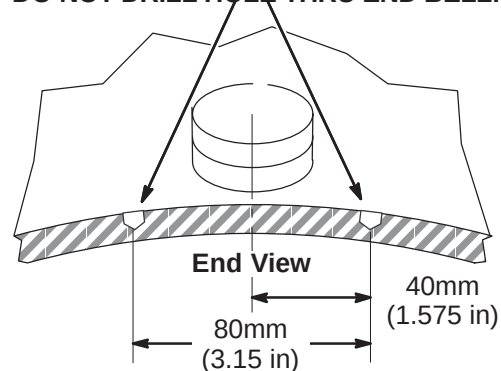
WARNING!

FERROUS MATERIAL HAZARD! DO NOT DRILL, HAMMER, CHISEL, ETC. OR USE ANY FERROUS TOOLS IN MAGNET ROOM UNLESS MAGNET IS RAMPED DOWN!

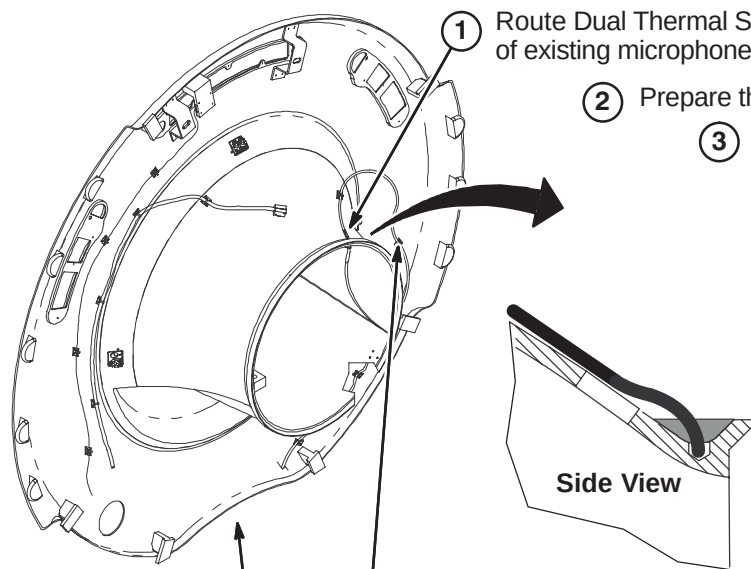


- ② Drill two $\Phi 7$ mm holes at indicated locations to a depth of 3mm (0.120 in).

DO NOT DRILL HOLE THRU END BELL!

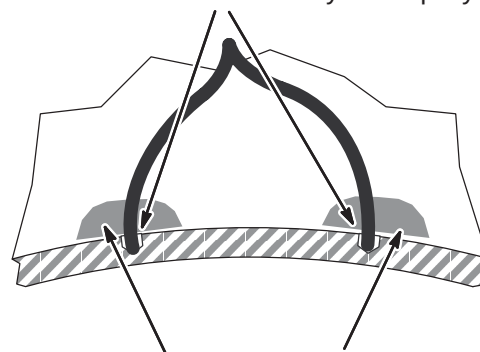


6-2-7-3 Attach Dual Thermal Sensor Cable



- ② Prepare the Oxybond Epoxy with nozzle and applicator.
- ③ Dispense a small amount of epoxy into each hole.

- ④ Push each sensor to bottom of hole and fill with Oxybond Epoxy.



- ⑤ If necessary, add extra Oxybond epoxy at each sensor location to ensure that it is securely mounted.

- ⑥ Allow time for the epoxy to set, then gently pull the cable to remove slack.

- ⑦ Use cable ties to secure Dual Thermal Sensor Cable to the Front End Bell.
- ⑧ Route cable thru bottom opening of end bell.

6-2-7-4 Install Front End Bell

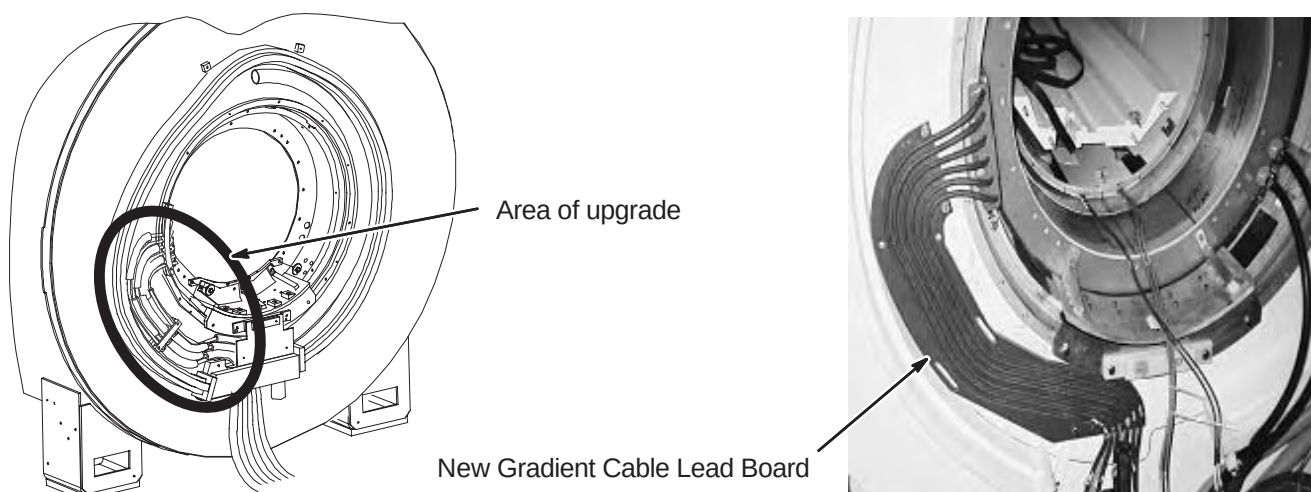
Reverse the Front End Bell Removal procedure in this section to reinstall the front end bell.

6-2-8 INSTALL BRM GRADIENT CABLE LEAD BOARD (MR/i or CV/i Systems Only)**Note**

For a system being upgraded, whether it is a fixed-site or a mobile, if the gradient lead restraint is using the 90 degree mobile van bracket or the IIQ “S” bracket assembly, do not modify or upgrade it. If the system being upgraded does not contain either of these, then install the IIQ “S” bracket assembly.

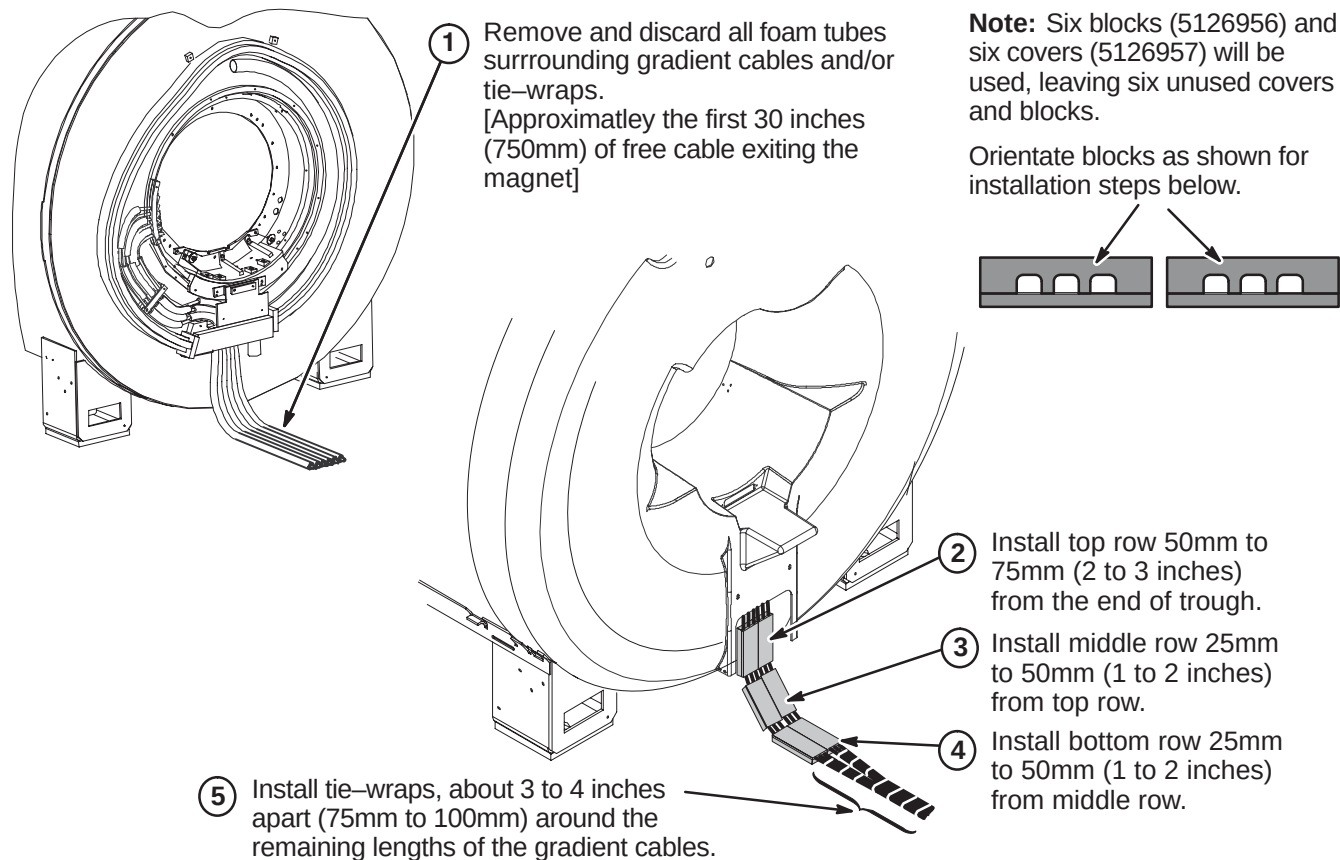
For fixed sites only, the mounting of the gradient cables at the rear of the magnet must use the Gradient Cable lead Board, also known as the “S” bracket assembly. If the cables are routed as shown in the illustration below, then the upgrade must be installed. This upgrade is part of M3335BD, *LCC WideOpen IIQ Upgrade*. Refer to Direction 2404642, provided with Gradient Cable Lead Board collector (5110336), for instructions on how to install the new design.

M3335BD also include parts and instructions for installing the Rear Pedestal Stabilization kit (5198961). Refer to Section 6-4-4.



6-2-9 REPLACE GRADIENT CABLE FOAM TUBES WITH CLAMP BLOCKS

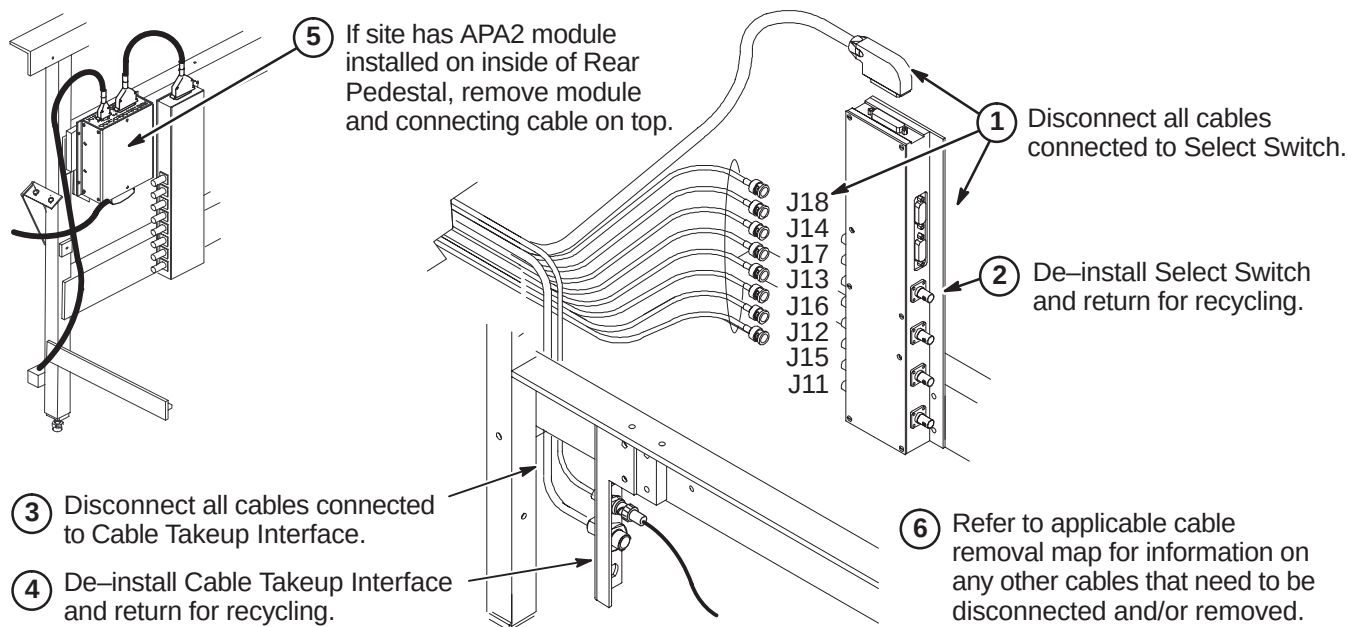
The existing foam tubes that are wrapped around the gradient cables at the rear of the magnet are to be removed and replaced with cable clamp blocks.



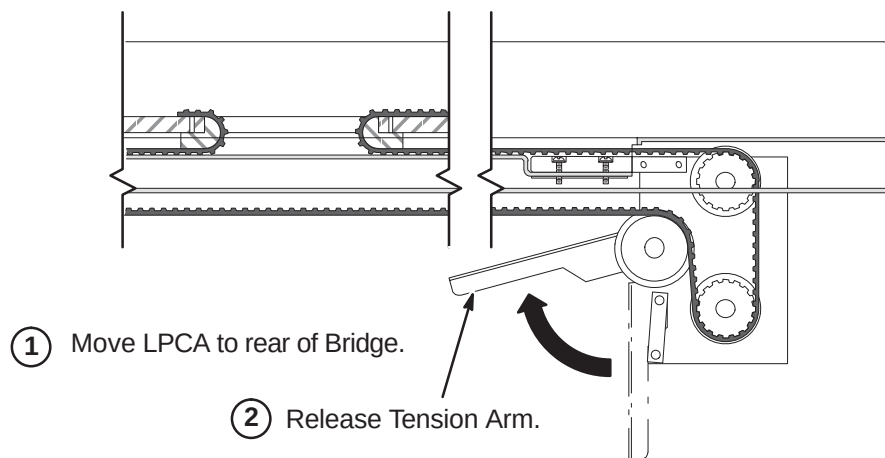
6-3 PREREQUISITE UPGRADE PROCEDURES FOR TwinSpeed ENCLOSURE

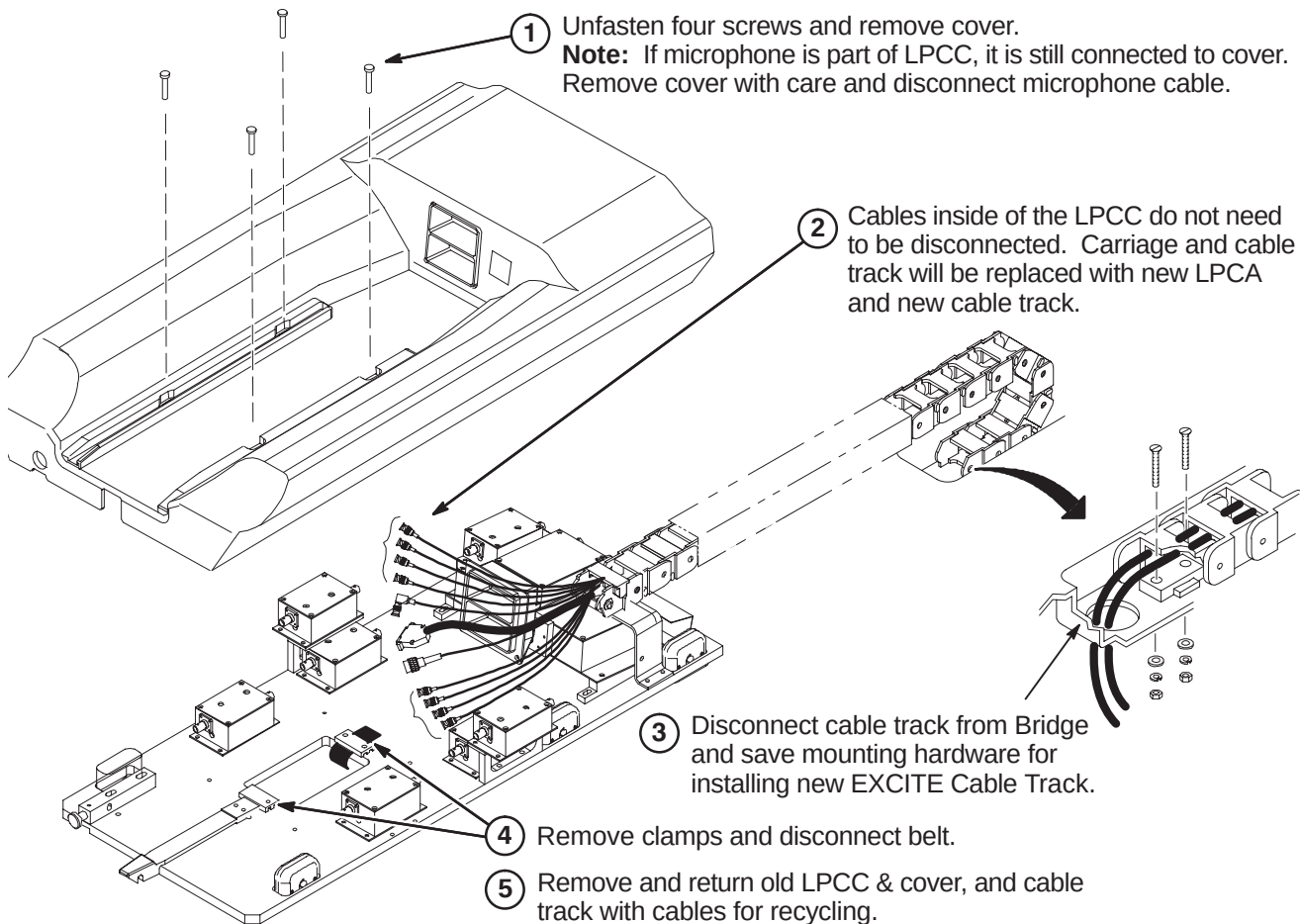
6-3-1 DE-INSTALL EXISTING LPCA & CABLE TRACK FOR TwinSpeed

6-3-1-1 Disconnect Cables In Rear Pedestal



6-3-1-2 Release Tension Arm Under Rear Pedestal



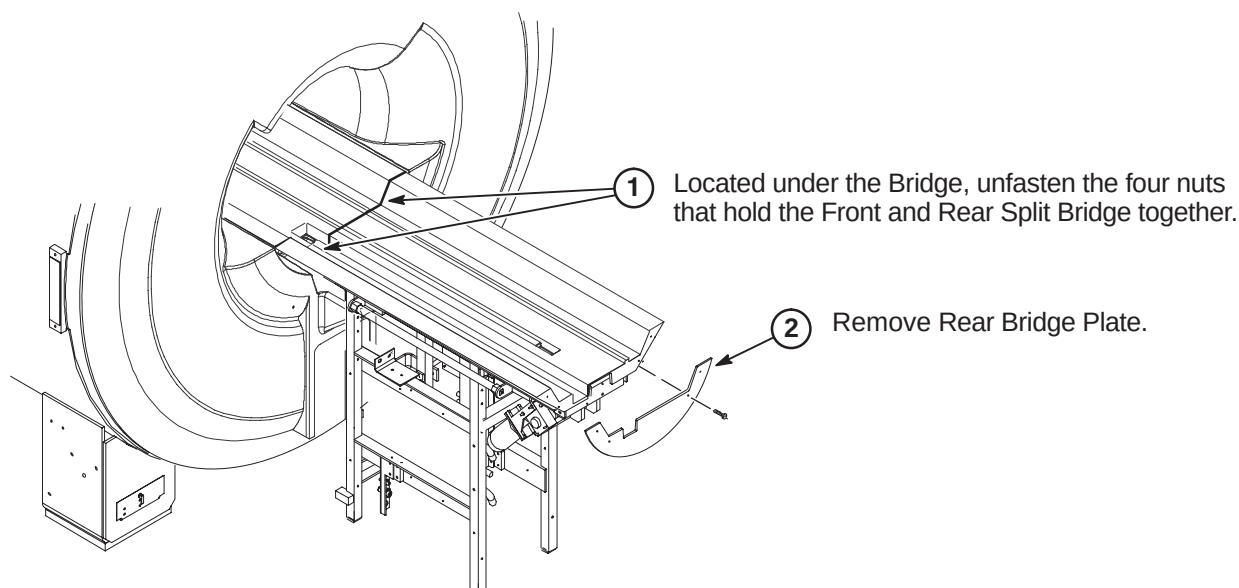
6-3-1-3 Disconnect Low Profile Carriage Cover & Cable Track

6-3-2 DE-INSTALL BRIDGE FOR TwinSpeed

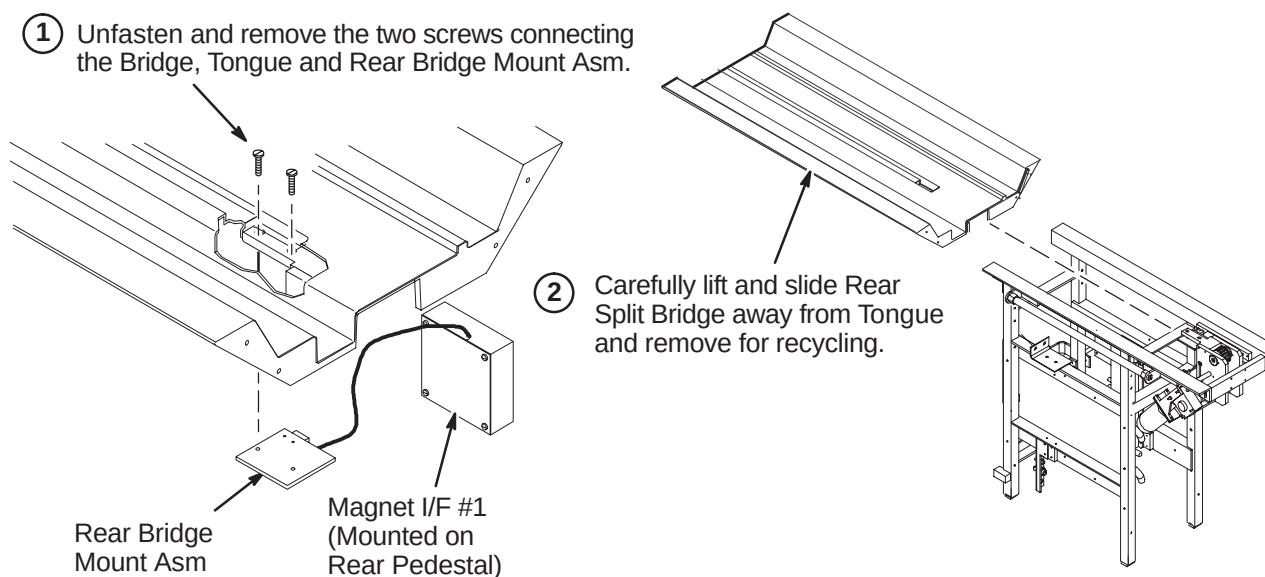
Upgrading to EXCITE TwinSpeed Release 12.x requires a new Front and Rear Split Bridge. The method of attachment at the front of the Bridge is different between a bridge that was installed for a forward production (new) installation and a Long bridge that was upgraded from a non-TwinSpeed WideOpen enclosure to TwinSpeed.

- Procedures 6-3-2-1 thru 6-3-2-3 shows the deinstallation for a forward production installed bridge.
- Procedure 6-3-2-4 thru 6-3-2-6 for the deinstallation of the upgraded bridge.

6-3-2-1 Unfasten Rear Split Bridge

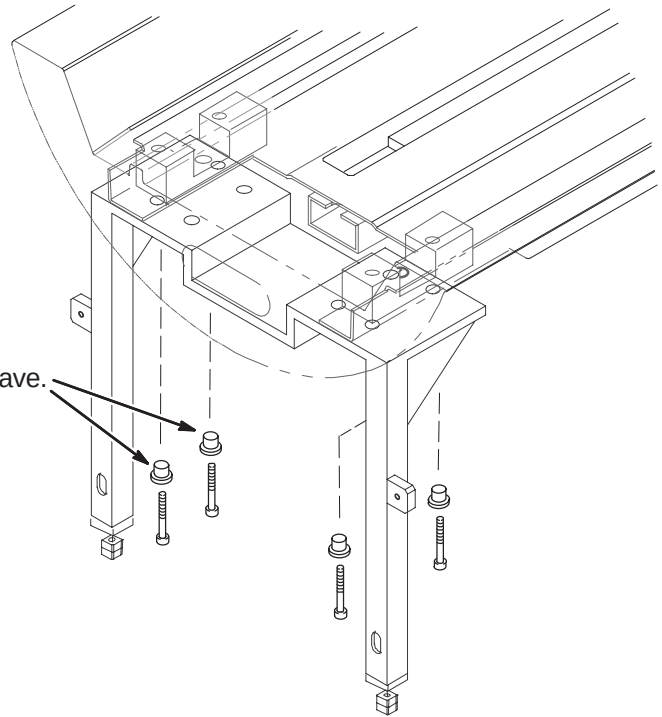


6-3-2-2 Remove Rear Split Bridge

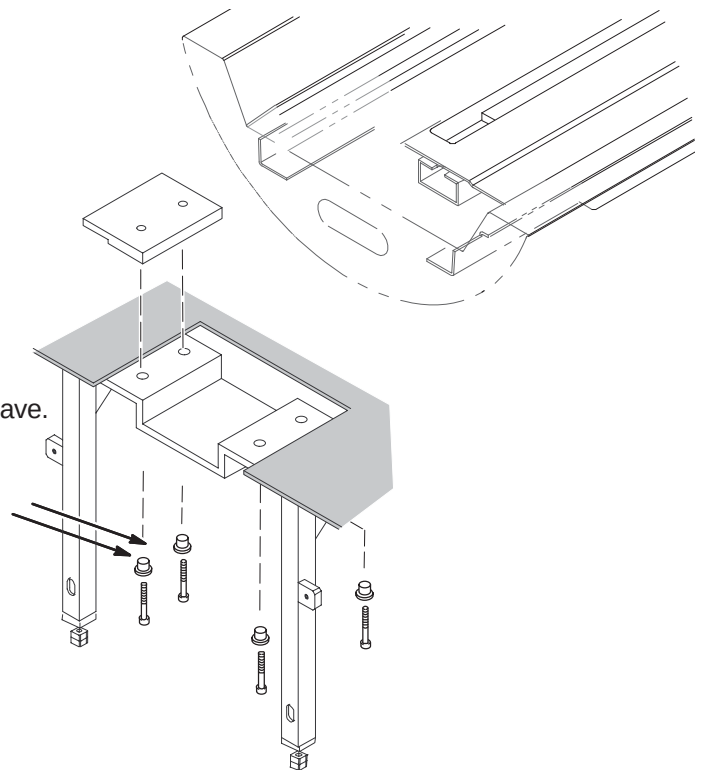


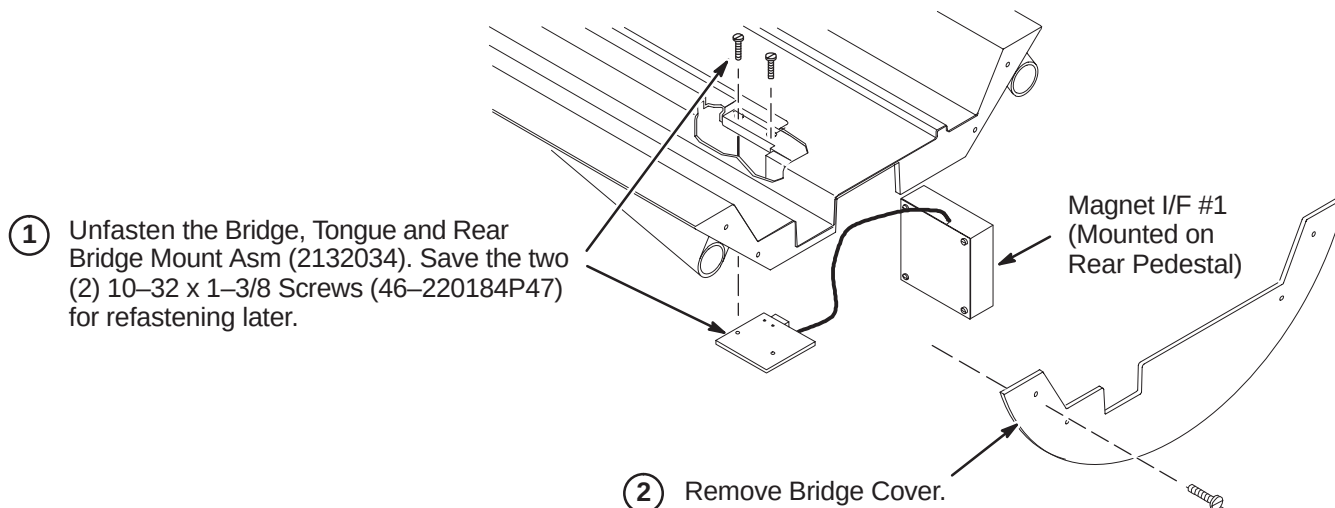
6-3-2-3 Remove Forward Production Installed Front Split Bridge

- ① Unfasten the four (4) M6 x 35mm long Socket Head screws (2109867-17) and Support Bushings and save. These will be reused for installing the new Bridge.
- ② Carefully lift and remove Front Split Bridge and return for recycling.

**6-3-2-4 Unfasten Front of Upgraded Bridge**

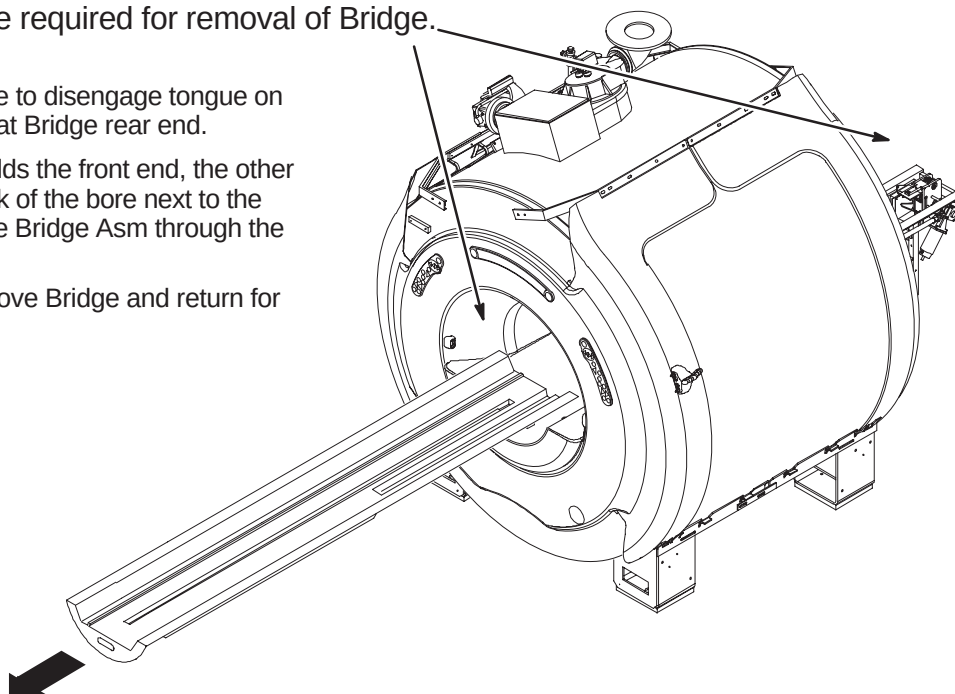
- ③ Unfasten the four (4) M6 x 35mm long Socket Head screws (2109867-17) and Support Bushings and save. These will be reused for installing the new Bridge.



6-3-2-5 Disconnect Bridge Mount Asm**6-3-2-6 Remove Upgrade Long Bridge**

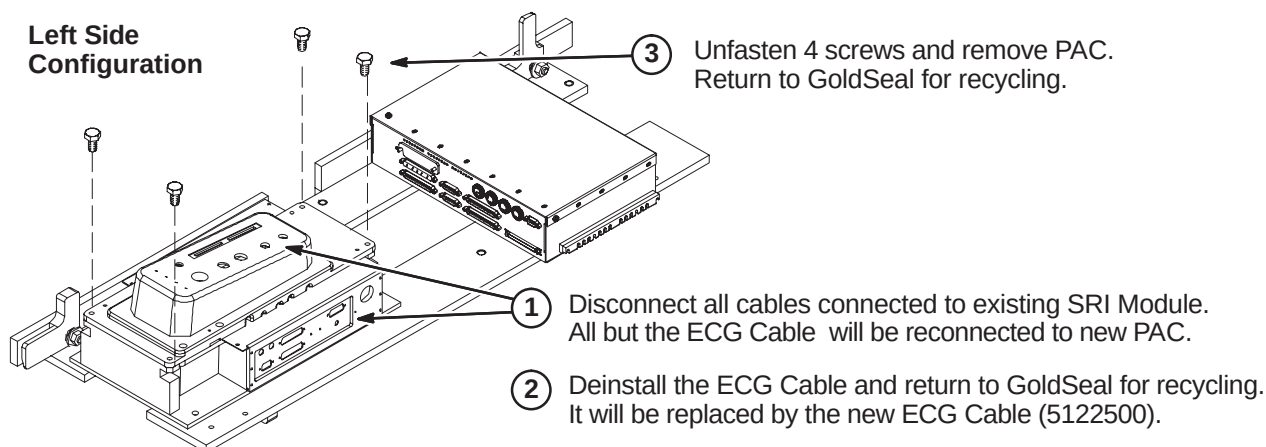
NOTE: Two people are required for removal of Bridge.

- ① Lift front end of bridge to disengage tongue on long drive assembly at Bridge rear end.
- ② While one person holds the front end, the other person stands in back of the bore next to the bridge and guides the Bridge Asm through the bore to the front end.
- ③ Carefully lift and remove Bridge and return for recycling.



6-3-3 DE-INSTALL PAC MODULE FOR TwinSpeed

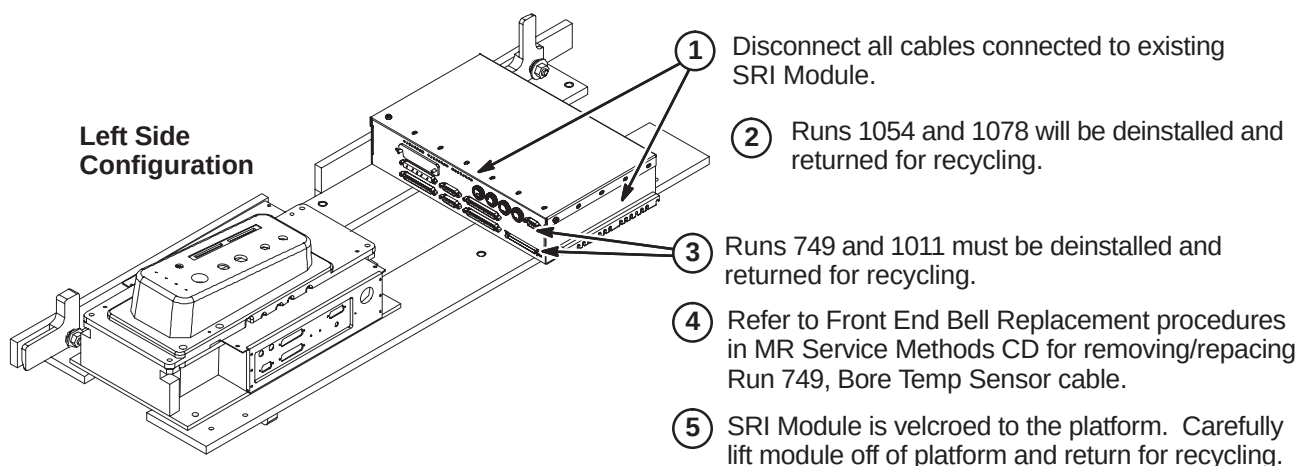
The existing PAC Module will be replaced by a new PAC Module with independent vector gating.



6-3-4 DE-INSTALL SRI MODULE FOR TwinSpeed

The existing SRI Module will be replaced by the SRI-3 Module.

NOTE: Existing Runs 749 and 1011 must be removed and replaced with new Runs 1260 and 1261. There is a functionality difference.



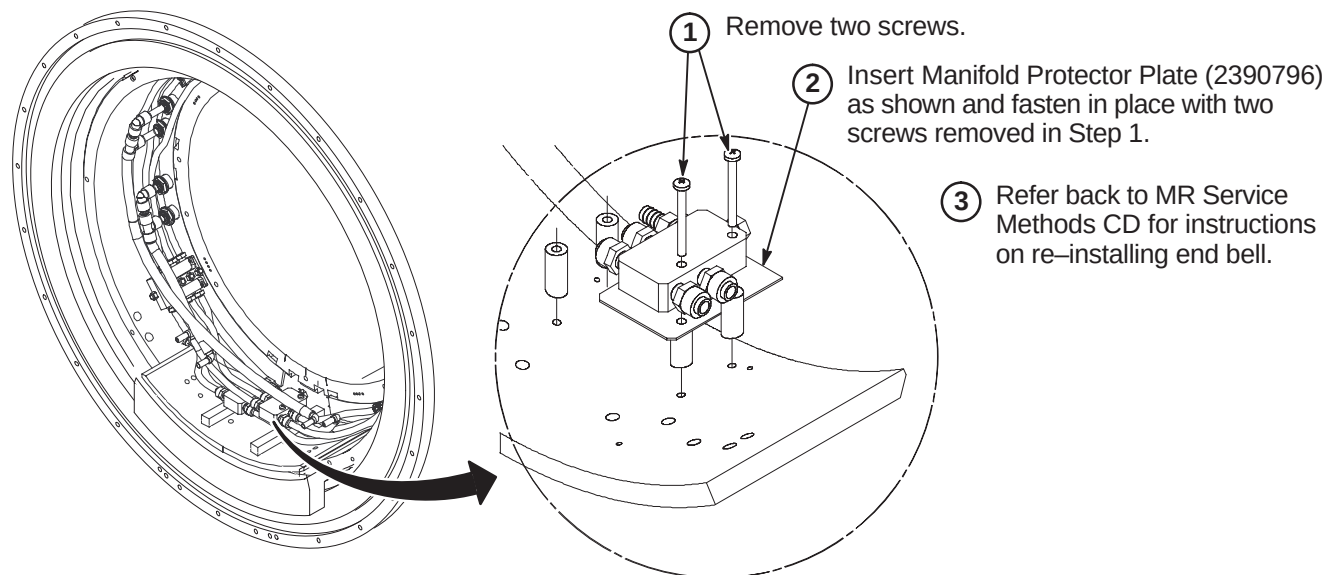
6-3-5 1.5T TwinSpeed RF BODY COIL REPLACEMENT

For replacement procedure, refer to applicable MR Service Methods CD-ROM and go to:

Replacements/1.5T Twin RF Body Coil Replacement Procedure.

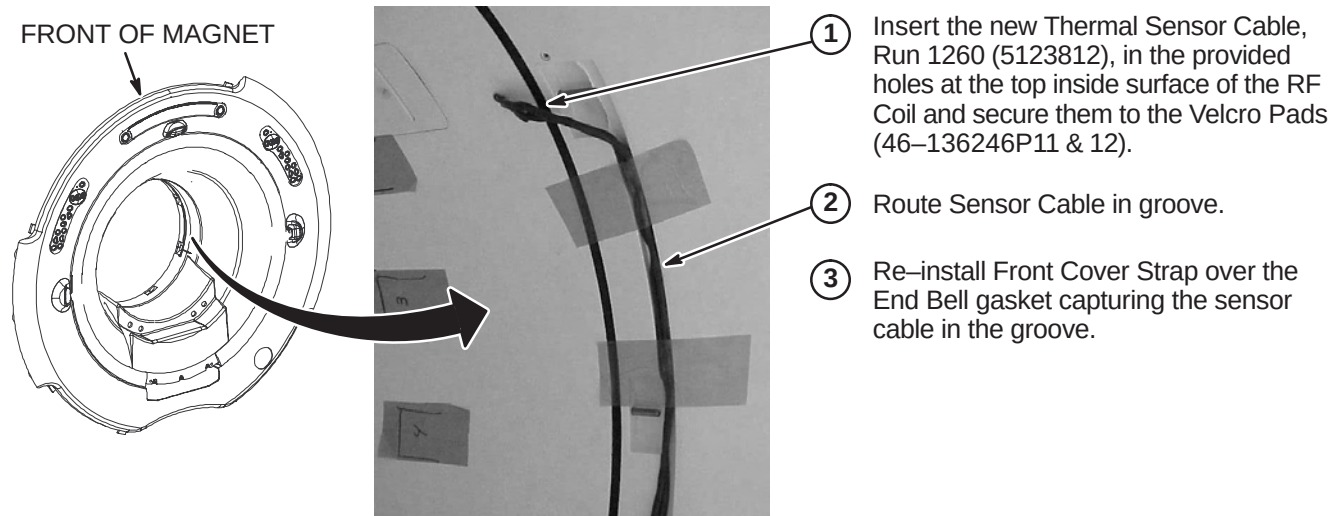
When front End Bell is removed, perform Section 6-3-6 for installing the Manifold Protector plate.

6-3-6 INSTALL MANIFOLD PROTECTOR PLATE



6-3-7 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE, RUN 1260

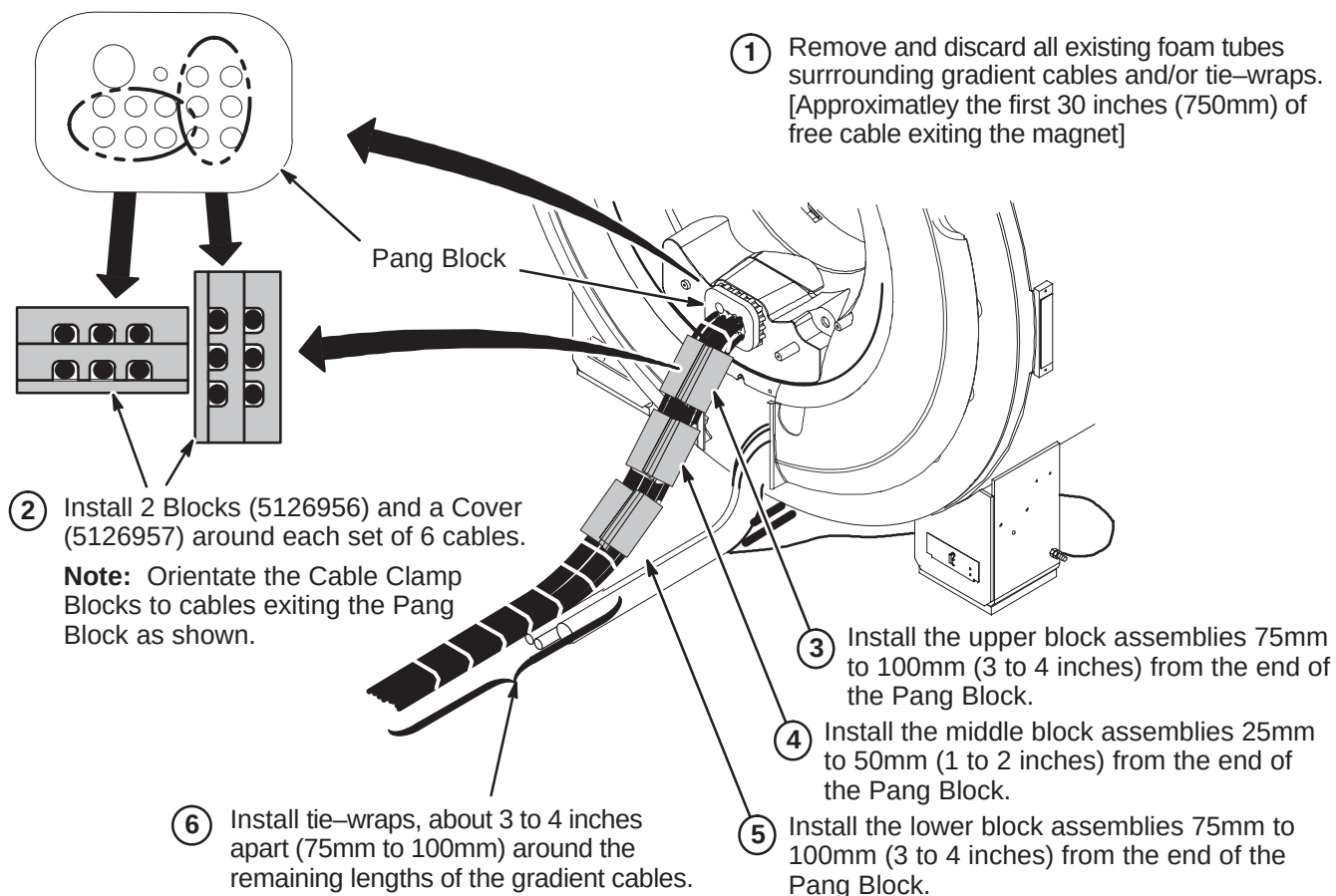
Install Run 1260 after the new RF Body Coil and front and rear end bells have been installed. Refer to the RF Body Coil Replacement procedure in Section 6-3-5 for more detailed steps.



6-3-8 REPLACE GRADIENT CABLE FOAM TUBES WITH CLAMP BLOCKS

The existing foam tubes that are wrapped around the gradient cables at the rear of the magnet are to be removed and replaced with cable clamp blocks.

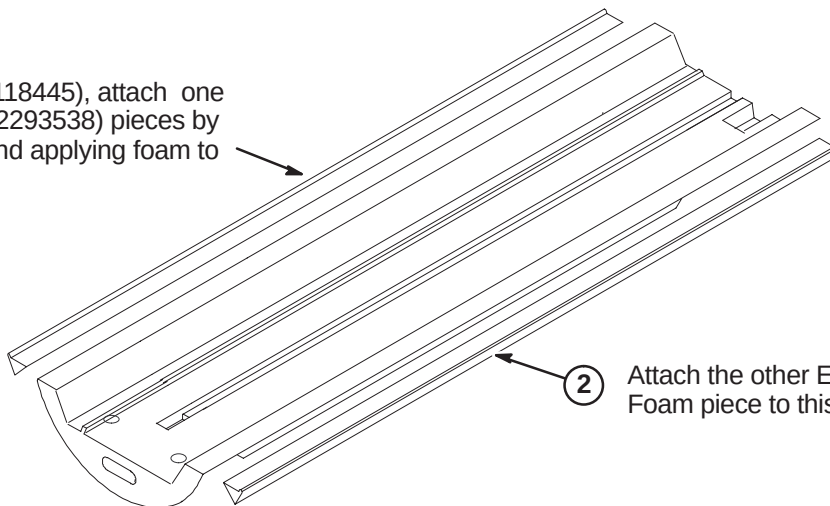
Note: Twelve blocks (5126956) and six covers (5126957) will be used, leaving six unused covers.



6-4 INSTALLATION OF NEW WideOpen AND TwinSpeed PARTS

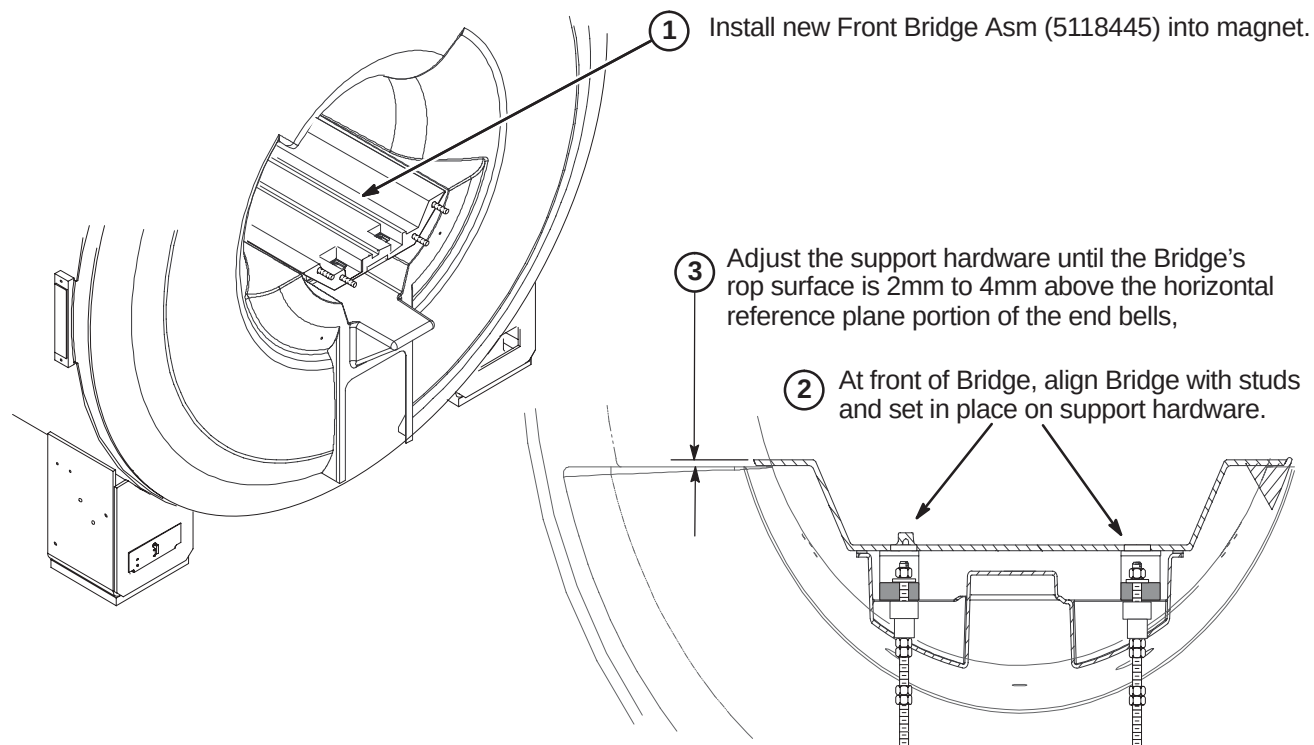
6-4-1 ATTACH EDGE FOAM TO BRIDGE

- ① Starting at the front of Bridge (5118445), attach one of the Twin Bridge Edge Foam (2293538) pieces by removing the protective cover and applying foam to bottom side of Bridge surface.



- ② Attach the other Edge Foam piece to this side.

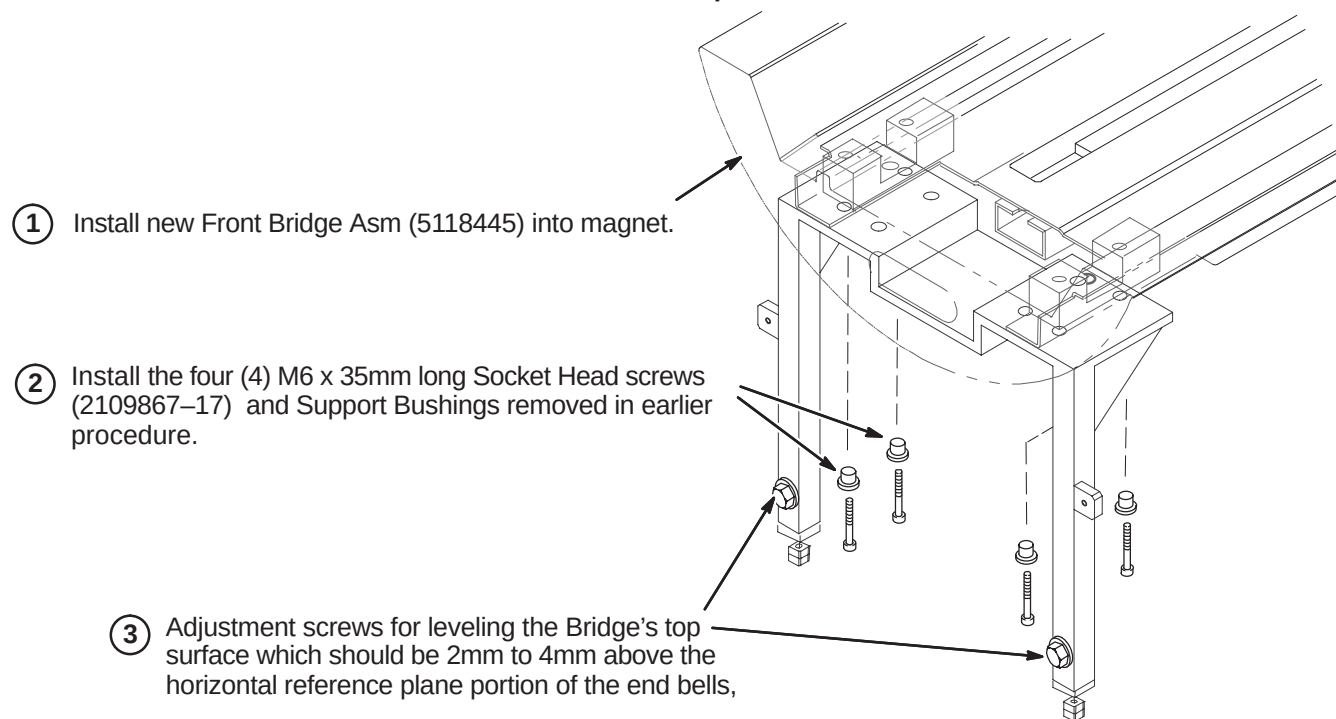
6-4-2 INSTALLATION OF FRONT BRIDGE For WideOpen



- ① Install new Front Bridge Asm (5118445) into magnet.

- ③ Adjust the support hardware until the Bridge's top surface is 2mm to 4mm above the horizontal reference plane portion of the end bells,

- ② At front of Bridge, align Bridge with studs and set in place on support hardware.

6-4-3 INSTALLATION OF FRONT BRIDGE For TwinSpeed

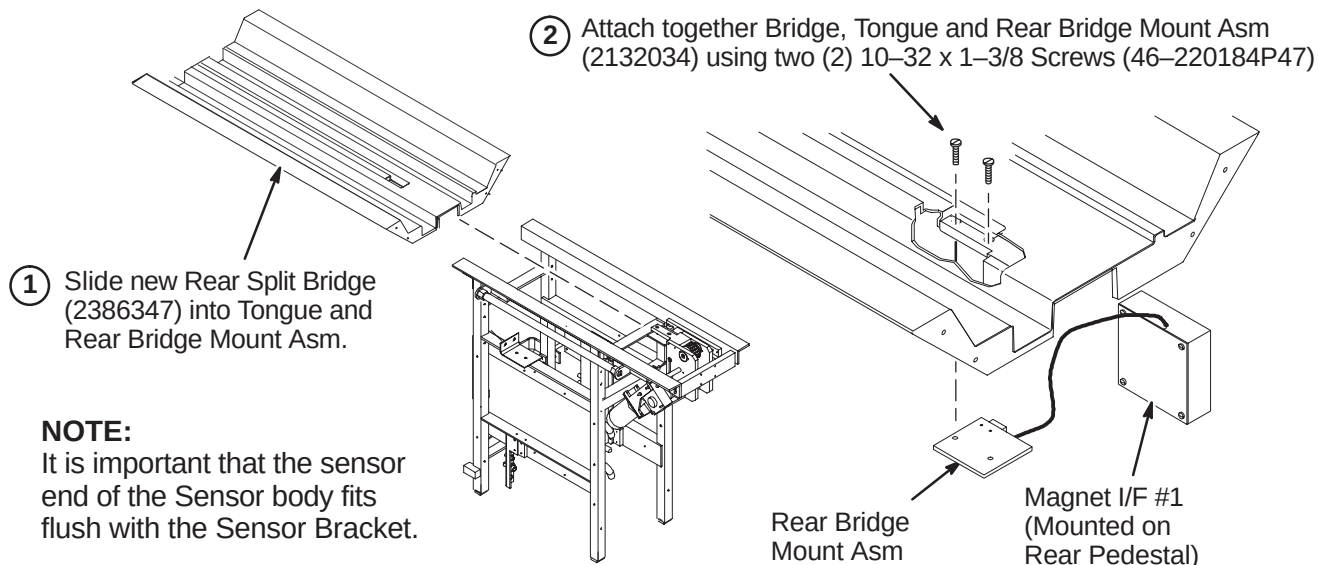
6-4-4 INSTALL REAR SPLIT BRIDGE

Prior to this upgrade, the site should have been audited to determine if the Rear Pedestal Stabilization Kit (5198961) has been installed. This kit includes parts and instructions for the installation of the kit. It would be required for earlier versions of the WideOpen style enclosure. If this upgrade has not been performed on the existing enclosure, then the applicable catalog should have been ordered and it must be installed at this point of the HDx upgrade.

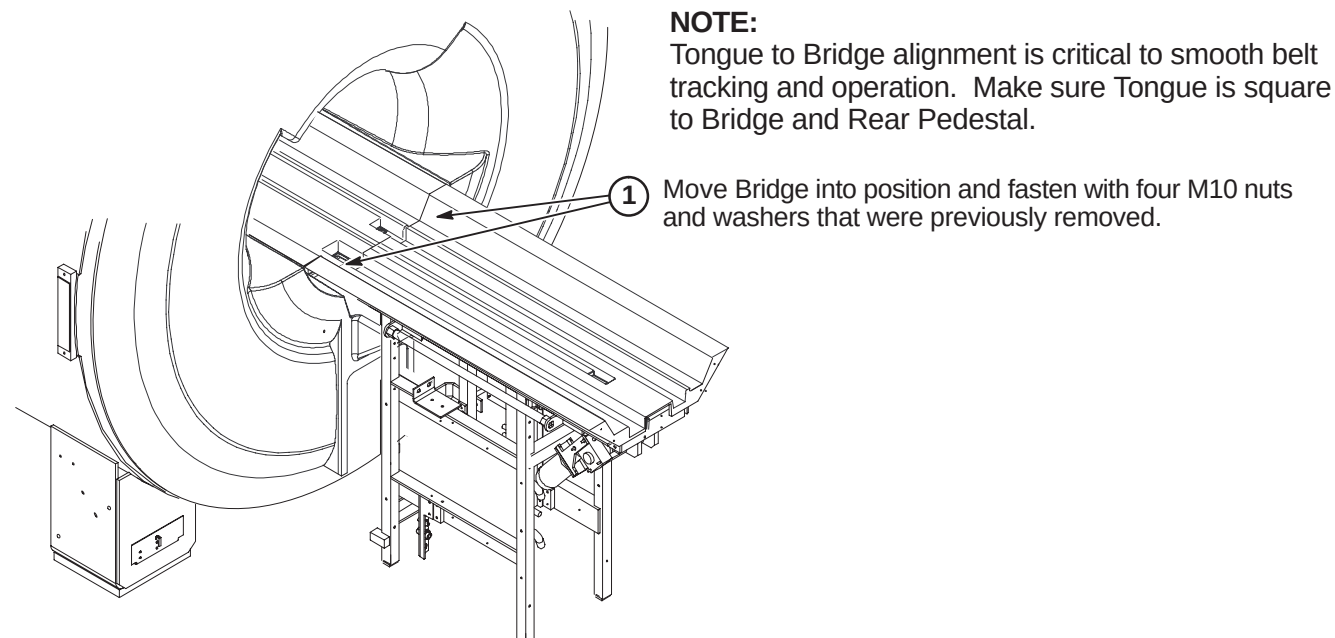
Refer to steering Direction 5244044, included in the kit, for guidance.

Note: If this procedure is not completed by Mechanical Installer or Van Partner, make sure that the GE Field Engineer has been notified that this upgrade procedure needs to be completed. For Van Partner, refer to the Design Review Checklist to report status of completion.

6-4-4-1 Install Rear Split Bridge on Rear Pedestal



6-4-4-2 Connect Rear Split Bridge to Front Split Bridge



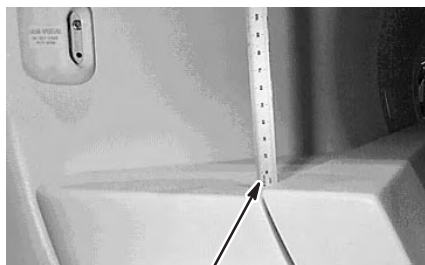
6-4-4-3 Level Bridge

NOTE:

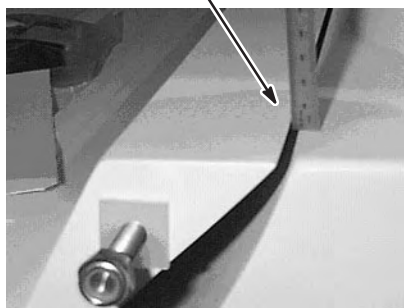
Make sure that surfaces between front and rear Bridge surfaces are even. This will provide smooth travel for the Carriage wheels.

NOTE:

Rear Pedestal must be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.

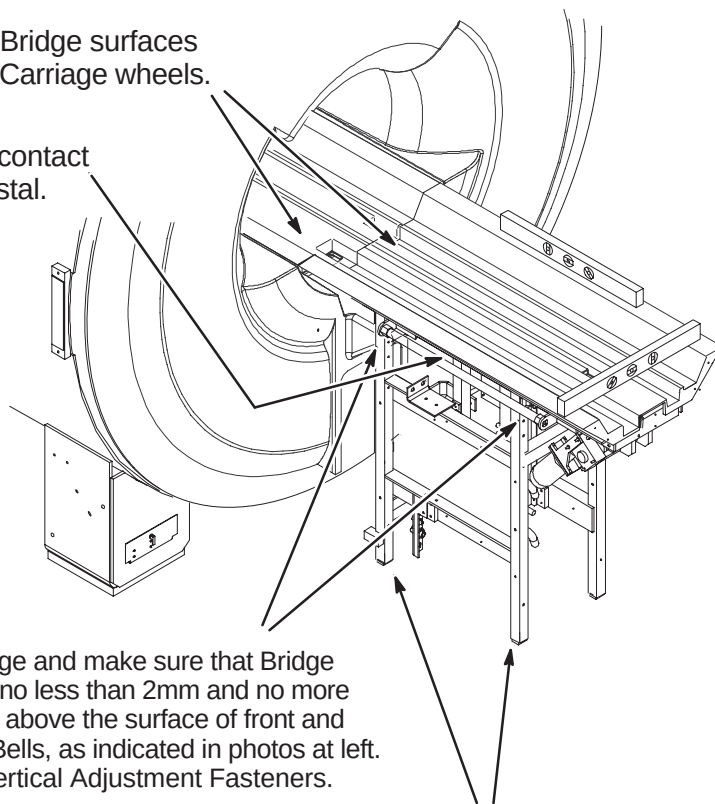


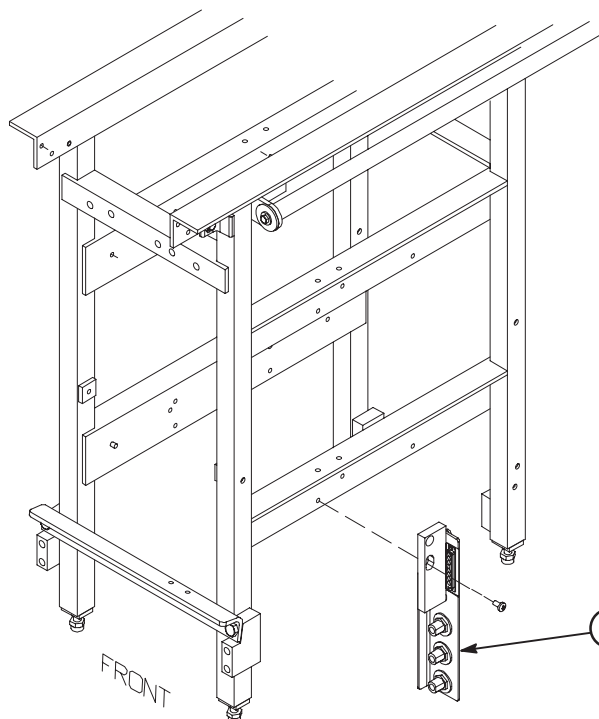
2mm to 4mm above End Bell surface.



- ① Level Bridge and make sure that Bridge surface is no less than 2mm and no more than 4mm above the surface of front and rear End Bells, as indicated in photos at left. Tighten Vertical Adjustment Fasteners.

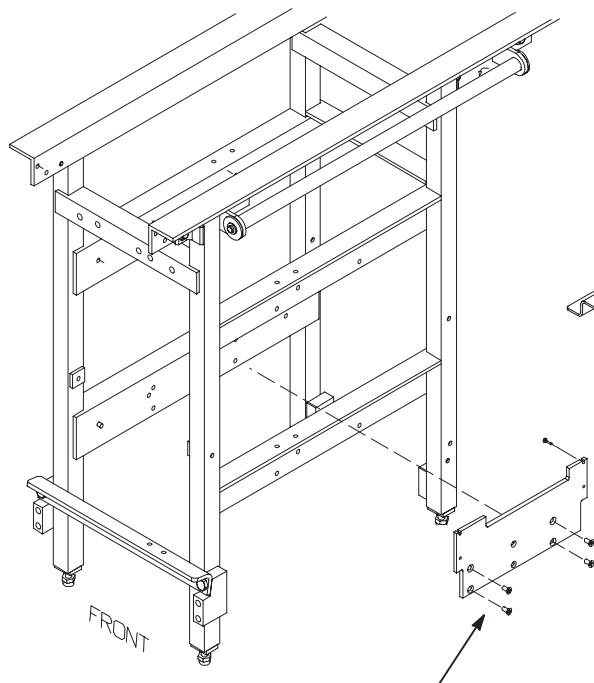
- ② Adjust leg screws on both sides as needed to level Rear Pedestal.



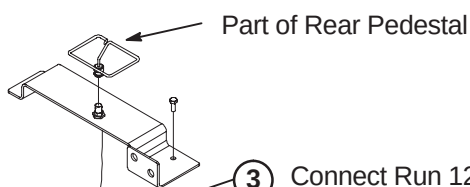
6-4-5 INSTALL BULKHEAD INTERFACE

NOTE: Components and cables not shown on Rear Pedestal for clarity.

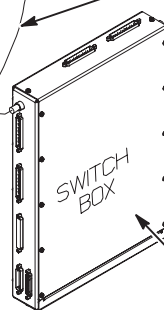
- ① Install new Bulkhead Interface (5113250) using Loctite 242 (46-170686P2) and two M6 x 12mm pane Head Screws (2109873-24).

6-4-6 INSTALL 16 CHANNEL SWITCH

NOTE: Components and cables not shown on Rear Pedestal for clarity.



- ① Install 16 Channel Switch Mounting Plate (5110428) using Loctite 242 (46-170686P2) and four M6 x 12mm Slotted Flat Head Screws (25120369-86) in the four larger outside holes.



- ③ Connect Run 1204 (5116105) as shown.
- ② Attach 16 Channel Switch (5109645) to Mounting Plate using Loctite 242 (46-170686P2) and four M3 x 12mm Pan Head screws (46-312342P24).

6-4-7 HYBRID SPLITTER UPGRADE

Depending on Rear Pedestal design, the Body Splitter may be in the horizontal or vertical position. Also:

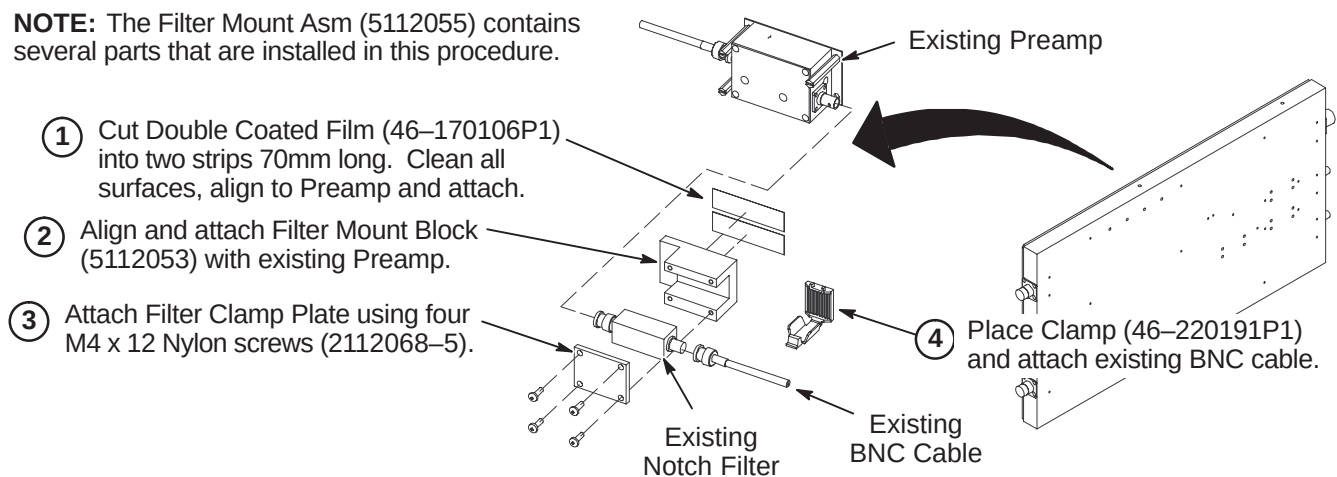
- Refer to Section 6-4-7-1 for systems that do not have mount assembly for the Notch Filter
- Refer to Section 6-4-7-2 for instructions on installing Isolation Filters on systems with Netcom Hybrid Splitter
- Refer to Section 6-4-7-3 for instructions on installing Isolation Filters on systems with Teledyne Splitter

The Ground Isolation Filter Modules are only installed on systems with BRM gradient coils. Do NOT install on systems with TRM gradient coils (TwinSpeed).

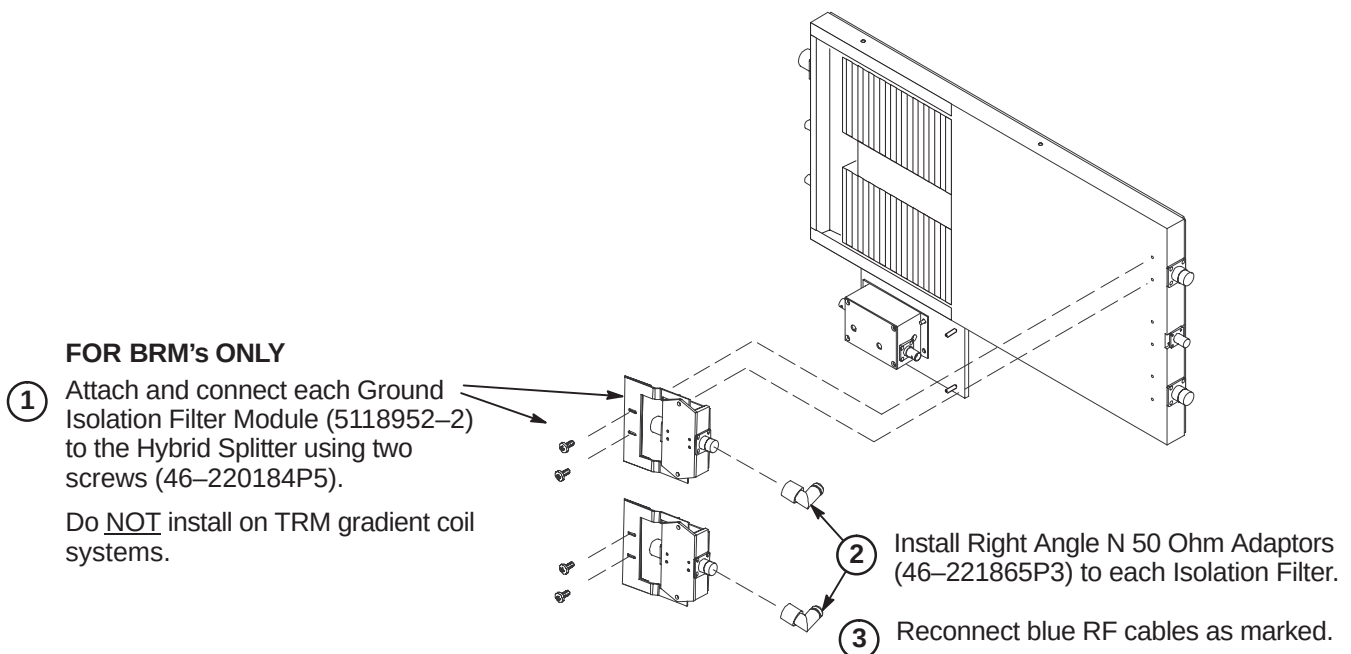
6-4-7-1 Install Filter Mount Asm to Body Hybrid Splitter

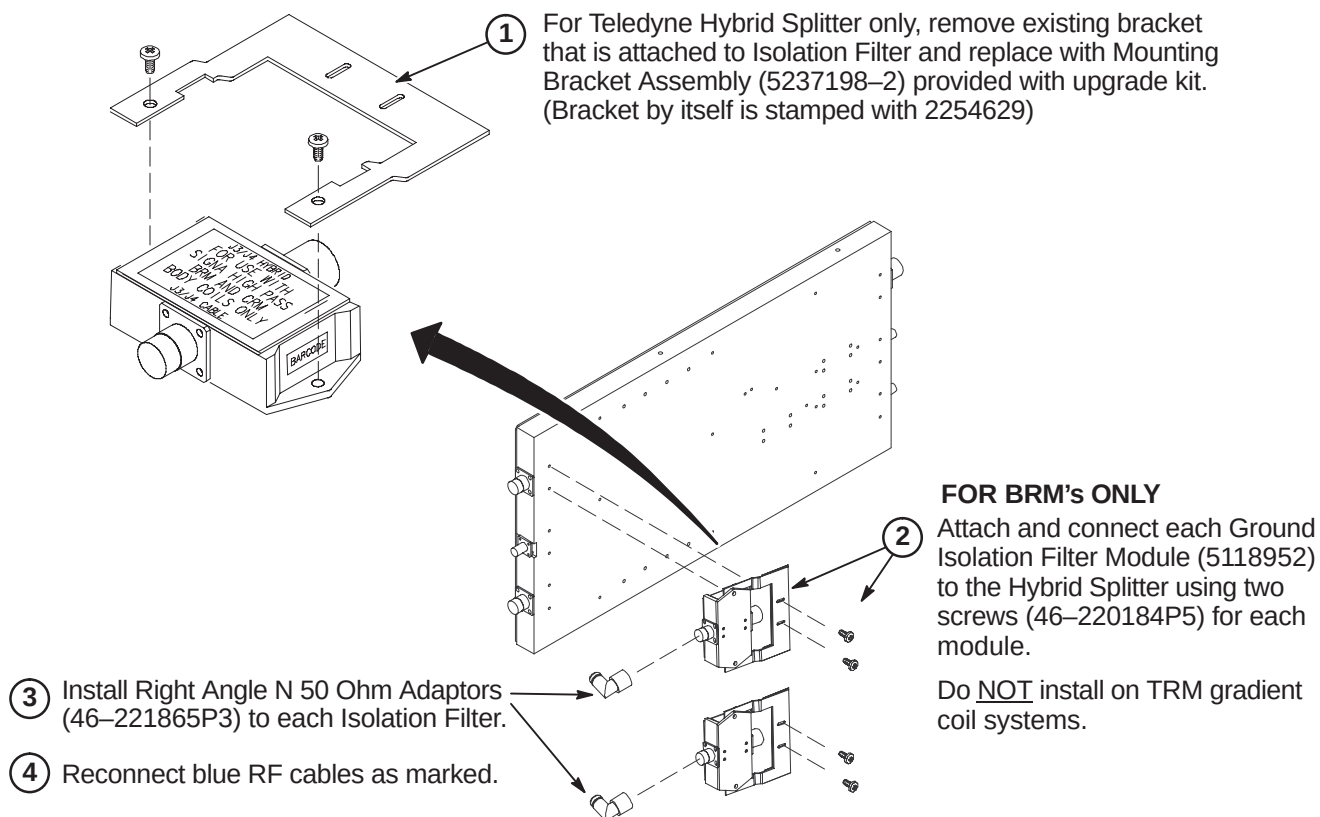
This Filter Mount assembly may not be required on newer Netcom Hybrid Splitter as it may already have a Notch Filter mounting clamp of a different style installed.

NOTE: The Filter Mount Asm (5112055) contains several parts that are installed in this procedure.



6-4-7-2 Install Isolation Filters to Netcom Body Hybrid Splitter (BRM Only)

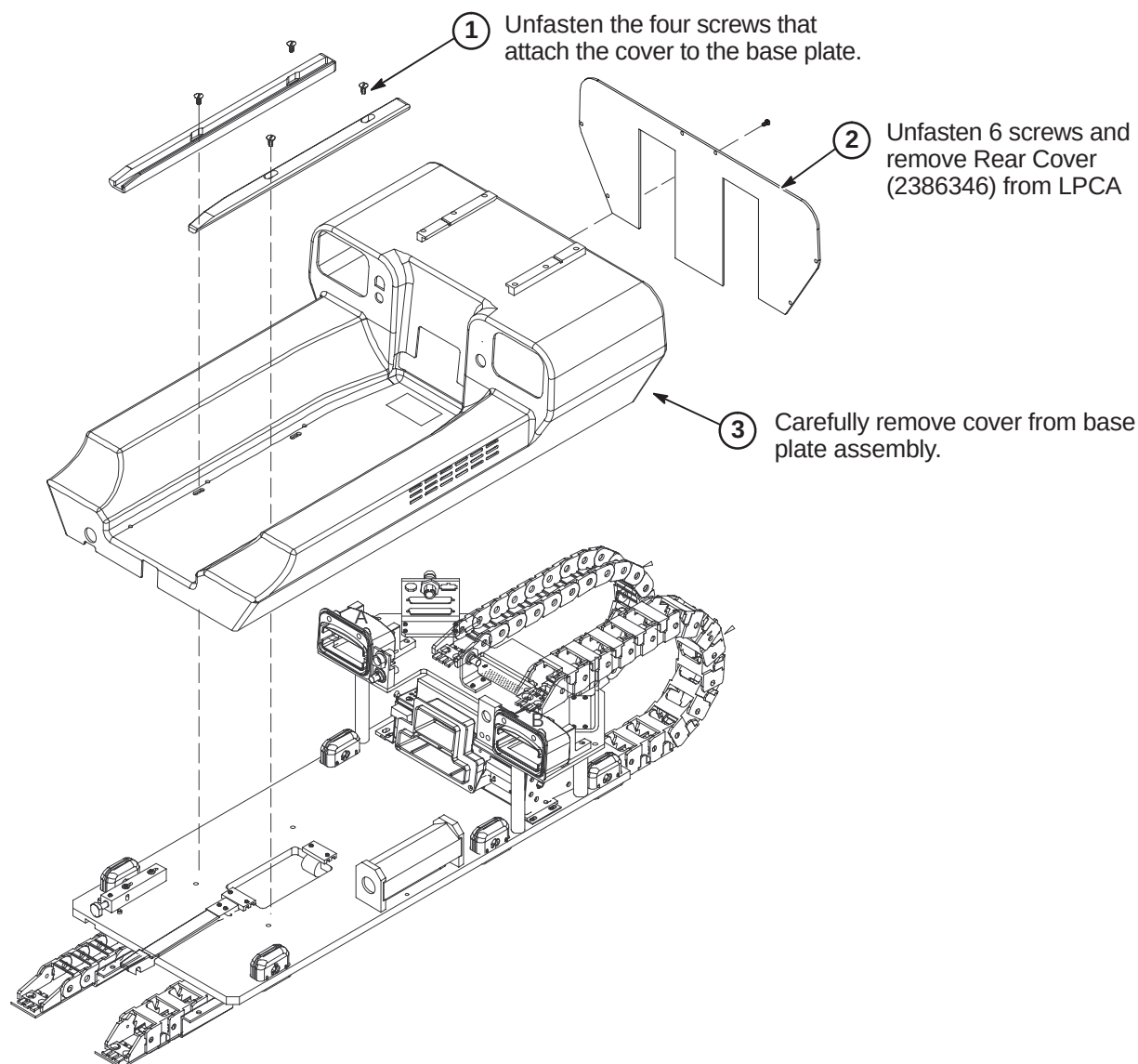


6-4-7-3 Install Isolation Filters to Teledyne Body Hybrid Splitter (BRM Only)

6-4-8 INSTALL LPCA AND CABLE TRACK(S)

The WideOpen style enclosure will be upgraded to an LPCA with a dual cable track. The S-Series and Cx magnet enclosures will install a new LPCA with only one cable track.

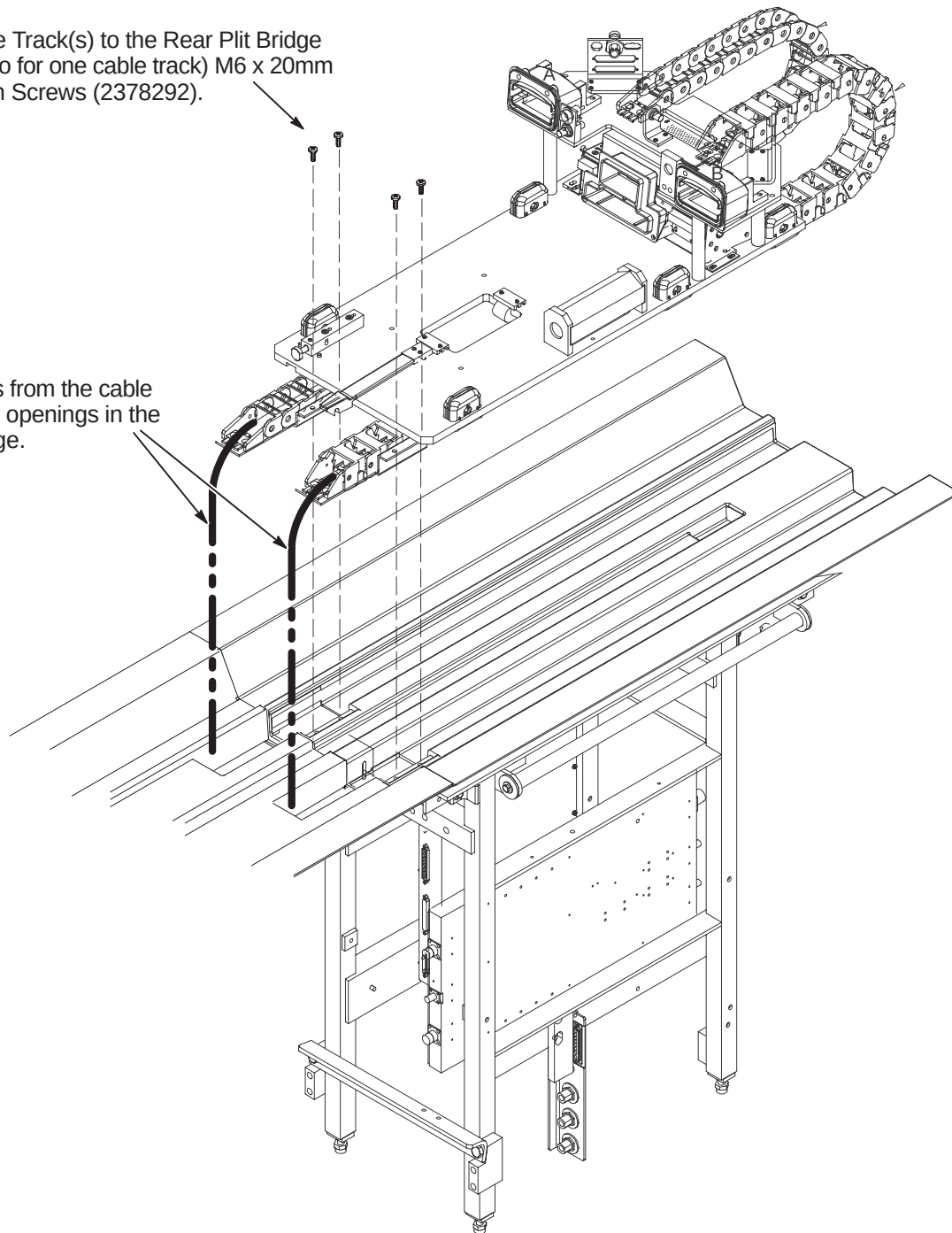
6-4-8-1 Remove LPCA Cover

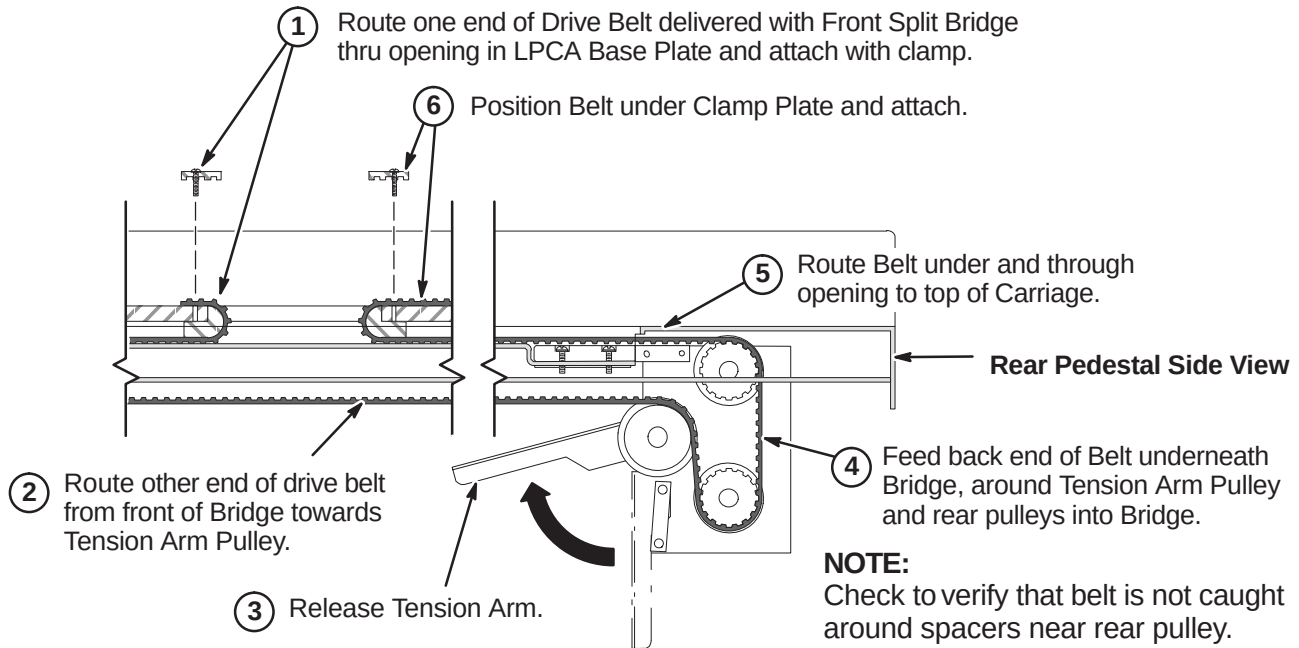
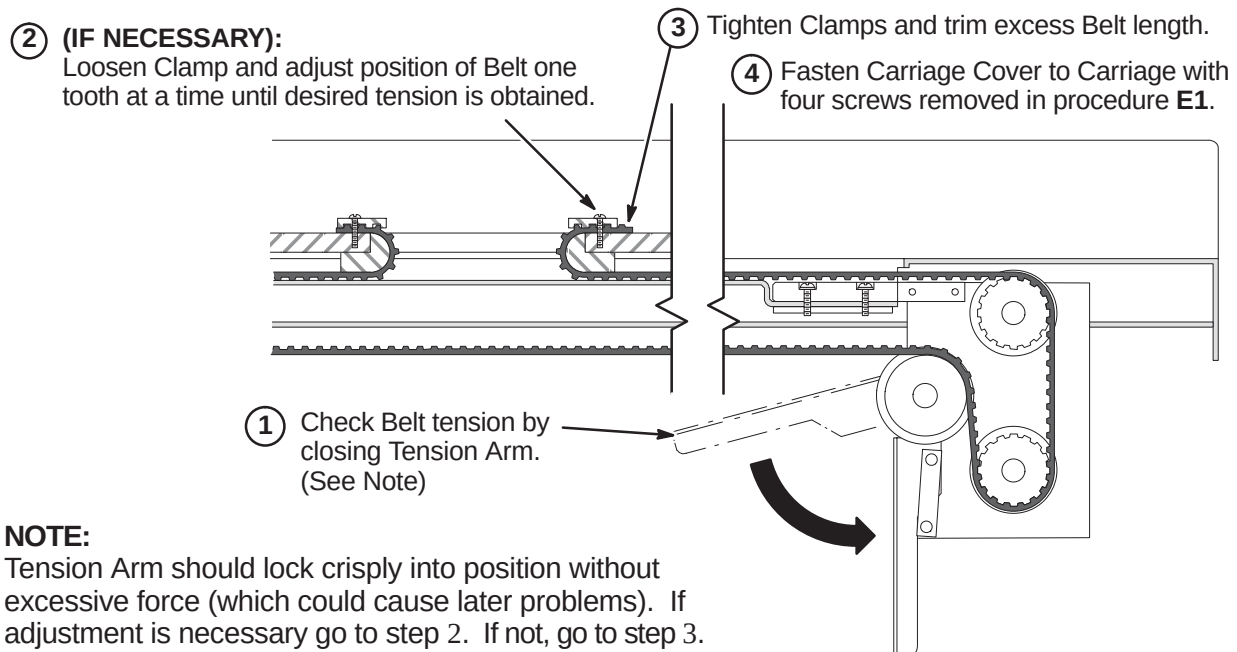


6-4-8-2 Attach Cable Track(s) to Bridge

- ② Attach the Cable Track(s) to the Rear Plit Bridge using four (or two for one cable track) M6 x 20mm Pan Head Nylon Screws (2378292).

- ① Feed the cables from the cable track(s) thru the openings in the Front Split Bridge.



6-4-8-3 Route and Attach Drive Belt**6-4-8-4 Final Adjustment of Drive Belt**

6-4-8-5 Installation of Sensor Flag

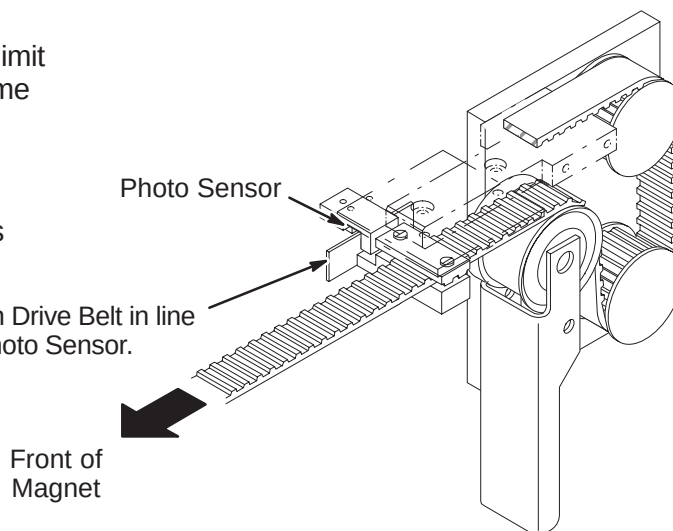
NOTE:

Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:

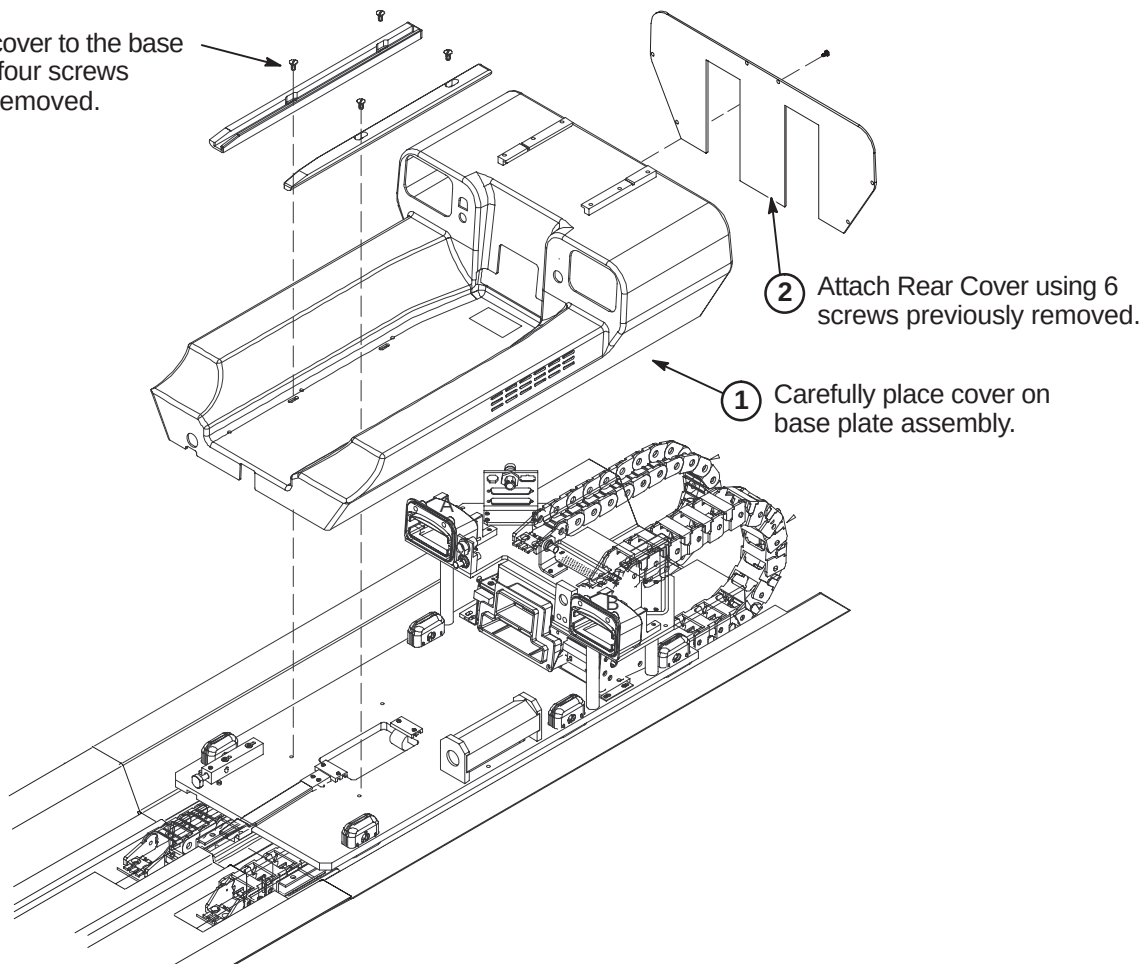
The following Step is needed if flag was removed during routing of drive belt.

- ① Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.



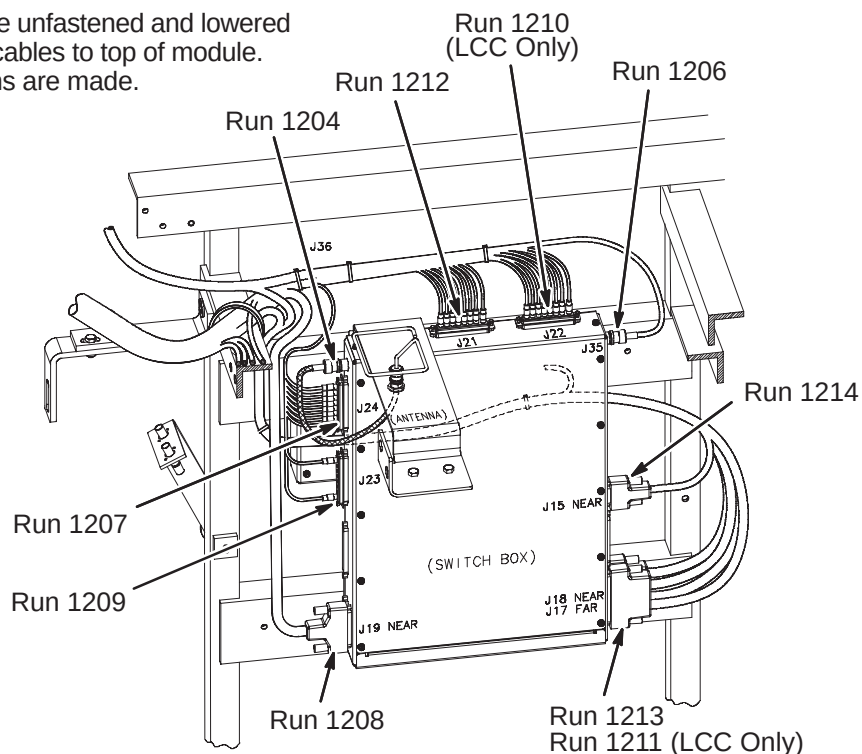
6-4-8-6 Installation of LPCA Cover

- ③ Attach the cover to the base plate using four screws previously removed.

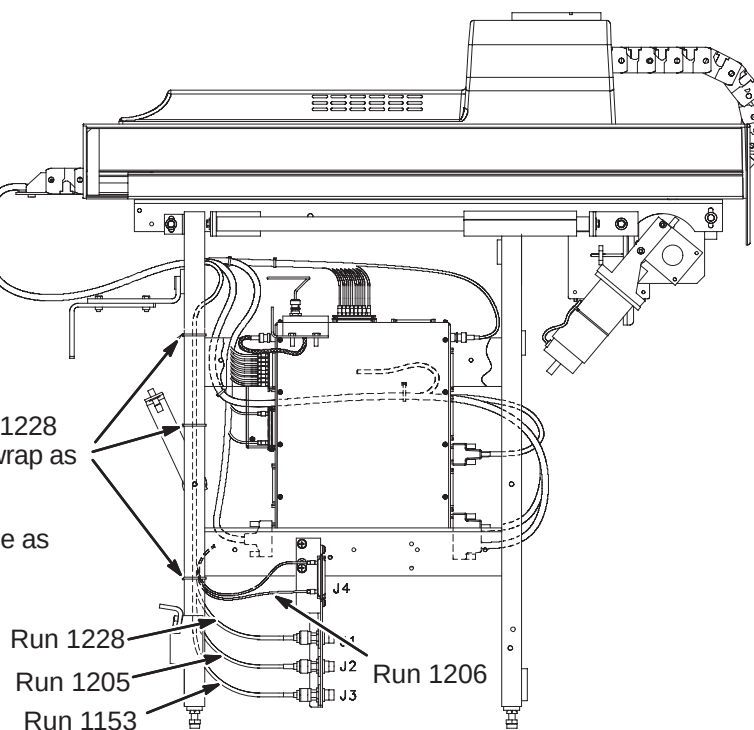


6-4-8-7 Connect Cables from Cable Track to 16 Channel Switch Module

- ① 16 Channel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re-mount module after connections are made.
- ② Connect cable Runs as indicated.

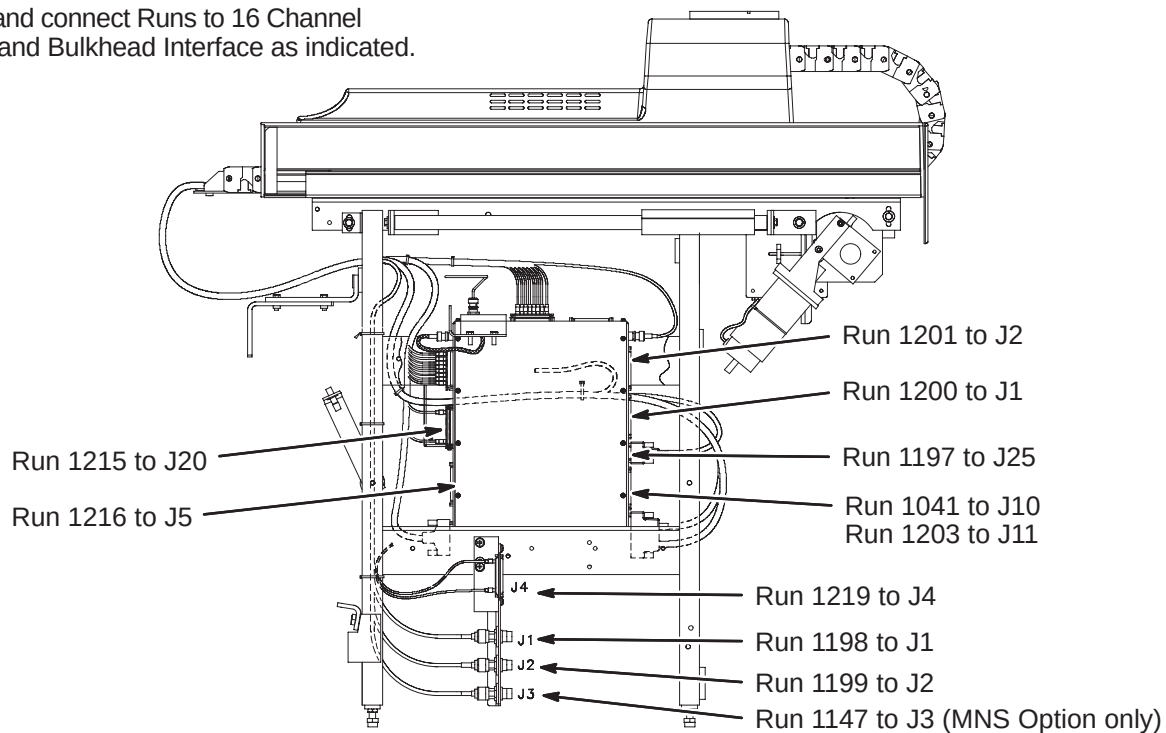
**6-4-8-8 Connect Cables from Cable Track to Bulkhead Interface**

- ① Route Runs 1153, 1205, 1206, and 1228 along leg of Rear Pedestal and ty-wrap as indicated to leg.
- ② Connect Runs to Bulkhead Interface as indicated.



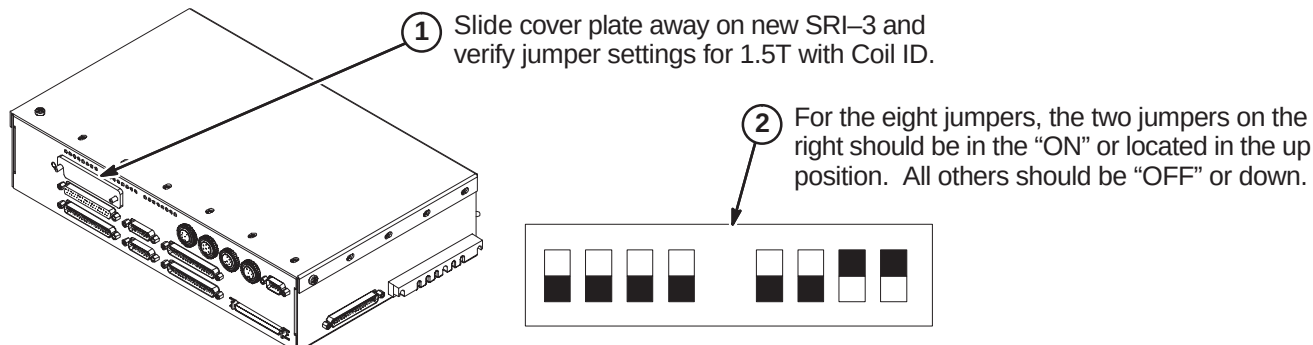
6-4-9 CONNECT REMAINING CABLES TO REAR PEDESTAL

- ① Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.

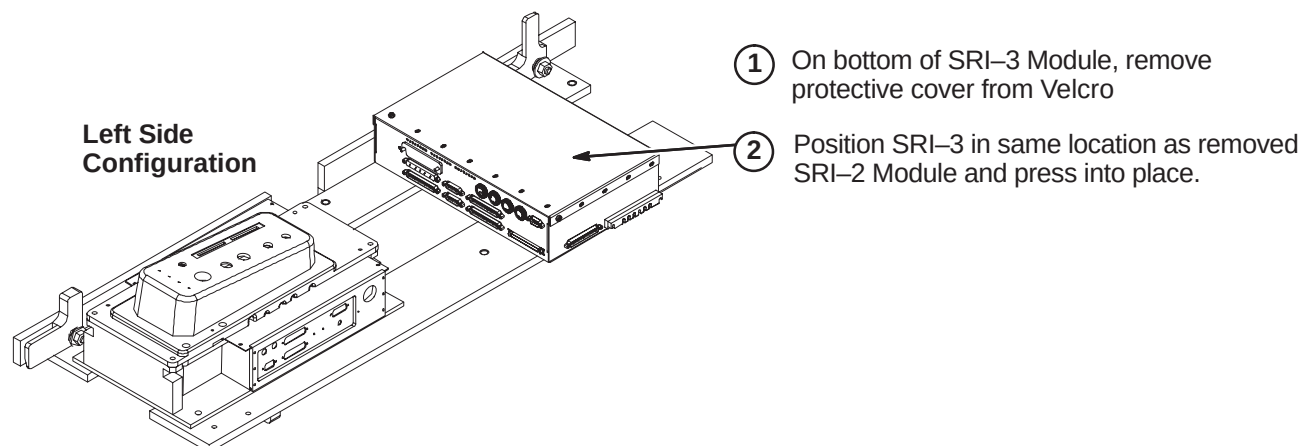


6-4-10 INSTALL SRI-3 MODULE

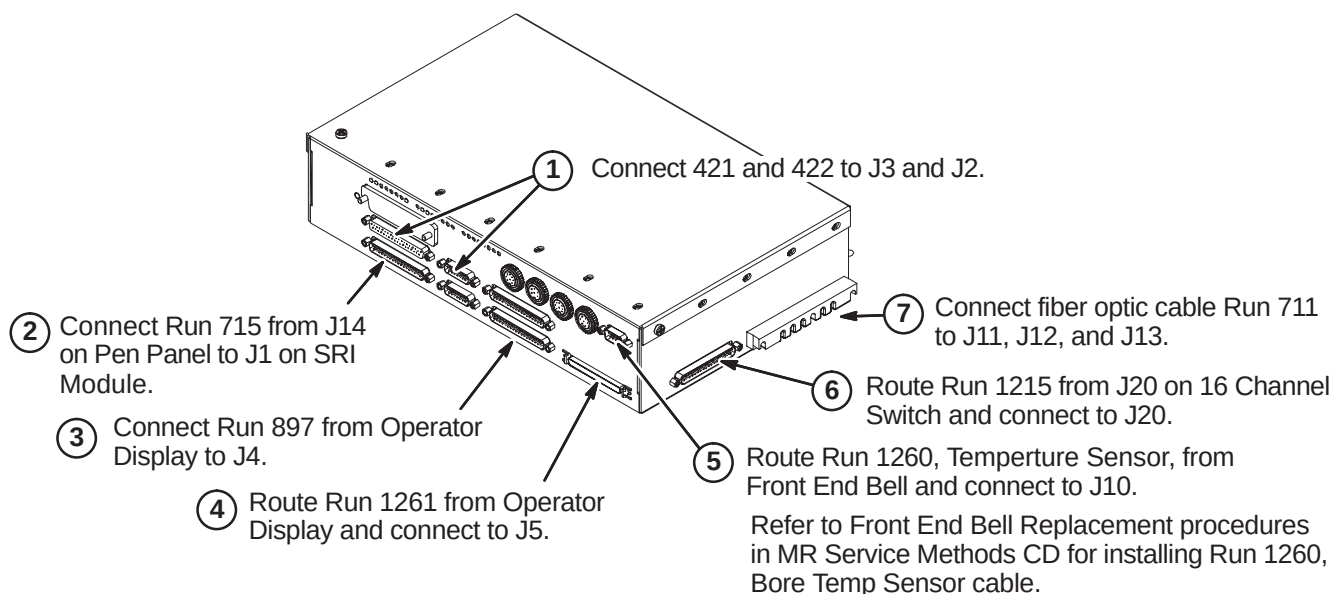
6-4-10-1 Verify SRI-3 Jumper Settings



6-4-10-2 Position Module on Platform

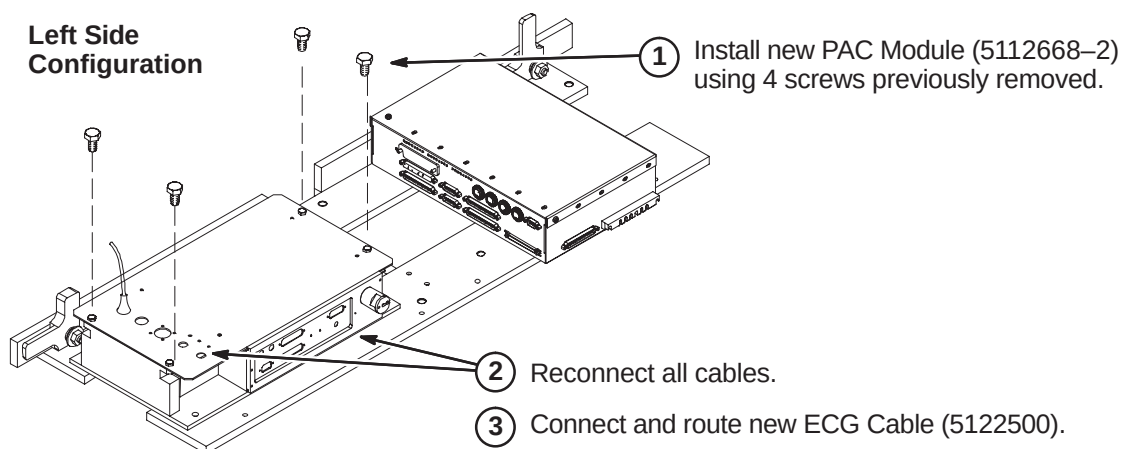
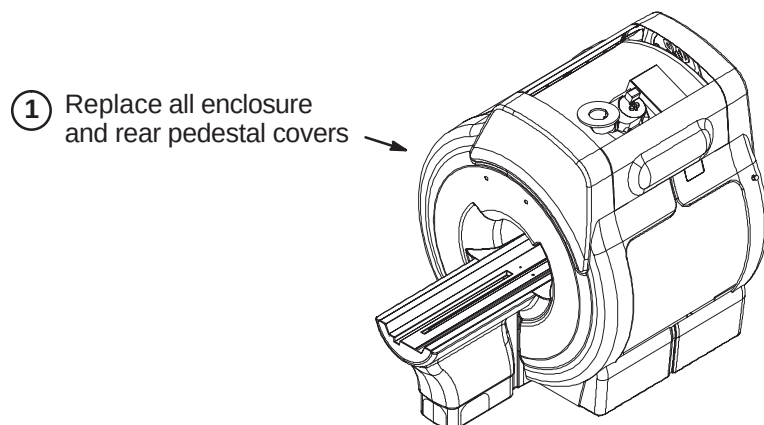


6-4-10-3 Connect Cables to SRI-3 Module



6-4-11 INSTALL PAC MODULE AND ECG CABLE

Reverse the deinstallation steps for installation of the new PAC Module. Also install the new ECG Cable.

**6-4-12 INSTALL COVERS**

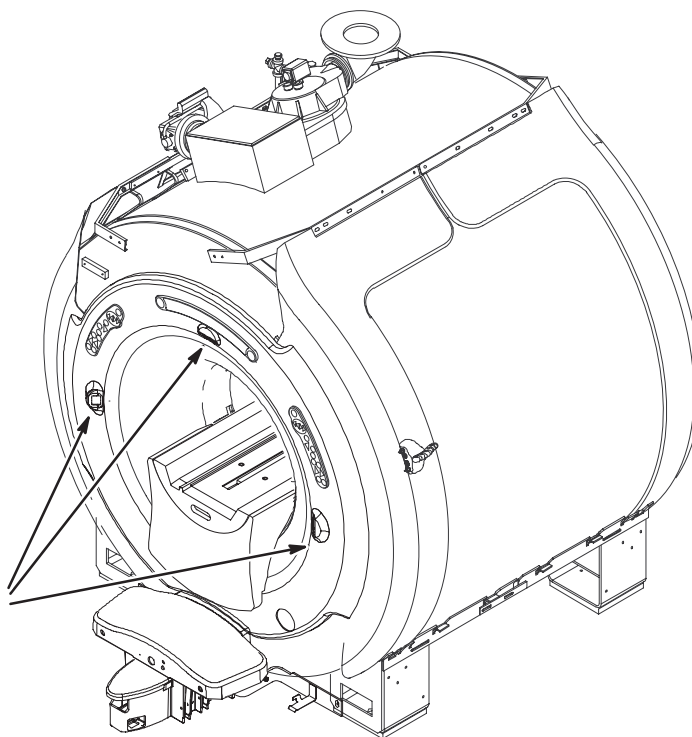
6-4-13 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS

A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below each of the three laser lights.



English



- ① For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.



French



German



Portuguese



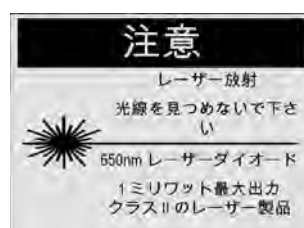
Spanish



Italian



Swedish

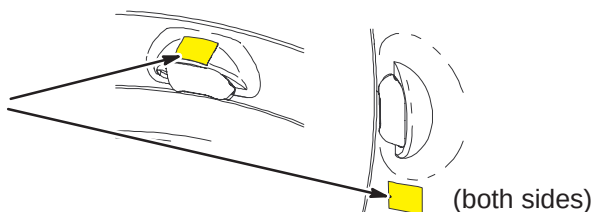


Japanese



Chinese

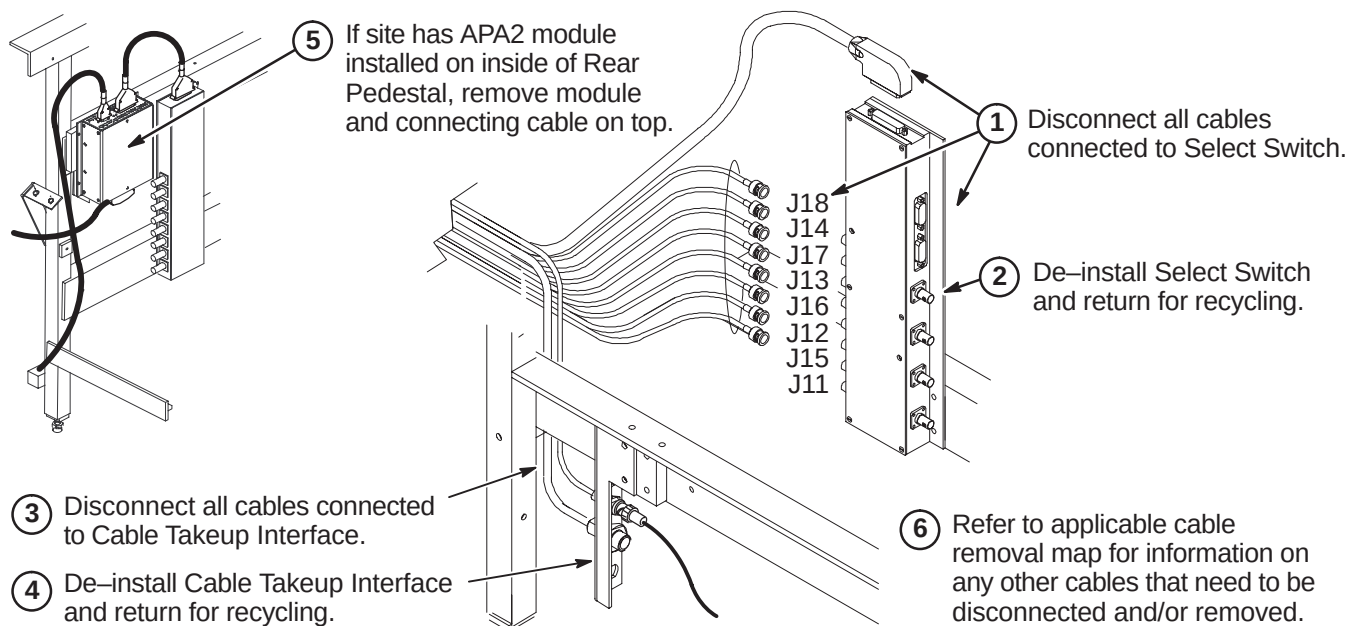
- ② If enclosure does not have labels attached, attach new labels as indicated.



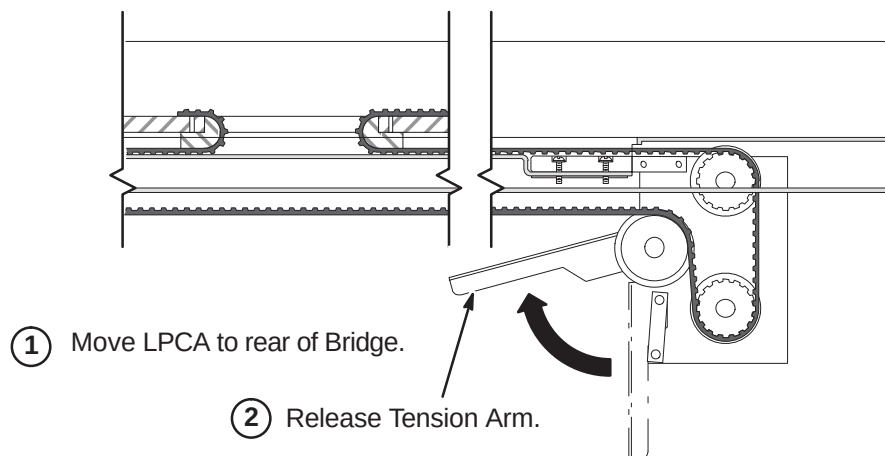
6-5 UPGRADE PROCEDURES FOR HORIZON STYLE MAGNET ENCLOSURES

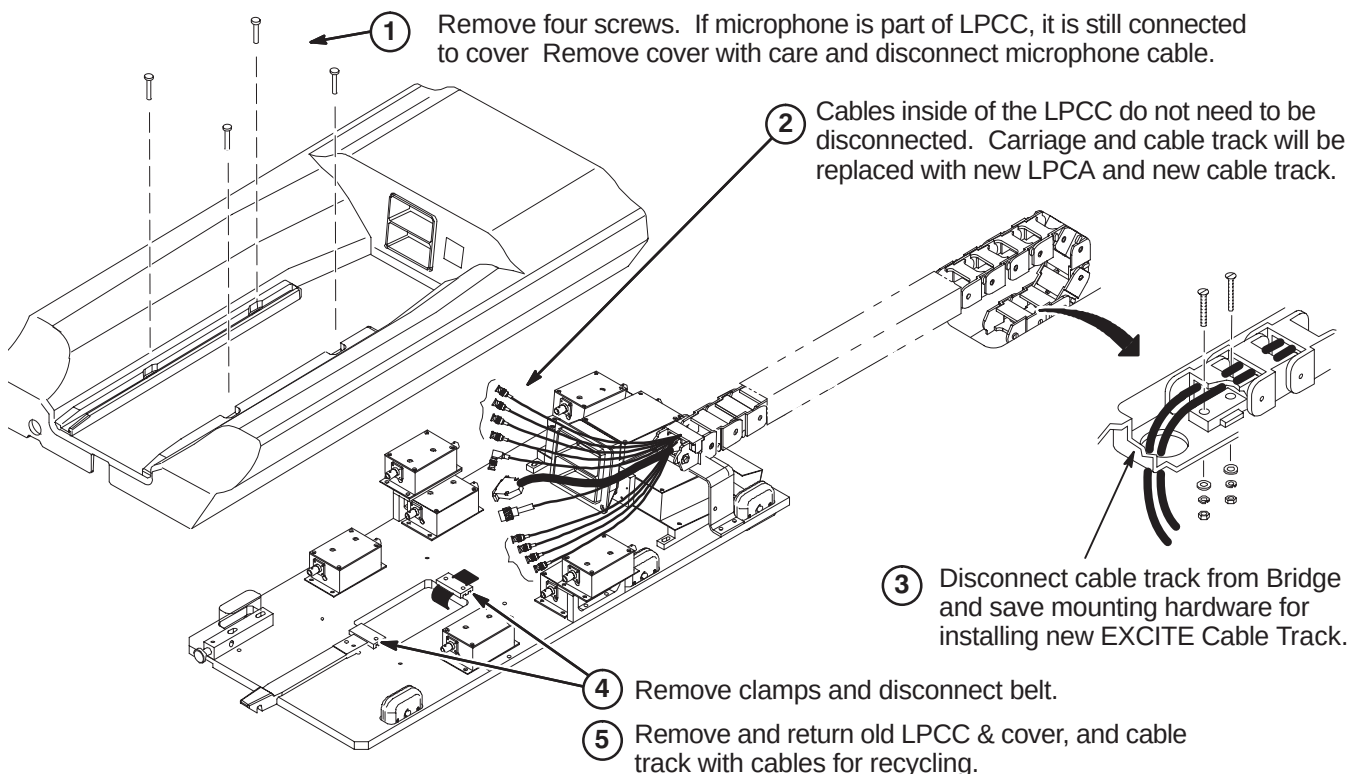
6-5-1 DE-INSTALL EXISTING LPCA & CABLE TRACK

6-5-1-1 Disconnect Cables In Rear Pedestal

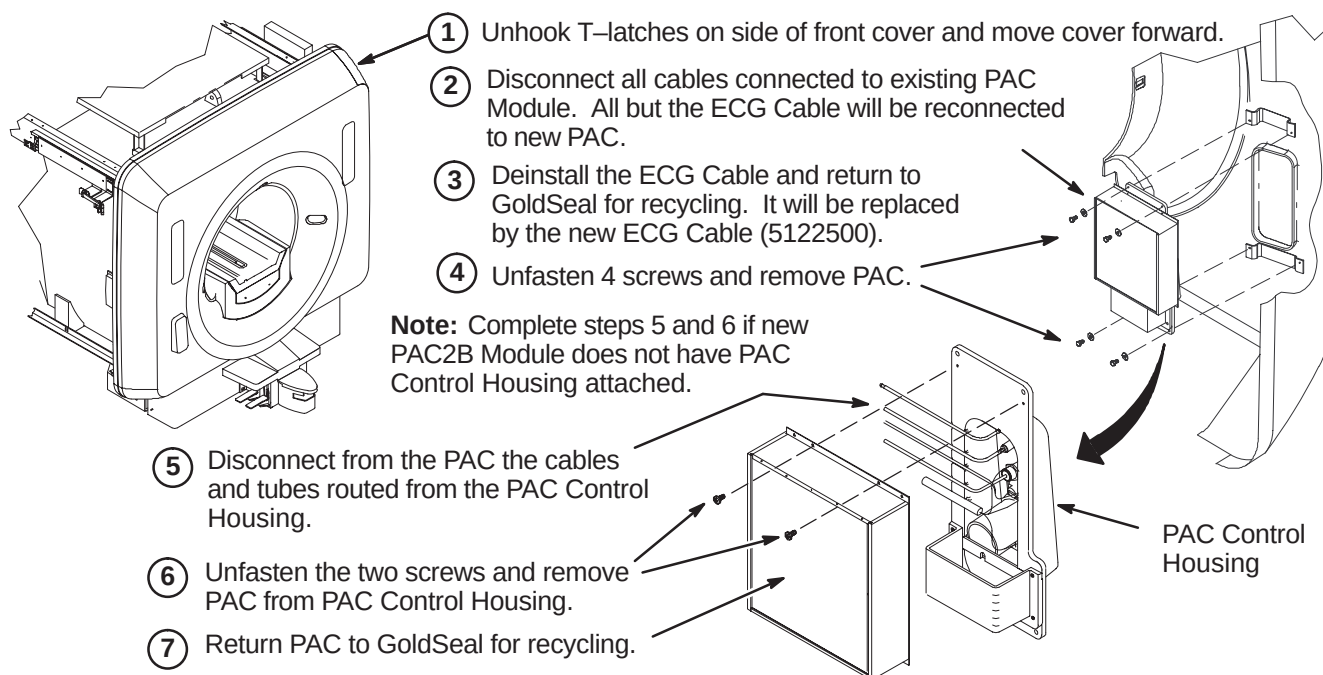


6-5-1-2 Release Tension Arm Under Rear Pedestal



6-5-1-3 Remove Existing Low Profile Carriage Cover & Cable Track**6-5-2 DE-INSTALL PAC MODULE**

The existing PAC Module will be replaced by a new PAC Module with independent vector gating.



6-5-3 DE-INSTALL SRI MODULE

Section 6-5-3-1 contains instructions for removing the existing SRI module on an S-Series magnet.

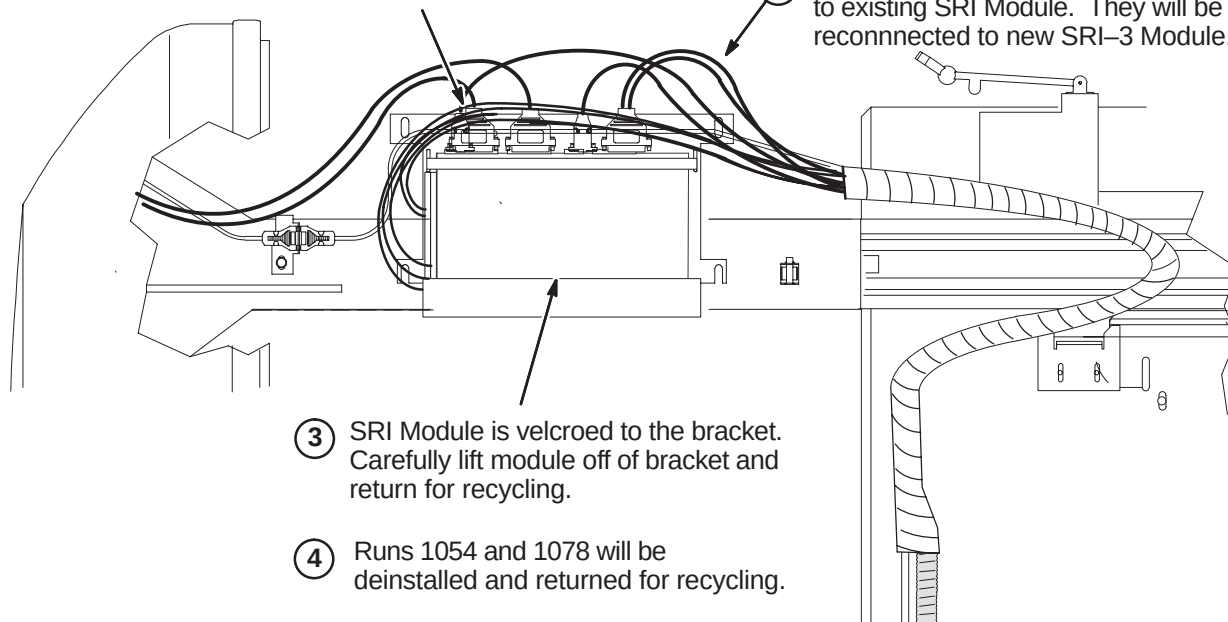
Section 6-5-3-2 contains instructions for removing the existing SRI module on an Cx magnet.

6-5-3-1 Remove SRI Module for S-Series Enclosure

- ① Runs 749 and 1011 must be deinstalled and returned for recycling.

NOTE: New Runs 1260 and 1261 will be installed.

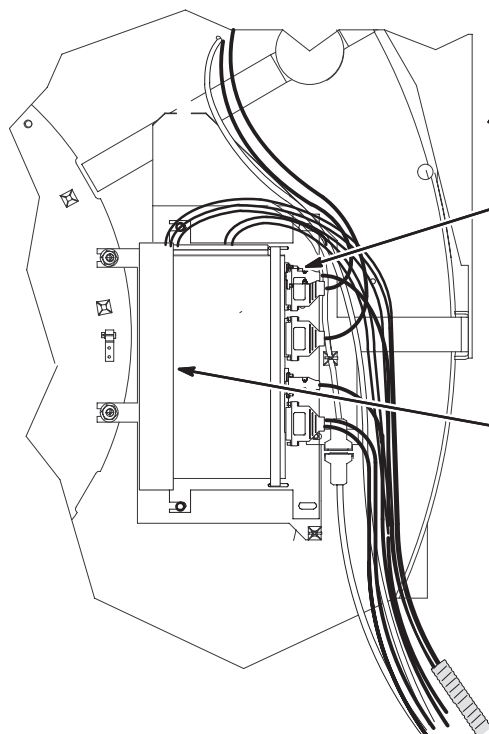
- ② Disconnect remaining cables connected to existing SRI Module. They will be reconnected to new SRI-3 Module.



- ③ SRI Module is velcroed to the bracket. Carefully lift module off of bracket and return for recycling.

- ④ Runs 1054 and 1078 will be deinstalled and returned for recycling.

6-5-3-2 Remove SRI Module for Cx or LCC Horizon Enclosure



- ① Disconnect all cables connected to existing SRI Module.

- ② Runs 749 and 1011 must be deinstalled and returned for recycling.

NOTE: New Runs 1260 and 1261 will be installed.

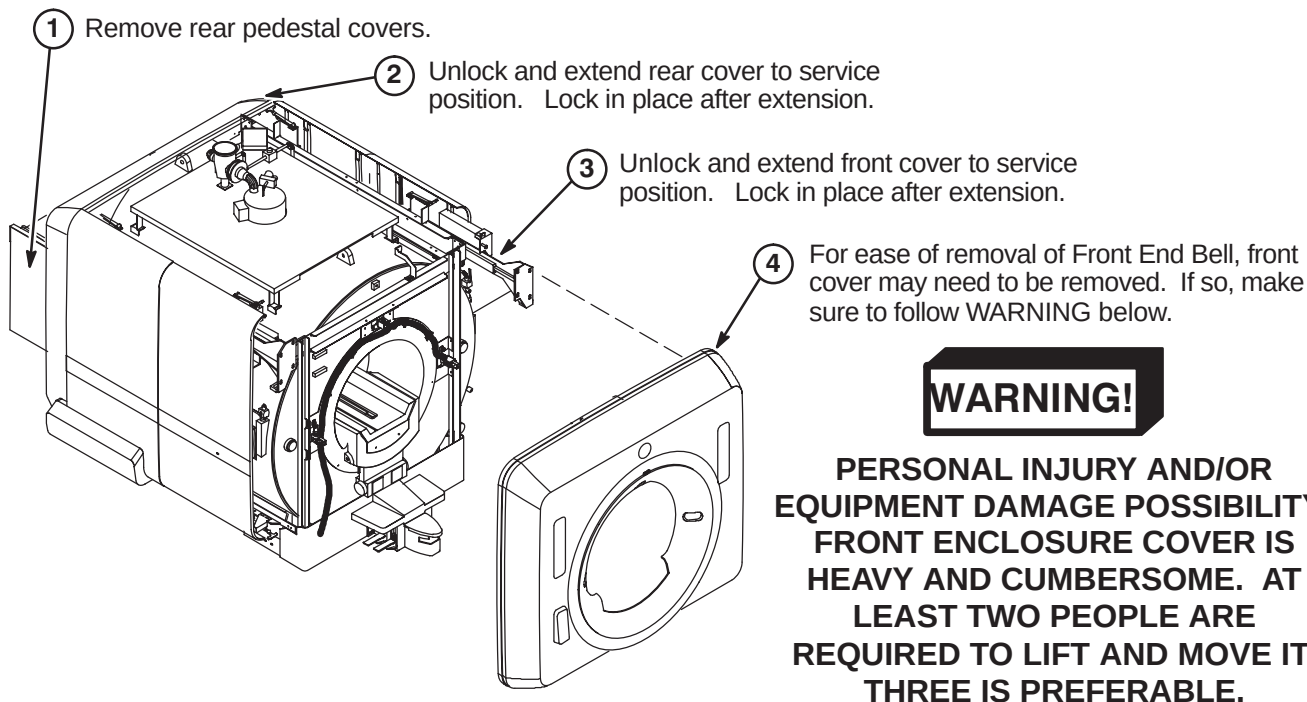
- ③ Runs 1054 and 1078 will be deinstalled and returned for recycling.

- ④ SRI Module is velcroed to the bracket. Carefully lift module off of bracket and return for recycling.

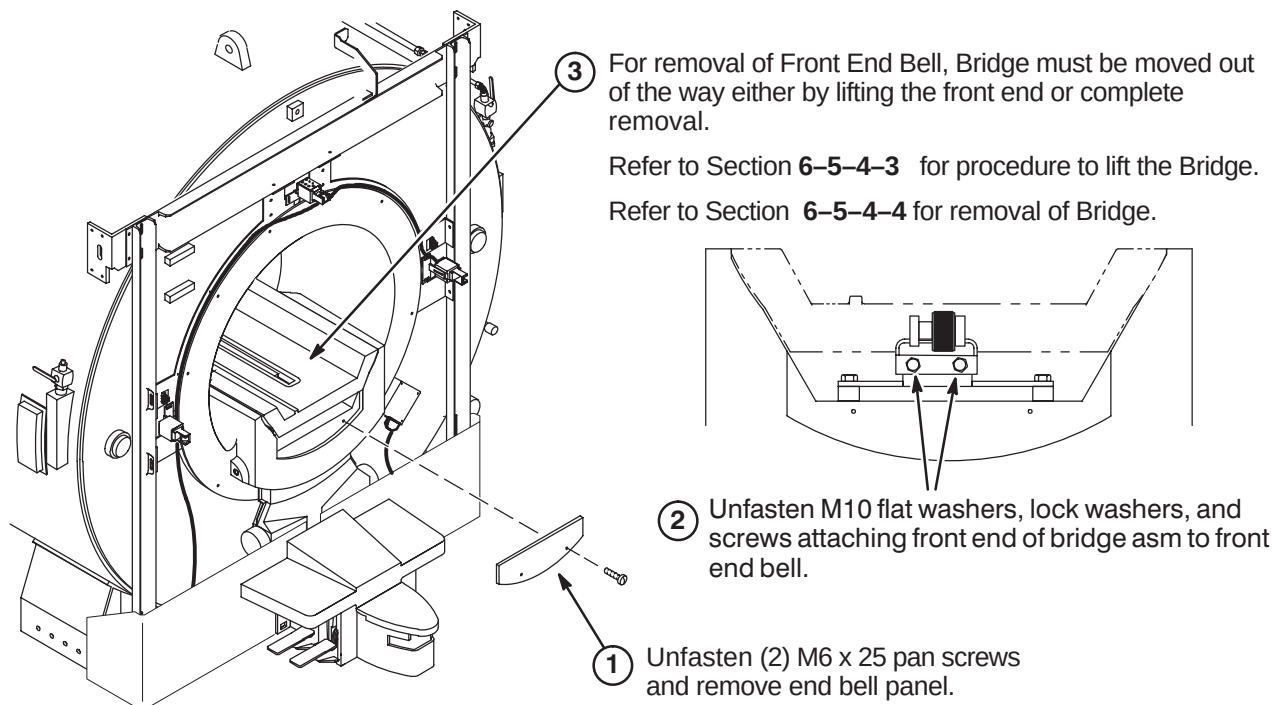
6-5-4 PREPARE END BELL FOR INSTALLATION OF NEW DUAL THERMAL SENSOR CABLE

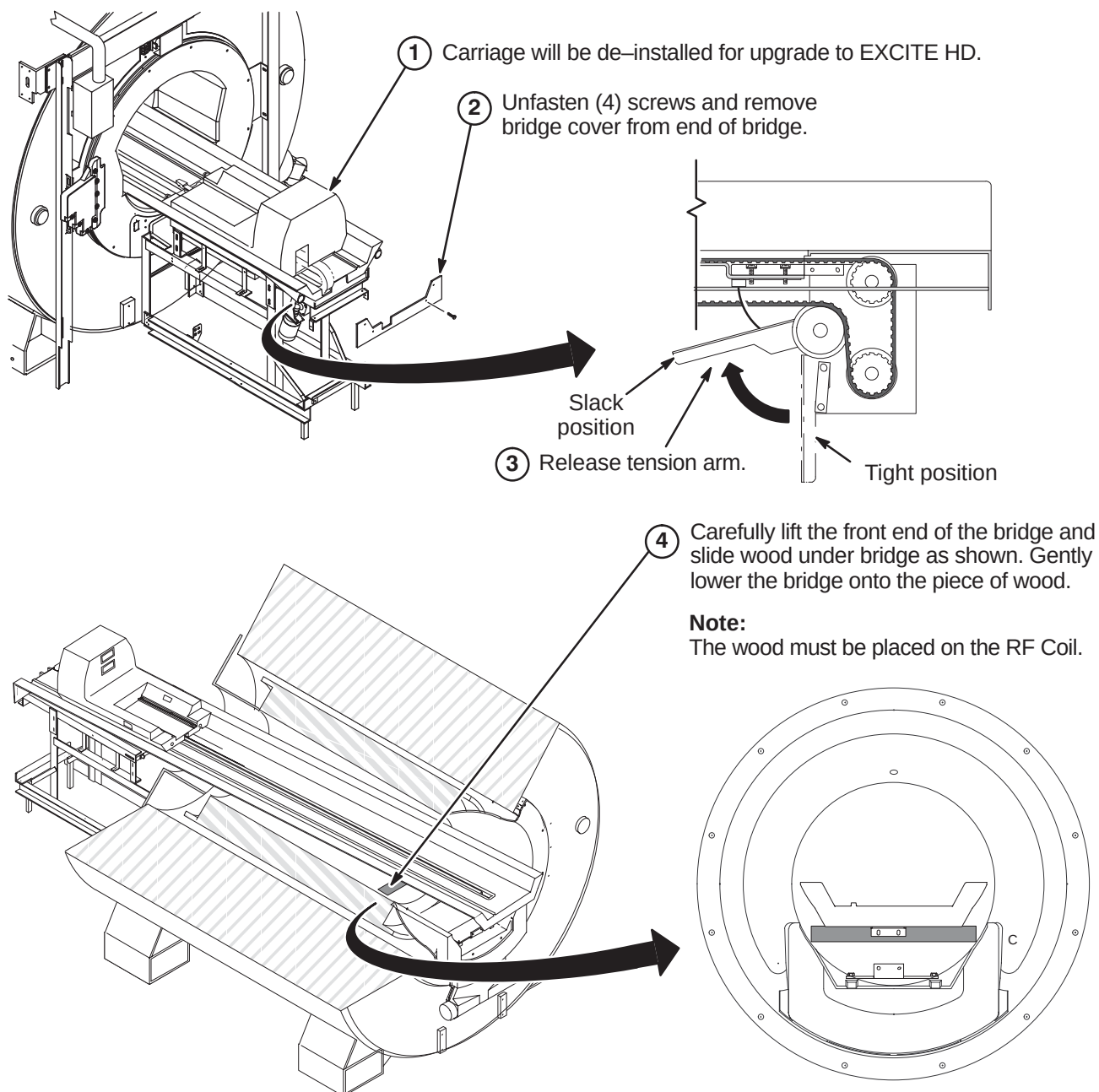
The Bridge needs to be repositioned or removed prior to removal of Front End Bell. Review sections 6-5-4-3 and 6-5-4-4. Determine which method would best fit the needs of site being upgraded.

6-5-4-1 Removal of Enclosure Covers



6-5-4-2 Removal of Front End Bell Panel

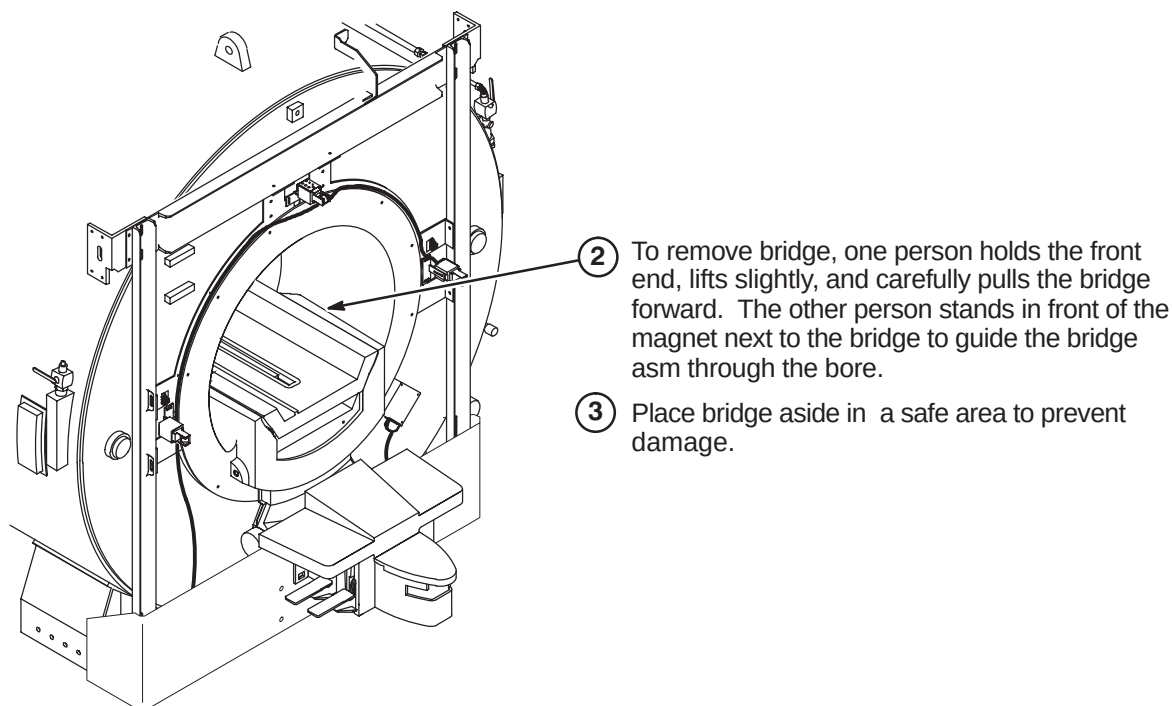
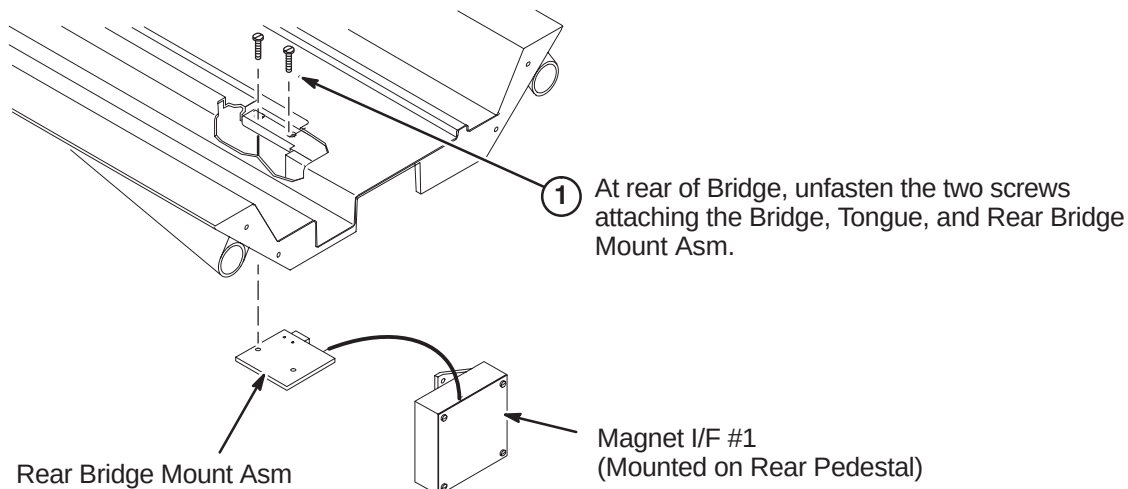


6-5-4-3 Lifting Front End Of Bridge Method

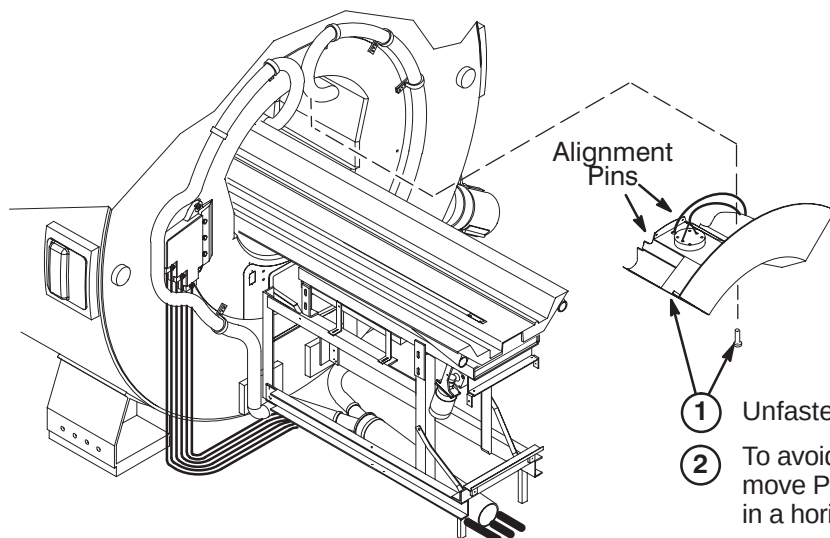
6-5-4-4 Removal of Bridge Method



Possible equipment damage! Be careful not to damage the home flag on the drive belt when removing or replacing the bridge. The home flag is fragile and can easily be damaged.



L

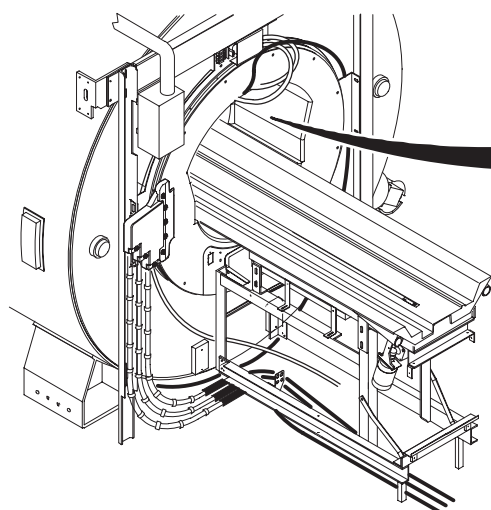
6-5-4-5 Remove Patient Comfort Module**Note:**

Hold Patient Comfort Module in place until mounting screws in Step 1 are removed. Alignment Pins can break off during removal if over-stressed.

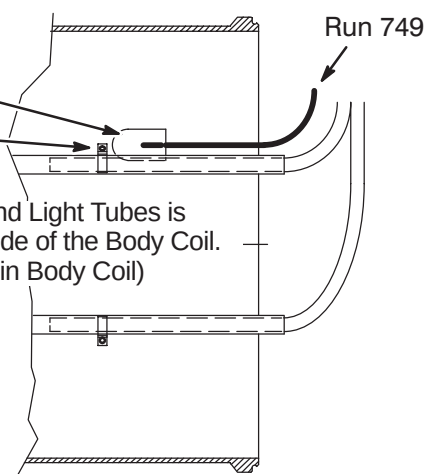
The site can have the existing Temperature Sensor Cable routed to the Rear end bell (earlier location) or Front end bell (later location). Either way, the cable needs to be removed and replaced with new cable, Run 1260. One of the two following procedures will be applicable to your site.

6-5-4-6 Remove Existing Single Temperature Sensor Cable at Rear End Bell Location

- ① Unfasten existing Temperature Sensor Cable (Run 749) from top inside of Body Coil. It may be attached with:
adhesive fastener (as shown)
OR
with the Light and Cable clamp



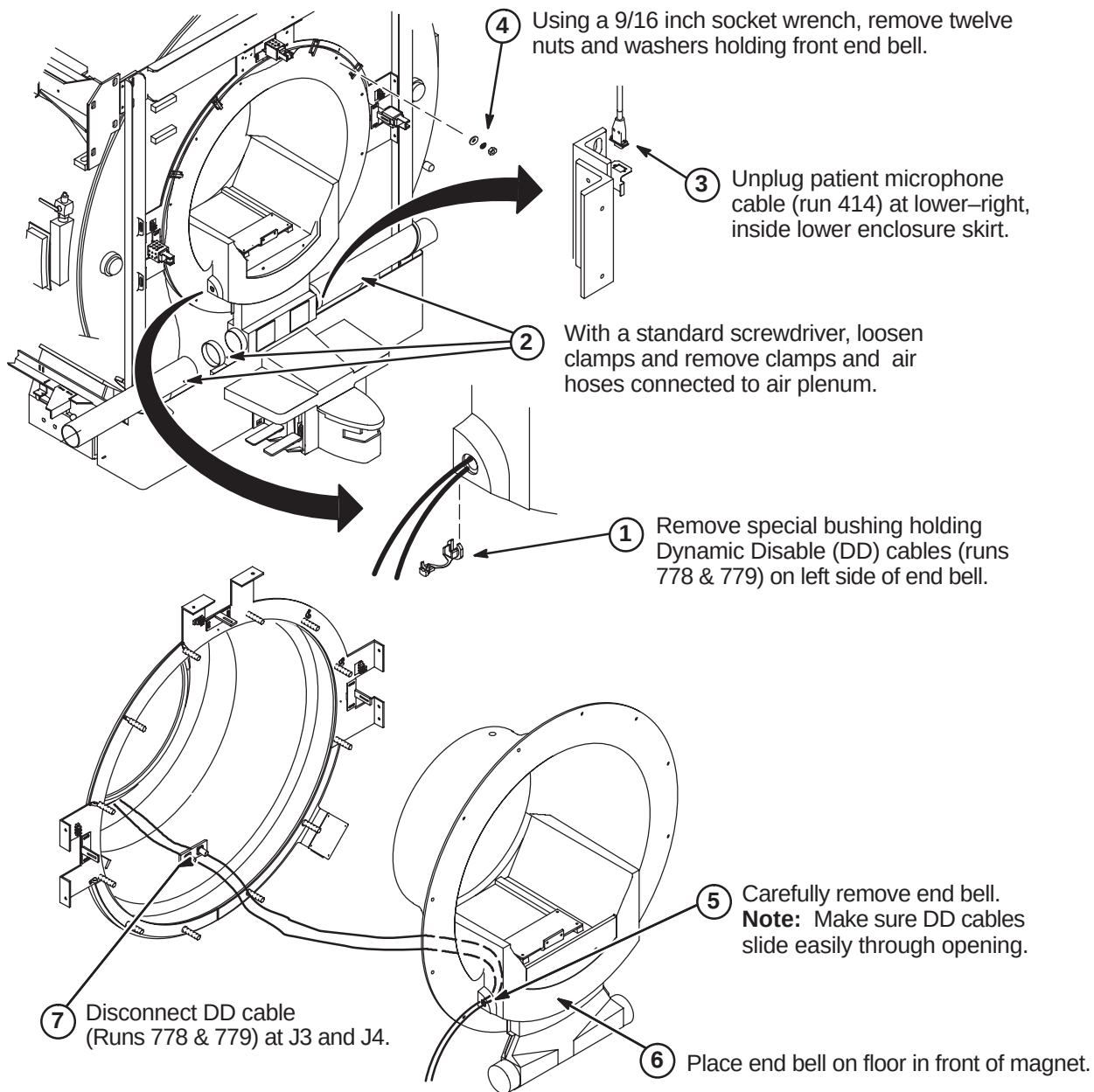
Note: View of Sensor and Light Tubes is looking up at the top inside of the Body Coil. (visible to patient laying in Body Coil)



- ② Disconnect other end of Run 749 from J10 on SRI module.
- ③ Remove and discard cable.

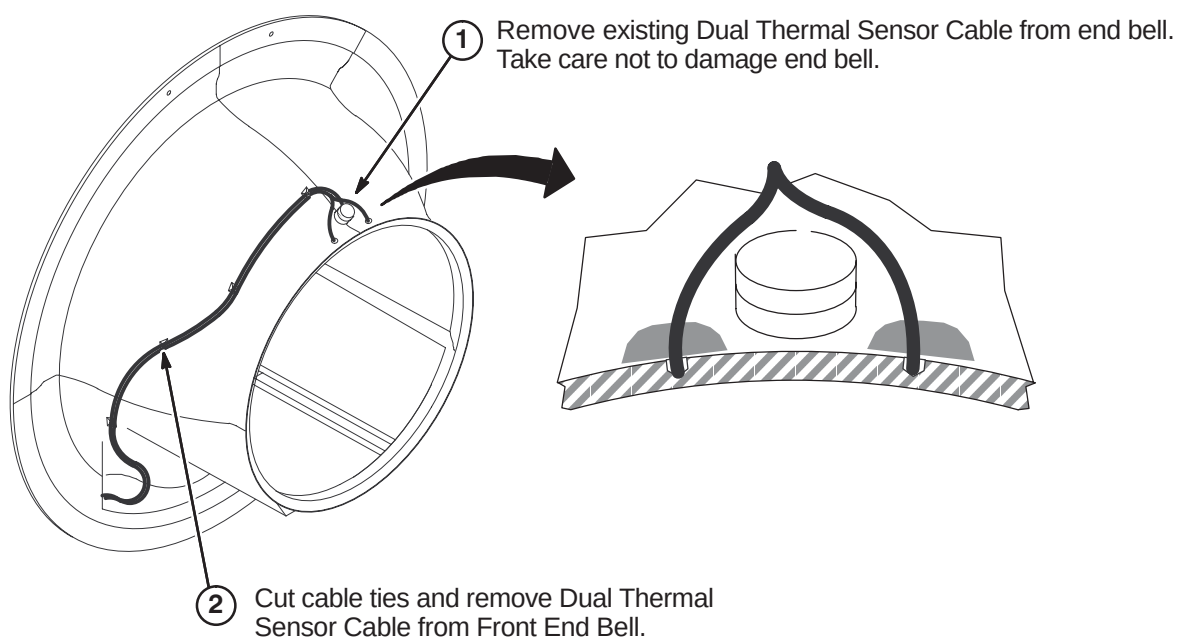
6-5-4-7 Unfasten Front End Bell

The site may have upgraded to a Dual Thermal Sensor Run 749 recently, but this cable has to be removed and replaced with new Dual Thermal Sensor cable, Run 1260 (5123812). EXCITE HD requires Run 1260 to be installed.



6-5-4-8 Existing Dual Temperature Sensor Cable Removal from Front End Bell (If applicable)**WARNING!**

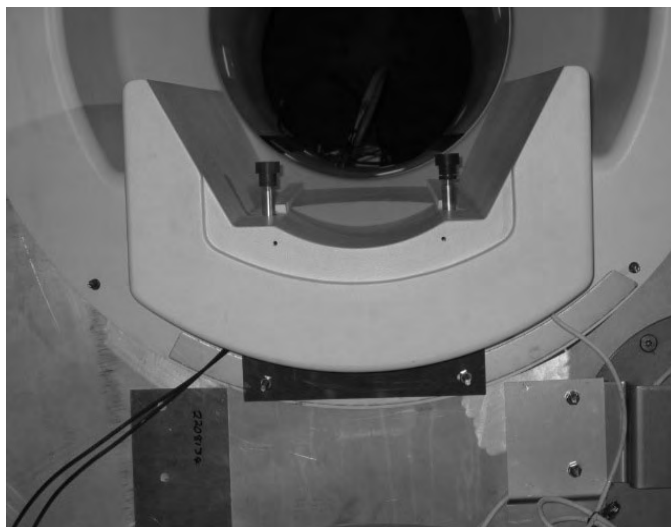
FERROUS MATERIAL HAZARD! DO NOT DRILL, HAMMER, CHISEL, ETC. OR USE ANY FERROUS TOOLS IN MAGNET ROOM UNLESS MAGNET IS RAMPED DOWN!



6-5-5 INSTALL IIQ KIT AND RF GROUND STRAP (If Required)

Prior to this upgrade, the site should have been audited to determine if the IIQ Kit (M3090NE) has been installed. The IIQ Kit includes parts and instructions for the installation of the Gradient Radial Support and the Bridge Support. It is required for all magnets with Horizon style enclosures. If this upgrade has not been performed on the existing enclosure, then M3090NE should have been ordered and must be installed at this point of the HDx upgrade.

Refer to steering Direction 2401983, included in M3090NE, for guidance. Directions 2401984, *S-Series Magnet Bridge Support* (shown below) and Direction 2319693, *Signa Gradient Radial Support Upgrade*, along with associated parts, are also included in M3090NE.



Also, the RF Ground Strap (M3090NF) must be installed if this upgrade hasn't been performed previously. This kit includes parts and instructions for installation of the RF Ground Strap. Refer to Direction 2319685, *BRM of CRM RF Shield Grounding Upgrade*, for instructions on installing this kit.

For fixed sites only, the mounting of the gradient cables at the rear of the magnet must be upgraded. This upgrade is part of M3335BD, *LCC WideOpen IIQ Upgrade*. Refer to Direction 2404642, provided with Gradient Cable Lead Board collector (5110336), for instructions on how to install the new design.

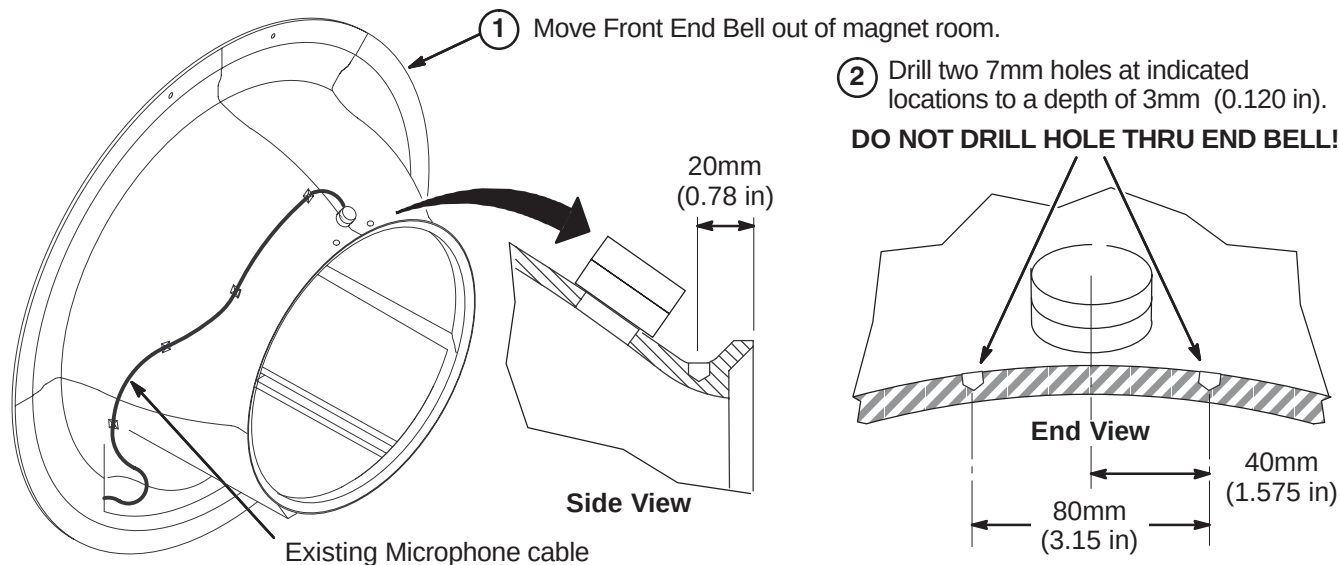
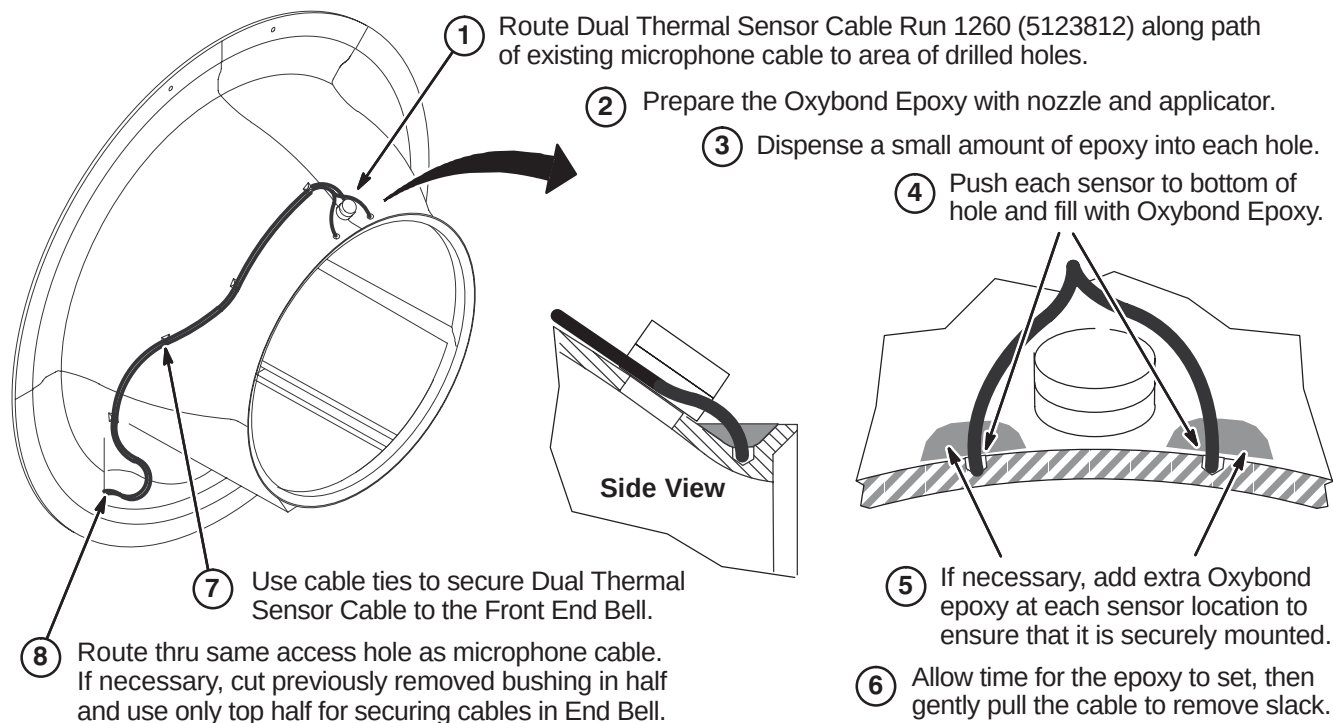
6-5-6 INSTALL GRADIENT LEAD BOARD AND RESTRAINT BLOCKS (If Required for Fixed-Sites Only)

Prior to this upgrade, the site should have been audited to determine if the Gradient Lead Board Kit has been installed. This kit includes parts and instructions for the installation of the Gradient Lead Board at the rear of the magnet.

Also, the gradient cable Restraint Blocks Kit (5186281-2) should have been installed.

These kits are required for all magnets with Horizon style enclosures. If these upgrades has not been performed on the existing enclosure, then the following should have been ordered and it must be installed at this point of the HDx upgrade.

- LCC Early WideOpen – M3335BC
- LCC Horizon – M3335BB
- Cx – M3335BB
- S-Series – M3335BA

6-5-7 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE, RUN 1260**6-5-7-1 Preparation of Front End Bell****6-5-7-2 Attach Dual Thermal Sensor Cable****6-5-8 RE-INSTALL FRONT END BELL, COMFORT MODULE, AND BRIDGE**

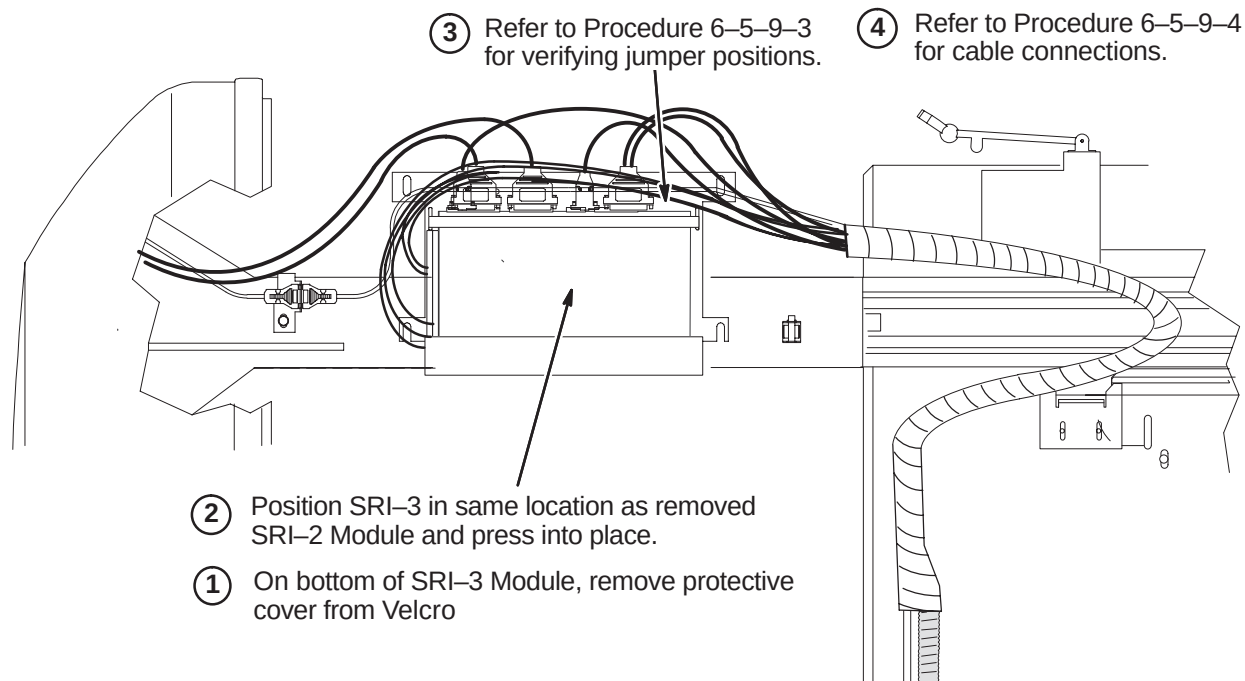
Reverse the applicable Front End Bell Removal procedures (Section 6-5-4) to reinstall the end bell, comfort module, and bridge.

6-5-9 INSTALL SRI-3 MODULE

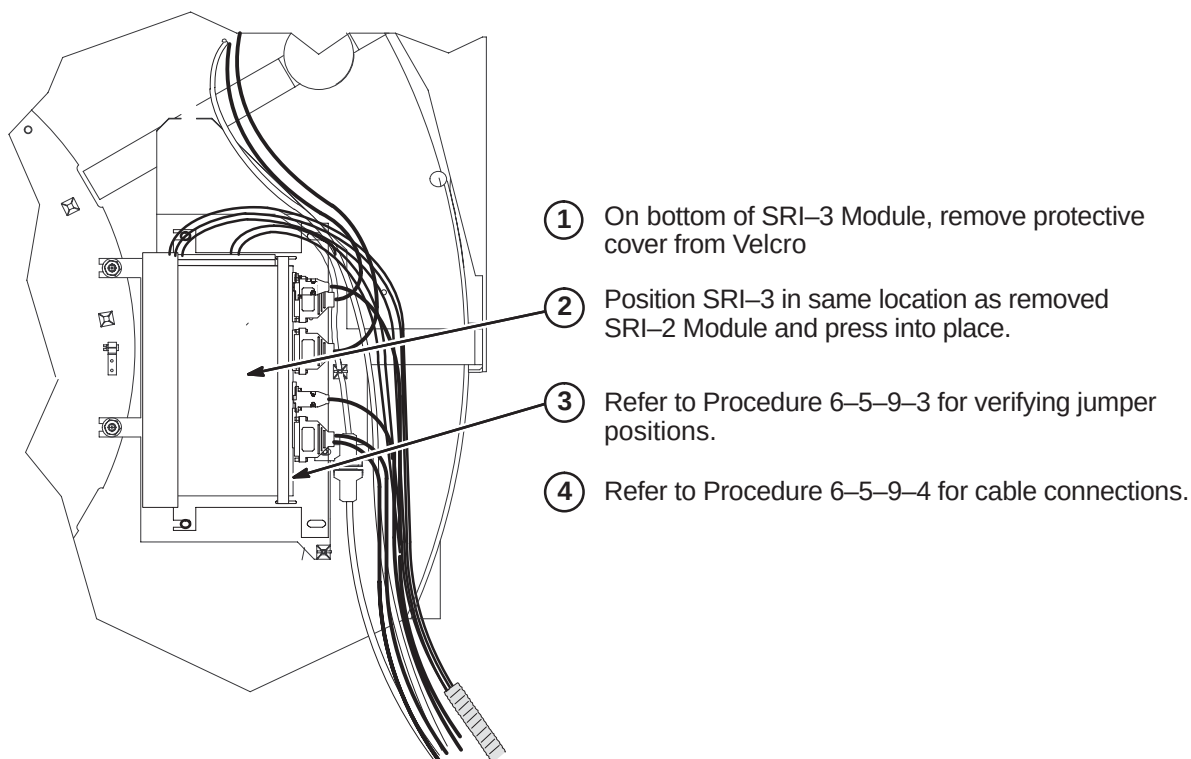
For site with S-series magnet, complete procedure in Section 6-5-9-1.

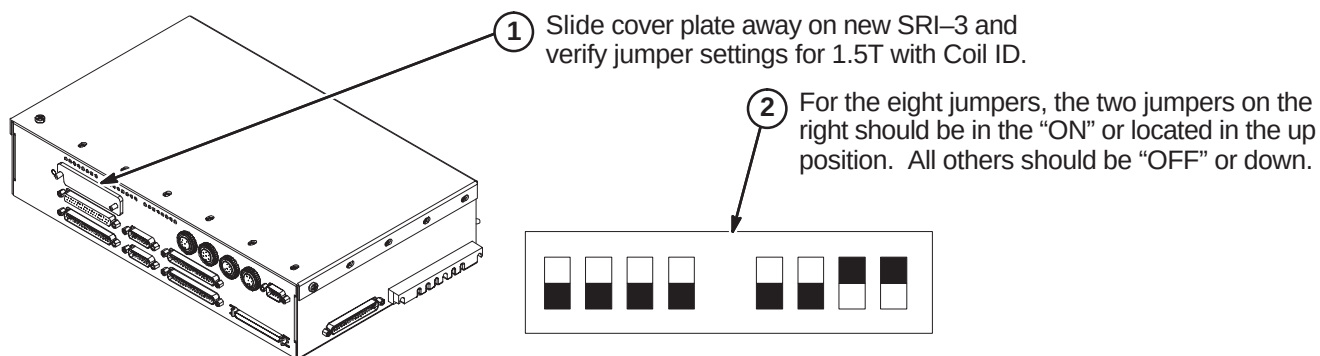
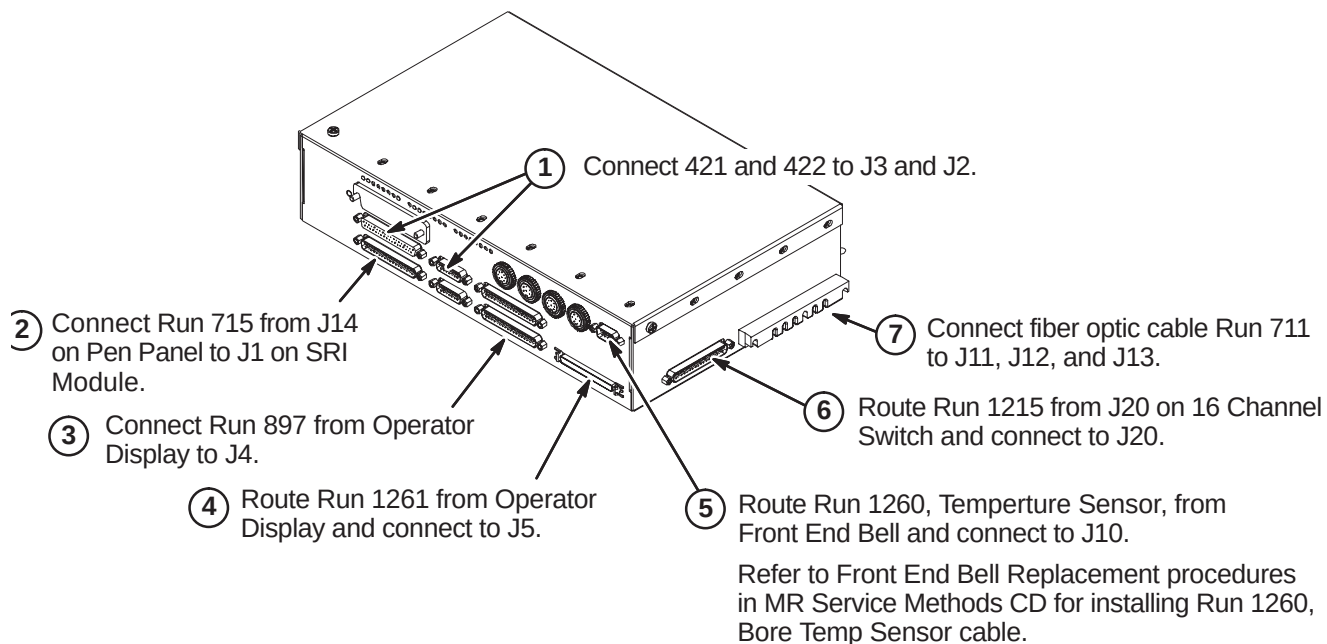
For site with Cx magnet, complete procedure in Section 6-5-9-2.

6-5-9-1 Install SRI-3 Module for S-Series Enclosure



6-5-9-2 Install SRI-3 Module for Cx or LCC Horizon Enclosure

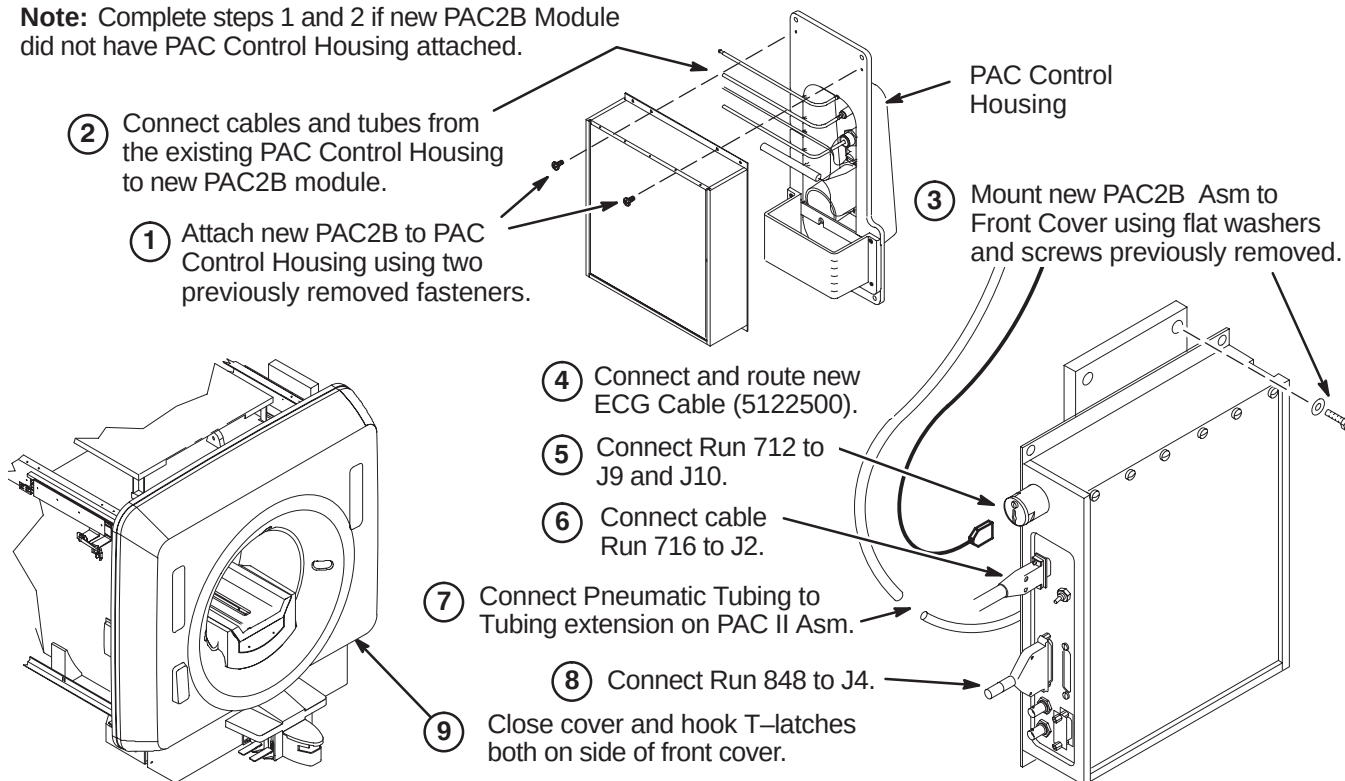
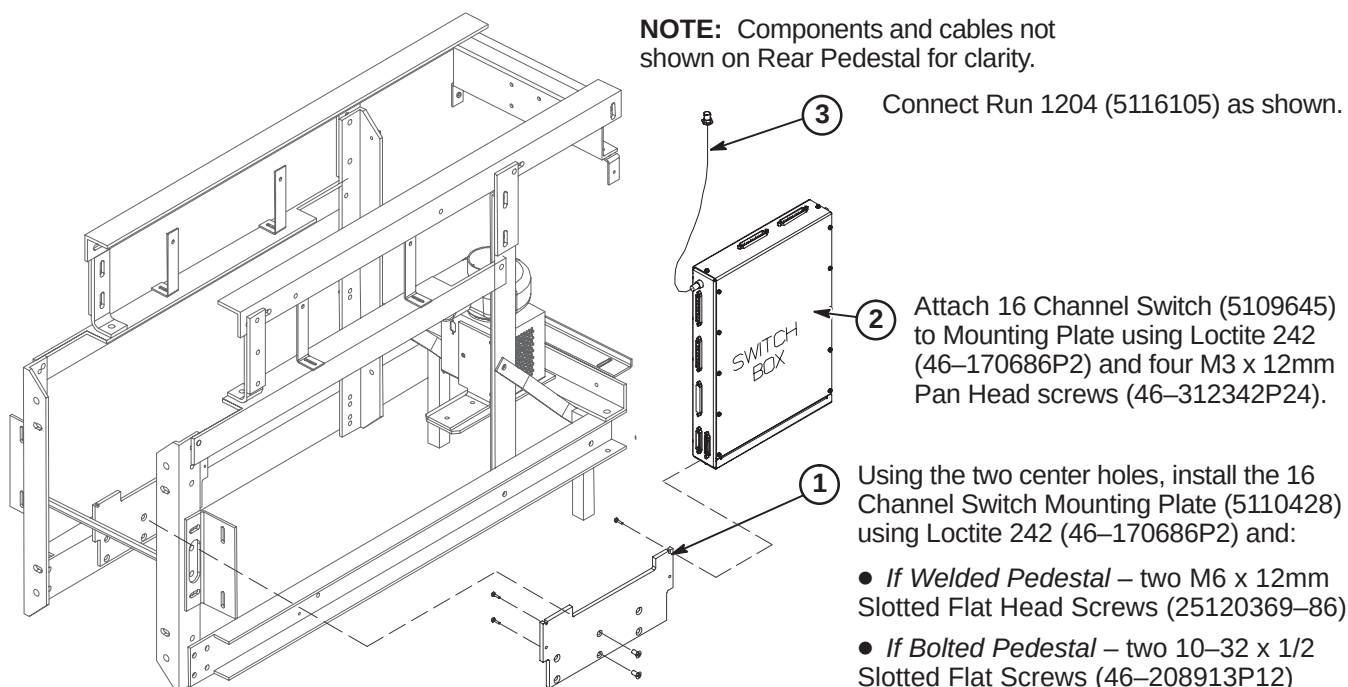


6-5-9-3 Verify SRI-3 Jumper Settings**6-5-9-4 Connect Cables to SRI-3 Module**

6-5-10 INSTALL NEW PAC MODULE AND ECG CABLE

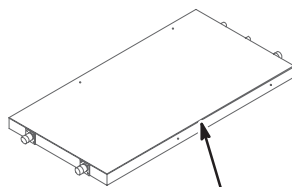
Reverse the deinstallation steps for installation of the new PAC Module. Also install the new ECG Cable.

Note: Complete steps 1 and 2 if new PAC2B Module did not have PAC Control Housing attached.

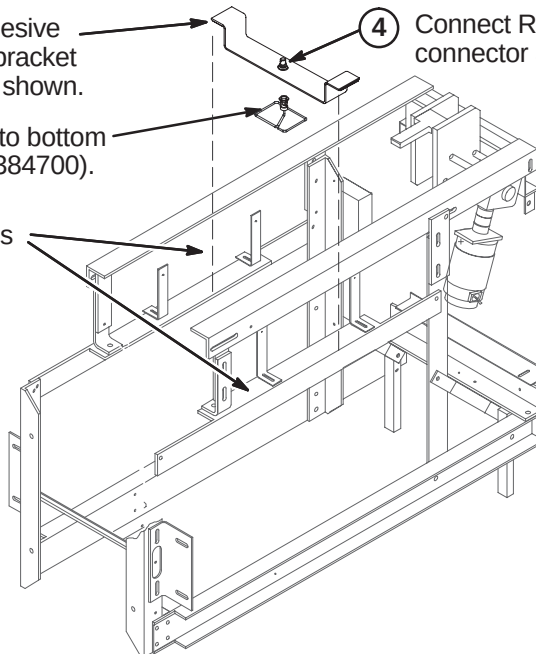
**6-5-11 INSTALL NEW COMPONENTS ON REAR PEDESTAL****6-5-11-1 Install 16 Channel Switch**

6-5-11-2 Install UTNS Antenna

- ③ Peel of liner from bottom of adhesive backed Velcro on each side of bracket and attach to Rear Pedestal as shown.
- ② Attach UTNS Antenna (2355200) to bottom side of UTNS Upgrade bracket (2384700).
- ① Make sure top surfaces of side braces are clean and dry for attachment of adhesive backed Velcro.
- ④ Connect Run 1204 to UTNS Antenna connector at top of bracket.



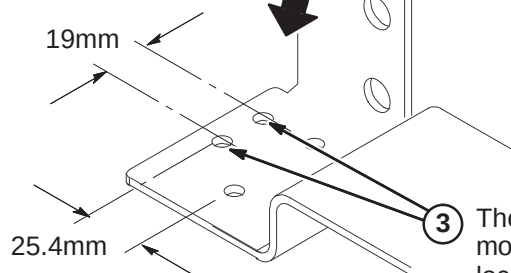
NOTE:
For clarity, Hybrid Splitter not shown in installed position.

**6-5-11-3 Install RF Drive Cable Mounting Bracket for LCC and Cx Magnets Only**

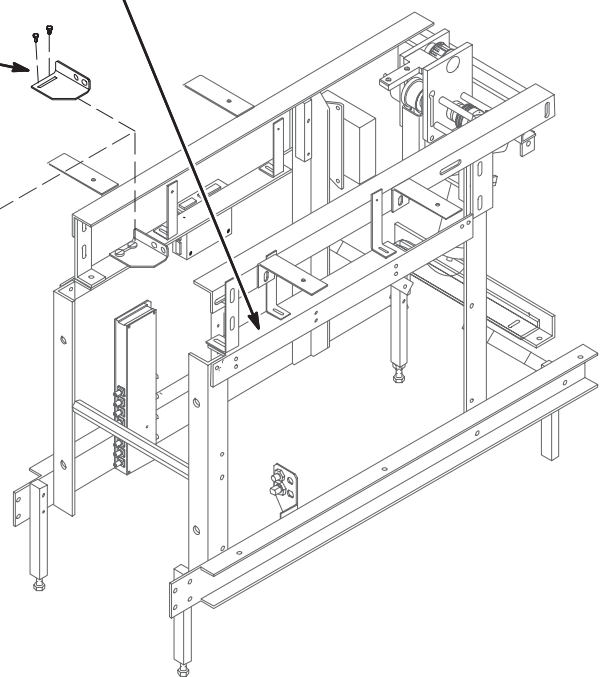
- ① Make sure top surfaces of side braces are clean and dry for attachment of adhesive backed Velcro.
- ② At existing Mounting Bracket, disconnect Runs 743 and 744 and RF Drive Input cables from Coil. Remove and save BNC cable connectors. Remove and discard bracket. Save mounting hardware for installation of new plastic bracket.

Note: Make sure bracket is de-installed. Leaving the metal bracket on frame may cause intermittent dynamic disable faults.

- ④ Attach Plastic Bracket to Rear Pedestal using mounting hardware removed in Step 5. Attach with velcro on other side of bracket. Install BNC connectors.
- ⑤ Reconnect Runs 743 & 744 and RF Drive cables.

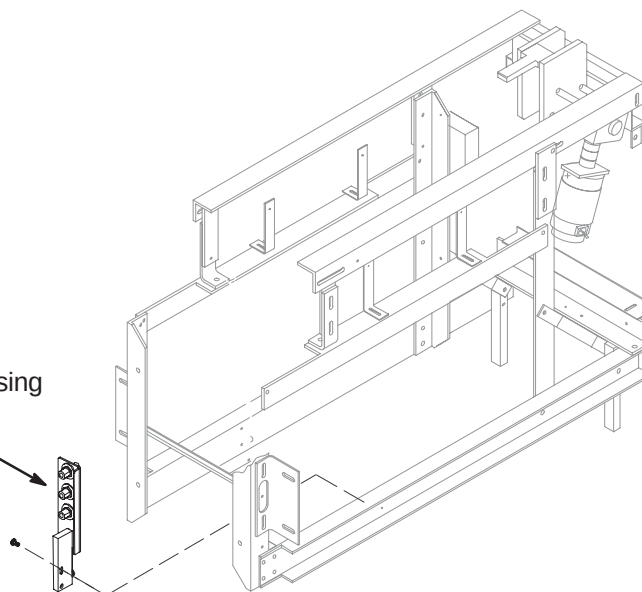


- ③ The new plastic bracket (2384445) requires two new mounting holes. Drill two 7mm diameter holes at locations dimensioned from the existing holes.



6-5-11-4 Install Bulkhead Interface

- ① Install new Bulkhead Interface (5113250) using Loctite 242 (46-170686P2) and two M6 x 12mm Pan Head Screws (2109873-24).

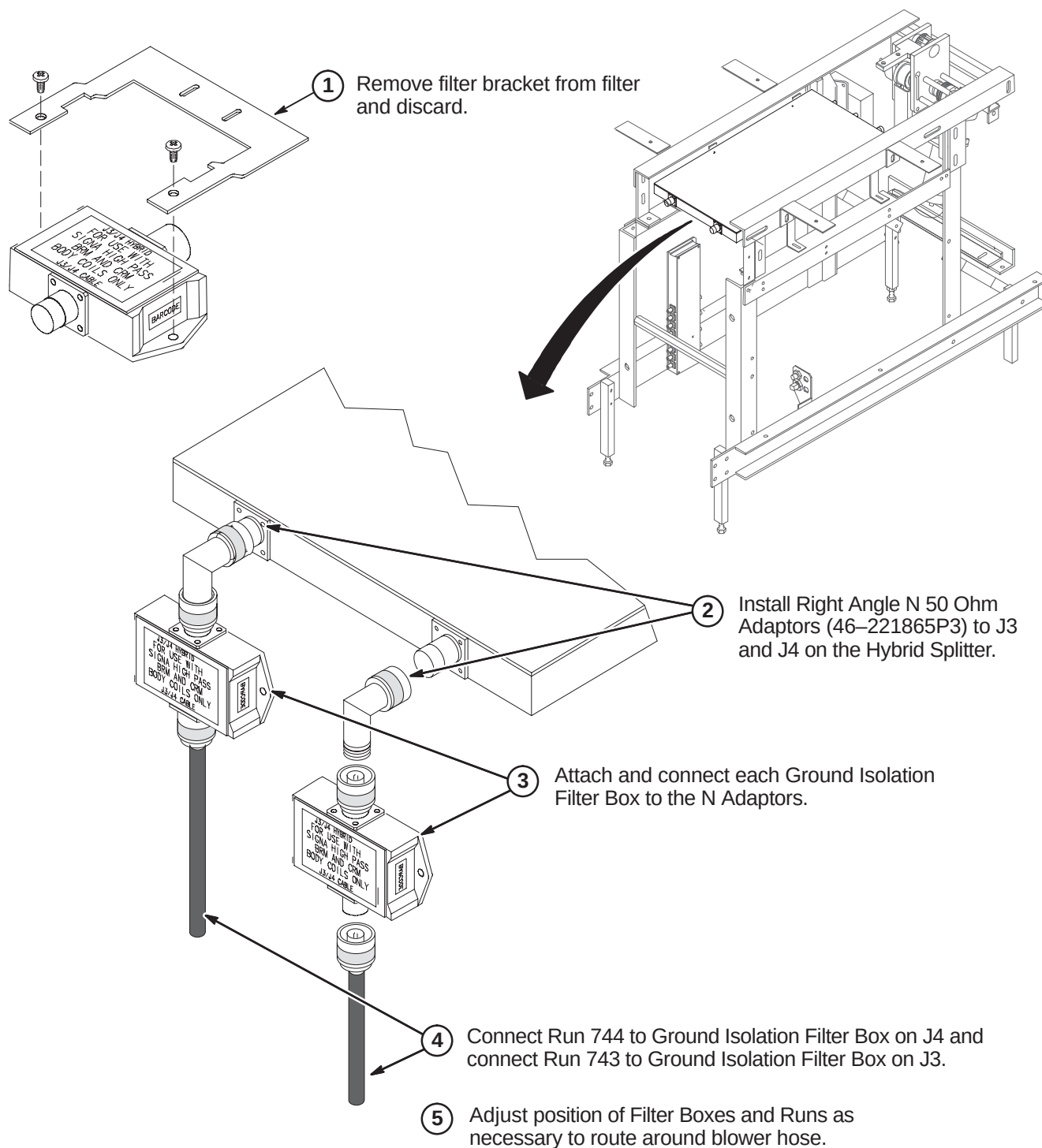


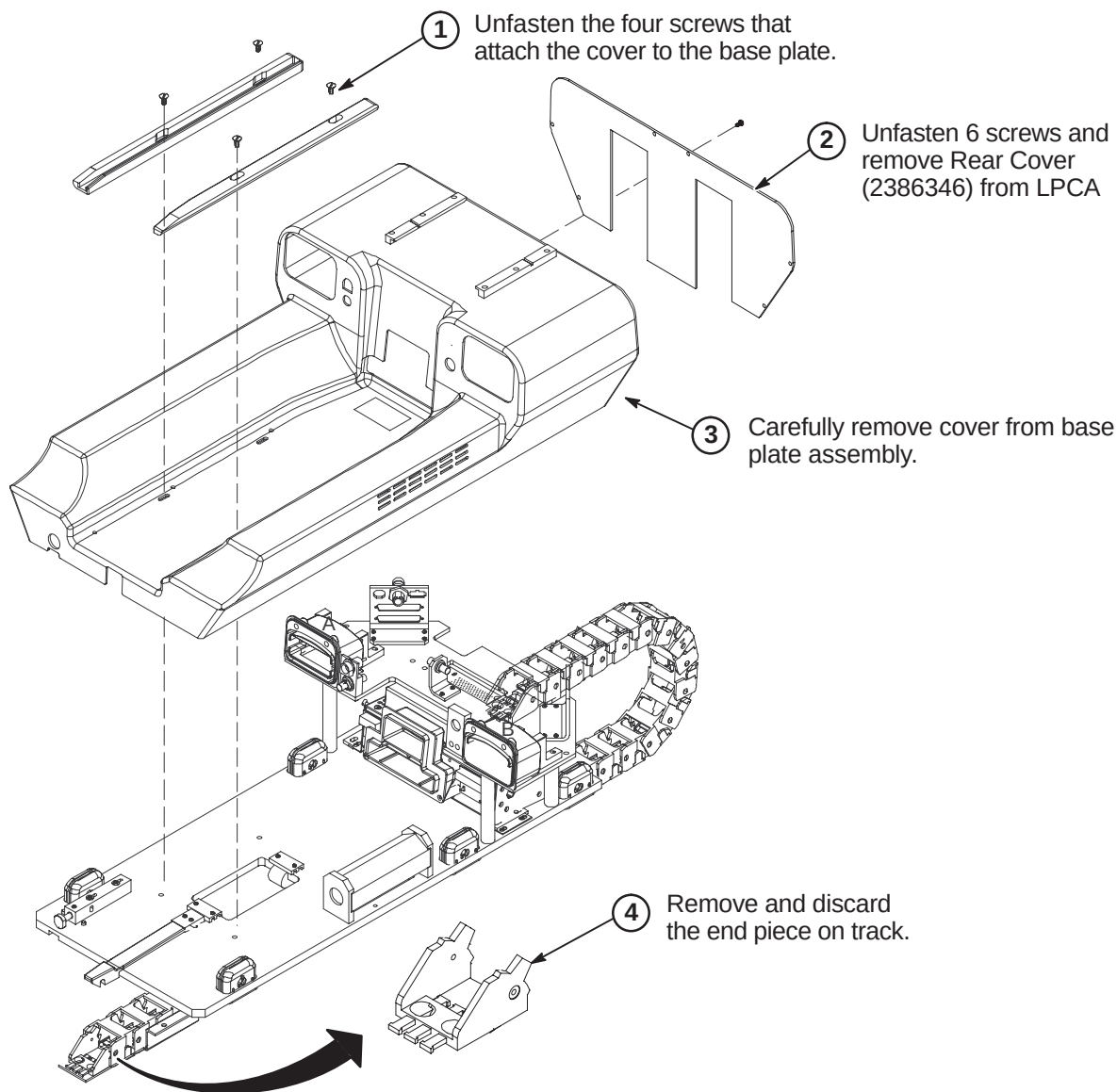
6-5-11-5 Install Filter Mount Asm and Isolation Filters to Body Hybrid Splitter

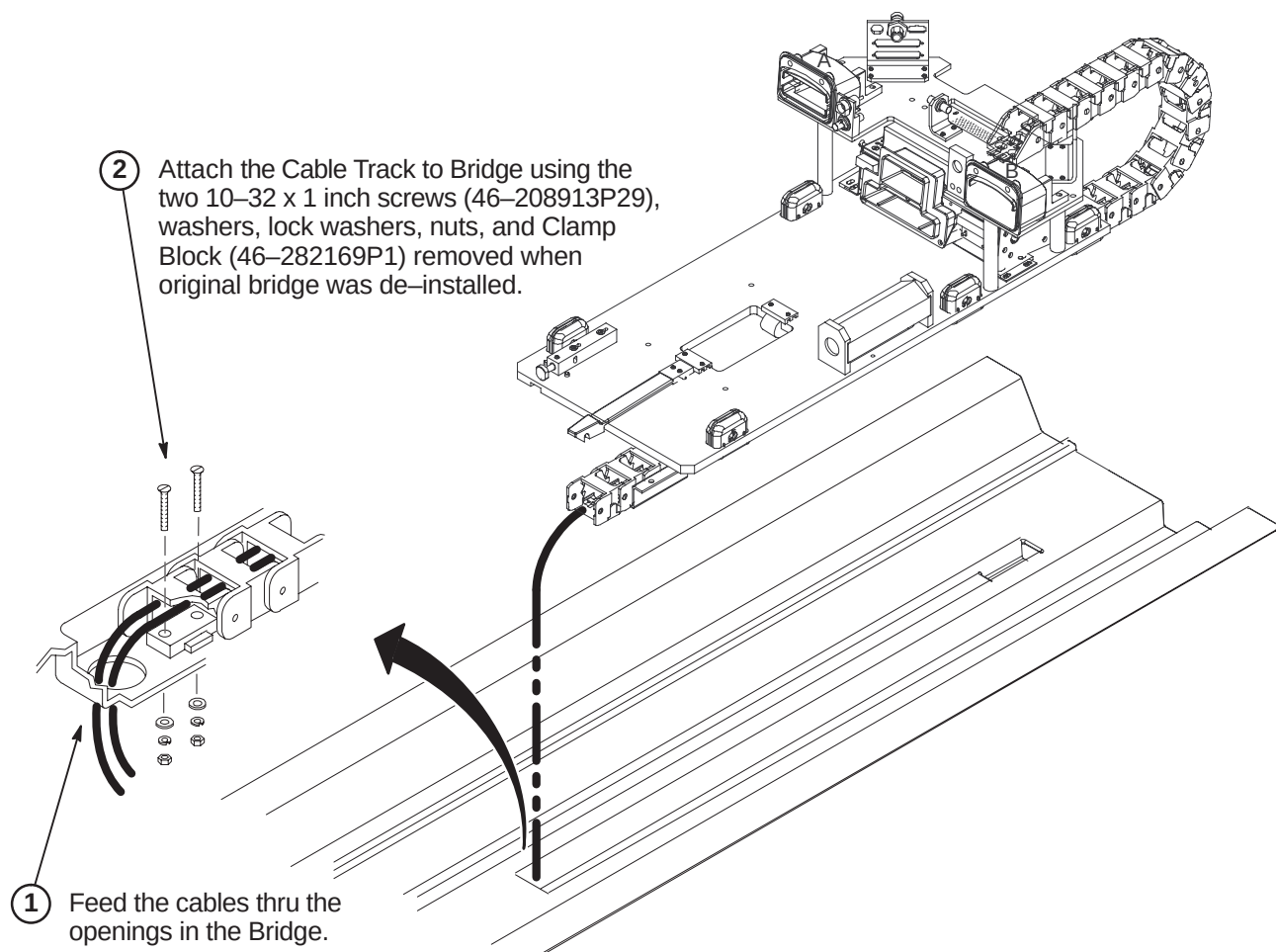
NOTE: The Filter Mount Asm (5112055) contains several parts that are installed in this procedure.

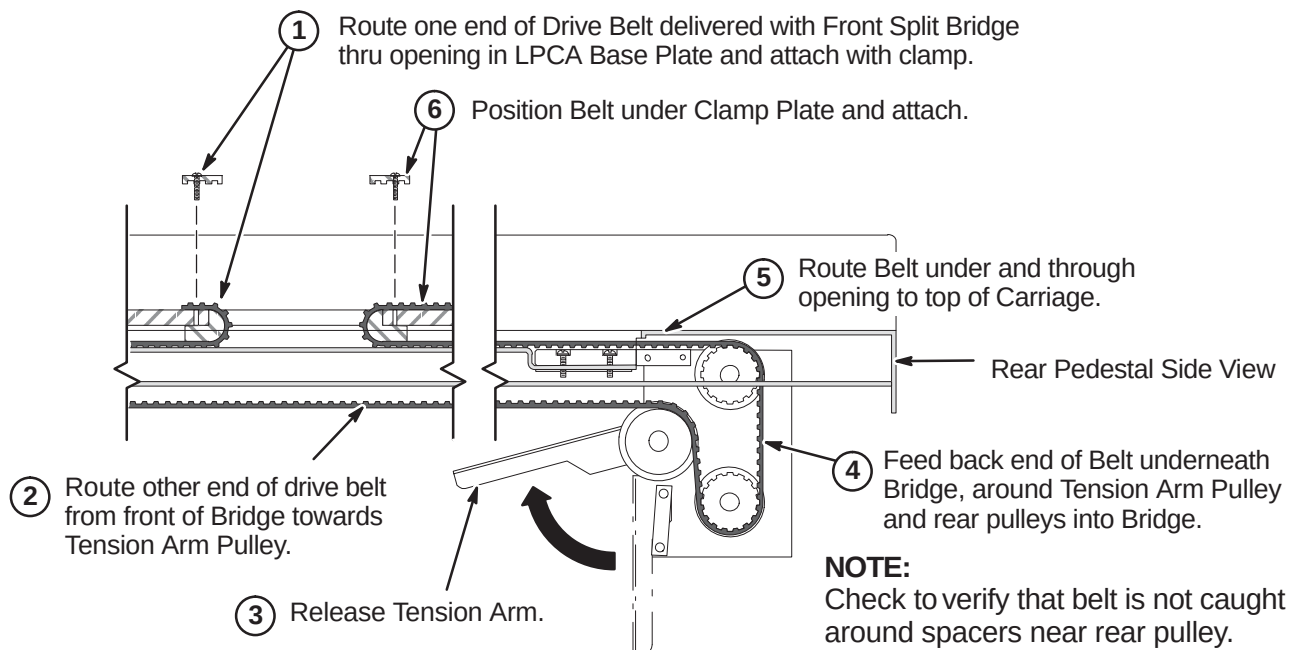
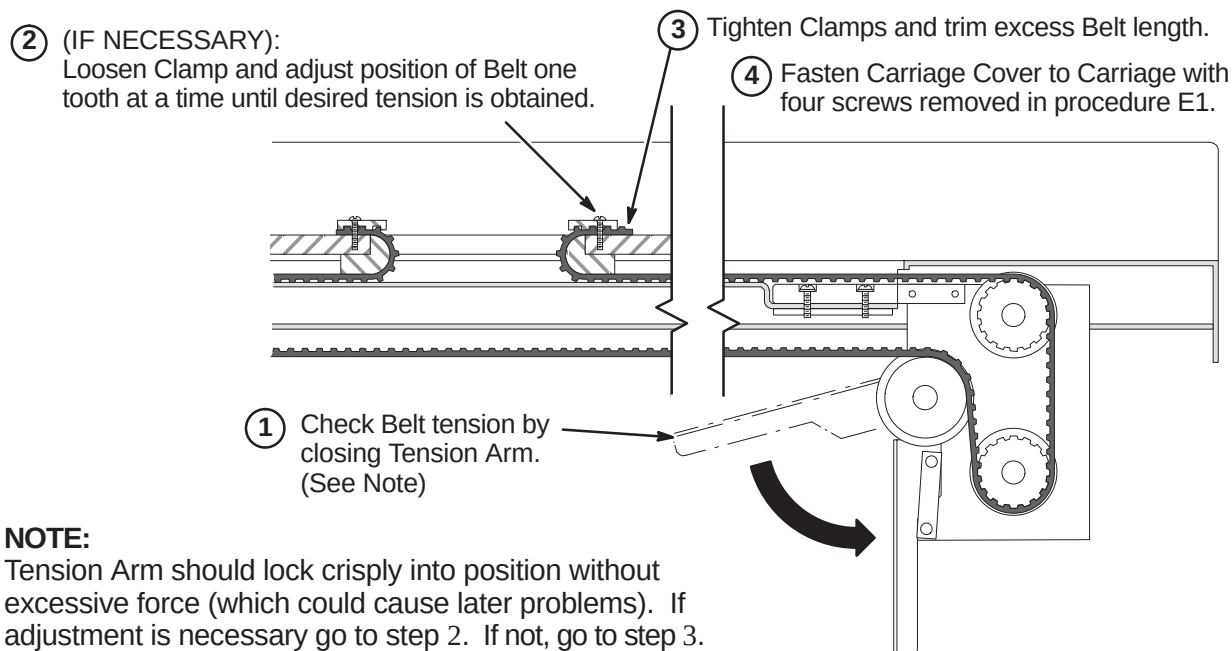
- ② Cut Double Coated Film (46-170106P1) into two strips 70mm long. Clean all surfaces, align to Preamp and attach.
- ③ Align and attach Filter Mount Block (5112053) with existing Preamp.
- ④ Position Notch Filter (2305086-2) in Filter Mount Block and connect to Preamp
- ⑤ Attach Filter Clamp Plate using four M4 x 12 Nylon screws (2112068-5).
- ⑥ Place Clamp (46-220191P1) at end of Notch filter and attach around cable.
- ⑦ Attach and connect each Ground Isolation Filter Module (5118952) to the Hybrid Splitter using two screws (46-220184P5).
- ⑧ Install Right Angle N 50 Ohm Adaptors (46-221865P3) to each Isolation Filter.
- ⑨ Reconnect blue RF cables after positioning Rear Pedestal.
- Existing Preamp on bottom of Hybrid Splitter
- Disconnect and discard existing BNC cable.
- New BNC Cable Run 1206.
- Do NOT install on TRM gradient coil systems.

6-5-11-6 Alternative Ground Isolation Filter Installation for LCC Magnet with Horizon Enclosure



6-5-12 INSTALL LPCA AND CABLE TRACK**6-5-12-1 Remove LPCA Cover**

6-5-12-2 Attach Cable Track to Bridge

6-5-12-3 Route and Attach Drive Belt**6-5-12-4 Final Adjustment of Drive Belt**

6-5-12-5 Installation of Sensor Flag

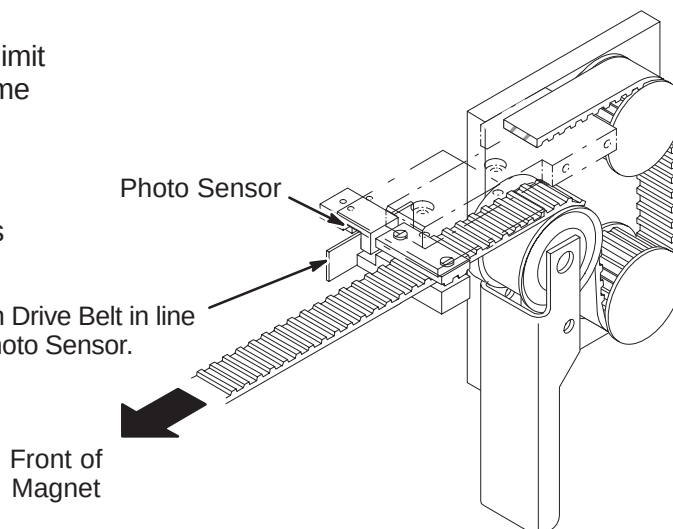
NOTE:

Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:

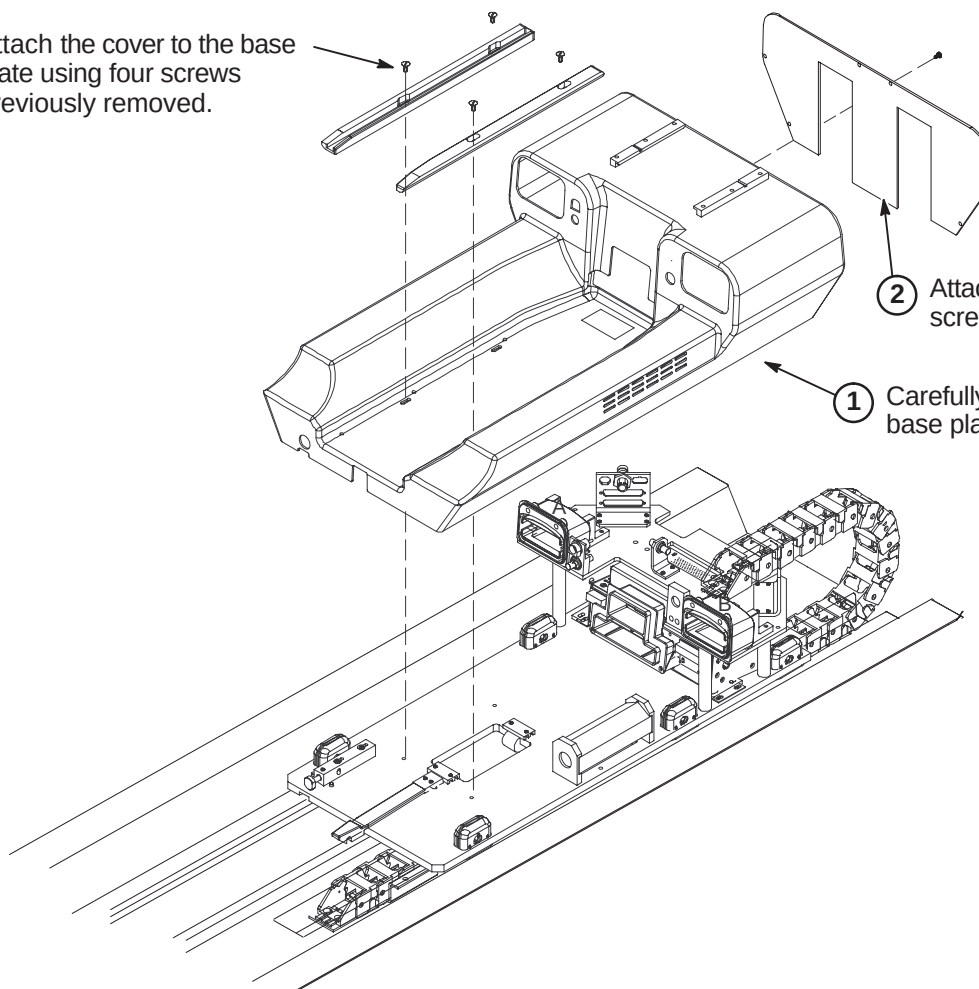
The following Step is needed if flag was removed during routing of drive belt.

- ① Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.



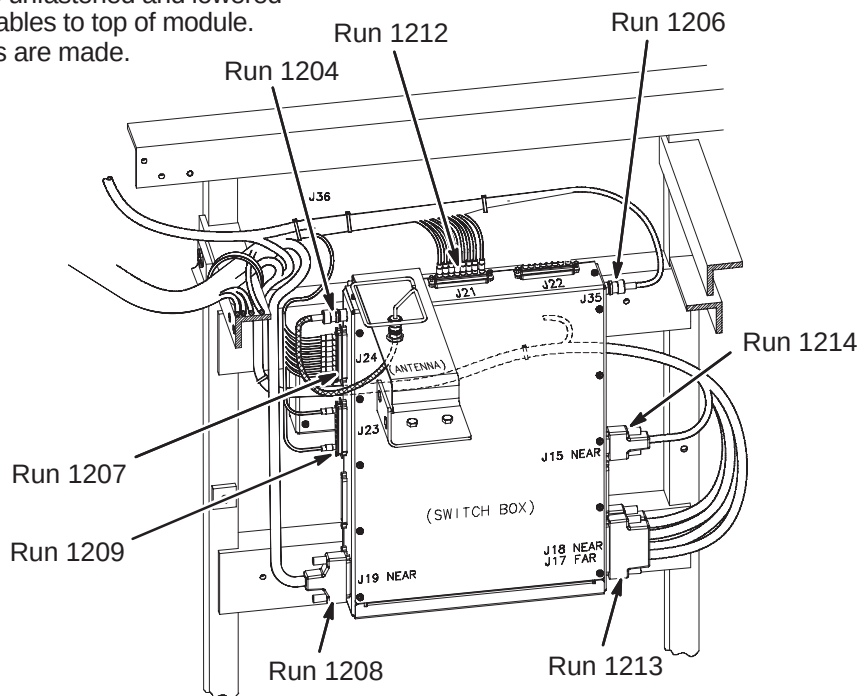
6-5-12-6 Installation of LPCA Cover

- ③ Attach the cover to the base plate using four screws previously removed.



6-5-12-7 Connect Cables from Cable Track to 16 Channel Switch Module

- ① 16 Channel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re-mount module after connections are made.
- ② Connect cable Runs as indicated.

**6-5-12-8 Connect Cables from Cable Track to Bulkhead Interface**

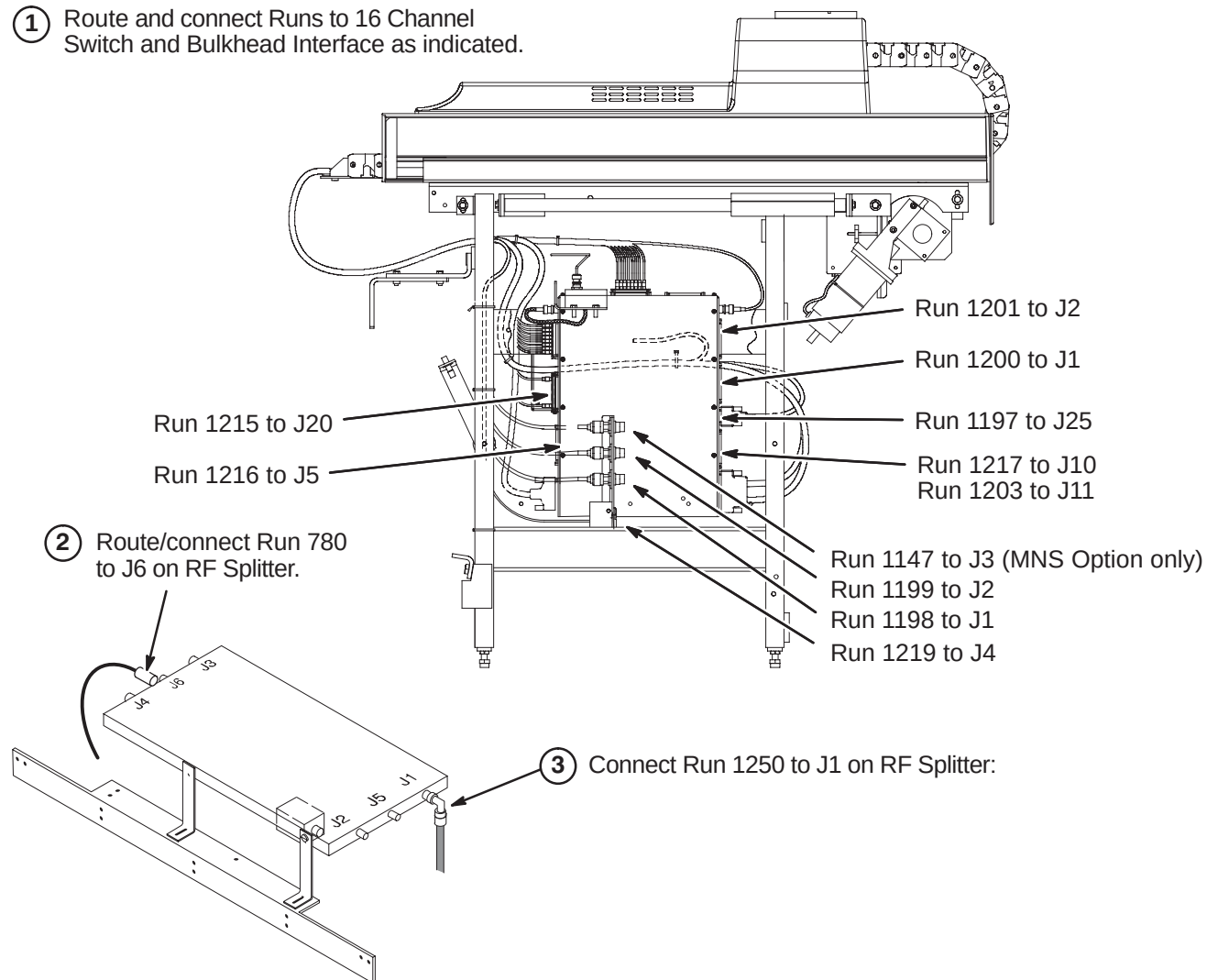
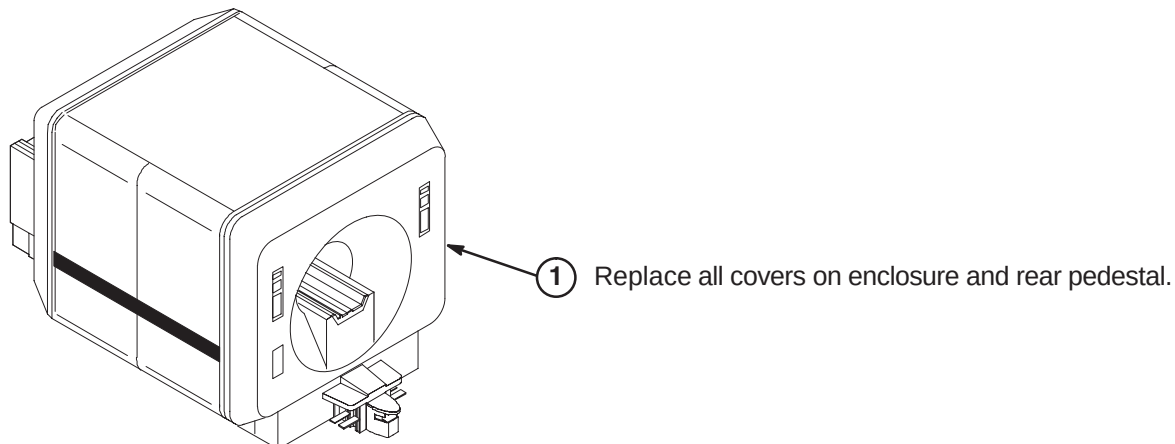
- ① Route Runs 1153, 1205, 1206, and 1228 along leg of Rear Pedestal and ty-wrap as indicated to leg.

- ② Connect Runs to Bulkhead Interface as indicated.

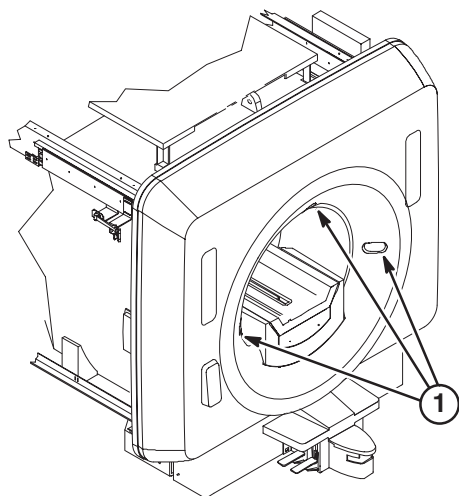
Run 1153 to J3
 Run 1205 to J2
 Run 1228 to J1
 Run 1206 to J4

6-5-13 CONNECT REMAINING CABLES TO REAR PEDESTAL

- ① Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.

**6-5-14 INSTALL COVERS**

6-5-15 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS



A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below each of the three laser lights.

For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.



French



German



Portuguese



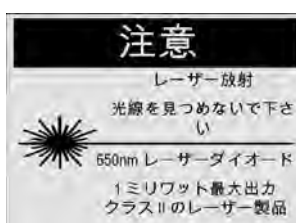
Spanish



Italian



Swedish



Japanese



Chinese

6-6 TIMES MICROWAVE RF CABLES INSTALLATION

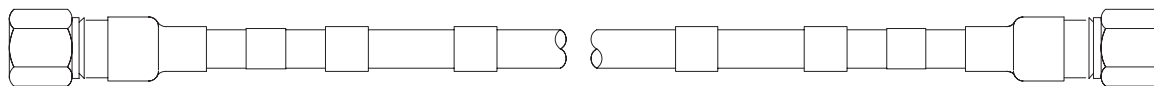
6-6-1 CABLE INFORMATION

Two cables are to be created and other associated parts to be installed are provided for this procedure. Also provided is a Times Microwave stripping tool for the LMR 600 (2352193), and a Times Microwave cutting tool (5111565).



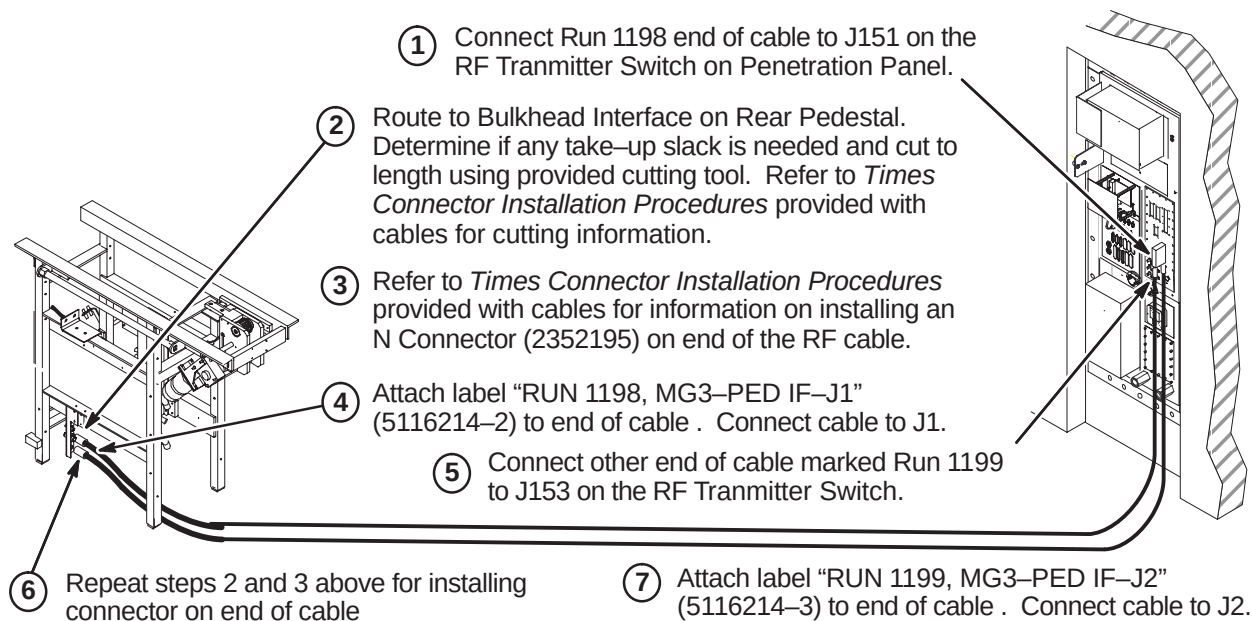
Do not subject LMR-600FR cable to a bend radius less than 3 inches (75mm). The cables will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

Cable 5115915, will be cut to length to create two cables, Runs 1198 and 1199, and terminated for routing in the scan room. The remaining connectors, fasteners, and labels will be installed as directed in the following steps.



Run 1198 and Run 1199

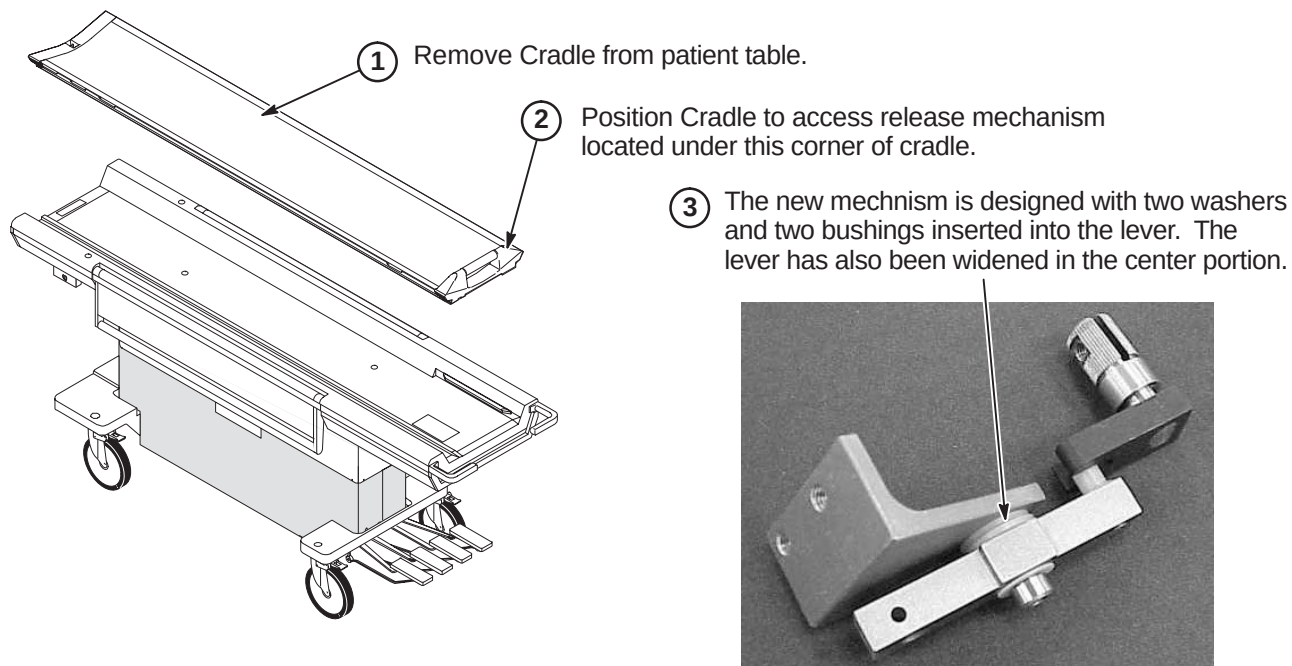
6-6-2 ROUTE, TERMINATE, AND CONNECT RF CABLES



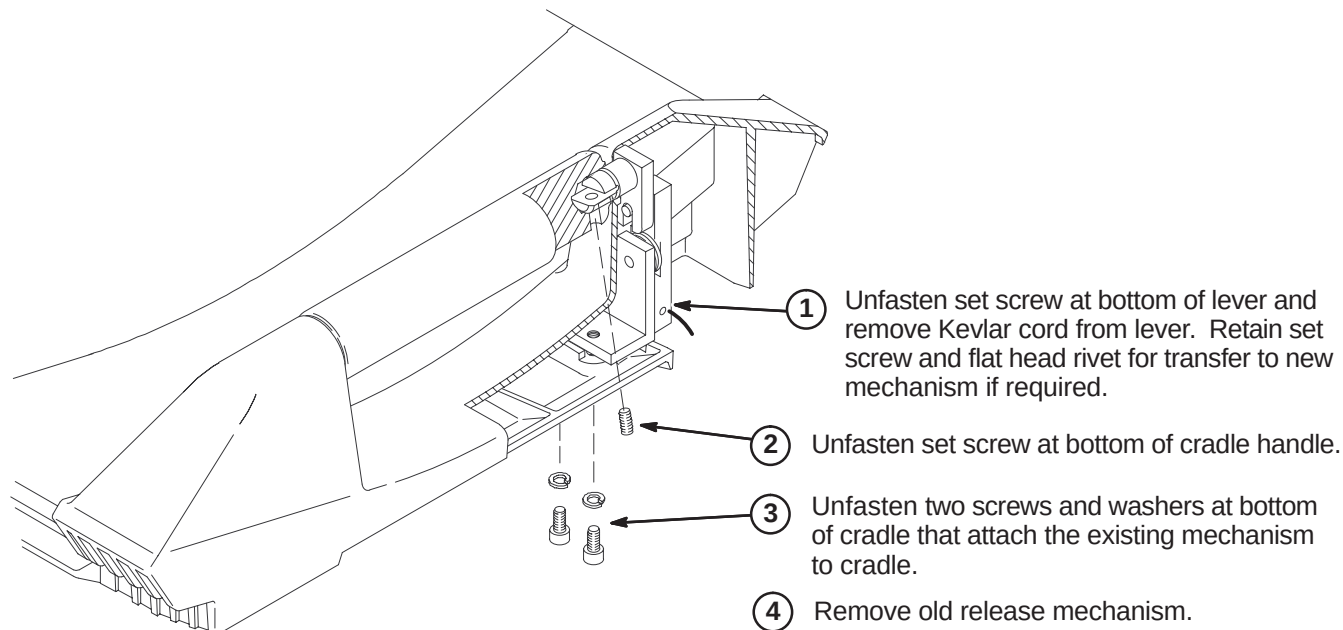
6-7 PATIENT TABLE CRADLE RELEASE UPGRADE

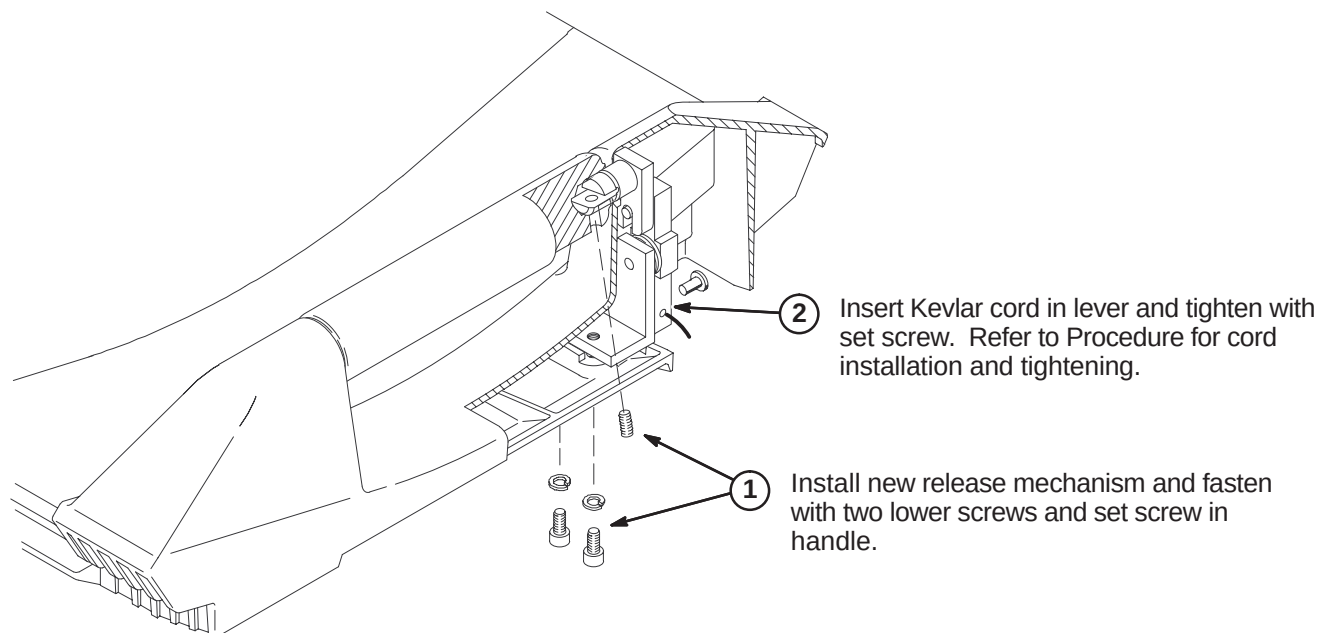
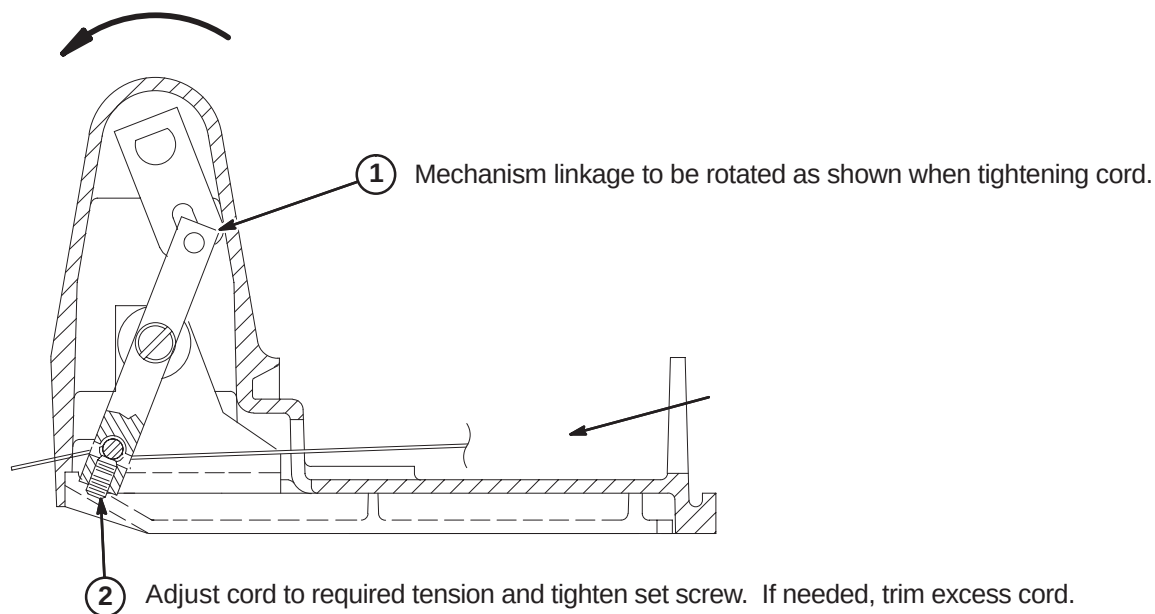
This upgrade is performed to improve the release mechanism and improve image quality.

6-7-1 PREPARE CRADLE FOR UPGRADE



6-7-2 REMOVE EXISTING RELEASE MECHANISM



6-7-3 INSTALL NEW RELEASE MECHANISM**6-7-4 ATTACH CORD**

SECTION 7 – OPERATOR WORKSPACE

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
7-1 INTRODUCTION	7-2
7-2 INSTALL NEW GOC CONTAINING Z400 COMPUTER	7-3
7-2-1 DE-INSTALL OPERATOR WORKSPACE COMPONENTS	7-3
7-2-2 POSITION COMPUTER CABINET	7-4
7-2-3 SEISMIC ANCHORING (IF REQUIRED)	7-4
7-2-4 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS	7-5
7-2-5 GOC CABINET COVER REMOVAL	7-6
7-2-6 INSTALL MODEM CABLE TO GOC CABINET (If Applicable)	7-7
7-2-7 GOC CABLE CONNECTIONS	7-8
7-2-8 POWER CABLE CONNECTIONS	7-9
7-3 INSTALL NEW Z400 COMPUTER IN EXISTING GOC	7-10
7-3-1 DE-INSTALL OPERATOR WORKSPACE COMPONENTS	7-10
7-3-2 GOC CABINET COVER REMOVAL	7-11
7-3-3 GOC COMPONENT REPLACEMENT	7-12
7-3-4 INSTALLATION OF NEW GOC COMPONENTS	7-12
7-3-5 GOC COMPUTER REPLACEMENT	7-14
7-3-5-A COMPUTER REPLACEMENT IN OLD STYLE GOC	7-14
7-3-5-B COMPUTER REPLACEMENT IN NEW STYLE GOC	7-15
7-4 TEMPERATURE AND HUMIDITY SENSOR KIT INSTALLATION	7-19
7-5 ROUTE RUN 1323	7-21
7-6 INSTALL NEW OW TABLE TOP COMPONENTS	7-22
7-7 CONNECT BAR CODE READER (If part of site configuration)	7-23
7-8 INSTALL SCIM/KEYBOARD	7-23
7-9 CONNECT CABLES TO Z400 COMPUTER	7-24
7-10 FINAL POSITIONING OF GOC CABINET	7-25
7-11 POWER-ON INFORMATION	7-26

7-1 INTRODUCTION

A new Monitor, SCIM, and Keyboard will replace existing components on the Operator Workspace Table. Also, the new Temperature and Humidity kit will be installed along with the new Z400 computer.

A new Z400 computer is also installed for this upgrade. Depending on site requirements, there are two options for this computer upgrade. They are:

- A replacement of the existing GOC with a new GOC containing the Z400 computer. Refer to Section 7-2.
- A replacement of only the existing computer with the new Z400 computer. Refer to Section 7-3.

After completing one of the above options, continue the Operator Workspace upgrade with Section 7-4.

7-2 INSTALL NEW GOC CONTAINING Z400 COMPUTER

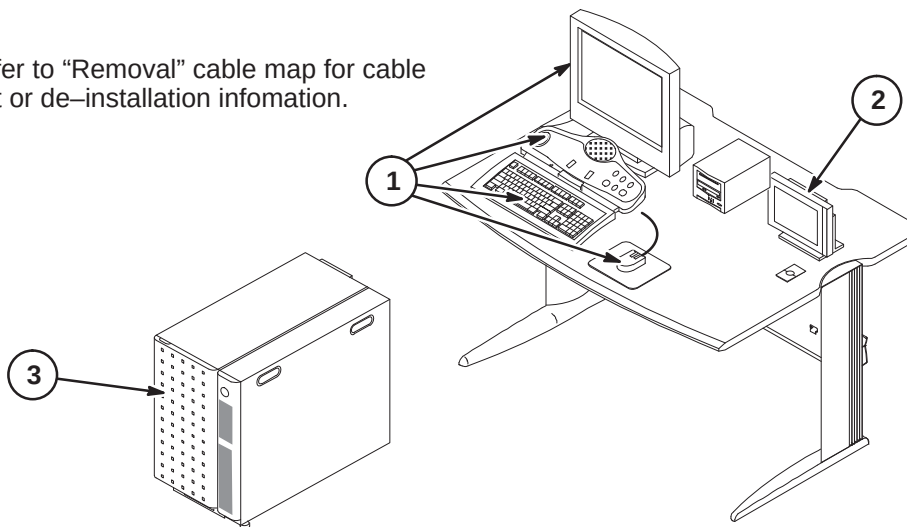
7-2-1 DE-INSTALL OPERATOR WORKSPACE COMPONENTS

CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

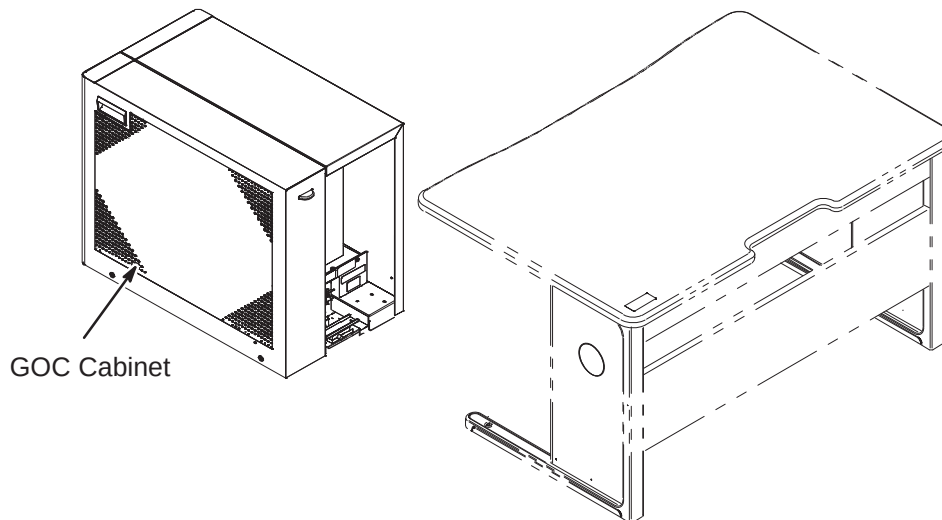
1. Existing Monitor including Runs 1063 and 1065, SCIM, Keyboard, and Mouse will be de-installed and returned for recycling.
Note: If Run 1062 is connected to the SCIM, remove and return for recycling. Install new Run 2053 which is provided with this upgrade (A few early versions of the cable upgrade kit provided a new Run 1256, without the short cables for connecting the keyboard and mouse. If received at this site, relabel as Run 2053). If existing SCIM cable is Run 1256, retain and reconnect to new SCIM.
2. De-install LCD Display, Runs 1068 and 1069, and return for recycling.
3. Existing GOC will be replaced by a new GOC containing the Z400 computer:

Note: Refer to "Removal" cable map for cable retainment or de-installation information.



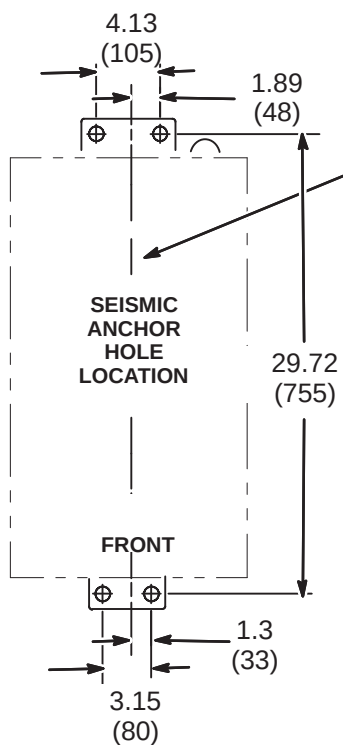
7-2-2 POSITION COMPUTER CABINET

Consult with Site customer for exact location of Global Operator Console (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.



7-2-3 SEISMIC ANCHORING (IF REQUIRED)

GOC Seismic Mount kit is shipped with GOC. Find kit inside of plastic bag.

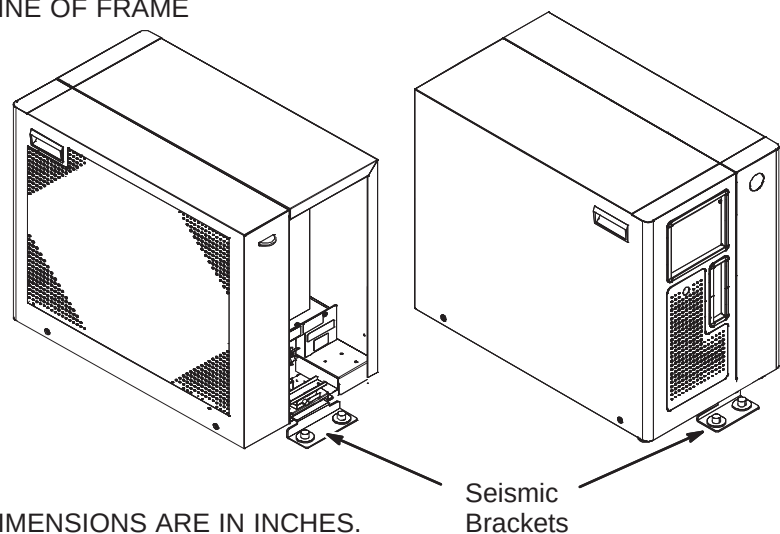


If seismic anchoring is required, refer to site architectural drawings for fastener type details or applicable Pre-installation manual. Illustration below indicates hole location for installing brackets

CENTERLINE OF FRAME

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



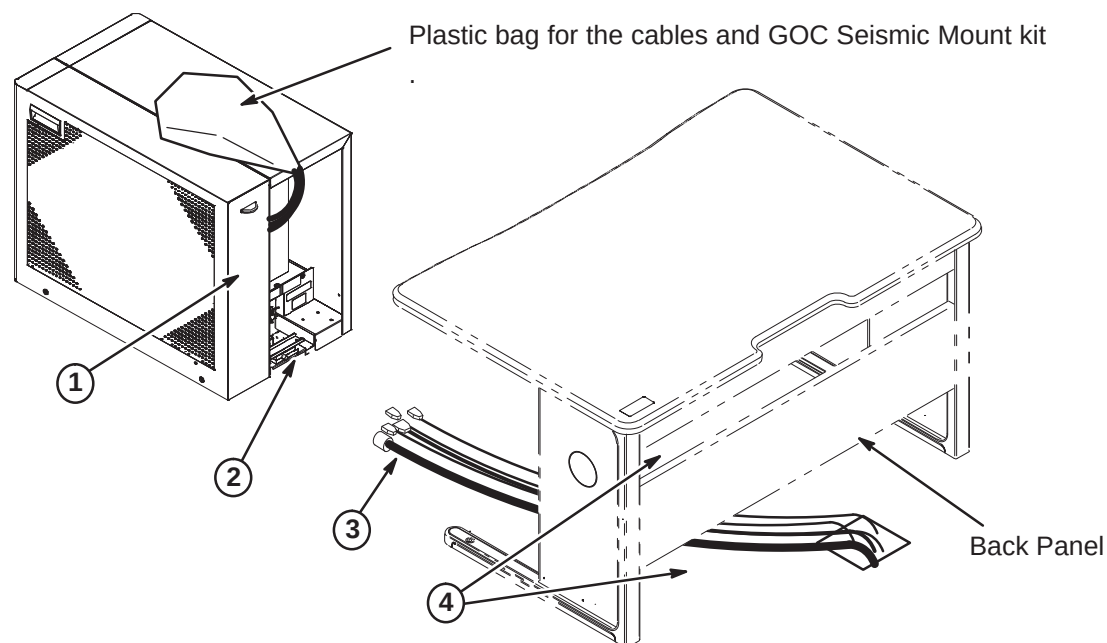
7-2-4 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS

The GOC (Computer) Cabinet is shipped with cables connected to the computer that will be routed and connected to equipment that will be installed on the top of the OW Table. The cables are:

- Two power cords
- One video output cable
- Keyboard output cable
- SCIM output cable
- SCSI output cable
- Modem output cable

These cables are already connected to the computer and routed thru the opening located at the top of the rear of the GOC cabinet as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will have to be routed thru the lower opening.

1. Runs 1064, 1070, E0015, E3045, E3046, E3047 and E3048 are pre-connected to computer and are routed at top of opening.
(These cables are inside of plastic bag)
2. Runs 1381, 1085, 1255, 1371, power cord for the Patient Alert system, and any other option cables will be routed thru lower opening.
3. Route all cables before connecting any cables. Refer to cable map for connection information.
4. If GOC Cabinet is to be located under OW Table, the cable coming out of the top opening of the cabinet will be routed above the Back Panel. The cables coming out of the lower opening will be routed below the Back Panel.



7-2-5 GOC CABINET COVER REMOVAL

The Computer Cabinet covers need to be removed to access the cable connection points.

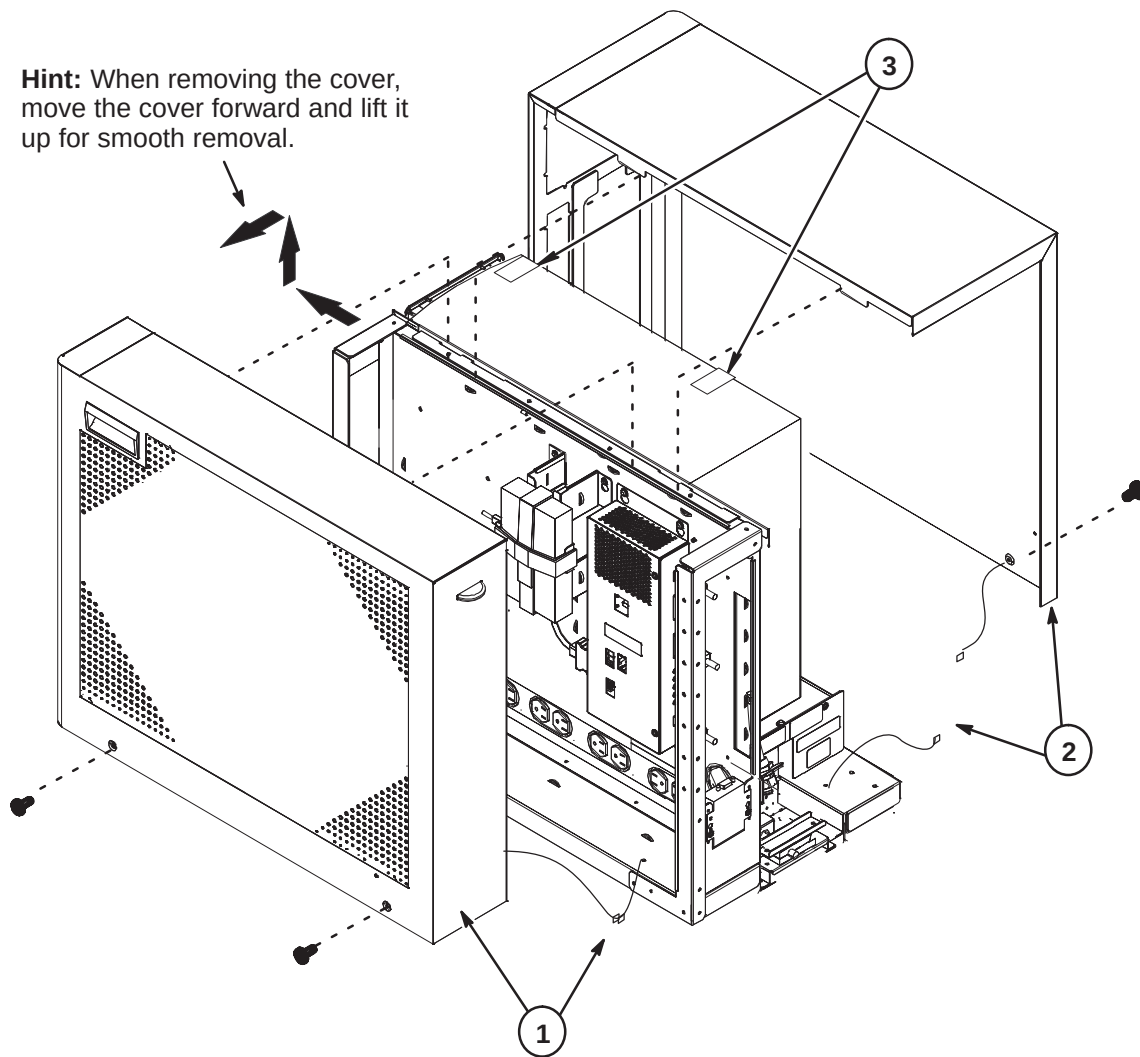
1. For right side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.
2. For left side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.

Note

A ground wire is connected to each side cover. Use care when removing side covers.

3. Peel the two shipping tapes off from PC.

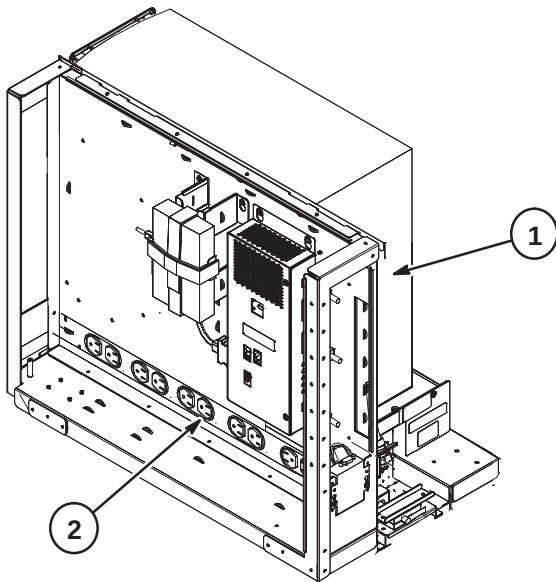
Hint: When removing the cover, move the cover forward and lift it up for smooth removal.



7-2-6 INSTALL MODEM CABLE TO GOC CABINET (If Applicable)

If Modem is part of system installation or upgrade, it will be placed on Operator Table or GOC.

1. Modem cable to the Linux PC is provided. Connect Run E3045 to Modem.
2. Modem power cord must be connected to J7 on GOC PDU Module. Route this cable to the rear top of the GOC Cabinet.



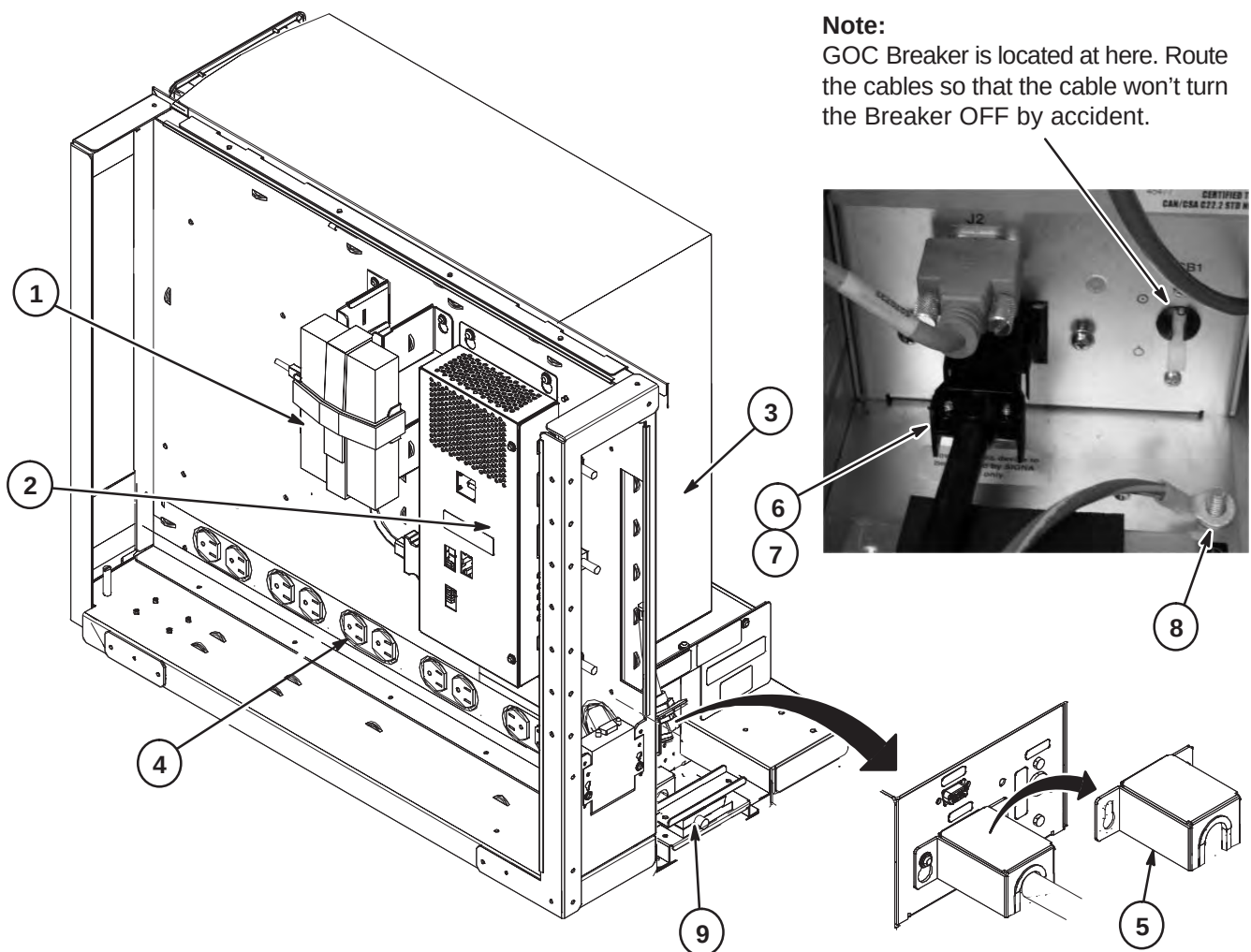
7-2-7 GOC CABLE CONNECTIONS

Refer to cable map for information on cables to be connected thru lower area of rear opening.

1. Route and connect Ethernet cables to Lan Hub.

Note Use ports 3, 7 or 8 for connecting site ethernet cables such as AW. Don not use any other ports for site ethernet cables.

2. Connect Run 1085 to J7 and Run 1255 to J18 on GOC AA.
3. Refer to Section 8-2-9 for all cables connected to Linux PC computer.
4. Refer to Section 7-2-8 for power cable connections to GOC PDU. Make sure Patient Alert power is connected to J3. If Modem is installed, power cable is connected to J7.
5. Before connecting GOC power cable Run 1381 to J1, loosen screws and remove cover.
6. Route and connect GOC power cable Run 1381 to J1 on PDU.
7. Confirm that GOC Power Cable is firmly connected to J1 by pulling the cable once. Re-install cover removed in Step 5. after connecting cable and tighten screws.
8. Connect Run 1382 to GND. Make sure nut securing GND cable is firmly tightened to GND stud.
9. Tighten clamp to secure cables in place.

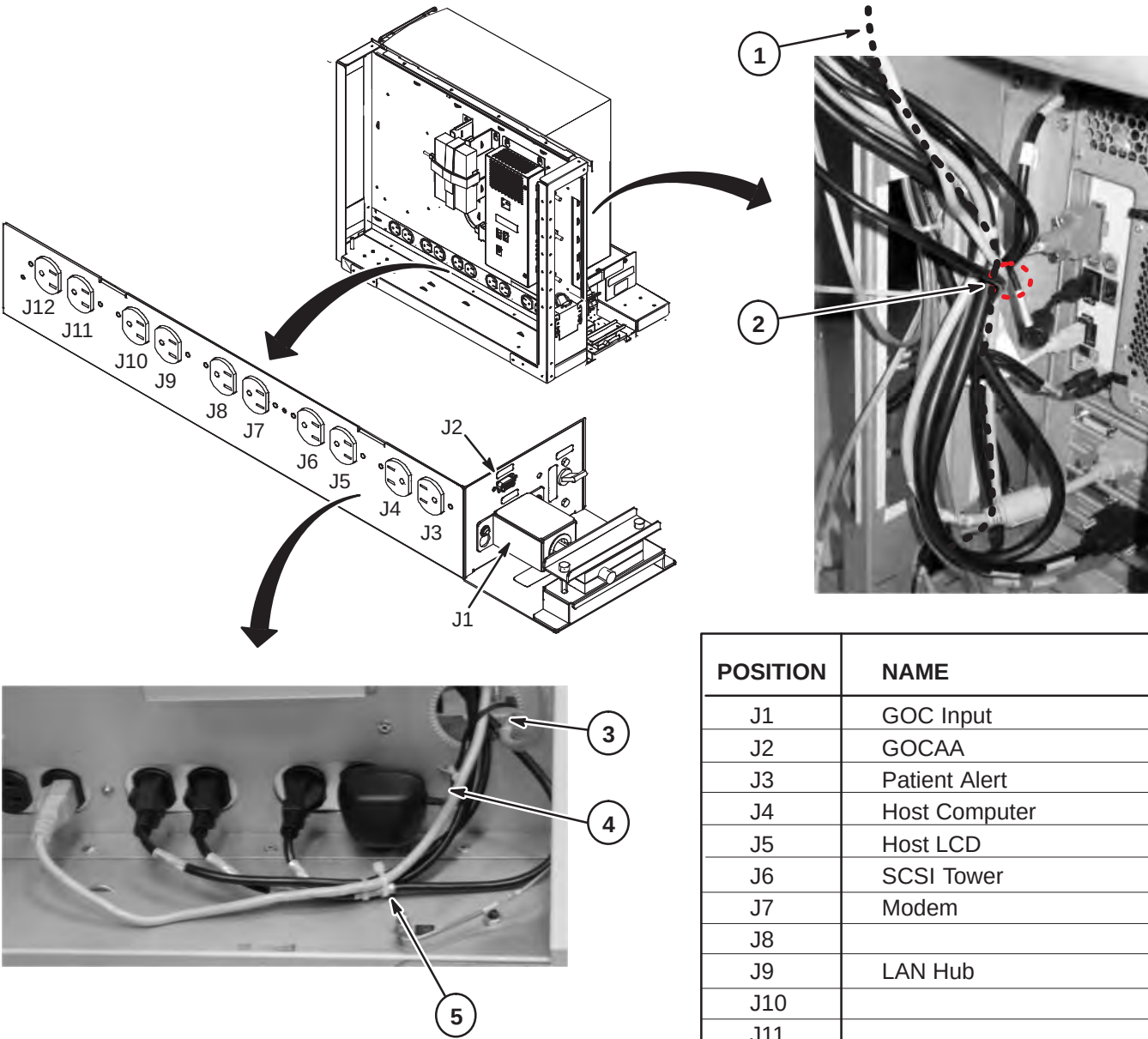


7-2-8 POWER CABLE CONNECTIONS

The illustration below show the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.

Also refer to cable map for information on power cable conections to OW1-PDM.

- 1. If Modem is installed, route power cable and connect to J7.
- 2. Tie-wrap point for Modem power cable.
- 3. Route power cables connected to J3 thru J7 through access hole.
- 4. route and connect Patient Alert power cable to J3 and tie-wrap at this point.
- 5. Tie-wrap power cables J4 to J7 at this point.



POSITION	NAME
J1	GOC Input
J2	GOCAA
J3	Patient Alert
J4	Host Computer
J5	Host LCD
J6	SCSI Tower
J7	Modem
J8	
J9	LAN Hub
J10	
J11	
J12	One Wire Hub

7-3 INSTALL NEW Z400 COMPUTER IN EXISTING GOC

7-3-1 DE-INSTALL OPERATOR WORKSPACE COMPONENTS

CAUTION

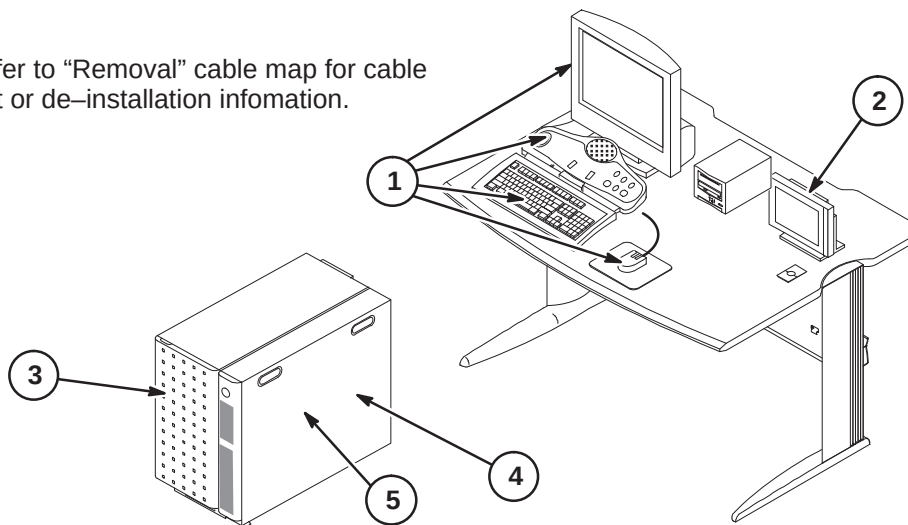
POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

1. Existing Monitor including Runs 1063 and 1065, SCIM, Keyboard, and Mouse will be de-installed and returned for recycling.

Note: If Run 1062 is connected to the SCIM, remove and return for recycling. Install new Run 2053 which is provided with this upgrade (A few early versions of the cable upgrade kit provided a new Run 1256, without the short cables for connecting the keyboard and mouse. If received at this site, relabel as Run 2053). If existing SCIM cable is Run 1256, retain and reconnect to new SCIM.

2. De-install LCD Display, Runs 1068 and 1069, and return for recycling.
3. Existing PC computer in the GOC will be replaced by a new computer:
 - If the existing GOC is old style (screws securing covers from rear), secure the new PC using straps.
 - If the existing GOC is new (screws securing covers from side), the new PC must be secured to the GOC using the conversion kit, part number 5389531.
4. The GOC AA module has been delivered as part of the upgrade kit.
 - If the existing GOC already has a GOC AA module installed, the new GOC AA is not required.
 - If the existing GOC has a WIM installed, it must be replaced by the new GOC AA module.
5. An 8-Port 10/100/1000 Mbps Gigabit Ethernet Switch has been delivered as part of the upgrade kit.
 - If the existing GOC already has this Switch installed, the new Switch is not required.
 - If the existing LAN Hub Switch is an 8-Port 10/100, then it must be replaced by the new 10/100/1000 Switch.

Note: Refer to "Removal" cable map for cable retainment or de-installation information.

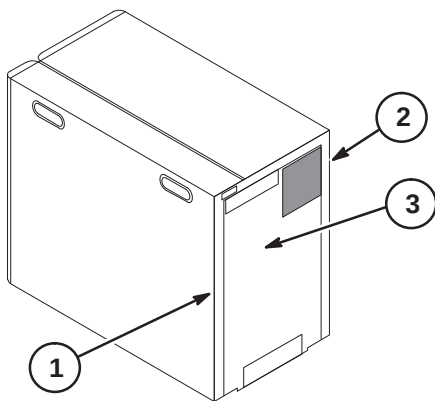


7-3-2 GOC CABINET COVER REMOVAL

The Computer Cabinet covers need to be removed to access the cable connection points.

7-3-2-A OLDER STYLE GOC COVERS

1. For right side cover, remove three screws at rear of cover, slide toward rear of cabinet and then lift up.
2. For left side cover, remove four screws at rear of cover, slide toward rear of cabinet and then lift up.
3. Loosen four screws holding the back cover. Lift and remove cover.

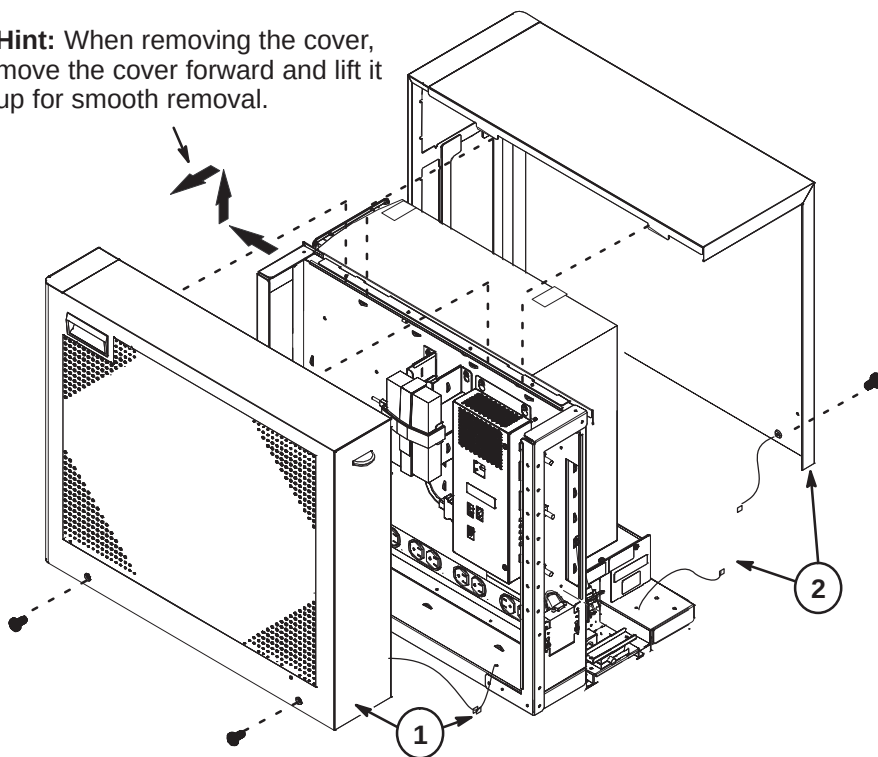


7-3-2-B NEWER STYLE GOC COVERS

Note: A ground wire is connected to each side cover. Use care when removing side covers.

1. For right side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.
2. For left side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.

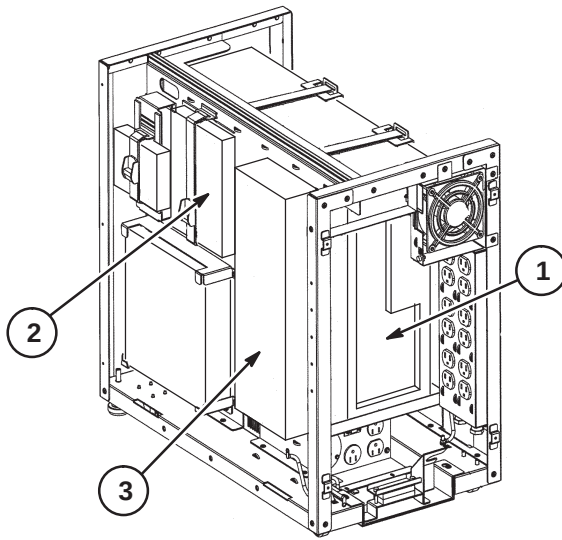
Hint: When removing the cover, move the cover forward and lift it up for smooth removal.



7-3-3 GOC COMPONENT REPLACEMENT

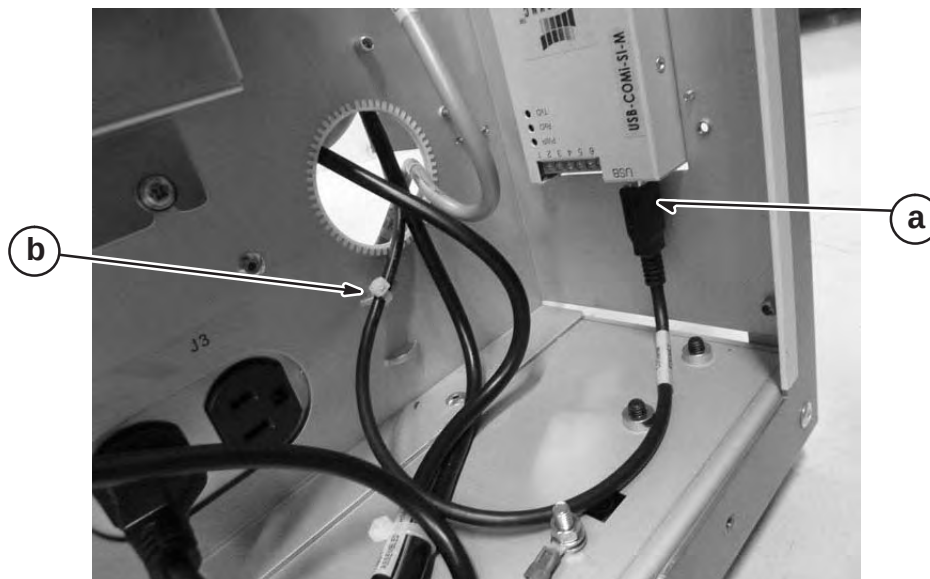
Refer to applicable cable removal map for de-installation of existing cables that will be replaced by new cables. The cable maps also show which cables will be disconnected from existing components and reconnected to new components.

1. Disconnect all cables from existing computer. Make sure they are properly labeled.
2. If required, replace existing LAN Hub Switch with new 10/100/1000 Mbps Gigabit Ethernet Switch (5139480-2).
3. If required, replace existing WIM with new GOC AA module (2395043).



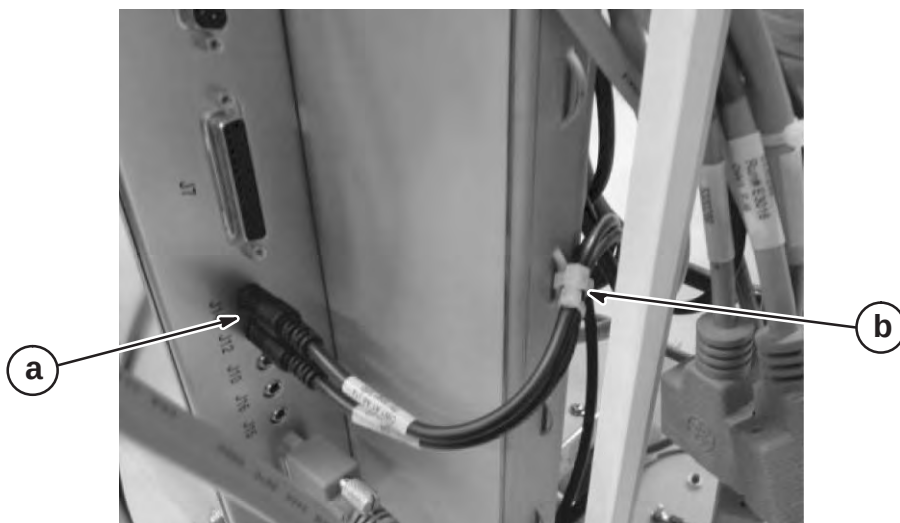
7-3-4 INSTALLATION OF NEW GOC COMPONENTS

1. Replace the Host to USB–Serial Converter USB Cable.
 - a. Replace the Host to USB–Serial Converter USB Cable with the new cable (5330295-2) supplied in the Conversion Kit.
 - b. Cut and restore tie-wrap(s) as required during cable replacement.

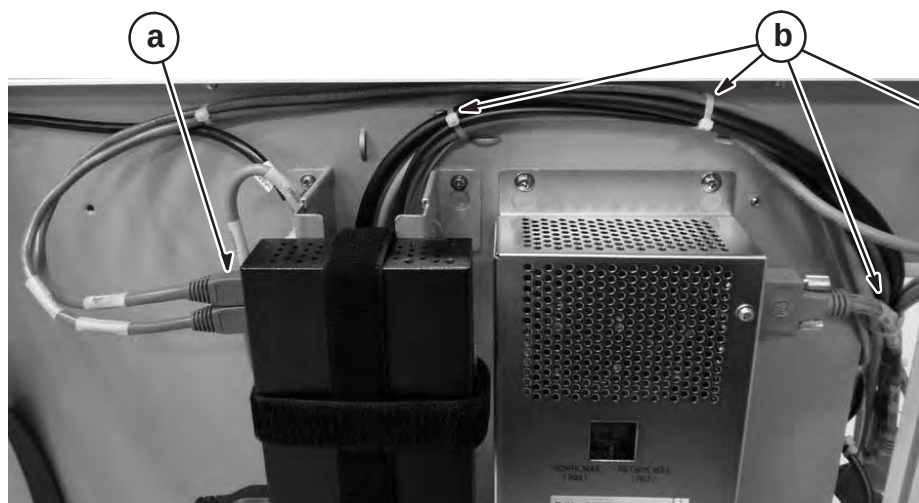


7-3-4 INSTALLATION OF NEW GOC COMPONENTS (Continued)**2. Replace the Audio Cable Assembly.**

- a. Replace the Audio Cable Assembly, connected to J12 and J14 on GOCAA with the new cable (5330301) supplied in the Conversion Kit.
- b. Cut and restore tie-wrap(s) as required during cable replacement.

**3. Replace the Host-Hub Ethernet Cable.**

- a. Replace the Ethernet Cable between the Host and Port 1 of Ethernet Switch Hub with the new Ethernet Cable (5330309) provided in the Conversion Kit.
- b. Cut and restore tie-wrap(s) as required during cable replacement.



View from GOC rear side.

7-3-5 GOC COMPUTER REPLACEMENT

Dependent on model of Host PC Computer being replaced, there are two different procedures for upgrading to a Z400 computer.

- For sites with an existing older style GOC, refer to Section 7-3-5-A for upgrade instructions.
- For sites with an existing newer style GOC, refer to Section 7-3-5-B for upgrade instructions.

IMPORTANT!

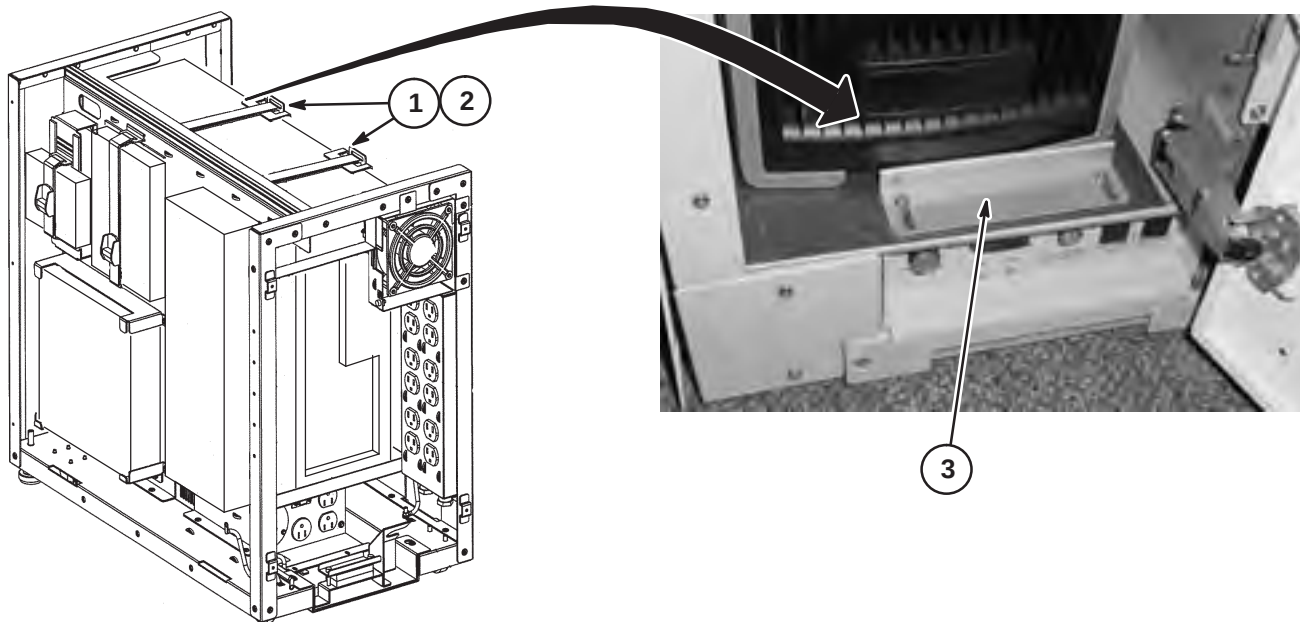
XW8000, 8200, 8400, or 9300 PC internal hardware, such as any hard drive, is not compatible with the Z400 computer. Do not attempt to transfer any internal hardware between any of these and the Z400 computer.

CAUTION

The computer weighs about 50 pounds (23Kg). This requires two people to handle the computer.

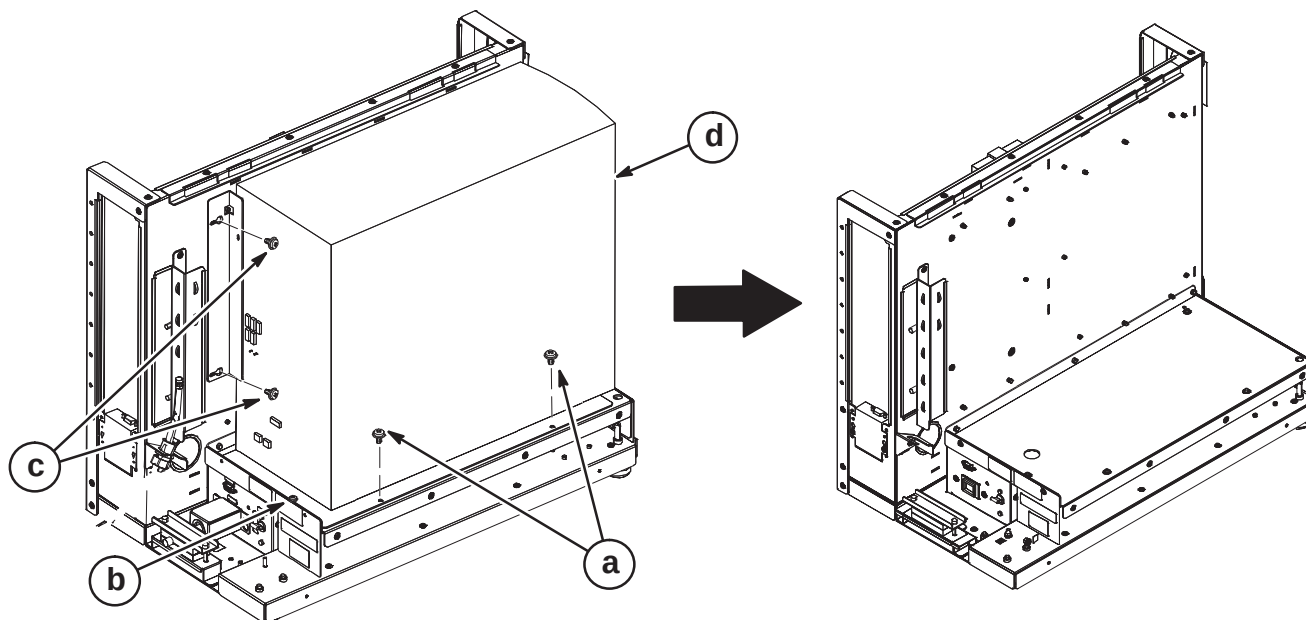
7-3-5-A COMPUTER REPLACEMENT IN OLD STYLE GOC

1. Unfasten straps and carefully remove computer.
2. Because the new Z400 computer is shorter than the original PC being replaced, install the new computer to the rear-most position and secure in place with straps.
3. If required, install PC Stopper Bracket (5127954) at front of computer using existing screws.

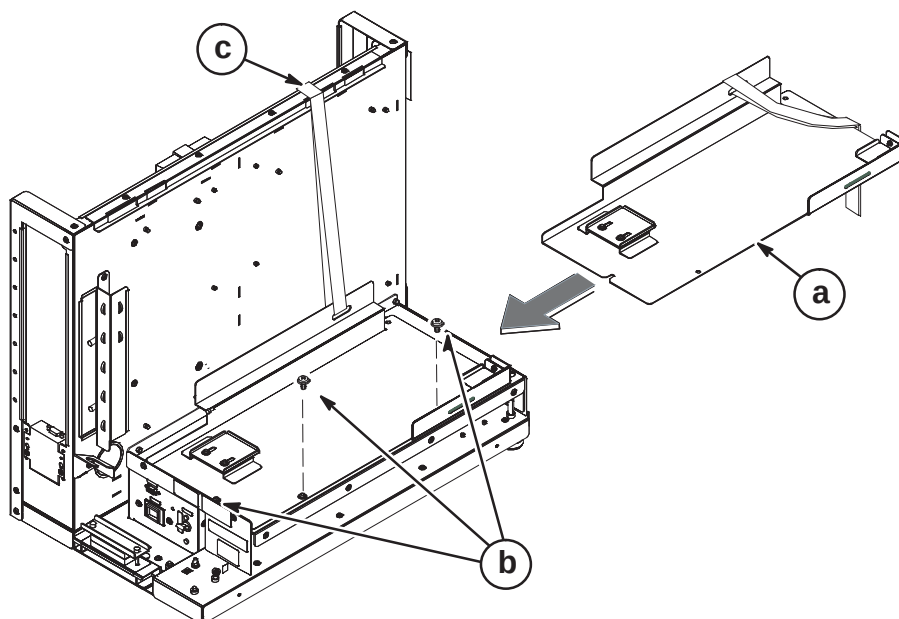


7-3-5-B COMPUTER REPLACEMENT IN NEW STYLE GOC

1. Remove the XW8400 Computer that is secured with back and bottom brackets as follows:
 - a. Remove the two screws securing the side of the bottom plate of the PC to the GOC frame.
 - b. Loosen the one screw securing the rear of the bottom plate of the PC to the GOC frame.
 - c. Remove the two screws from the back bracket.
 - d. Carefully remove the computer with back and bottom brackets from the GOC

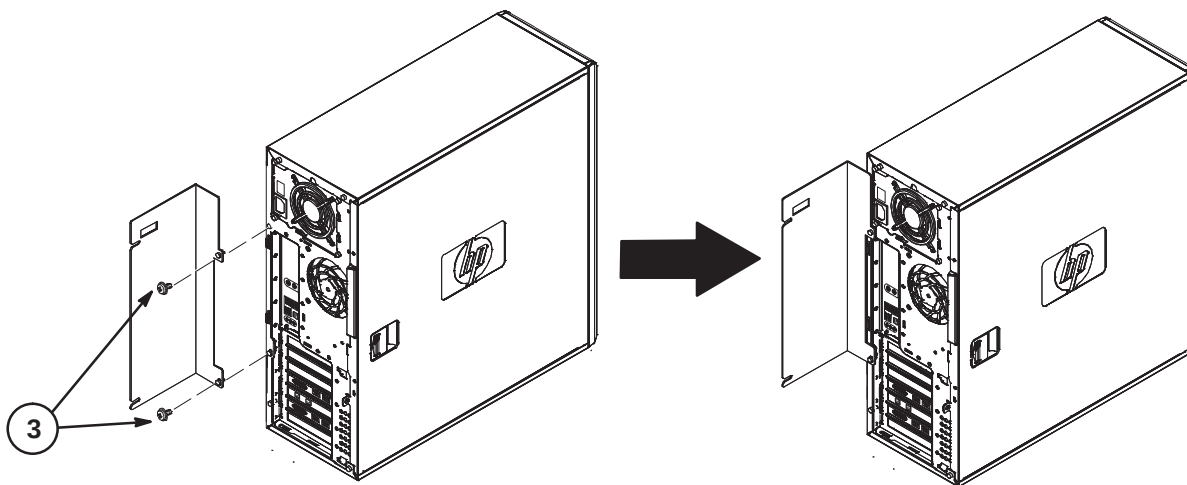


2. Install the Host Mounting Plate for the Z400 computer as follows:
 - a. Insert the Host Mounting Plate so that the GOC chassis screw holes are visible through the plate.
 - b. Tighten the one screw at the rear and the two screws on the left as shown below.
 - c. Temporarily hang the strap on the opposite side.



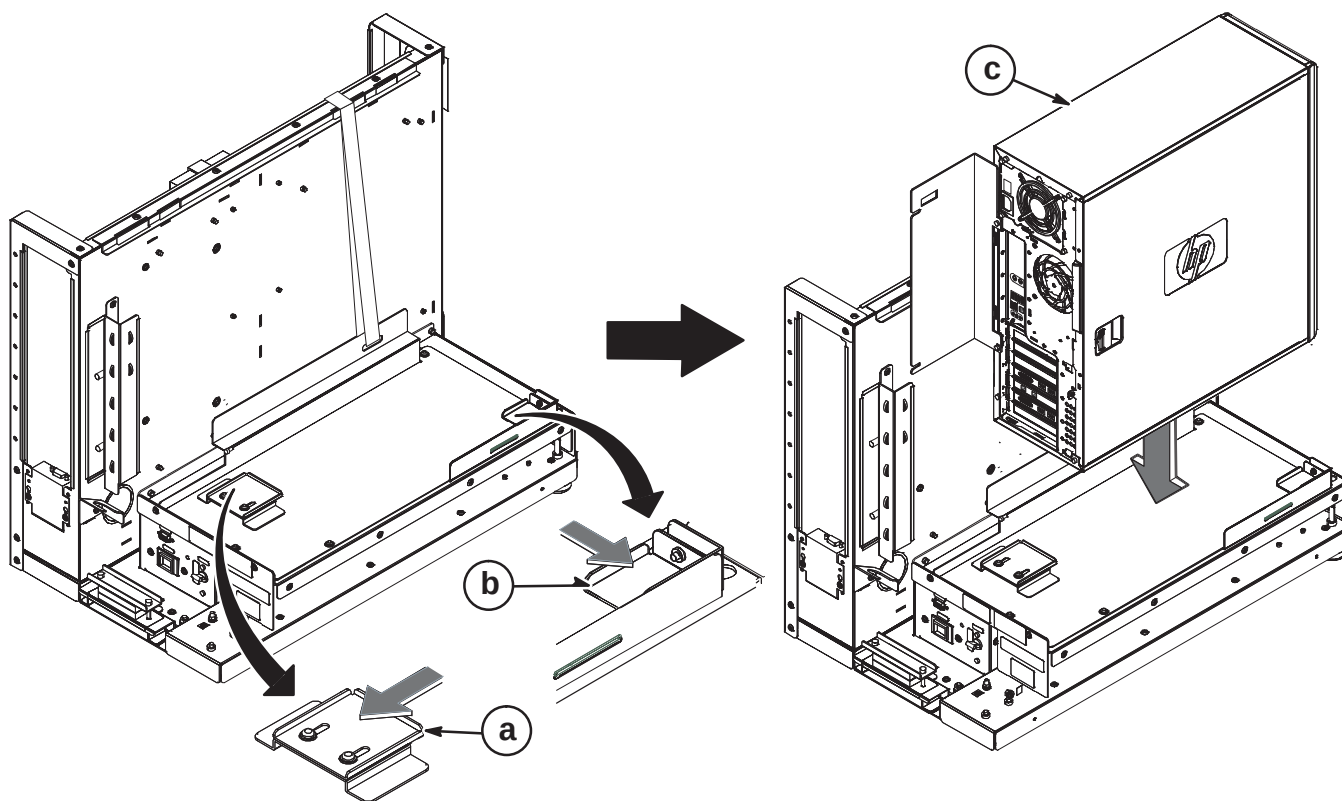
7-3-5-B COMPUTER REPLACEMENT IN NEW STYLE GOC (Continued)

3. Install the back bracket to the computer with the two screws.



4. Position the PC as follows:

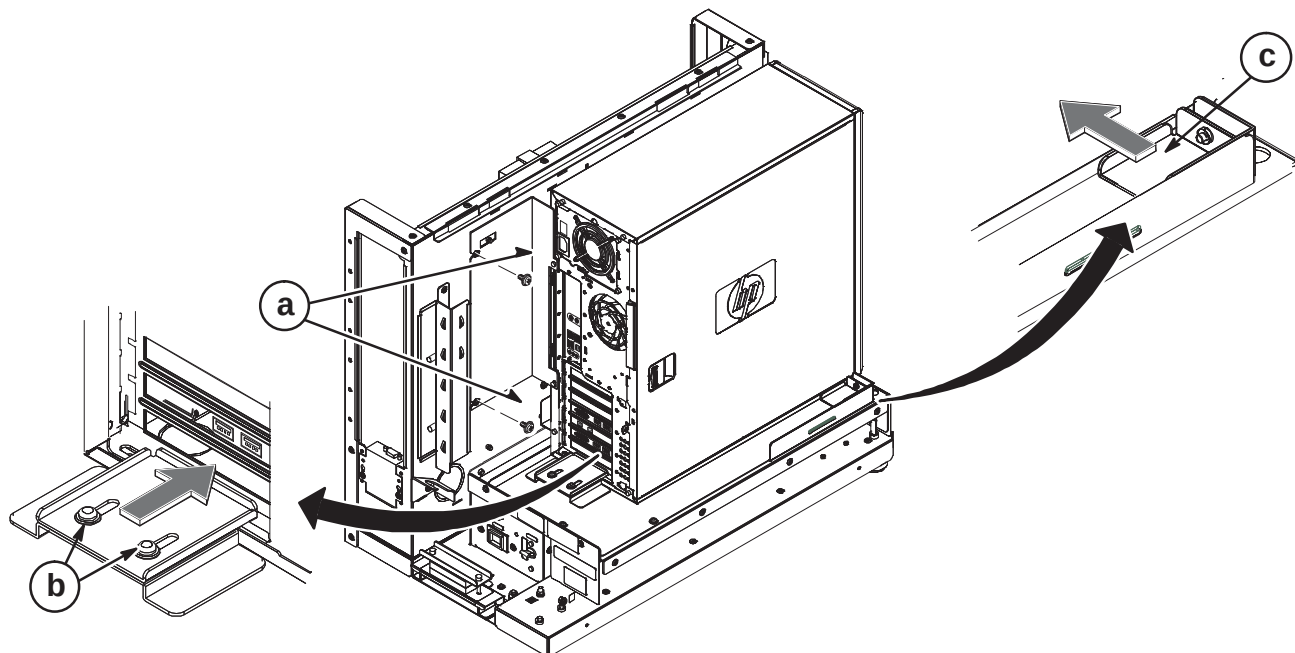
- Loosen the two screws on the back slide plate, and slide the plate backward.
- Loosen the two screws on the side slide plate, and slide the plate in the direction of the arrow as shown.
- Carefully place the Z400 Computer on the mounting plate.



7-3-5-B COMPUTER REPLACEMENT IN NEW STYLE GOC (Continued)

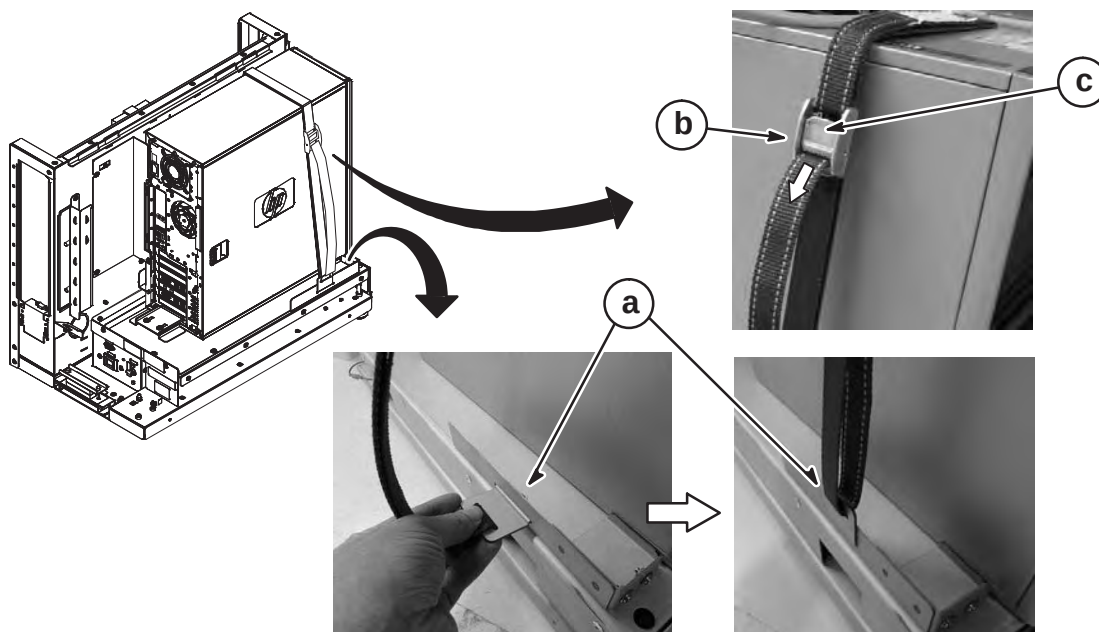
5. Secure the computer as follows:

- a. Secure the back bracket to GOC frame with two screws.
- b. Slide the back slide plate forward until it makes contact with the computer, and tighten the two screws.
- c. Slide the side slide plate until it makes contact with the computer, and tighten the two screws.



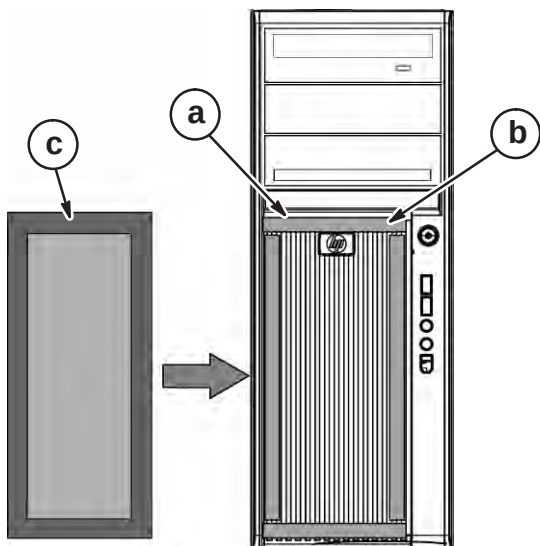
6. Tighten the strap around the computer as follows:

- a. Hook the bracket onto the slot of bottom plate.
- b. Pull and tighten the strap to secure the computer.
- c. **Note:** To loosen the strap, press the release lever on the strap.



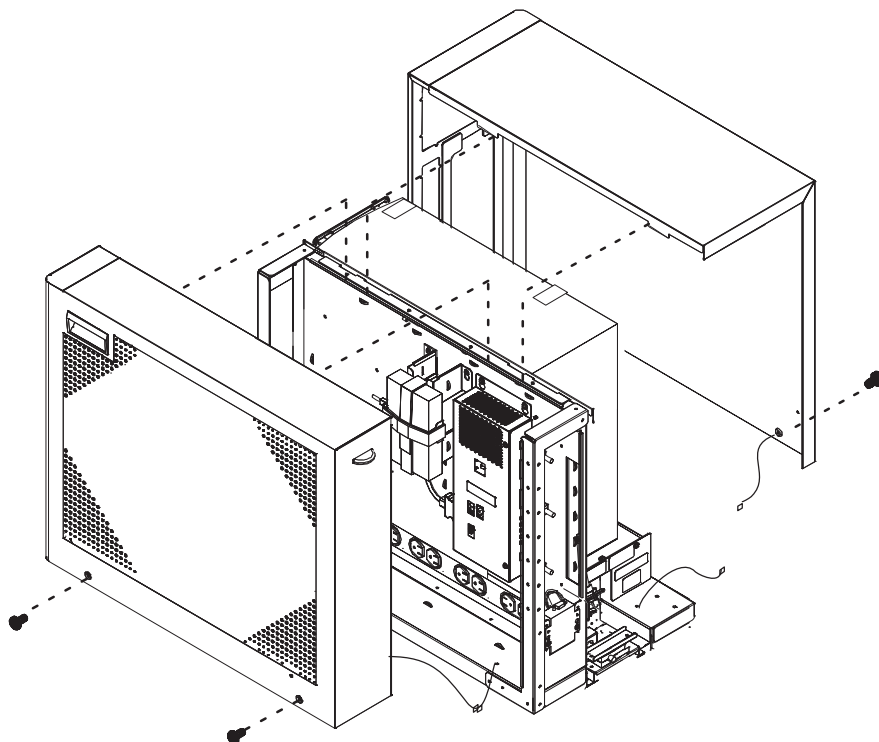
7-3-5-B COMPUTER REPLACEMENT IN NEW STYLE GOC (Continued)

7. Attach the Dust Filter to the front panel after following the cleaning instructions below.
 - a. Clean the velcro area of front panel with alcohol and cloth. Peel the velcro from backside of filter.
 - b. Attach the velcro around front panel.
 - c. Attach the filter to front panel.



8. Replace the GOC covers, using the new left cover for the Z400 computer that shipped with the GOC Conversion Kit.

IMPORTANT! Make sure the Z400 computer is installed at the rear-most position before installing covers. This position will aid in connecting existing cables to rear of Z400 computer.



7-4 TEMPERATURE AND HUMIDITY SENSOR KIT INSTALLATION

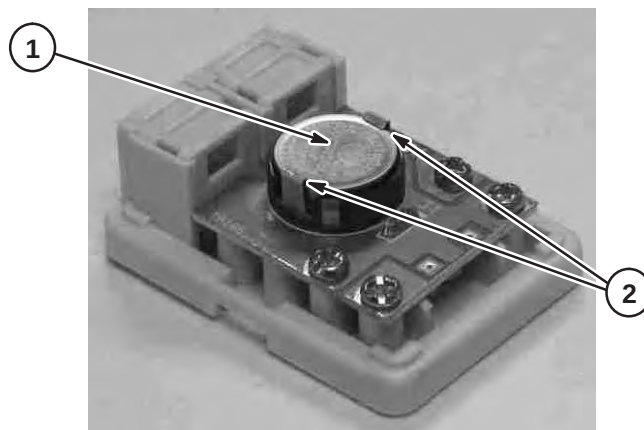
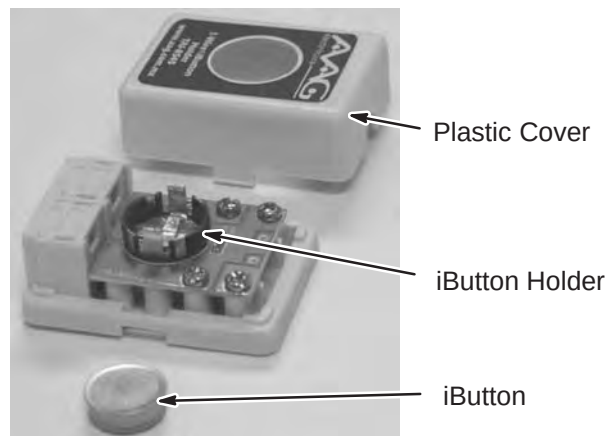
The components of the Temperature and Humidity Sensor Kit (5147136-2) will be installed on the back of the GOC cabinet and on the back of the System Cabinet with one cable routed from the GOC and connected to it.

7-4-1 INSTALL iBUTTON INTO HOLDER

Install the iButton into the holder in the orientation shown in the illustration.

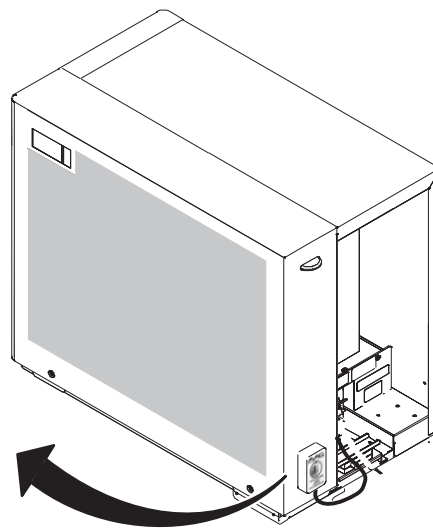
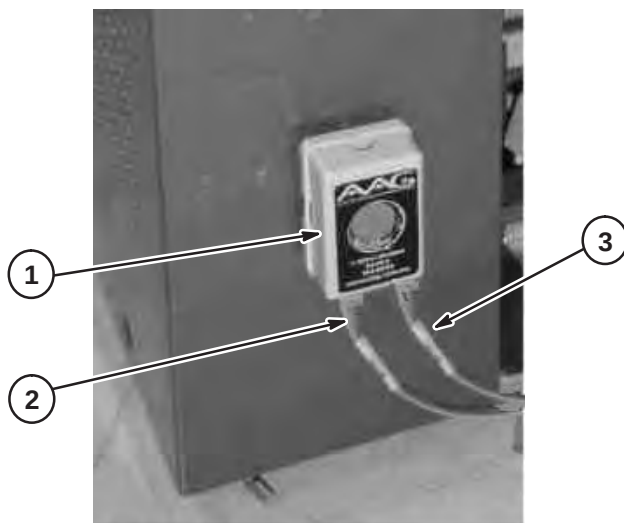
Note: The pictures below show the plastic cover removed although this is not necessary to install the iButton.

1. Install the iButton in the holder so that the solid part of the case is exposed.
2. Make sure the “fingers” latch over the top of the iButton and secure it in the holder.
3. If removed, reinstall the Plastic Cover.



7-4-2 INSTALL TEMPERATURE & HUMIDITY SENSOR ON GOC

1. Attach iButton Sensor assembly (Holder, 5141856, and iButton 5141857-5), to back of GOC Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779-2).
2. Route and connect Run 1322 (5143794-2) from System cabinet to Sensor at back of GOC.
3. Connect cable Run 1321 (5143794-2) to Sensor and route other end thru bottom opening of GOC and connect to USB to 1-Wire Adapter (5307784) installed in GOC USB port. Refer to step 10 in Section 7-3-5-B.



7-4-3 INSTALL TEMPERATURE & HUMIDITY SENSOR ON SYSTEM CABINET

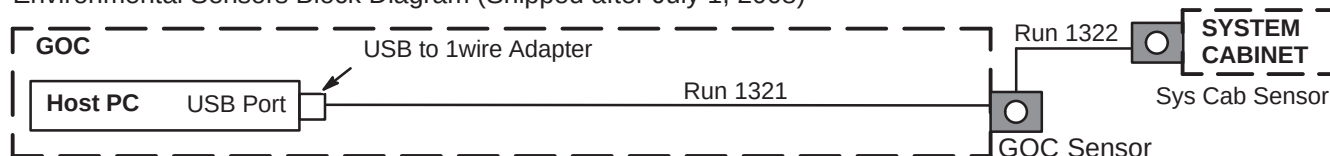
Note: iButton Holders are unique to each location. Make sure to install 5141857-4 to System Cabinet and 5141857-5 to GOC.

1. Attach iButton Sensor assembly (Holder, 5141856, and Humidity iButton (5141857-4), to back of System/RFS Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779-2).
2. Route and connect cable Run 1322 (5143794) from GOC rear cover.

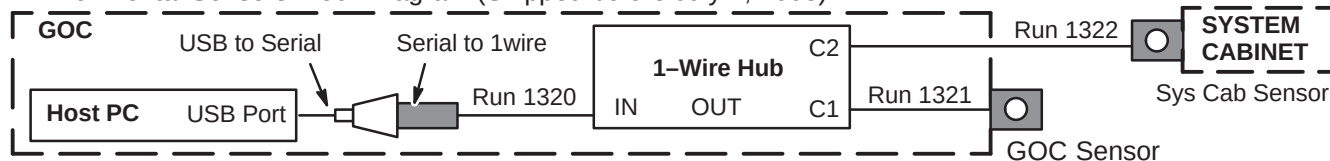


7-4-4 TEMPERATURE & HUMIDITY SENSOR WIRING SCHEMATICS

Environmental Sensors Block Diagram (Shipped after July 1, 2008)



Environmental Sensors Block Diagram (Shipped before July 1, 2008)

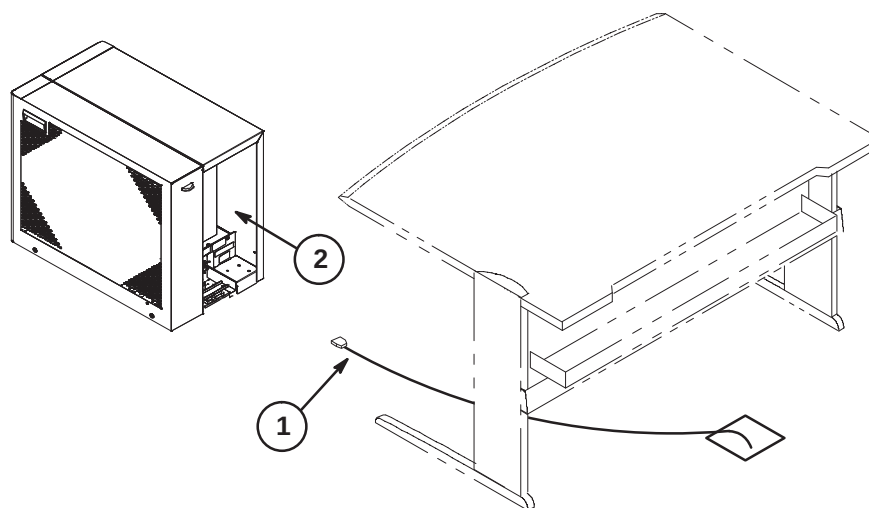


7-5 ROUTE RUN 1323

1. Route Run 1323 from RFS or System Cabinet to location of GOC. This is a new CAT6 cable that must be installed. Existing CAT5 cables cannot be reused.

NOTE: For optimal efficiency, route Run 1322, Section 7-4-3, at the same time as Rune 1323. It is also routed between RFS/System Cabinet and GOC.

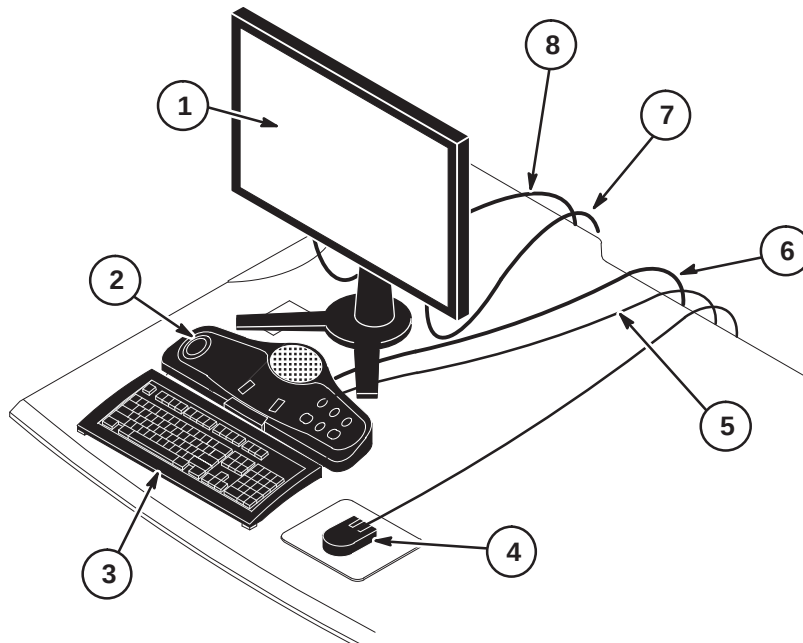
2. Cable will be routed thru the lower rear opening.



7-6 INSTALL NEW OW TABLE TOP COMPONENTS

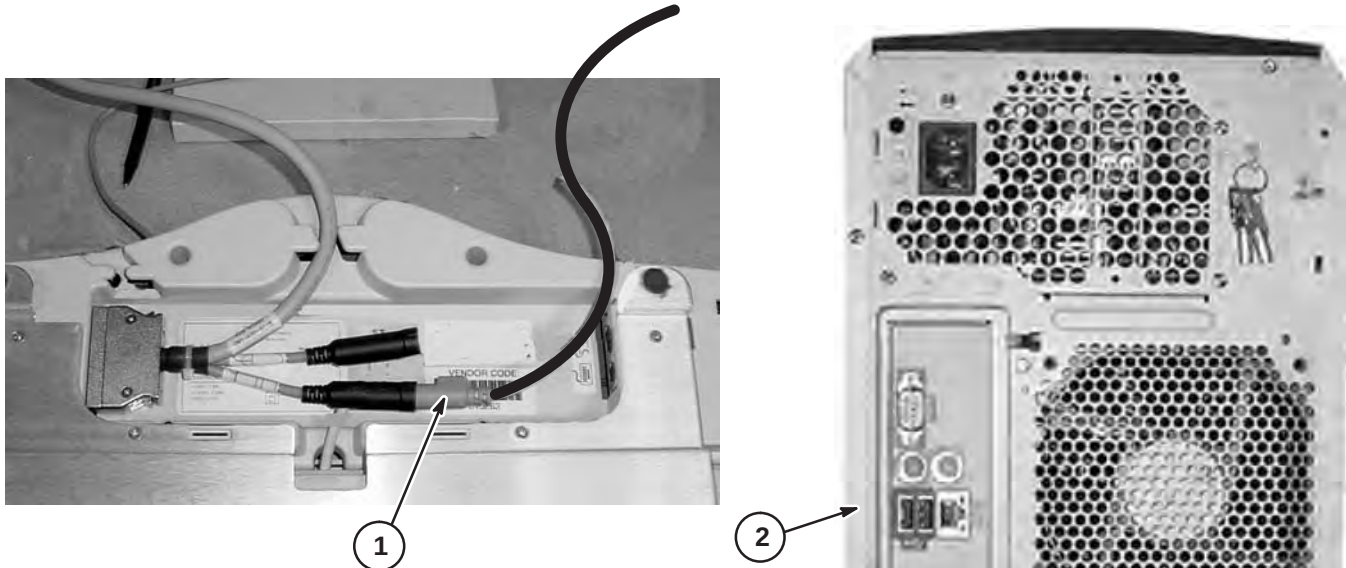
1. Install new Monitor on table.
2. Place new SCIM (5148388-x) on table.
3. Place new Keyboard (5143798-x) on table.
4. Install new Mouse (5143798-6) on table and route USB cable to computer in GOC. Attach provided extension USB cable if necessary.
5. Connect extension cable Run 2038 to USB Keyboard cable and route to computer in GOC.
6. Connect Run 2053 (or 1256) to SCIM and route to rear of GOC.
7. Connect provided monitor power cable to monitor. Connect other end of cable to Run 1065, which is connected to OW power strip.
8. Connect DVI cable, Run 2037, to monitor and route to rear of computer in GOC.

Note: Do not use or connect provided VGA cable.

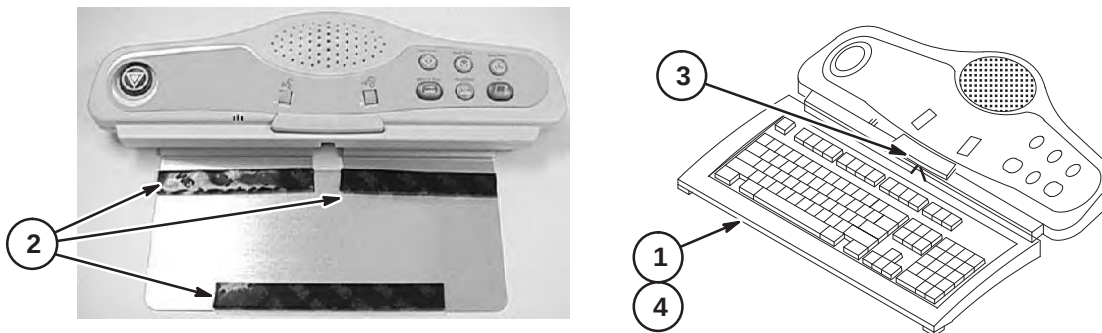


7-7 CONNECT BAR CODE READER (If part of site configuration)

1. If Run1256 is present – Disconnect the Bar Code Reader cable from Run 1256 Y-connector under SCIM.
2. Connect cable Run 1379 to rear of computer and Bar Code Reader.

**7-8 INSTALL SCIM/KEYBOARD**

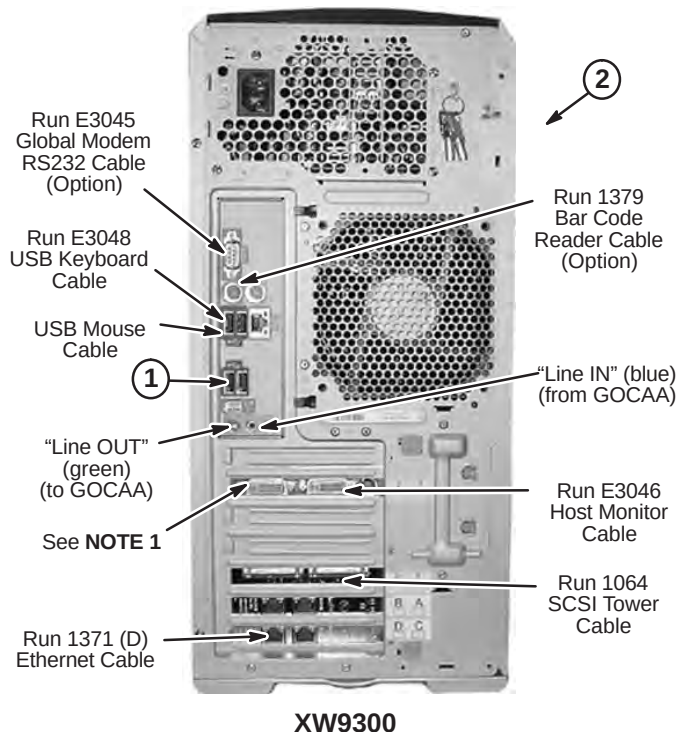
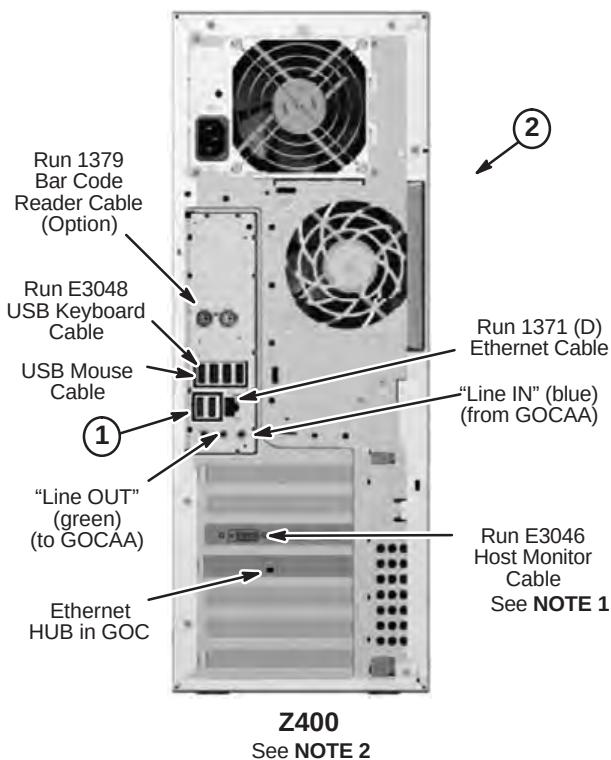
1. Place keyboard in front of SCIM, pull the keyboard forward, flip it, and place on top of the SCIM up side down.
2. Remove the plastic strips from the velcro located on SCIM plate, exposing the adhesive surface.
3. Guide the cable back into the center notched opening until all the cable is under the SCIM base.
4. Place the keyboard tight against the SCIM, Center it as best as possible, and drop into place. The Velcro pads will now be sticking to the bottom of the keyboard.

**Note**

The adhesive on the velcro takes 24 hours to completely adhere to the bottom of the keyboard. Allow the adhesive to dry completely before removing keyboard from SCIM base.

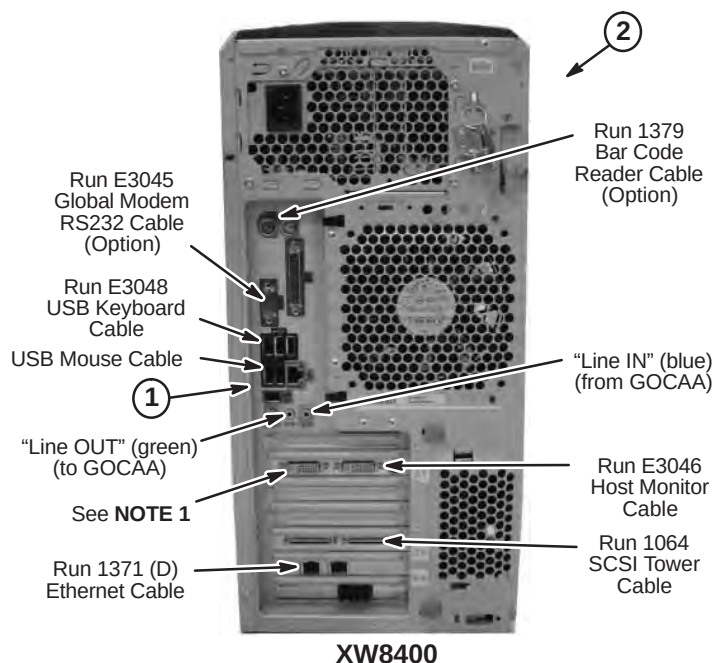
7-9 CONNECT CABLES TO Z400 COMPUTER

1. Refer to Section 7-4 for installing the USB to 1-Wire Adapter (5307784) to USB port.
2. Connect remaining cables as marked at rear of computer.



NOTE 1:
Do not connect any additional monitors prior to loading operating system software. Doing so will cause boot-up to stop.

NOTE 2:
SCSI Tower and DASM are not supported for Z400PC.



7-10 FINAL POSITIONING OF GOC CABINET

Complete positioning of GOC Cabinet.

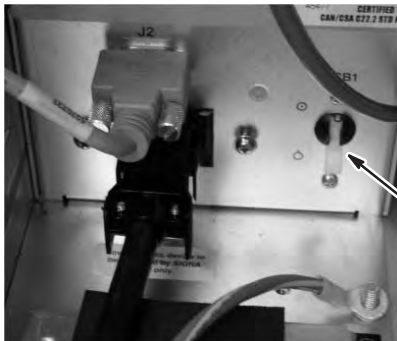
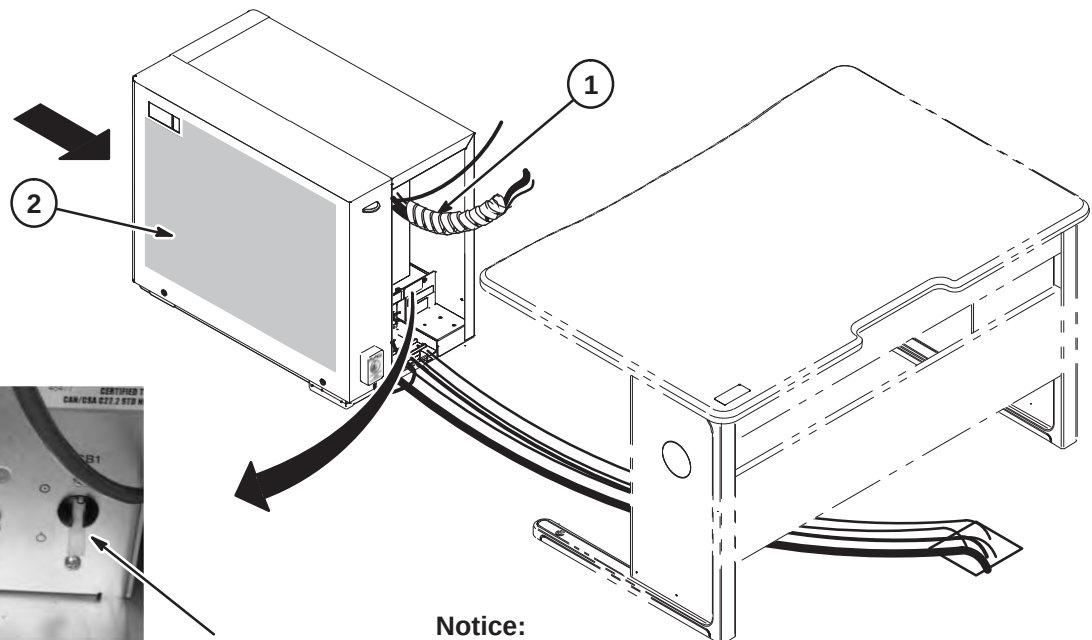
1. Use the spiral tube to bind the following cables. (Length of spiral tube is about 1 m).

Notice

Do Not include Modem power cable in Spiral Tube.

- Two power cords
- One video cable
- Keyboard cable
- SCIM cable
- Modem cable (if applicable)

2. Place cabinet in final position as determined by customer.



Notice:

GOC circuit breaker location. To avoid turning the Breaker OFF by accident, do not place the cables inside of GOC Cabinet near the circuit breaker.

7-11 POWER-ON INFORMATION

1. Remove LOTO from the PDU.

IMPORTANT

When turning on the circuit breaker in the PDU, both the PC and Host circuit breakers must be energized.

Note

When powering up the GOC, there is a Main breaker located in the front of the GOC. There are also two switches for the GOC AA and Modem power.

2. Restore power to the GOC, boot software, and ensure proper system operation.

NOTE: If necessary, reconnect the optional UPS accessory and ensure power is restored.

3. Perform the applicable Loading Host System Software procedure to load software and applications onto the operating system.

NOTE: The Load from Cold must be performed to properly transfer customer options

4. Update the elicense with the new Host ID.
5. Repackage the replaced computer for return
6. Perform a scan to ensure the system and applications are working properly.

SECTION 8 – SYSTEM CABINET

TABLE OF CONTENTS

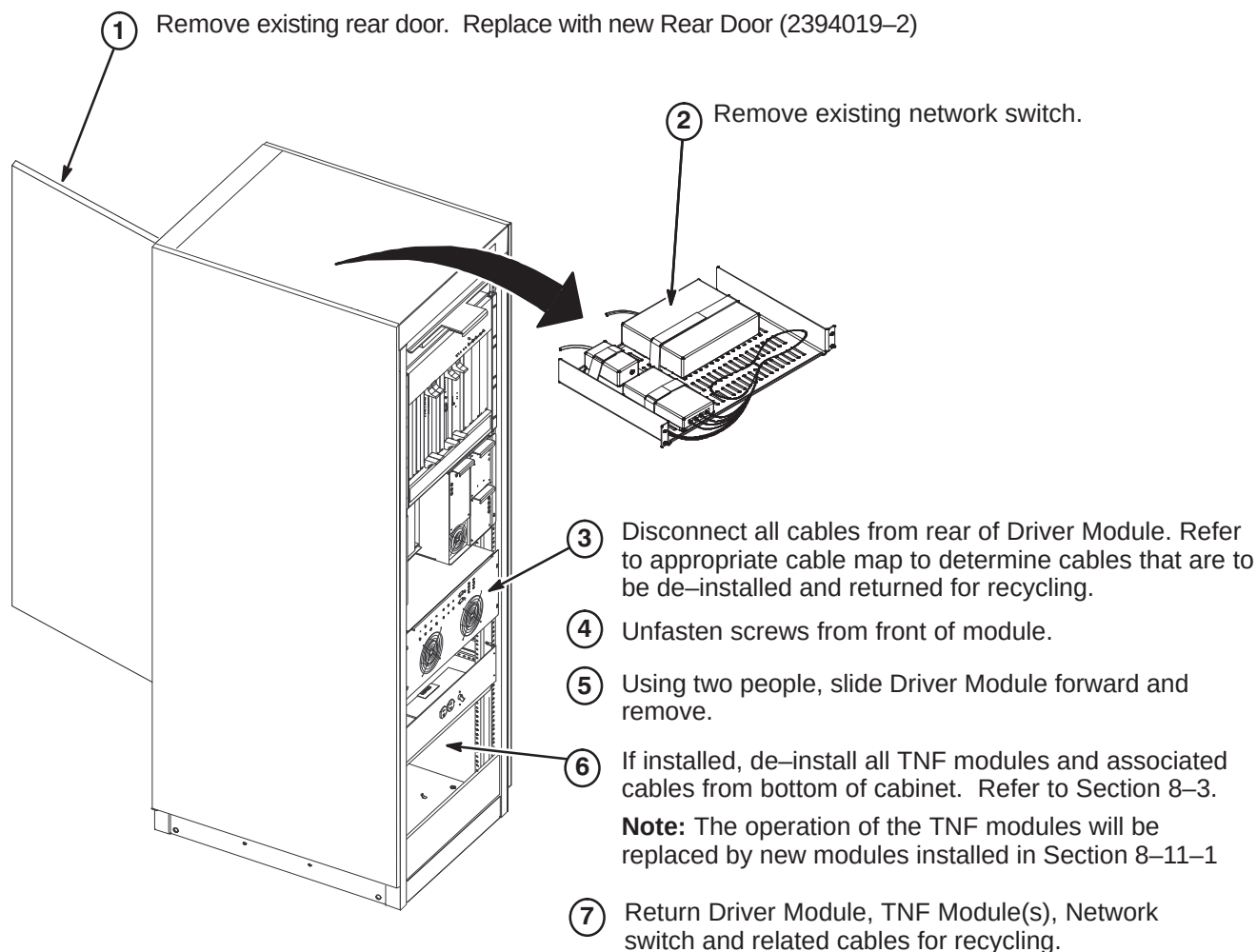
<u>SECTION</u>	<u>PAGE</u>
8-1 INTRODUCTION	8-2
8-2 PREPARE SYSTEM CABINET FOR UPGRADE	8-2
8-3 DE-INSTALL TNF MODULES AND CABLES	8-3
8-4 INSTALL COAX POWER SPLITTER	8-3
8-5 RELOCATE SERVICE POWER PANEL	8-4
8-6 INSTALL NEW DRIVER MODULE	8-5
8-7 INSTALL NEW INTERFACE PANEL	8-5
8-8 CONNECT CABLES FROM INTERFACE PANEL AND DRIVER MODULE	8-6
8-9 UPGRADE BOARDS IN MGD CHASSIS	8-7
8-10 INSTALL VRE COMPONENTS	8-8
8-10-1 INSTALL VRE MODULE RAILS	8-8
8-10-2 INSTALL VRE MODULES	8-9
8-10-3 INSTALL VRE CABLES	8-10
8-11 PROCEDURES FOR UPGRADING TO 16 CHANNEL (M3000PE)	8-11
8-11-1 REMOVE EXISTING COMPONENTS FROM RRF CHASSIS	8-11
8-11-2 INSTALL NEW MODULES IN RRF CHASSIS	8-11
8-11-3 INSTALL CABLES AT REAR OF RRF CHASSIS	8-12
8-11-4 INSTALL CABLES AND POWER SUPPLY AT FRONT OF RRF CHASSIS	8-12
8-12 INSTALL NEW REAR DOOR AND RATING PLATE	8-13

CAUTION

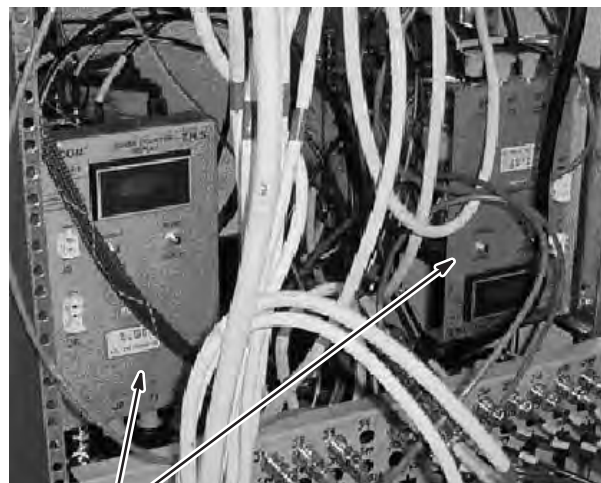
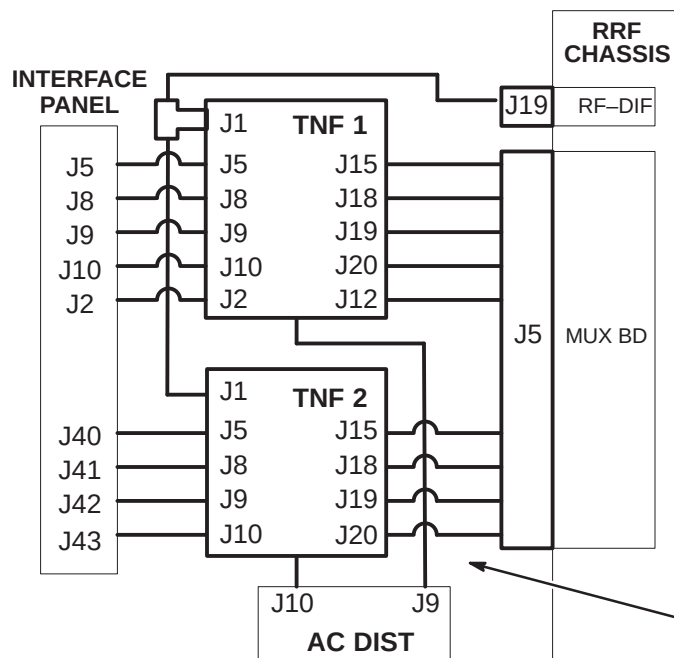
POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

8-1 INTRODUCTION

The System cabinet will be modified for the upgrade to HDx Release 14.x. Refer to cable maps in Section 4 for disconnection and removal of existing cables and installation of new cables. Refer to the following steps for the upgrade procedures that pertain to your site.

8-2 PREPARE SYSTEM CABINET FOR UPGRADE

8-3 DE-INSTALL TNF MODULES AND CABLES



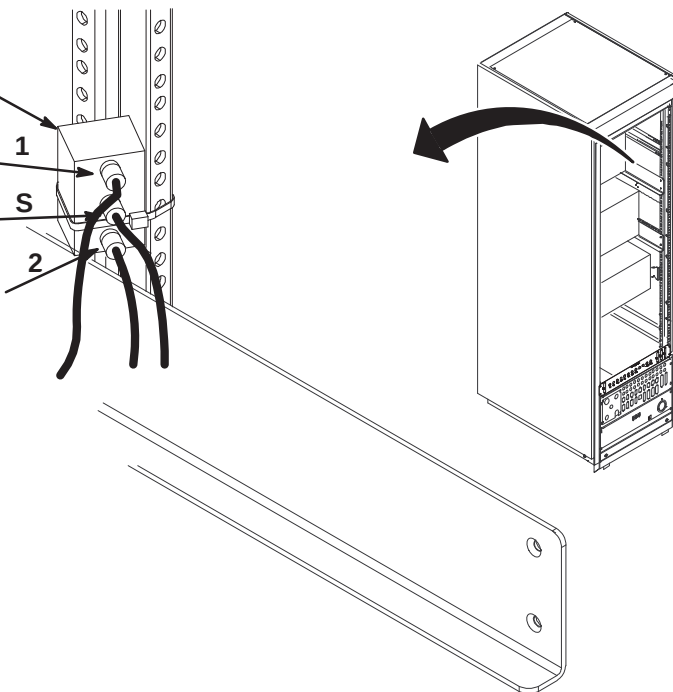
- ① TNF 1 and, if installed, TNF 2 modules and associated cables will be removed and returned for recycling.

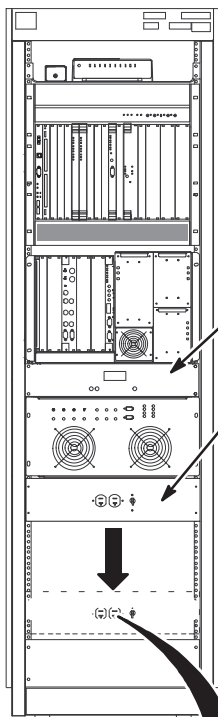
② — = Cables and modules to be de-installed.

8-4 INSTALL COAX POWER SPLITTER

- ① Install BNC Coax Power Splitter (5125176) on vertical rail.
- ② Connect cable from slot 12R J31 at rear of RRF Chassis to "1" on Splitter.
- ③ Connect cable from Interface Panel J202 to "S" on Splitter.
- ④ Install 50 Ohm BNC Terminator (46-296817P1) on "2" of Splitter if site is NOT being upgraded to a 16 channel system.

If 16 Channel is installed, connect cable from slot 3R J31 at rear of RRF Chassis to "2" on Splitter.



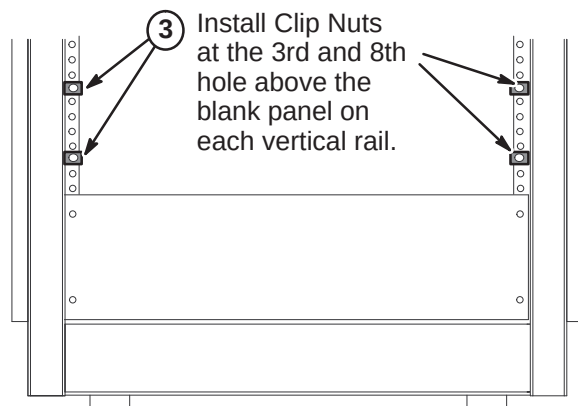
8-5 RELOCATE SERVICE POWER PANEL

① If Cryogen Meter is part of cabinet, de-install and return for recycling.

② Service Power Panel has to be relocated down to new location.

⑤ Cut tie-wraps holding Service Power Panel wires to vertical rail, re-position cables and tie-wrap as shown.

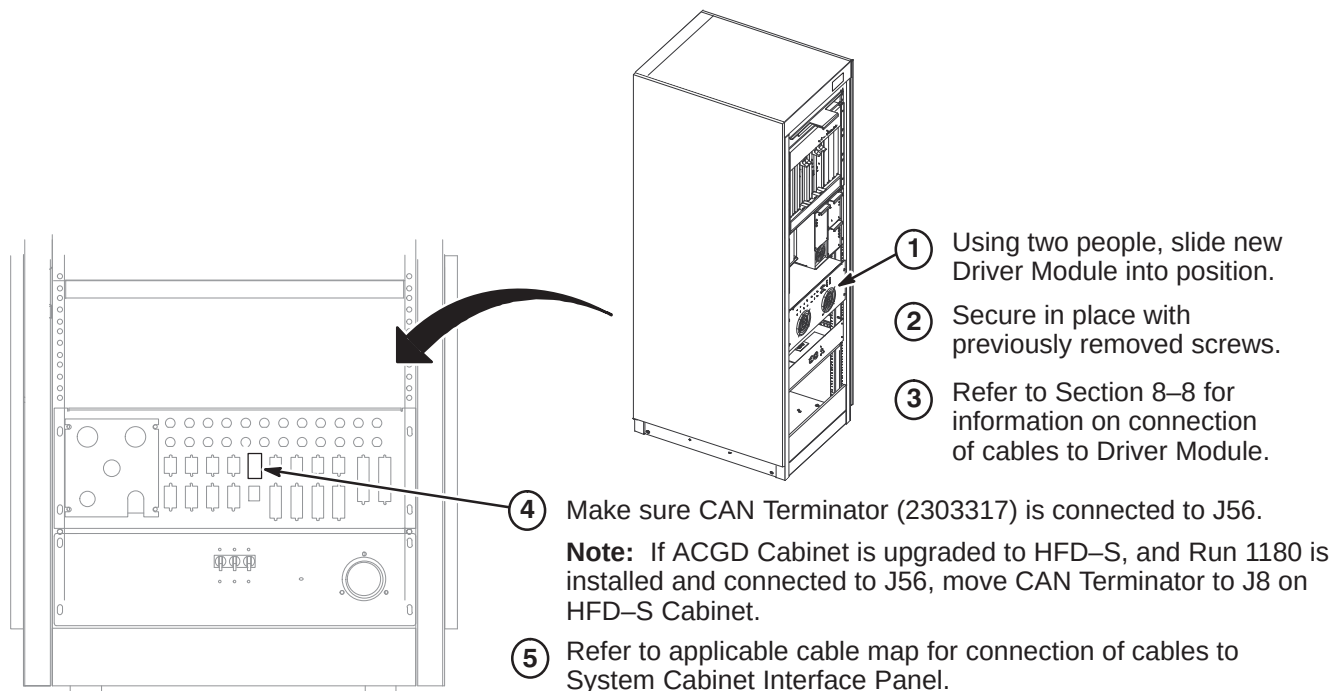
④ Fasten the Panel in place using the same mounting hardware. There will be a gap between Service Power Panel and the blank panel below.



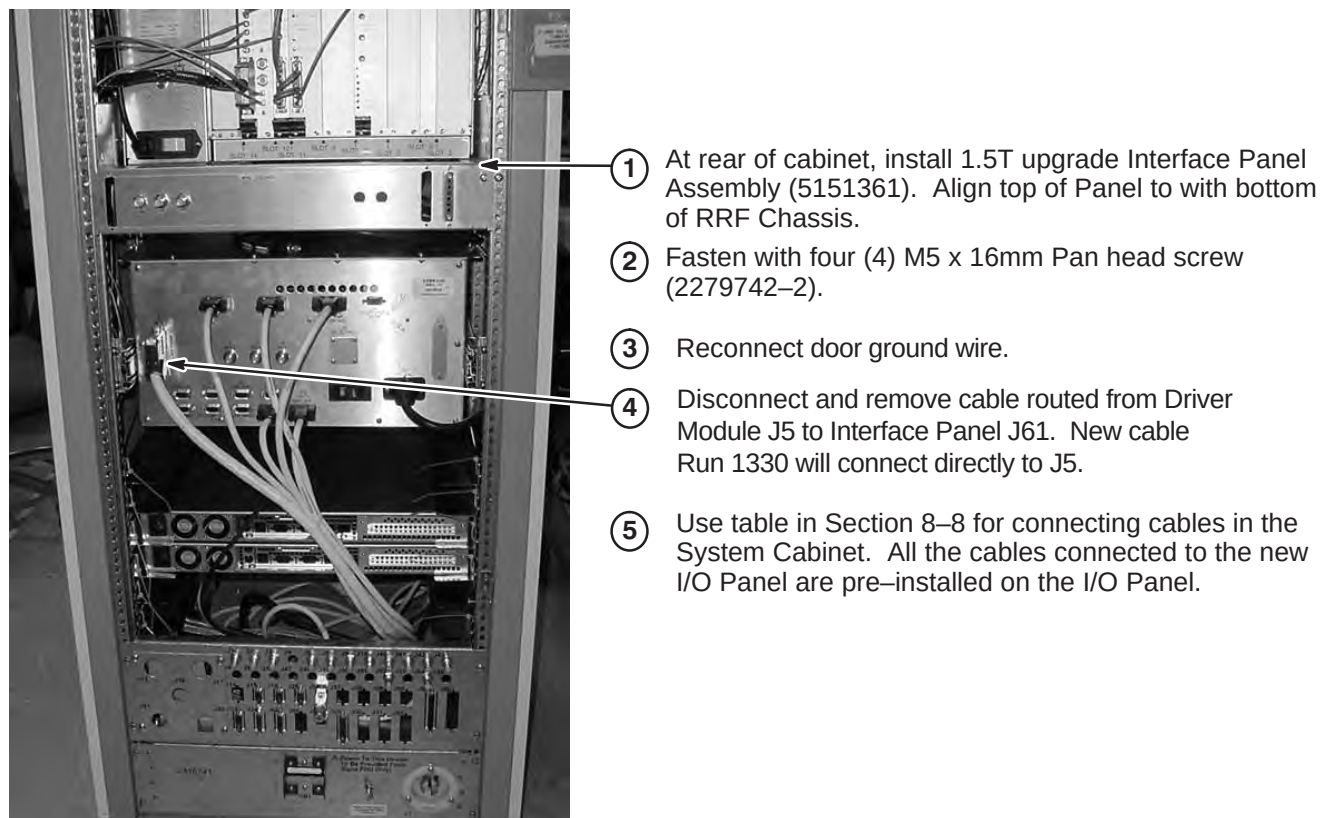
③ Install Clip Nuts at the 3rd and 8th hole above the blank panel on each vertical rail.



8-6 INSTALL NEW DRIVER MODULE



8-7 INSTALL NEW INTERFACE PANEL



8-8 CONNECT CABLES FROM INTERFACE PANEL AND DRIVER MODULE

The following table provides information for connecting existing and new cables within the System Cabinet.

CABLE PART #	ONE END	OTHER END
2363363	Driver Module J1	System Cabinet Interface Panel J56
2323878-3	Driver Module J2	RF Slot 10F J23
5117295	Driver Module J4	MGD Slot 11R J21
2374675	Driver Module J8	I/O Panel J4
2374675-2	Driver Module J7	I/O Panel J53
2374675-3	Driver Module J6	I/O Panel J55
2394870-3	RF Slot 14R J5	I/O Panel J200
2394870-4	RF Slot 5R J5	I/O Panel J201
5115416	Splitter S and RF Slot 14R J13	I/O Panel J202

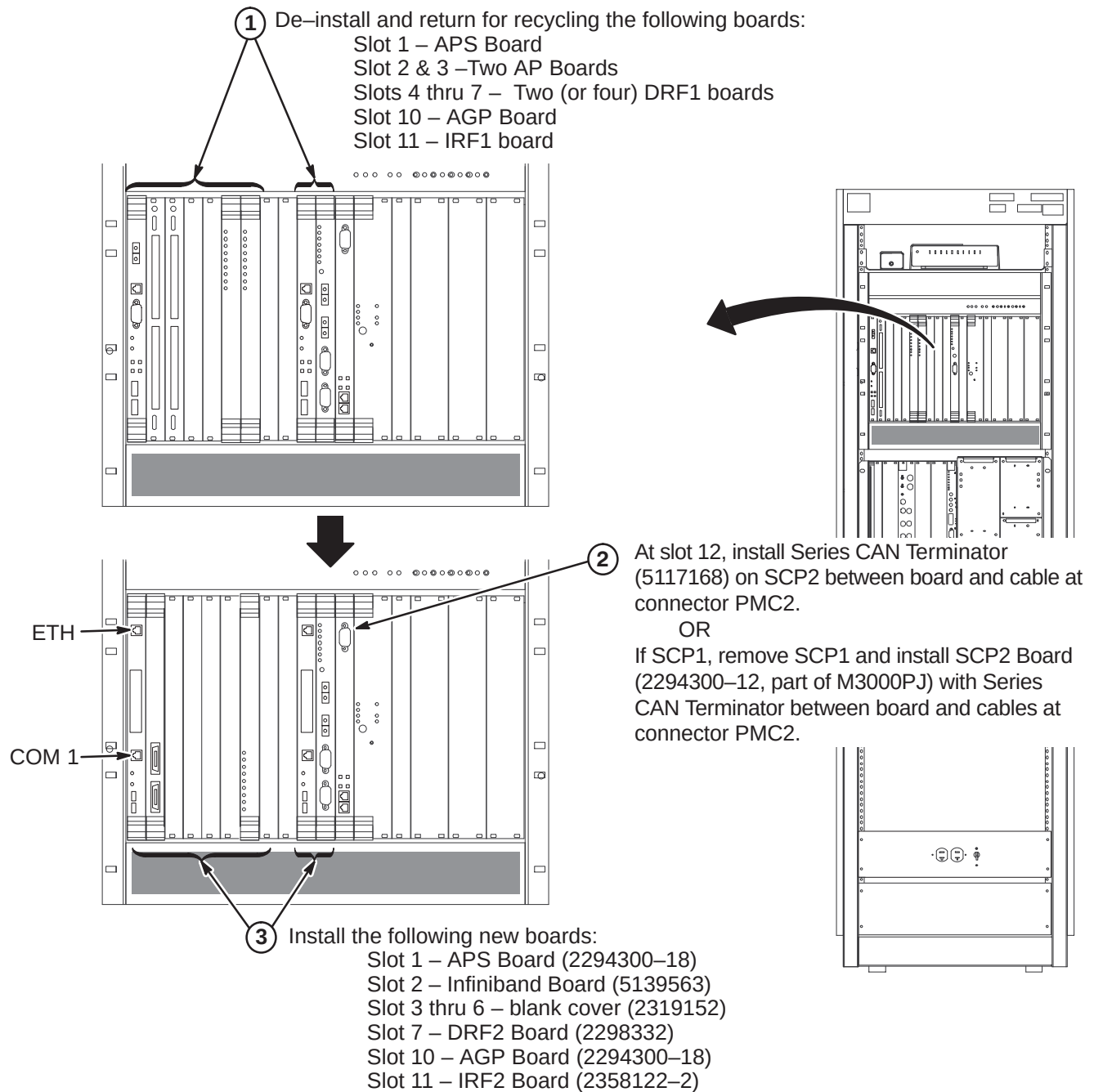
8-9 UPGRADE BOARDS IN MGD CHASSIS

After upgrade, all systems will have one DRF2 board and IRF2 board.



Note: Modules are static sensitive components and need to be handled as such (wrist strap).

Note: to remove modules – loosen the screws in the ejector pins, then push the top and bottom tabs to eject the boards from the chassis.



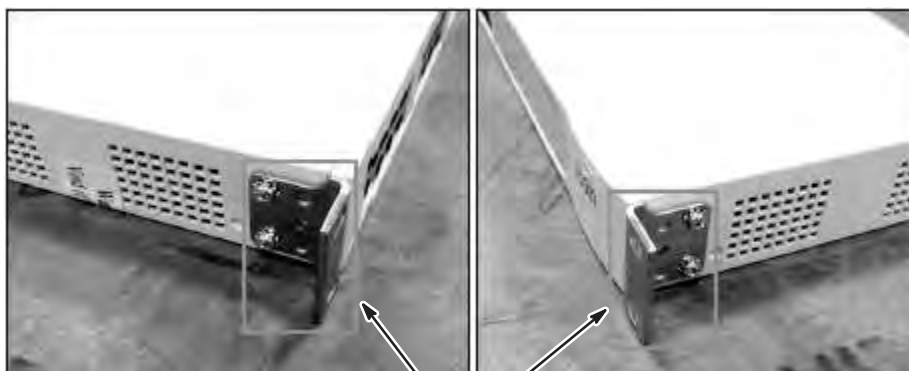
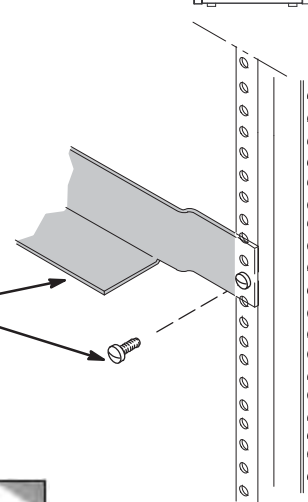
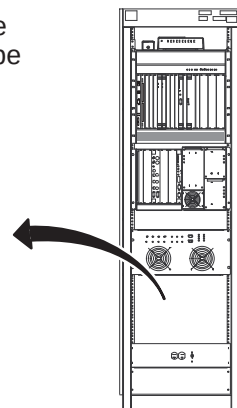
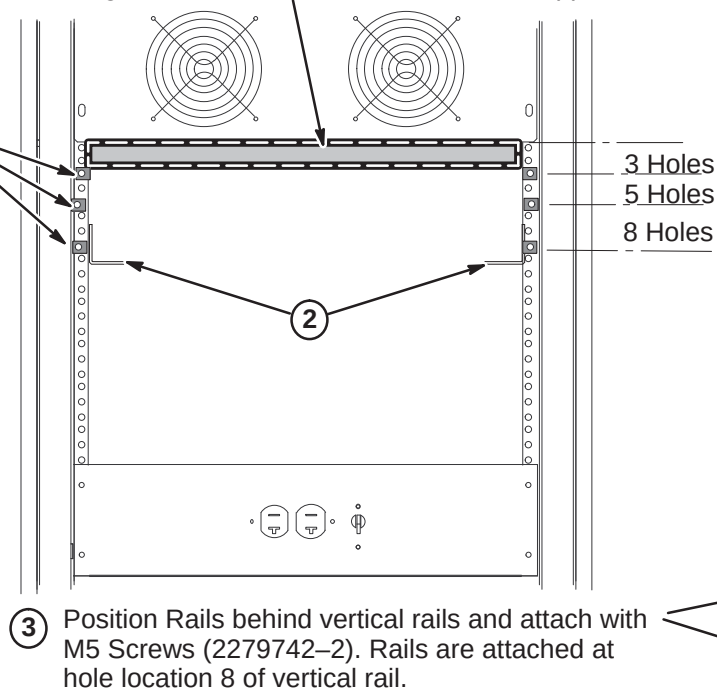
8-10 INSTALL VRE COMPONENTS

8-10-1 INSTALL VRE MODULE RAILS

- ① Make sure that the clip nuts are located below two mounting holes on the vertical rail. This area between Driver Module and Ethernet Switch will be used for routing cables from the Ethernet Switch to applicable modules.

- ④ Install 4 Clip Nuts (2180698-2) on each side at locations indicated.

- ② Location of rails (5158758), one on each side at the position shown for the ICN modules.



- ⑤ Install brackets on front corners of Ethernet switch as shown. Hardware kit comes with unit.

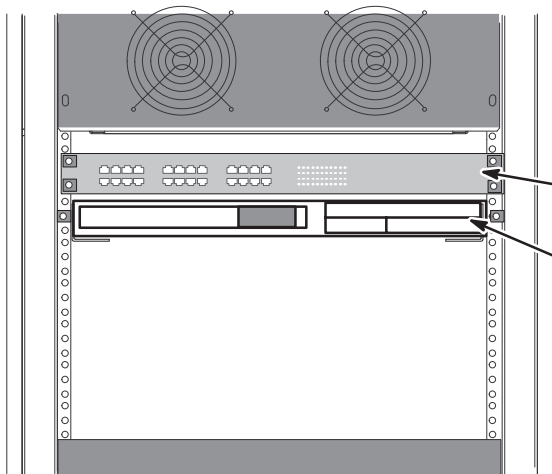
8-10-2 INSTALL VRE MODULE

- ① Install 2 (5306485) Brackets on the ICN as shown.

On the left side, the bracket is installed downward and snapped into place.



On the right side, the bracket must be installed upward at a 45 degree angle and then snapped into place.

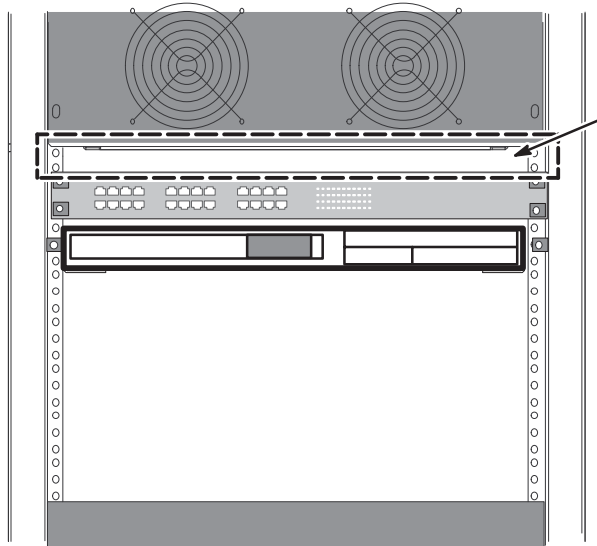


- ② Install Ethernet Switch at this location using M5 Screws (2279742-2).

- ③ Install ICN Module on Rail at the location shown. Tighten PEM screws on the ICN brackets to secure.

8-10-3 INSTALL VRE CABLES

Route and connect new VRE Cables as shown below. Cables 5145040-4 and 5145040-5 are replacing existing cables.



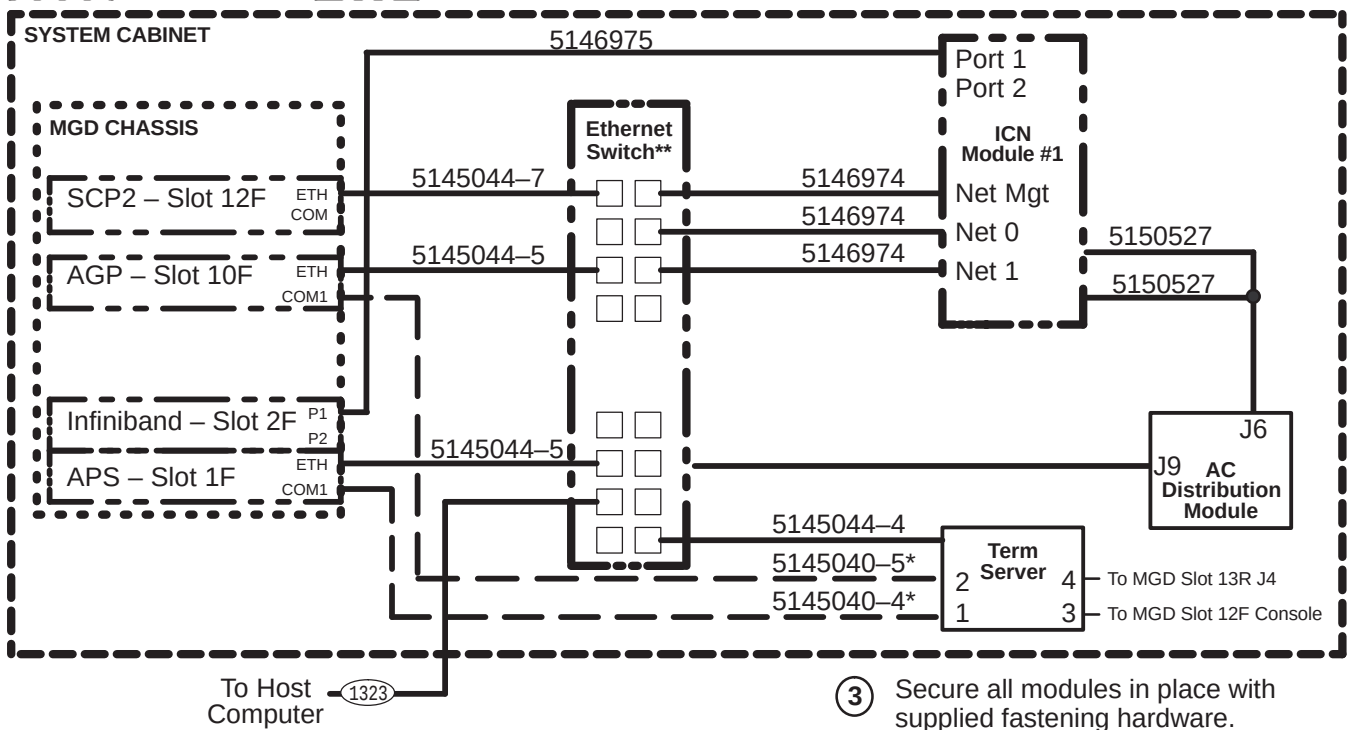
- ① All cables connecting between Ethernet Module and ICN Modules must be routed thru area above Ethernet Switch. Ethernet cables routed to Ethernet Switch can be connected to any ethernet port.

NOTE: Any existing cables connected to the MGD Chassis, or any other modules, that are not identified below should remain connected. They are not affected by this upgrade.

- ② Route and connect new VRE Cables as shown below. Cables 5145040-4 and 5145040-5 are replacing existing cables.

- ③ Use power cable supplied with Ethernet switch, connect to J9.

LEGEND:



* = New cable replacing existing cable

** = Ethernet cables routed to Ethernet Switch can be connected to any port.

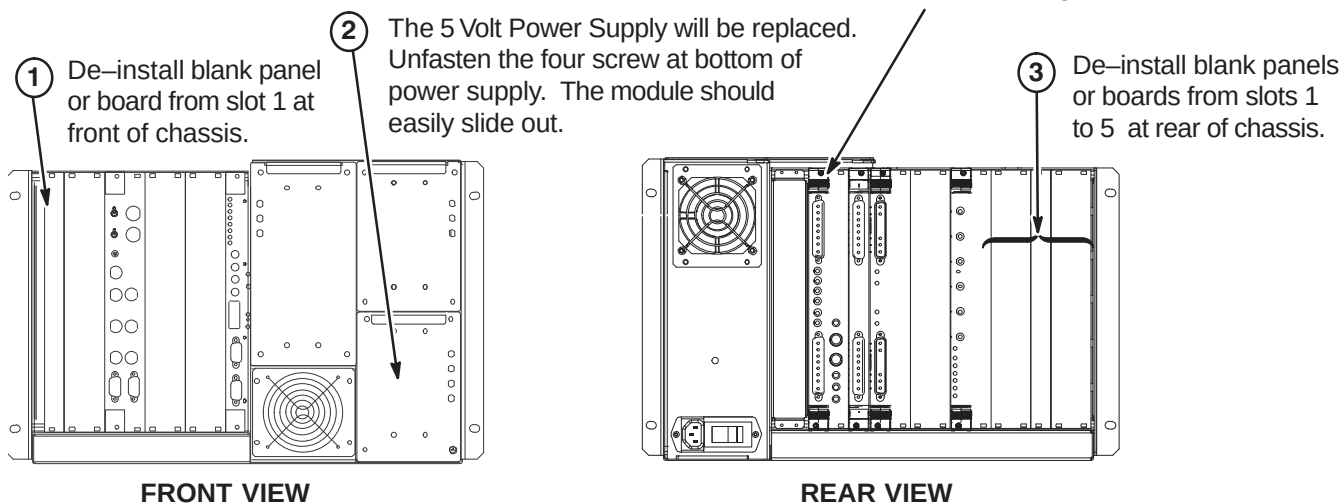
8-11 PROCEDURES FOR UPGRADING TO 16 CHANNEL (M3000PE)

8-11-1 REMOVE EXISTING COMPONENTS FROM RRF CHASSIS



Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

Note: It is recommended that the existing MUX board be at the same part number and revision level as the new MUX board being installed in slots 4 & 5.

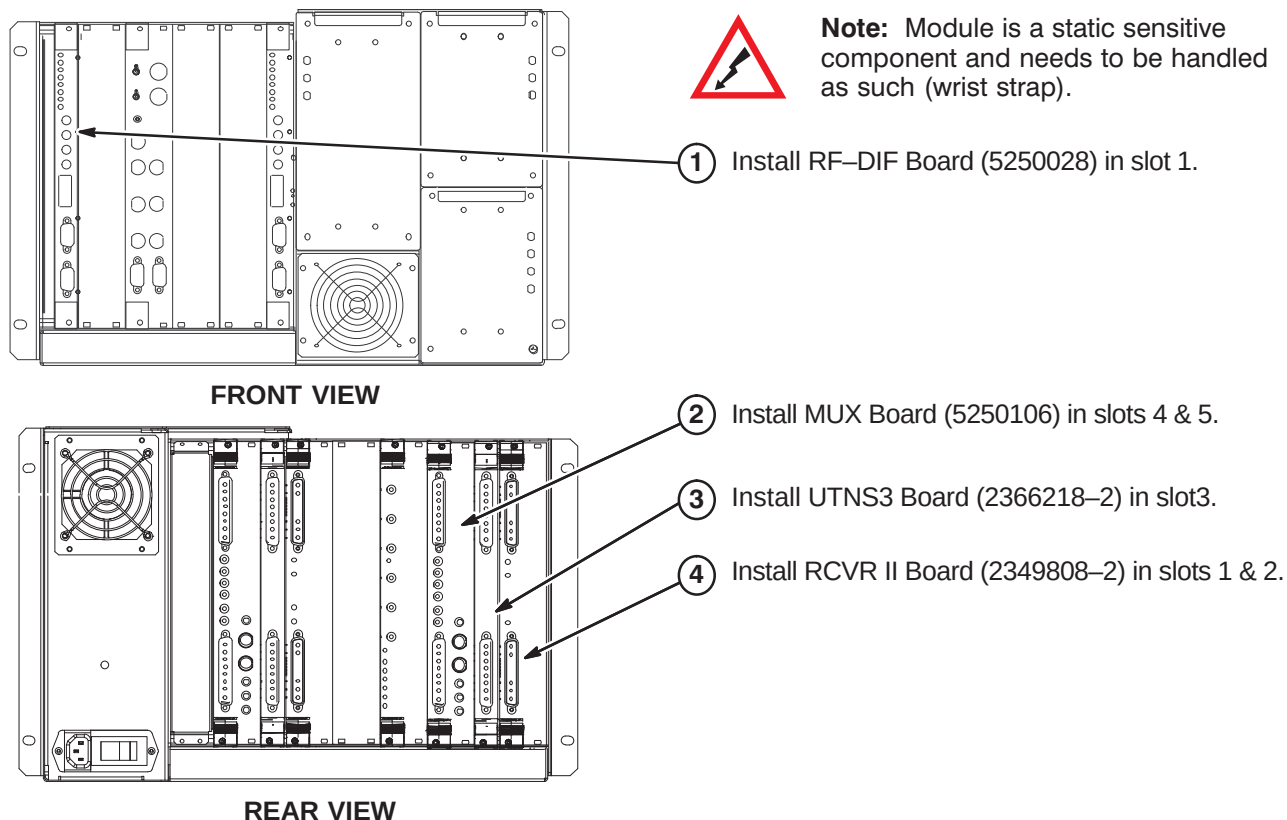


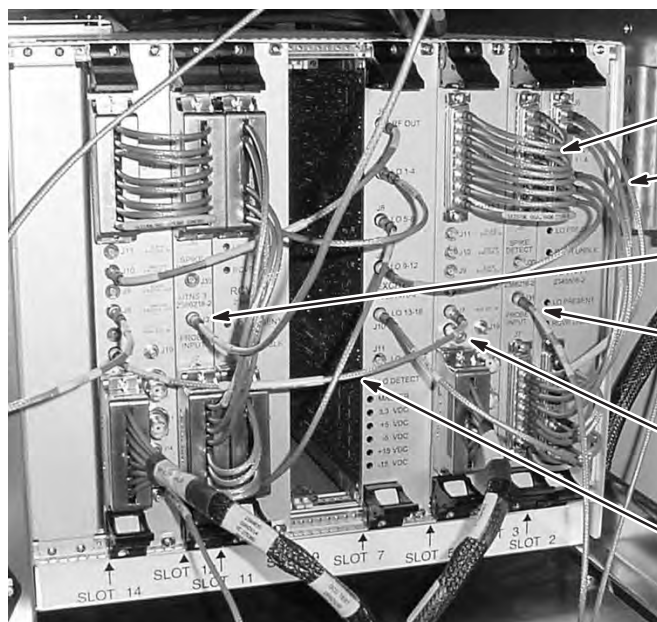
Note: All de-installed boards are to be returned for recycling.

8-11-2 INSTALL NEW MODULES IN RRF CHASSIS



Note: Module is a static sensitive component and needs to be handled as such (wrist strap).



8-11-3 INSTALL CABLES AT REAR OF RRF CHASSIS

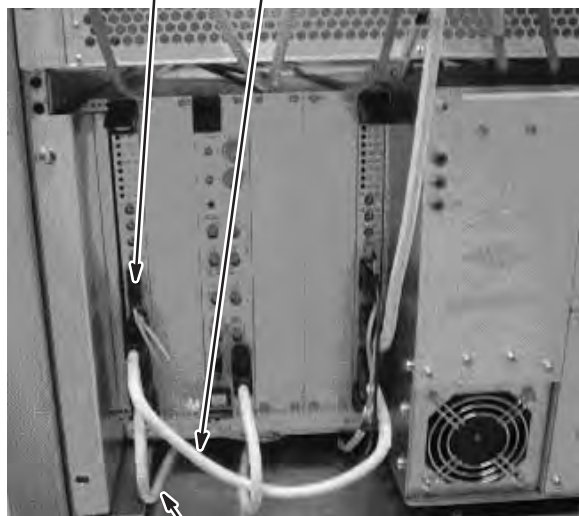
- ① Connect cable 5123706 as marked.
- ② Connect cable 5113193-2 as marked.
- ③ Make sure cable 5115417 slot 12R J31 is connected as marked. it should be routed and connect to Coax Power Splitter "1".
- ④ Connect cable 5115417-2 as marked. Route and connect to Coax Power Splitter "2". Remove 50 Ohm terminator if installed at "2".
- ⑤ Move 50 Ohm Terminator from Mux1 J6 (slots 13 & 14) to Mux2 J6 (slots 4 & 5).
- ⑥ Connect cable 5115630 as marked.

8-11-4 INSTALL CABLES AND POWER SUPPLY AT FRONT OF RRF CHASSIS

- ③ Connect cable 5121122-2 as marked to RF Slot 1F J21. Route and connect other end to MGD Slot 11F J7.

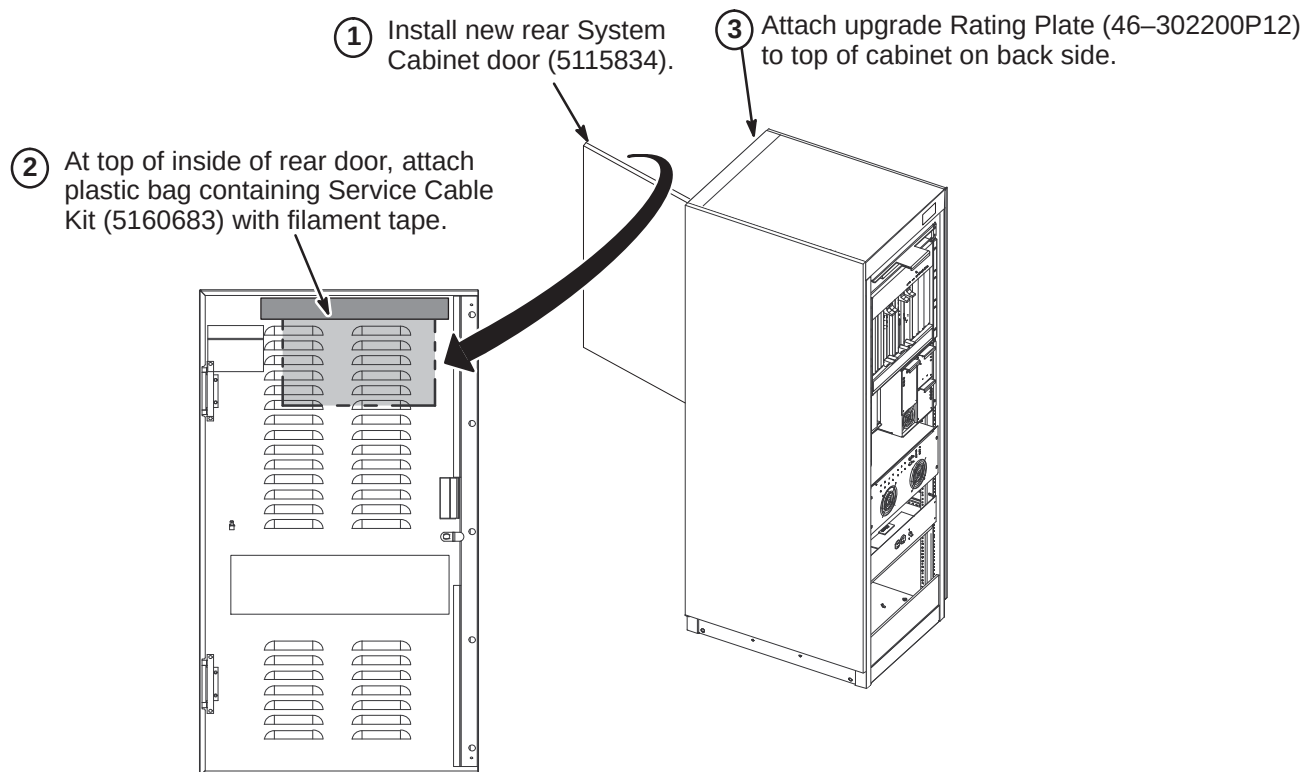
- ② Connect cable 5113182-2 as marked.

- ④ Slide in new 5V Power Supply (2292267-2). Make sure that it is properly seated before tightening the four screws.



- ① Disconnect cable (2323878-3) from Slot 10F J23 and connect to Slot 1F J23.



8-12 INSTALL NEW REAR DOOR AND RATING PLATE

SECTION 9 – SRF OR SRFD2 CABINET

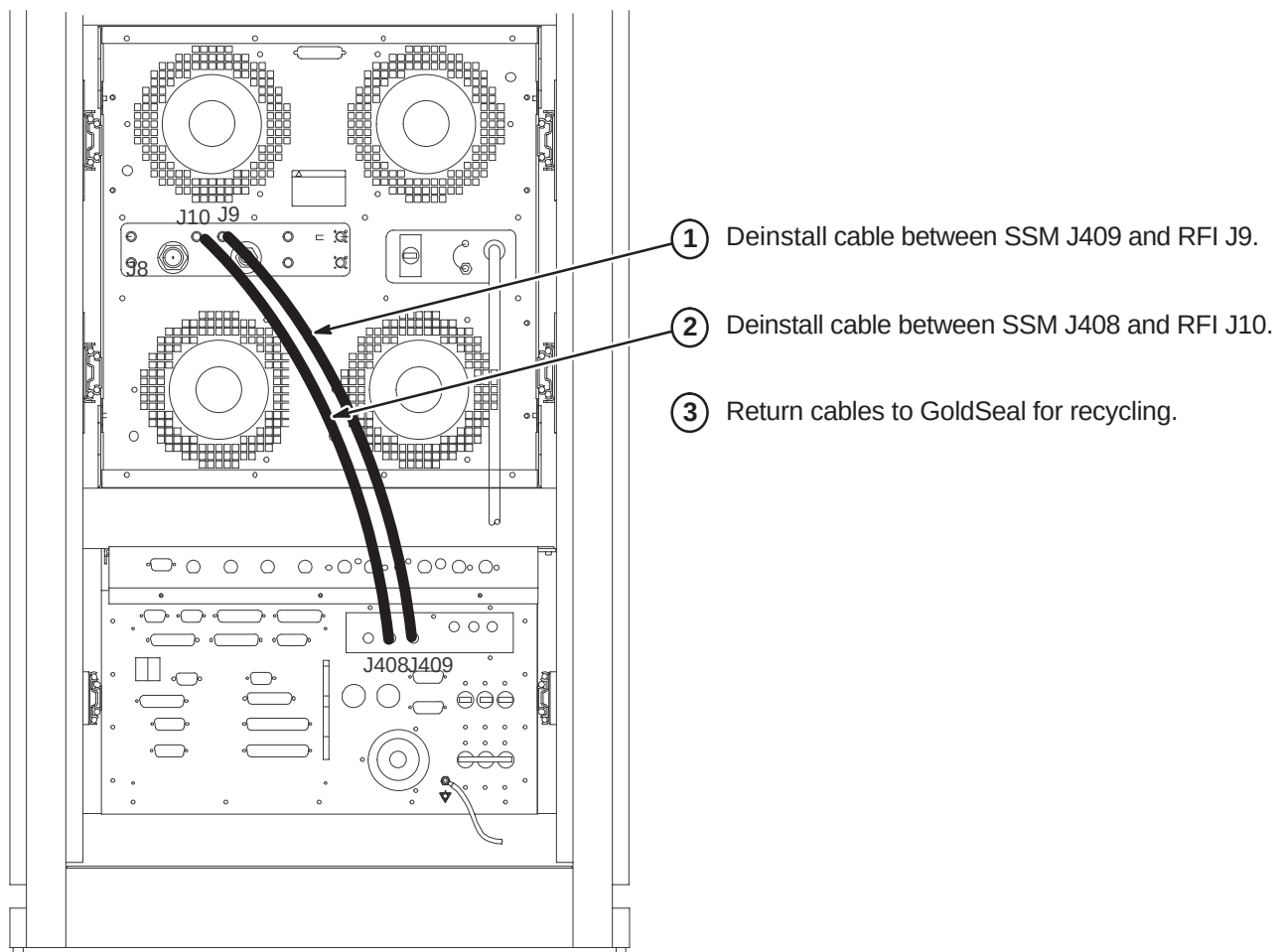
CAUTION

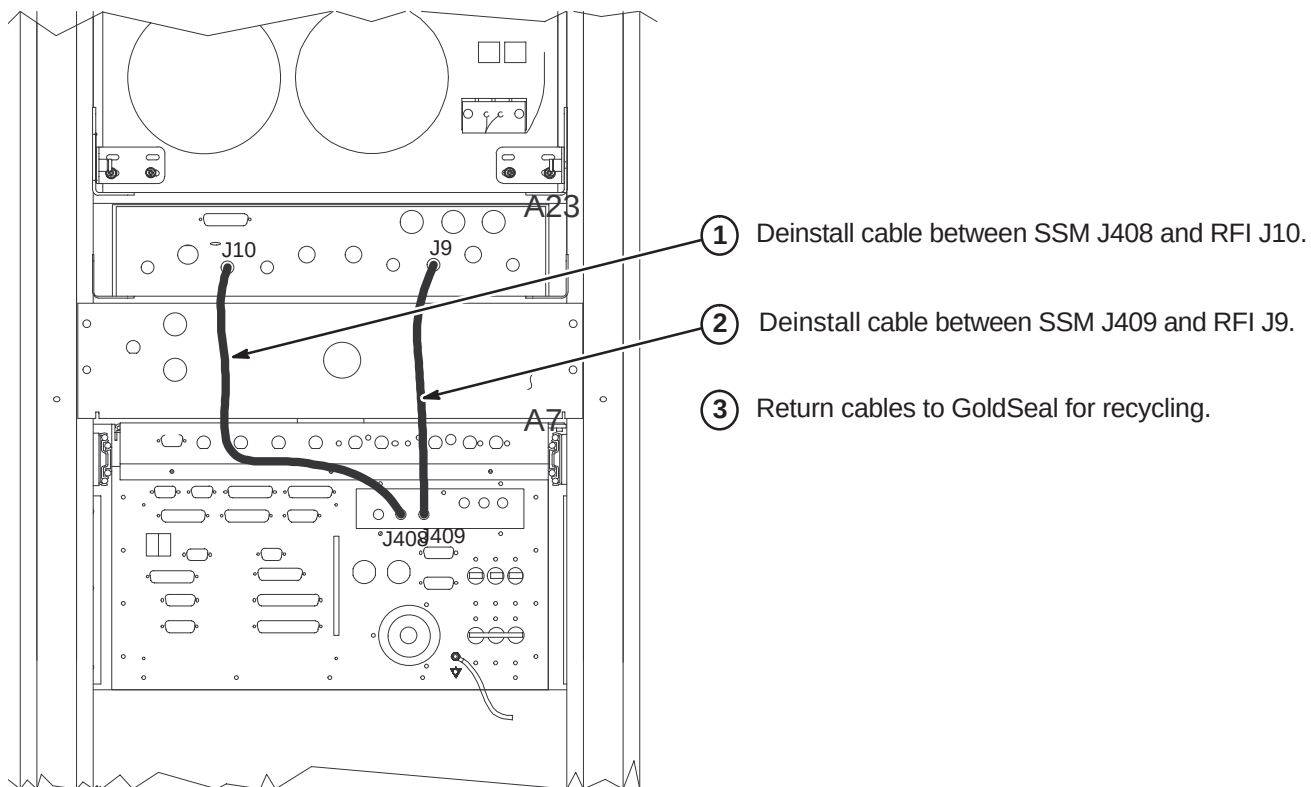
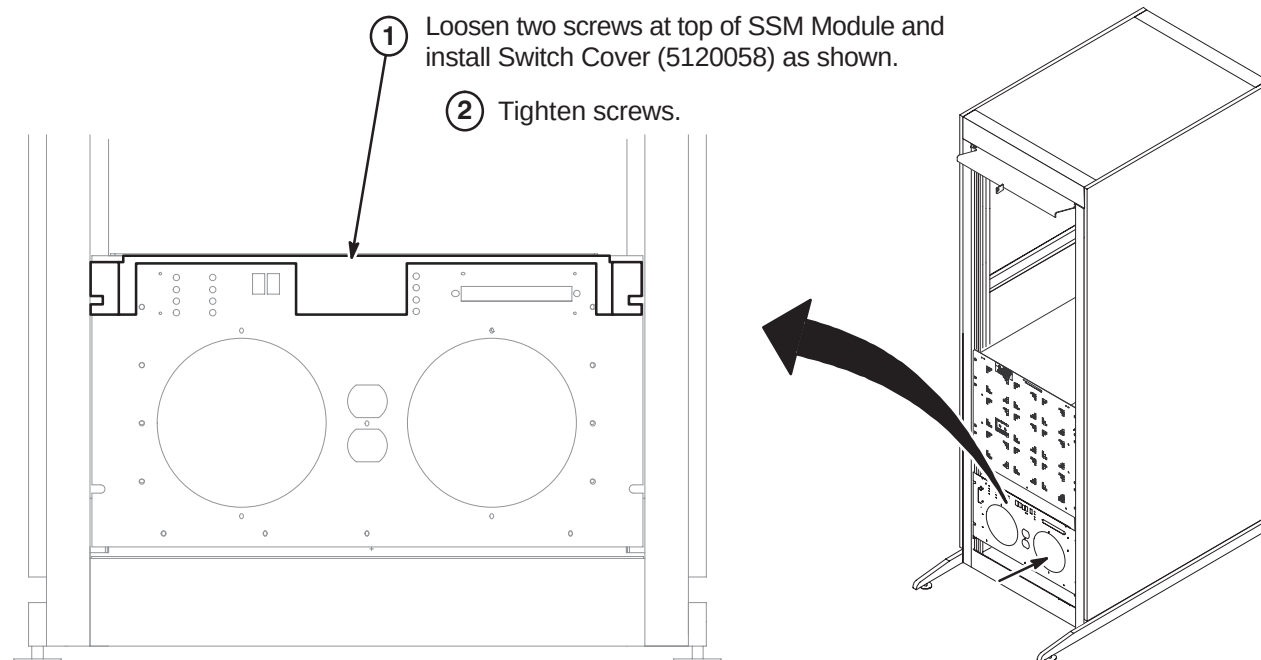
POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

9-1 INTRODUCTION

The existing SRF Cabinet will have only a few changes. Some of the functionality of the SSM Module is being transferred to the new Driver Module in the System Cabinet. The following procedures describe the steps needed to accomplish this change.

9-2 DEINSTALL CABLES IN SRFD2 CABINET



9-3 DEINSTALL CABLES IN SRF CABINET**9-4 INSTALL SWITCH COVER ON FRONT OF SSM MODULE**

SECTION 10 – ACGD/HFD CABINET

CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

10-1 INTRODUCTION

The ACGD/PDU Cabinet **WILL NOT BE** replaced but **WILL BE** upgraded to HFD-S when upgrading a site from EXCITE 11.x to 12.x. Instruction for the upgrade are found in Direction 5131778, *ACGD to HFD-S Upgrade Installation*, which is shipped with the M3000PT or M3000PW collector of parts.

The ACGD-Lite/PDU cabinet **WILL BE** replaced by an HFD Cabinet if the site is an MR/i SmartSpeed with an ACGD-Lite Cabinet. Refer to Section 4 for additional information and instructions on disconnecting, removing, relabeling, and/or reconnecting of cables when upgrading to EXCITE.

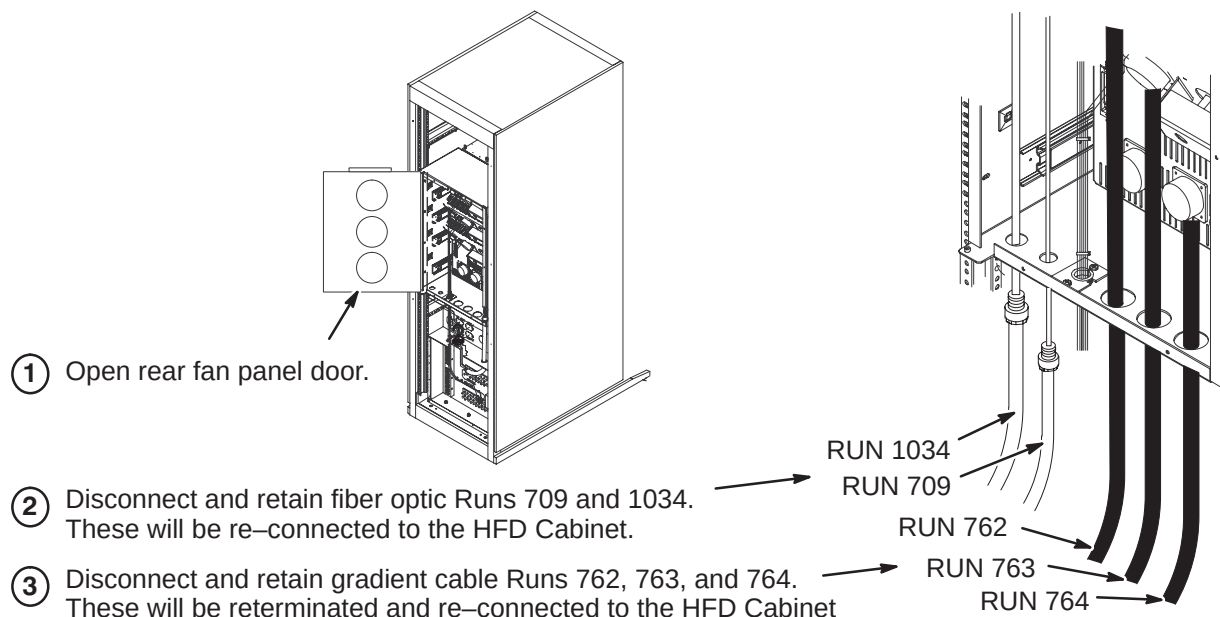
Refer to the following sub-sections for removal of existing ACGD-Lite Cabinet and installation of the new HFD Cabinet.

10-2 DEINSTALL ACGD-Lite CABINET

DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

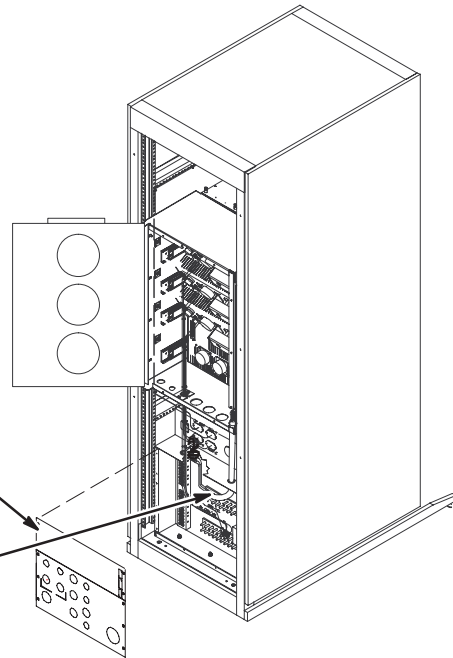
10-2-1 Disconnect Fiber Optic and Gradient Cables



10-2-2 Disconnect Power and Data Cables

NOTE: Refer to Section 4 for cable deinstallation information.

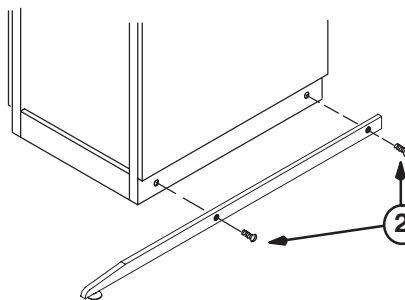
- ① Disconnect data cables at top Interface panel. Loosen clamps around power cables at bottom Interface panel. Unfasten Interface panel screws and remove panels.
- ② Disconnect and retain all PDU power cables. These will be re-connected to the HFD Cabinet.
- ③ Reinstall Interface panel and close all doors.



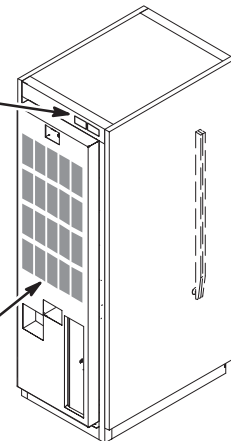
10-2-3 Remove ACGD-Lite Cabinet

WARNING!

DUE TO WEIGHT OF CABINET, AT LEAST TWO PEOPLE ARE REQUIRED TO MOVE CABINET



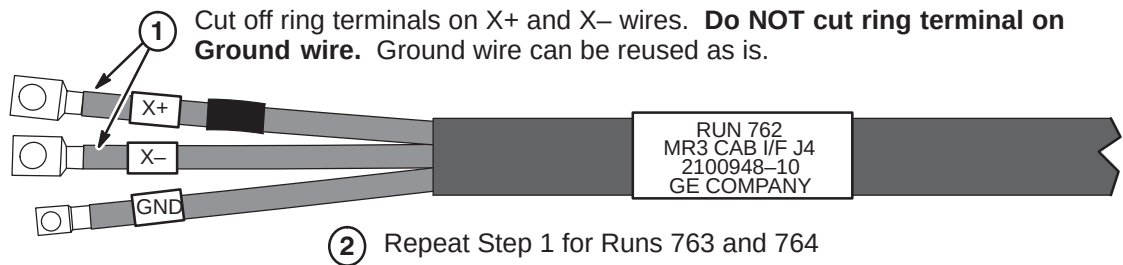
- ① Remove and save cabinet magnets for transfer to new HFD Cabinet.
- ② Unfasten Anti-tip bars (if installed) and place inside cabinet.
- ③ Return cabinet to GoldSeal. Refer to instructions in Section 1.



10-3 RETERMINATE GRADIENT CABLE RUNS 762, 763, AND 764

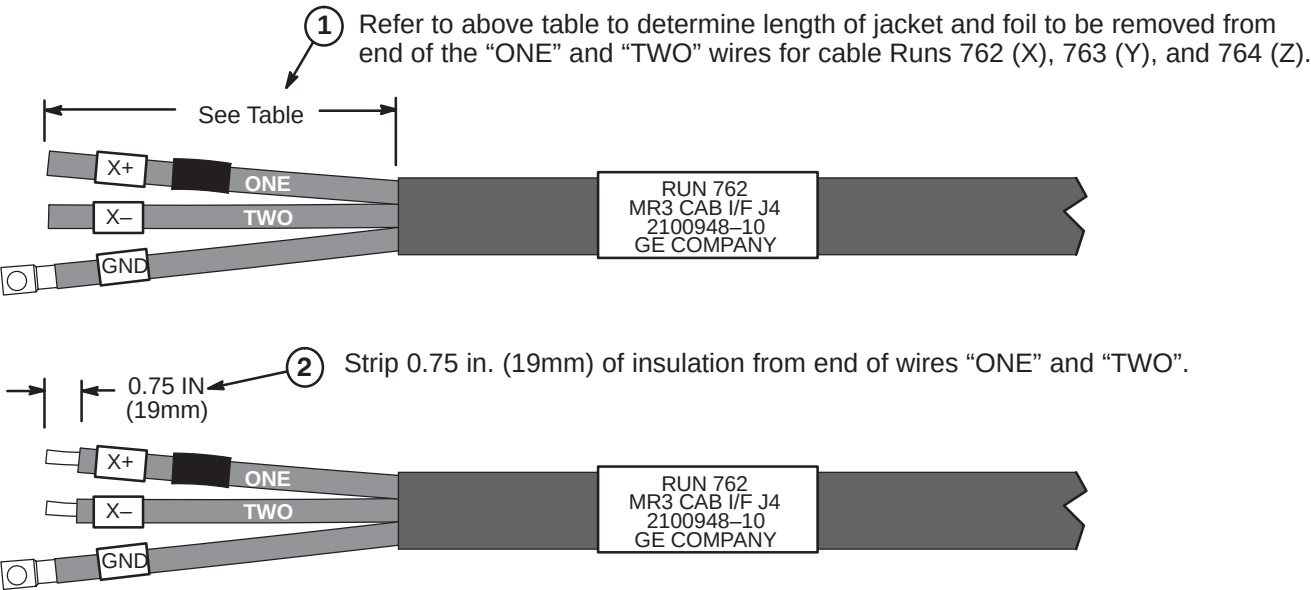
The Gradient cables will be reused but have to be reterminated. The HFD Cabinet uses pressure connectors. Refer to the following steps for the alteration to the “ONE” and “TWO” wires.

10-3-1 Remove Existing Ring Terminals



10-3-2 Terminate Runs 762, 763, and 764 for HFD Cabinet

	Run 762	Run 763	Run 764
Recommended Jacket Length to be Removed	8.0 IN (203mm)	11.0 IN (280mm)	14.0 IN (356mm)



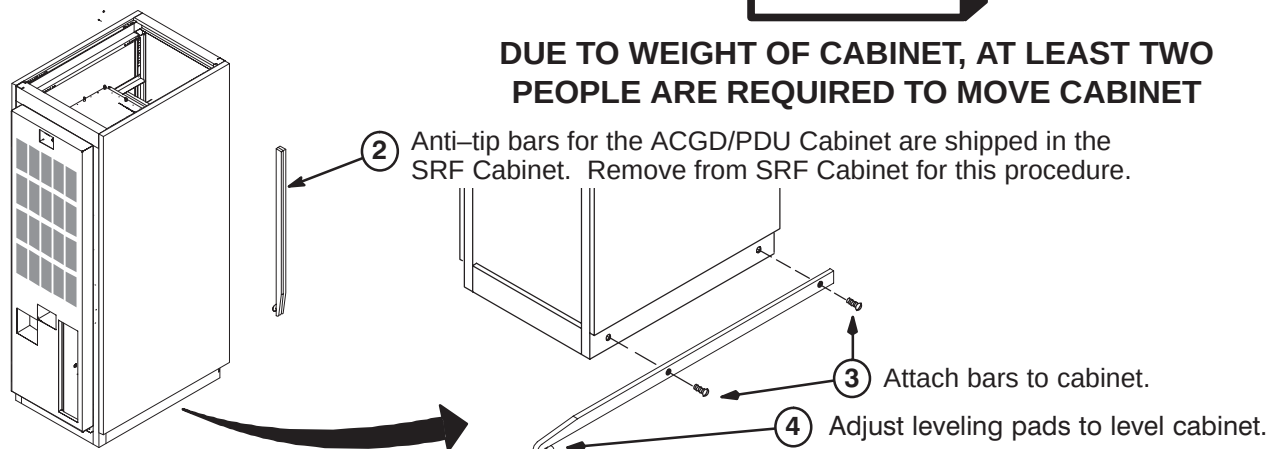
10-4 INSTALLATION OF HFD CABINET

10-4-1 POSITION HFD CABINET

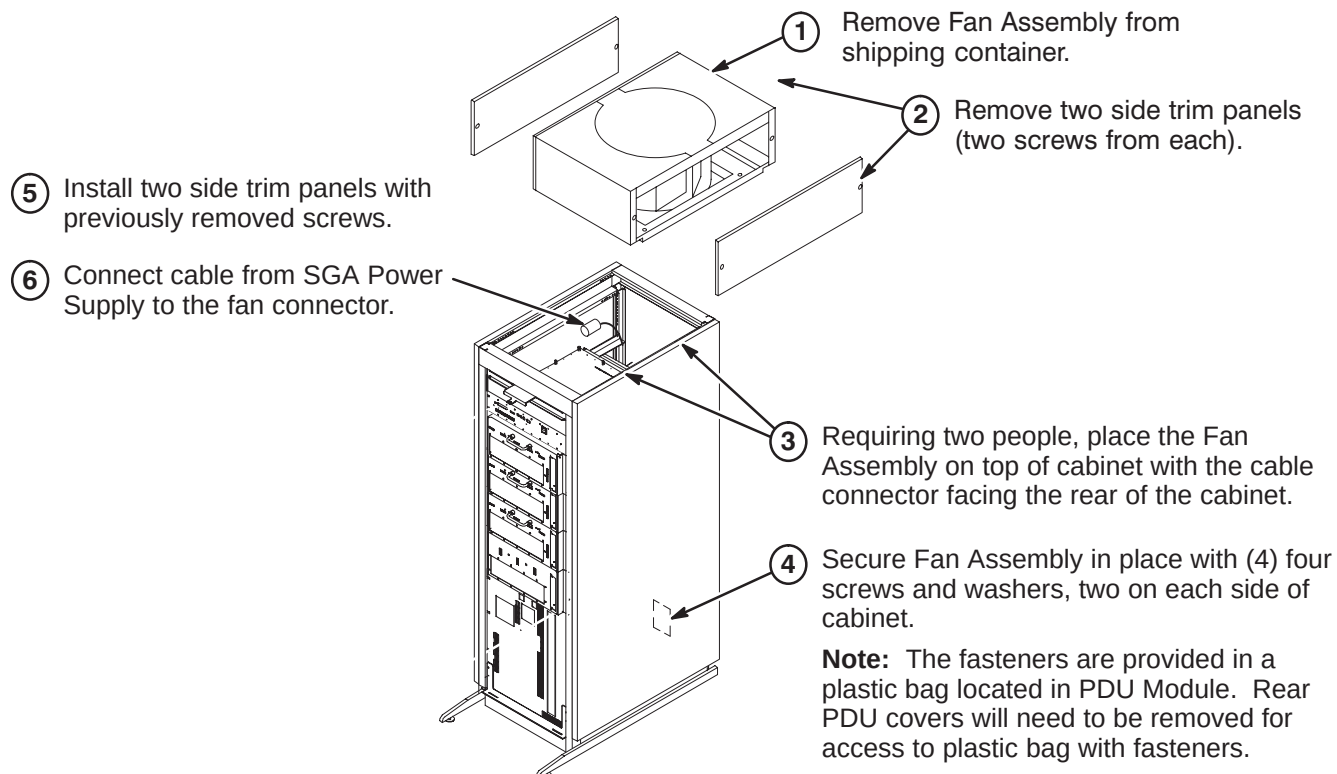
- ① Move cabinet to final position.

WARNING!

DUE TO WEIGHT OF CABINET, AT LEAST TWO PEOPLE ARE REQUIRED TO MOVE CABINET



10-4-2 INSTALL FAN ASSEMBLY AND ATTACH INSITE MAGNET





LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

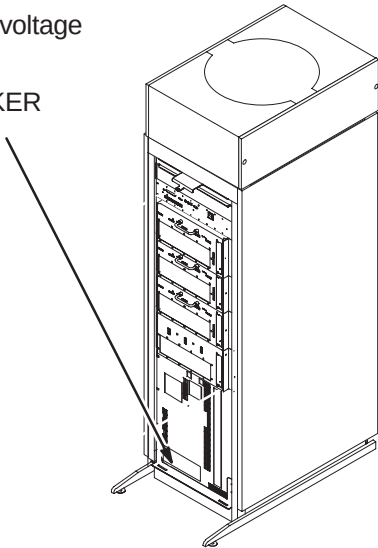
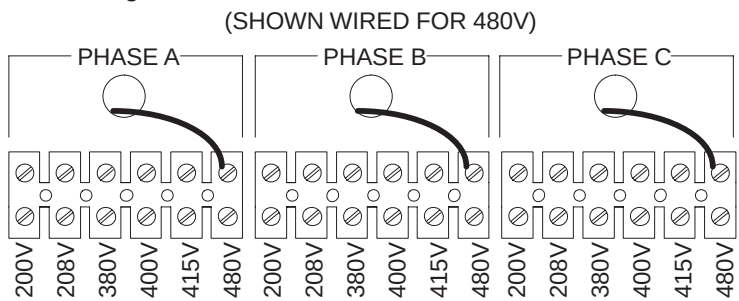
10-4-3 INPUT VOLTAGE SELECTION

- 1

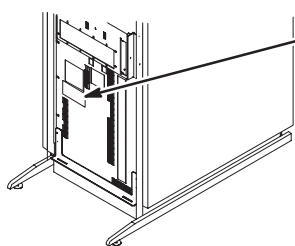
Using a voltmeter or other voltage indicating device, check to be sure that no voltage is being applied to the input of the PDU.
- 2

Remove the plate from the front of the PDU marked INPUT CIRCUIT BREAKER CURRENT SELECT / INPUT VOLTAGE SELECT.
- 3

For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.



10-4-4 CIRCUIT BREAKER DIP SWITCH SETTINGS



- 1

Remove small cover plate under the main circuit breaker to access DIP switches controlling the circuit breaker operation.
- 2

Table below shows the position of the DIP switches for each input voltage selection. The switches immediately under the letter “L” and “I” are the only ones that should be changed.

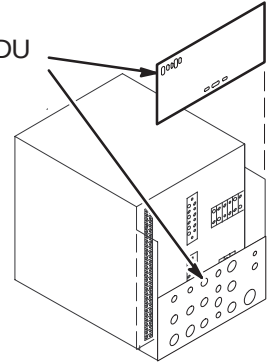
INPUT VOLTAGE	DIP SWITCH “L” SETTING	L	I	DIP SWITCH “I” SETTING	INPUT VOLTAGE
200V					200V
208V		Set these switches according to the input voltage shown at the left.	Set these switches according to the input voltage shown at the right.		208V
380V					380V
400V		 If these switches are present, set both of them in the DOWN position as shown.			400V
415V					415V
480V					480V

10-4-5 FACILITY POWER AND GROUND CONNECTION

WARNING!

IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT IS USED THE NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT CONTROL AND NOT BROUGHT TO THE ACGD/PDU CABINET.

- ① Remove rear PDU module covers.



- ⑥ Connect RF screen room ground conductor to Ground Bus Bar.



- ⑤ Connect facility input dedicated ground conductor to Ground (GRN/YEL) Bus Bar.

- ④ Connect L1–L3 conductors to terminal block.

- ③ **NOTE:** Air flow thru cabinet must exit thru top of cabinet. If not, fan at top of cabinet is rotating in wrong direction. To reverse rotation of fan, switch any two of the three incoming L1–L3 power wires.

DANGER!!!

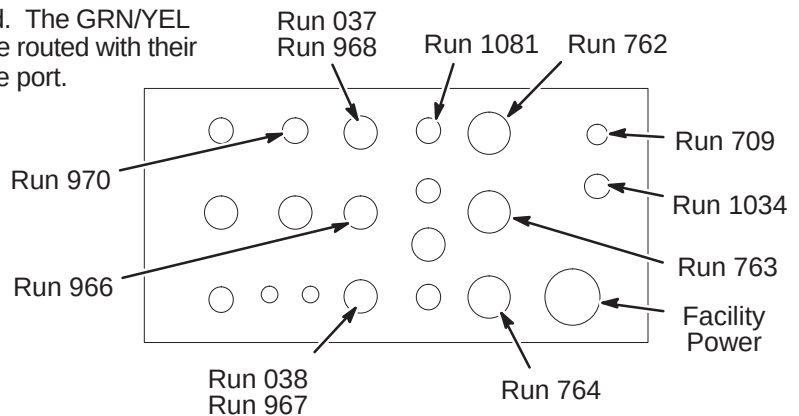
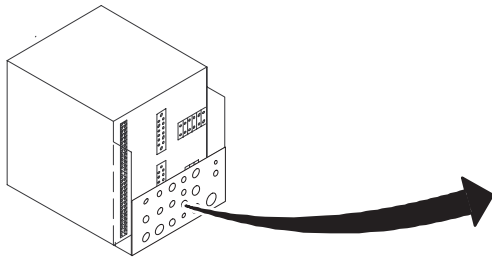
MAKE SURE ALL POWER TO CABINET IS OFF, LOCKED, & TAGGED BEFORE SWITCHING ANY ABOVE WIRES.

- ② Route facility input power cable and ground wire thru interface panel.

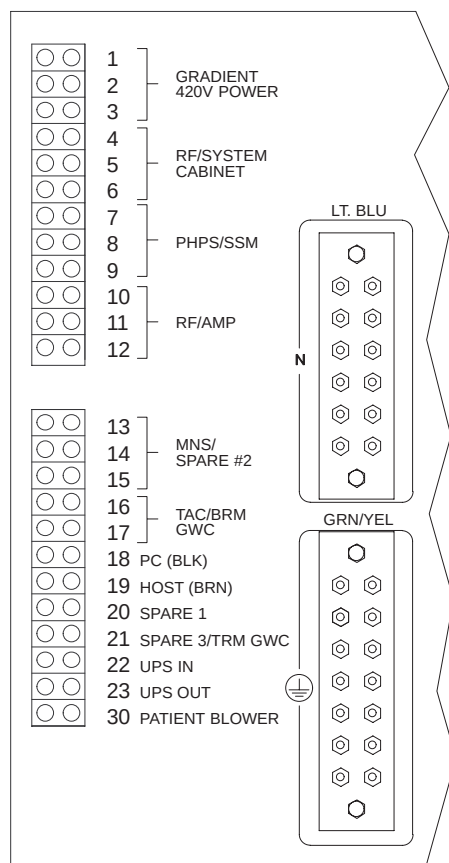
10-4-6 ROUTE POWER CABLES THRU INTERFACE PANEL

- ① Route Runs thru interface panel as marked. The GRN/YEL ground wires (Runs 037 and 038) **MUST** be routed with their respective power cable Runs thru the same port.

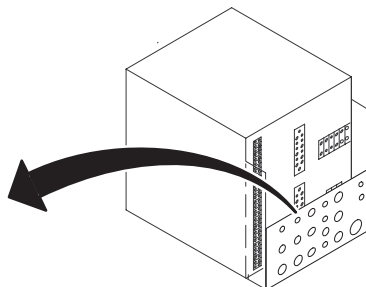
- ② Route any option Runs as marked.



10-4-7 CONNECT POWER CABLES TO PDU MODULE

**WARNING!**

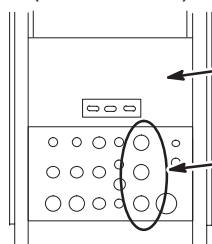
ALL POWER CABLES LISTED BELOW MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN ACGD/PDU CABINET.



- 1 Route/connect power cables to pressure connectors and studs on back of PDU module as shown in Table below.
- 2 Trim and connect ground wire cables (Runs 037 and 038) to ground bus connectors.
- 3 Tighten clamps of interface panel to secure cables in place.

CONNECT TO	RUN	PHASE	DESTINATION
4,5,6	968	ϕ A, B, C	System Cabinet 3 ϕ , Neutral, Ground
7,8,9	966	ϕ A, B, C	SSM 3 ϕ Neutral, Ground
10,11,12	967	ϕ A, B, C	RFI/AMP 3 ϕ , Ground
16,17	970*	ϕ A,B	Water Chiller Single ϕ , Ground
18,19	1081	ϕ C,A	Operator Workspace Single ϕ , Neutral, GND
20	973*	ϕ A	Heat Exchanger Single ϕ , Neutral, GND
GRN/YEL	037		System Cabinet Ground Wire
GRN/YEL	038		SRF Cabinet Ground Wire
COLOR CODE: for remainder of cables: 3 ϕ = Black, Red, Orange Single ϕ = Brown, Black * Non-WideOpen sites Neutral = Light blue Gnd = Grn/Yel			

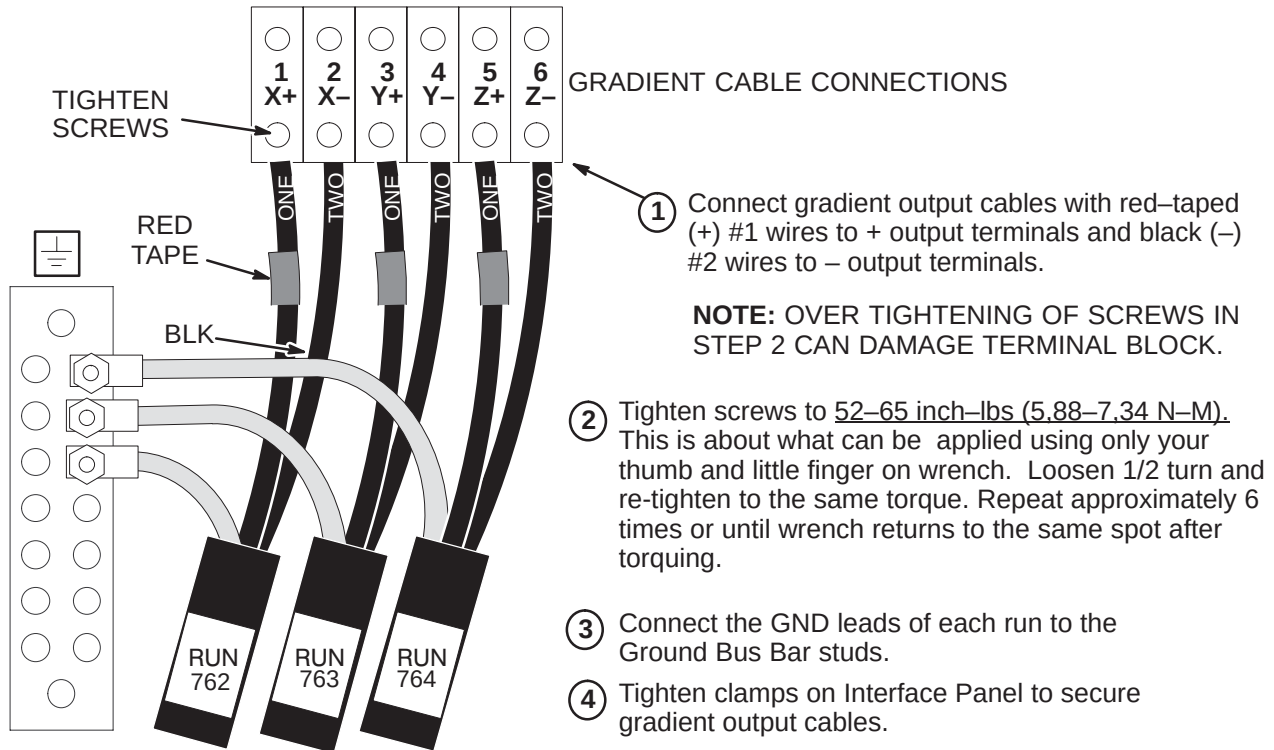
10-4-8 GRADIENT OUTPUT CABLE INTERFACE PANEL ACCESS

HFD CABINET
(REAR VIEW)

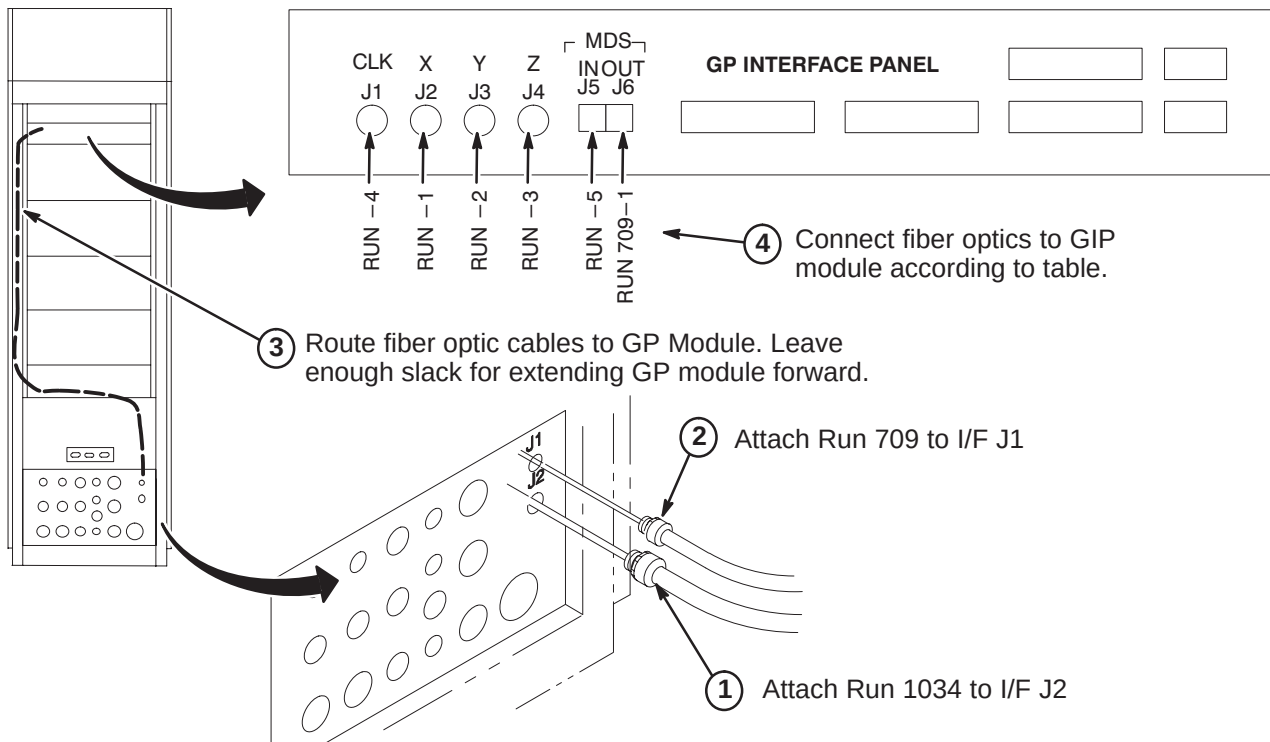
NOTE: Output cables (Runs 762, 763, and 764) were previously routed, cut to length, and terminated.

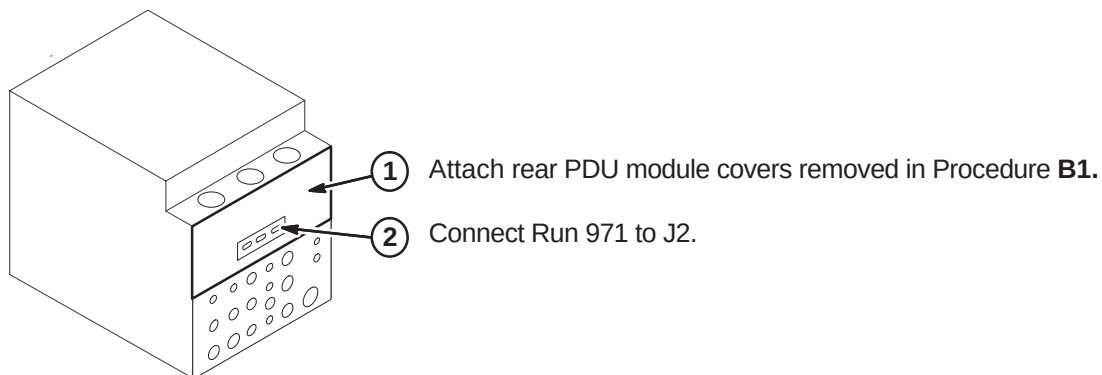
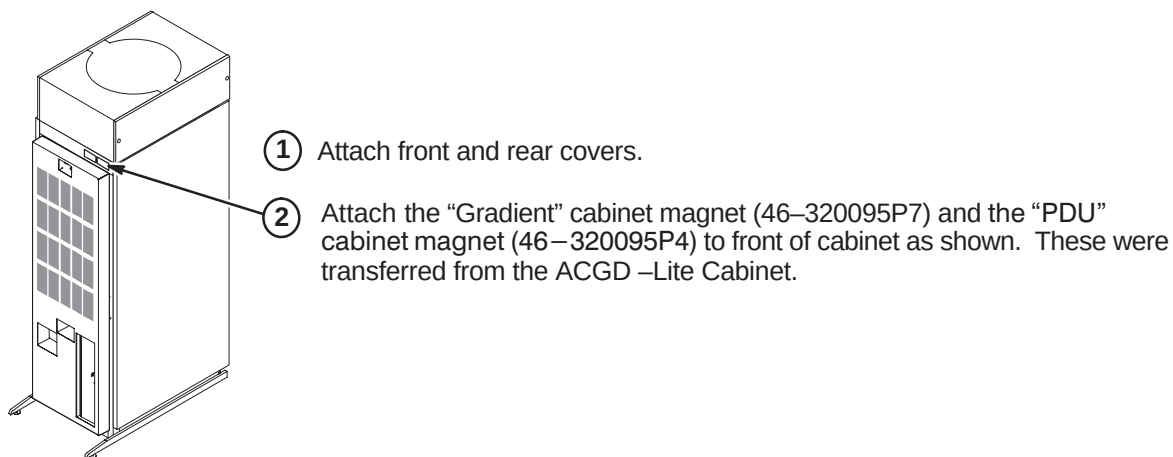
- 1 Remove Rear Cover Plate by removing seven screws that hold it in place.
- 2 Loosen screws on clamps for gradient output cables. Insert ends of gradient output cables into Cabinet Interface as marked. (Runs 762 into J3, Run 763 into J4, and Run 764 into J5)

10-4-9 GRADIENT OUTPUT CABLE ROUTING



10-4-10 CONNECT FIBER OPTICS TO GP MODULE



10-4-11 CONNECT RUNS 971 TO PDU MODULE**10-4-12 ATTACH INSITE CABINET MAGNETS**

SECTION 11 – COMPLETION

11-1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non-GE personnel. Check off boxes at the front of each item below when complete.

- ☐ Check mechanical flowchart to make sure all steps are complete (Section 2)
- ☐ Resolve shipment shortages
- ☐ Resolve omissions by mechanical contractors
- ☐ Make sure all rating plates are installed
- ☐ Recheck all wiring connections using cable map (Section 4)
- ☐ Correct for wiring errors if necessary
- ☐ Complete Sections 11-2, 11-3, and 11-4.

11-2 RETURNING SHIPPING EQUIPMENT

The shipping material must be returned **as soon as possible**. Prompt return is required so that production shipments can continue with these labor saving dollies. These can be included with the obsolete equipment and materials being returned in Section 11-3. Recycling Operation will route them to appropriate location.

11-3 MATERIAL DISPOSITION GUIDANCE

Refer to Section 1-7. The equipment or material being disposed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

11-4 FINAL COORDINATION

1. Replace cabinet covers that have not been previously replaced.
2. Replace Magnet Enclosure covers and Rear Pedestal covers
3. Dock Patient Transport in position at front of bore.
4. Record and enter applicable data into applicable site configuration files and records.
5. Complete Product Locator information. See Section 1-8.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

6. Leave the site's set of Manuals and CD-ROM at site. Organize and set up reference cabinet for the Manuals.
7. Locate any new Material Safety Data Sheets (MSDS). They must be retained on site. Customer is to be informed that material with MSDS information was brought into site and customer should know/decide where MSDS be retained.
8. Verify that coordination of application support for instruction of site users on Software Release 14.x has been made before returning system to the users. Turn site over to applications who will instruct users.

11-5 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD-ROM, Installation Calibration Flow, and perform the following:
 - PDU Voltage Checks Compact > PDU E-Stop Check
 - Startup Checks Compact
 - Transformer Taps Compact
4. Press the FULL ON button on the PDU front control panel.
5. Except for the ACGD Gradient power modules, switch all cabinet circuit breakers to ON. The ACGD Gradient power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.
6. **Make sure the UPS circuit breaker is switched to the ON position.** This is required to feed power to the Operator Workspace.